Statistics for Data Analytics / Engineering Statistics

IV Semester SEE Question Papers

Course codes: IM241AT, 21IM41, 18IM41, 16IM43 / 12IM41, 16IM43, 12IM41, 12IM41 / 10IM43 / 07IM42

- Jun/Jul 2025 (IM241AT)
- Oct/Nov 2024 Supplementary (21IM41)
- Sep/Oct 2024 (IM241AT)
- Oct 2023 (21IM41)
- Oct/Nov 2022 (18IM41)
- Grade Improvement Oct 2021 (18IM41)
- Grade Improvement Nov 2020 (18IM41)
- Fast Track Oct 2020 (16IM43 / 12IM41)
- Fast Track Jul 2019 (16IM43 / 12IM41)
- Apr/May 2019 (16IM43 / 12IM41)
- Fast Track Jul 2018 (16IM43)
- Apr/May 2018 (16IM43)
- Fast Track Jul 2017 (12IM41)
- Apr/May 2017 (12IM41)
- Fast Track Jul 2016 (12IM41 / 10IM43 / 07IM42)
- Apr/May 2016 (12IM41)

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B.E. Regular/Supplementary Examinations June/July – 2025

Industrial Engineering and Management

STATISTICS FOR DATA ANALYTICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	In what way is a stem and leaf diagram better than a histogram?	02	2	1
	1.2	A = {1, 2, 3, 4}; B = {2, 4, 6, 8}; C = {6, 7, 8, 9}; D = {4, 5, 6, 7}. If the sample space S = A ∪ B ∪ C ∪ D, and each element of S is equally likely to happen, then P(A′∩B) = _____	02	2	2
	1.3	For a continuous random variable, the probability that it takes an exact value is _____.	01	2	2
	1.4	A negative binomial distribution counts the number of trials needed to get 3 successes with p = 0.4. What is the probability that the 3rd success occurs on the 5th trial?	02	2	2
	1.5	A random variable X is uniformly distributed over the interval [2, 6]. Find the probability that X lies between 3 and 5.	02	2	2
	1.6	For a normal distribution with $\mu = 70$, $\sigma = 8$, find P(62 < X < 78).	02	2	2
	1.7	If a population has a mean μ and standard deviation σ, the standard error of the mean is given by $\sigma/\sqrt{n}$, where n is the _____.	01	2	2
	1.8	From Minitab output: Coefficients: Intercept = 2.5, Slope = 1.8. If X = 6, what is the predicted value of Y?	02	2	2
	1.9	What does a correlation coefficient of −0.85 indicate?	02	2	3
	1.10	In linear regression the error term is distributed as _____ distribution (include its parameters also).	02	2	3
	1.11	The test statistic in a test for population variance based on a given sample of size n is _____ (also state what the terms are).	02	2	4

PART-B

2	a	The no of defects when 25 parts were analyzed were recorded as: 15, 18, 20, 9, 11, 8, 16, 14, 13, 15, 17, 21, 7, 9, 11, 8, 18, 12, 6, 19, 15, 22, 19, 12, 10 a) Draw the box plot for the above data b) Plot the stem and leaf diagram and comment on the data	08	3	1
	b	Suppose it is known that 1% of the population suffers from a particular disease. A blood test has a 97% chance of a positive blood test (identifying the disease) for diseased individuals, but also has a 6% chance of falsely indicating that a healthy person has the disease. (i) What is the probability that a person will have a positive blood test? (ii) If your blood test is positive, what is the chance that you have the disease?	08	4	1

3	a	The probability mass function of a discrete random variable X is defined as: (pmf table not legible in the source copy) a) Find the expected value E(X) b) Find the probability that X takes a value greater than 3.	08	3	2
	b	The random variable X has a binomial distribution with n = 10 and p = 0.01. Determine the following probabilities. a) P(X = 5) b) P(X $\leq$ 2) c) P(X $\geq$ 9) d) P(3 $\leq$ X $<$ 5)	08	2	2
		OR			
4	a	If a random variable X has its cumulative distribution as F(x) = 0, $-\infty < x < 1$ 0.4, $1 \leq x < 3$ 0.5, $3 \leq x < 8$ 1.0, $8 \leq x < \infty$ Determine the following: (i) P(X at most 3) (ii) E(X) (iii) P(X between 1 and 2, both inclusive) (iv) P(X at least 2) (v) the median of the distribution	10	3	1
	b	A carton of washers contains 5% whose variability in thickness around the circumference is unacceptable. (i) What is the probability that none of the unacceptable washers is found in a sample of 10 washers? (ii) What is the probability that at least one unacceptable washer is found in a sample of 20 washers?	06	3	2
5	a	Describe the characteristics and applications of the Normal distribution.	04	2	2
	b	What are the key differences between population and sample? Why is sampling important in statistics?	06	2	2
	c	What is the Central Limit Theorem (CLT)? State its significance in statistical inference.	06	3	2
		OR			
6	a	An electrical firm manufactures light bulbs that have a life, before burn-out, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find a) the probability that a bulb burns between 778 and 834 hours. b) Probability that a bulb has a life of at least 1000 hours. c) What percent of bulbs have their life within $\pm 2\sigma$ from their own mean life, where σ is the standard deviation?	08	3	2
	b	The life of a semiconductor laser at a constant power is normally distributed with a mean of 7000 hours and a standard deviation of 600 hours. a) What is the probability that a laser fails before 6000 hours? b) What is the life in hours that 95% of the lasers exceed? c) If three lasers are used in a product and they are assumed to fail independently, what is the probability that all three are still operating after 7000 hours?	08	3	2

7	a	Regression methods were used to analyze the data from a study investigating the relationship between roadway surface temperature (x) and pavement deflection (y). Summary quantities were n = 20. $\sum y_i = 12.75$, $\sum y_i^2 = 8.86$, $\sum x_i = 1478$, $\sum x_i^2 = 143{,}215.8$, and $\sum x_i y_i = 1083.67$ a) Calculate the least squares estimates of the slope and intercept. Graph the regression line. Estimate σ^2. b) Use the equation of the fitted line to predict what pavement deflection would be observed when the surface temperature is 85°F. c) What is the mean pavement deflection when the surface temperature is 90°F? d) What change in mean pavement deflection would be expected for a 1°F change in surface temperature?	12	3	3
	b	Define the correlation coefficient and its range.	04	2	3
		OR			
8	a	The summarized data on 10 observations are given below. $\sum(y_i - \hat{y}_i)^2 = 3.267$, $\sum y_i = 63.3$, $\sum y_i^2 = 423.49$, $\sum x_i = 139$, $\sum x_i^2 = 2239$ a) Estimate β_1 and β_0 assuming positive correlation between X and Y. b) Obtain an estimate of the underlying variance. c) What is the proportion of variability explained by the regression model? d) Assuming normal distribution of the errors, how will a residual plot of errors vs predicted values appear in this case? (Show an indicative plot).	10	3	3
	b	What are the assumptions of simple linear regression? How do you validate them using residual plots?	06	2	3
9	a	Explain the steps involved in conducting a hypothesis testing.	08	2	4
	b	a) What are the hypothesis statements and the statistic to be used in a two-sided test of variance of a population? What are the assumptions to use this statistic and what is the rejection region for this test? b) State the probability of a t-random variable with 25 degrees of freedom taking on any value between 2.06 and 3.45. c) What is the value x such that a chi-square random variable with 15 degrees of freedom takes on at the most a value of x with 95% likelihood?	08	4	4
		OR			
10	a	When is a z-test used for population mean inference?	04	2	4
	b	Cloud seeding has been studied for many decades as a weather modification. The rainfall in acre-feet from 20 clouds that were selected at random and seeded with silver nitrate are as follows: 18.0, 30.7, 19.8, 27.1, 22.3, 18.8, 31.8, 23.4, 21.2, 27.9, 31.9, 27.1, 25.0, 24.7, 26.9, 21.8, 29.2, 34.8, 26.7, and 31.6. a) Construct a 95% one-sided lower bound on the mean diameter. b) Construct a 90% two-sided confidence interval on the mean diameter. What is the width of this interval? c) If the width of the interval in part (b) were to be halved, how much should be the sample size? Use trial and error method if needed.	12	4	4

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IV Semester B.E. Supplementary Examinations Oct/Nov – 2024

Industrial Engineering and Management

STATISTICS FOR DATA ANALYTICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of statistical tables and formulae handbook permitted.

		PART-A	**M**	**BT**	**CO**
1	1.1	Differentiate between descriptive and inferential statistics.	02	2	1
	1.2	What is the central limit theorem in statistics?	02	1	2
	1.3	State any two assumptions of Binomial Distribution.	02	2	2
	1.4	State any two properties of Normal Distribution.	02	2	1
	1.5	What is the standard error of an estimator in statistics?	02	2	2
	1.6	Define Chi-squared test in statistics.	02	1	2
	1.7	Differentiate between correlation and causation.	02	1	1
	1.8	What is the significance of correlation?	02	2	1
	1.9	Define Type I and Type II error.	02	2	2
	1.10	What is critical value?	02	1	2

PART-B

2	a	A part selected for testing is equally likely to have been produced on any one of six cutting tools. (i) What is the sample space? (ii) What is the probability that the part is from tool 1? (iii) What is the probability that the part is from tool 3 or tool 5? (iv) What is the probability that the part is not from tool 4?	08	2	2
	b	A lot of 100 semiconductor chips contains 20 that are defective. (i) Two are selected, at random, without replacement, from the lot. Determine the probability that the second chip selected is defective. (ii) Three are selected, at random, without replacement, from the lot. Determine the probability that all are defective.	08	2	1
3	a	The phone lines to an airline reservation system are occupied 40% of the time. Assume that the events that the lines are occupied on successive calls are independent. Assume that 10 calls are placed to the airline. (i) What is the probability that for exactly three calls the lines are occupied? (ii) What is the probability that for at least one call the lines are not occupied? (iii) What is the expected number of calls in which the lines are all occupied?	08	2	1

	b	Printed circuit cards are placed in a functional test after being populated with semiconductor chips. A lot contains 140 cards, and 20 are selected without replacement for functional testing. (i) If 20 cards are defective, what is the probability that at least 1 defective card is in the sample? (ii) If 5 cards are defective, what is the probability that at least 1 defective card appears in the sample?	08	2	2
		OR			
4	a	The number of failures of a testing instrument from contamination particles on the product is a Poisson random variable with a mean of 0.02 failure per hour. (i) What is the probability that the instrument does not fail in an 8-hour shift? (ii) What is the probability of at least one failure in a 24-hour day?	08	3	3
	b	The number of telephone calls that arrive at a phone exchange is often modeled as a Poisson random variable. Assume that on the average there are 10 calls per hour. (i) What is the probability that there are exactly 5 calls in one hour? (ii) What is the probability that there are 3 or less calls in one hour? (iii) What is the probability that there are exactly 15 calls in two hours? (iv) What is the probability that there are exactly 5 calls in 30 minutes?	08	3	2
5	a	The fill volume of an automated filling machine used for filling cans of carbonated beverage is normally distributed with a mean of 12.4 fluid ounces and a standard deviation of 0.1 fluid ounce. (i) What is the probability a fill volume is less than 12 fluid ounces? (ii) If all cans less than 12.1 or greater than 12.6 ounces are scrapped, what proportion of cans is scrapped? (iii) Determine specifications that are symmetric about the mean that include 99% of all cans.	12	4	4
	b	List any four characteristics of normal distribution curve.	04	2	3
		OR			
6	a	The time between the arrival of electronic messages at your computer is exponentially distributed with a mean of two hours. (i) What is the probability that you do not receive a message during a two-hour period? (ii) If you have not had a message in the last four hours, what is the probability that you do not receive a message in the next two hours? (iii) What is the expected time between your fifth and sixth messages?	08	3	4
	b	The time between calls to a corporate office is exponentially distributed with a mean of 10 minutes. (i) What is the probability that there are more than three calls in one-half hour? (ii) What is the probability that there are no calls within one half hour? (iii) Determine x such that the probability that there are no calls within x hours is 0.01.	08	4	4
7		The following data for the dependent variable, y, and the independent variable, x, have been collected using simple random sampling: (i) Construct a scatter plot for these data. Based on the scatter plot, how would you describe the relationship between the two variables? (ii) Compute the correlation coefficient.	16	3	3

x	10	14	16	12	20	18	16	14	16	18
y	120	130	170	150	200	180	190	150	160	200

OR

8 An article in the Journal of Sound and Vibration (Vol. 151, 1991, pp. 383–394) described a study investigating the relationship between noise exposure and hypertension. The following data are representative of those reported in the article. | 16 | 3 | 3

y	1	0	1	2	5	1	4	6	2	3
x	60	63	65	70	70	70	80	90	80	80
y	5	4	6	8	4	5	7	9	7	6
x	85	89	90	90	90	90	94	100	100	100

(i) Draw a scatter diagram of y (blood pressure rise in millimeters of mercury) versus x (sound pressure level in decibels). Does a simple linear regression model seem reasonable in this situation?
(ii) Fit the simple linear regression model using least squares. Find an estimate of σ^2.
(iii) Find the predicted mean rise in blood pressure level associated with a sound pressure level of 85 decibels.

9 a Explain the general procedure for Hypothesis Tests. | 08 | 2 | 2

b The mean water temperature downstream from a power plant cooling tower discharge pipe should be no more than 100°F. Past experience has indicated that the standard deviation of temperature is 2°F. The water temperature is measured on nine randomly chosen days, and the average temperature is found to be 98°F. | 08 | 3 | 3

(i) Should the water temperature be judged acceptable with $\alpha = 0.05$?
(ii) What is the P-value for this test?
(iii) What is the probability of accepting the null hypothesis at $\alpha = 0.05$ if the water has a true mean temperature of 104°F?

OR

10 Grades in a statistics course and an operations research course taken simultaneously were as follows for a group of students. | 16 | 3 | 4

Statistics Grade	Operations Research Grade			
	A	B	C	Other
A	25	6	17	13
B	17	16	15	6
C	18	4	18	10
Other	10	8	11	20

Are the grades in statistics and operations research related? Use $\alpha = 0.01$ in reaching your conclusion. What is the P-value for this test?

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IV Semester B.E. Examinations Sept./Oct. – 2024

Industrial Engineering and Management

STATISTICS FOR DATA ANALYTICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of statistical tables and formula handbook permitted.

PART-A

			M	BT	CO
1	1.1	Give an example for ordinal data type.	01	1	1
	1.2	Give one distinction between Histogram and Box plot.	01	1	1
	1.3	What are mutually exclusive events?	01	2	1
	1.4	State Bayes' Theorem.	01	2	1
	1.5	Define Random variable.	01	1	2
	1.6	What is a probability mass function?	01	2	2
	1.7	Sketch the shape of a discrete uniform distribution.	01	1	1
	1.8	Define a Poisson random variable.	01	2	1
	1.9	What is the theoretical range of a Normal distribution?	01	1	2
	1.10	What is a Z score?	01	2	2
	1.11	State the central limit theorem.	01	1	3
	1.12	Sketch the relative spread of a population which is normally distributed and the sampling distribution of means. (Mean μ, standard deviation σ, sample size n).	01	2	3
	1.13	What is the meaning of an Empirical model?	01	2	3
	1.14	List any one of the properties of Least square estimation.	01	1	3
	1.15	Define correlation coefficient.	01	1	4
	1.16	Mention any one of the common abuses of regression.	01	2	4
	1.17	Give the correct structure of a pair of hypothesis for any one of the population parameter.	01	1	4
	1.18	Define prediction interval.	01	1	4
	1.19	What is the meaning of the term Goodness of fit?	01	2	4
	1.20	When do you use the F statistic?	01	2	4

PART-B

2	a	Given below is a frequency distribution of the length of phone calls made by 175 people during a weekend. Construct a histogram for these data.	08	3	1

Length (min)	1–7	8–14	15–21	22–28	29–35	36–42	43–49	50–56
Frequency	45	32	34	22	16	12	9	5

(i) Describe the general shape of the histogram. Does there appear to be a pattern?

(ii) Suppose all the people were making their calls from a room that had 10 different phones and each person knew which time class the call would belong to. Suggest an ordering so that all calls can be completed as fast as possible.

	b	The university's library has been randomly surveying patrons over the last month to see who is using the library and what services they have been using. Patrons are classified as undergraduate student, graduate or faculty. Services are classified as reference, periodicals or books. The data for 350 people are given below. Assume a patron uses only one service per visit. Find the probability that a randomly chosen patron (i) Is a graduate student. (ii) Visited the periodicals section, given that the patron is a graduate student. (iii) Is a faculty member given a reference section visit? (iv) Is an undergraduate who visited the book section?	08	2	1

Patron	Reference	Periodicals	Books
Undergraduate	44	26	72
Graduate	24	61	20
Faculty	16	69	18
Total	84	156	110

3	a	The probability is 0.80 that a randomly selected technician will finish his or her project successfully. Let X be the number of technicians among a randomly selected group of 10 technicians who will finish their project successfully. (i) Find the probability distribution of the random variable X. (ii) Find the probabilities that at least three technicians will finish their projects successfully. (iii) Find the probabilities that at most five technicians will finish their projects successfully. (iv) Find the probability that between four and six (inclusive) technicians will finish their project successfully.	08	3	2
	b	A total of 75 parts are received from the suppliers. We are informed that 8 defective parts were shipped by mistake and 5 parts have already been installed on machines. (i) What is the probability that exactly 1 defective part was installed on a machine? (ii) What is the probability of finding less than 2 defective parts?	08	3	2
		OR			
4	a	The number of surface flaws in plastic panels used in the interior of automobiles has a Poisson distribution with a mean of 0.05 flaws per square foot of plastic panel. Assume an automobile interior contains 10 square feet of plastic panel. (i) What is the probability that there are no surface flaws in an auto's interior? (ii) If 10 cars are sold to a rental company, what is the probability that none of the 10 cars has any surface flaws? (iii) If 10 cars are sold to a rental company, what is the probability that at most one car has any surface flaws?	08	3	2
	b	Let X denote the number of bits received in error in a digital communication channel. Assume that X is a binomial random variable with $p = 0.001$. If 1000 bits are transmitted, determine the following: (i) $P(X = 1)$ (ii) $P(X \geq 1)$ (iii) $P(X \leq 2)$ (iv) Mean and Variance of X.	08	3	3

5	a	Suppose the time it takes a data collection operator to fill out an electronic form for a database is uniformly distributed between 1.5 and 2.2 mins. (i) What is the mean and variance of the time it takes an operator to fill out the form? (ii) What is the probability that it will take less than two minutes to fill out the form?	08	4	3
	b	The reaction time of a driver to visual stimulus is normally distributed with a mean of 0.4 seconds and a standard deviation of 0.05 seconds. (i) What is the probability that a reaction requires more than 0.5 seconds? (ii) What is the probability that a reaction requires between 0.4 and 0.5 seconds? (iii) What is the reaction time that is exceeded 90% of the time?	08	4	3
		OR			
6	a	The thickness of the flange on an aircraft component is uniformly distributed between 0.95 and 1.05 millimeters. (i) Determine the cumulative distribution function of flange thickness. (ii) Determine the proportion of flanges that exceed 1.02 millimeters. (iii) What thickness is exceeded by 90% of the flanges? (iv) Determine the mean and variance of the flange thickness.	08	3	3
	b	The sick leave time of employees in a firm in a month is normally distributed with a mean of 100 hours and a standard deviation of 20 hours. (i) What is the probability that the sick leave time for the next month will be between 50 and 80 hours? (ii) How much time should be budgeted for sick leave if the budgeted amount should be exceeded with a probability of only 10%?	08	4	2
7	a	Gasoline consumption of a test automobile was recorded at speeds (x) ranging from 56 to 112 km/hr. The observed gasoline consumption were converted to distance travelled per liter of gasoline (y). The following information was compiled: n = 12, $\sum x = 964$, $\sum x^2 = 84116$, $\sum xy = 13418.4$, $\sum y = 165.3$, $\sum y^2 = 2282.45$, $\bar{x} = 80.333$, $\bar{y} = 13.755$ (i) Find the regression equation for y on x. (ii) Find the correlation coefficient.	12	4	3
	b	Give the benefits of initial visualization of data pattern using a scatter plot. Give an example to highlight the visualization of patterns in data.	04	2	3
		OR			
8	a	For a sample of 12 trees the volume of lumber (in m^3) and diameter (in cm) at a fixed height above ground level was measured. The results were as follows:	14	4	3

Diameter	35.1	48.4	47.9	35.3	47.3	26.4	33.8	45.3	25.2	28.5	30.1	30
Volume	0.81	1.39	1.31	0.67	1.46	0.47	0.8	1.69	0.3	0.19	0.63	0.64

(i) Construct a scatter plot of volume (y) versus diameter (x).
(ii) Compute the parameters of the least square line for predicting volume from diameters.
(iii) If two trees differ in diameter by 8 cm, how much would you predict their volumes to differ?
(iv) Predict the volume for a tree whose diameter is 44 cm.
(v) For what diameter would you predict a volume of 1 m^3?

	b	How is causation different from correlation?	02	2	3

9	a	A manufacturer produces piston rings for an automobile engine. It is known that the ring diameter is normally distributed with σ = 0.001 and $\bar{x}$ = 74.036 millimeter based on a random sample of 15 rings. (i) Construct a 99% two sided confidence interval on the mean piston ring diameter. (ii) Construct a 99% lower confidence band on the mean piston ring diameter.	08	3	4
	b	A post mix beverage machine is adjusted to release a certain amount of syrup into a chamber, where it is mixed with carbonated water. A random sample of 25 beverages was found to have a mean syrup content of $\bar{x}$ = 1.10 fluid ounces and a standard deviation of s = 0.015 fluid ounces. Find a 95% confidence interval on the mean volume of syrup dispensed.	08	3	4
		OR			
10	a	A consumer product company is formulating a new shampoo and is interested in foam height (in mm). Foam height is approximately normally distributed and has a standard deviation of 20 mm. The company wants to test H_0: μ = 175 mm versus H_1: $\mu > 175$ mm using the results of n = 10 samples. Suppose that the sample data result in $\bar{x}$ = 190 mm. (i) What conclusions would you reach? (ii) How "unusual" is the sample value in mm if the true mean is really 175 mm? That is, what is the probability that you would observe a sample average as large as 190 mm (or larger) if the true mean foam height was really 175 mm?	10	3	4
	b	Outline the 8 steps procedure for Testing of hypothesis. Give an example for a null hypothesis and alternate hypothesis.	06	4	4

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2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	**M**	**BT**	**CO**
1	1.1	Differentiate between descriptive and inferential statistics.	02	–	–
	1.2	What is the central limit theorem in statistics?	02	–	–
	1.3	State any two assumptions of Binomial Distribution.	02	–	–
	1.4	State any two properties of Normal Distribution.	02	–	–
	1.5	What is the standard error of an estimator in statistics?	02	–	–
	1.6	Define Chi-squared test in statistics.	02	–	–
	1.7	Differentiate between correlation and causation.	02	–	–
	1.8	What is the significance of correlation?	02	–	–
	1.9	Define Type I and Type II error?	02	–	–
	1.10	What is critical value?	02	–	–

PART-B

2	a	A part selected for testing is equally likely to have been produced on any one of six cutting tools. (i) What is the sample space? (ii) What is the probability that the part is from tool 1? (iii) What is the probability that the part is from tool 3 and tool 5? (iv) What is the probability that the part is not from tool 4?	08	–	–
	b	A lot of 100 semiconductor chips contains 20 that are defective. (i) Two are selected, at random, without replacement, from the lot. Determine the probability that the second chip selected is defective. (ii) Three are selected, at random, without replacement, from the lot. Determine the probability that all are defective.	08	–	–
3	a	The phone lines to an airline reservation system are occupied 40% of the time. Assume that the events that the lines are occupied on successive calls are independent. Assume that 10 calls are placed to the airline. (i) What is the probability that for exactly three calls the lines are occupied? (ii) What is the probability that for at least one call the lines are not occupied? (iii) What is the expected number of calls in which the lines are all occupied?	08	–	–

	b	Printed circuit cards are placed in a functional test after being populated with semiconductor chips. A lot contains 140 cards, and 20 are selected without replacement for functional testing. (i) If 20 cards are defective, what is the probability that at least 1 defective card is in the sample? (ii) If 5 cards are defective, what is the probability that at least 1 defective card appears in the sample?	08	–	–
		OR			
4	a	The number of failures of a testing instrument from contamination particles on the product is a Poisson random variable with a mean of 0.02 failure per hour. (i) What is the probability that the instrument does not fail in an 8-hour shift? (ii) What is the probability of at least one failure in a 24-hour day?	08	–	–
	b	The number of telephone calls that arrive at a phone exchange is often modeled as a Poisson random variable. Assume that on the average there are 10 calls per hour. (i) What is the probability that there are exactly 5 calls in one hour? (ii) What is the probability that there are 3 or less calls in one hour? (iii) What is the probability that there are exactly 15 calls in two hours? (iv) What is the probability that there are exactly 5 calls in 30 minutes?	08	–	–
5	a	The fill volume of an automated filling machine used for filling cans of carbonated beverage is normally distributed with a mean of 12.4 fluid ounces and a standard deviation of 0.1 fluid ounce. (i) What is the probability a fill volume is less than 12 fluid ounces? (ii) If all cans less than 12.1 or greater than 12.6 ounces are scrapped, what proportion of cans is scrapped? (iii) Determine specifications that are symmetric about the mean that include 99% of all cans.	12	–	–
	b	List any four characteristics of normal distribution curve.	04	–	–
		OR			
6	a	The time between the arrival of electronic messages at your computer is exponentially distributed with a mean of two hours. (i) What is the probability that you do not receive a message during a two-hour period? (ii) If you have not had a message in the last four hours, what is the probability that you do not receive a message in the next two hours? (iii) What is the expected time between your fifth and sixth messages?	08	–	–
	b	The time between calls to a corporate office is exponentially distributed with a mean of 10 minutes. (i) What is the probability that there are more than three calls in one-half hour? (ii) What is the probability that there are no calls within one half hour? (iii) Determine x such that the probability that there are no calls within x hours is 0.01.	08	–	–
7		The following data for the dependent variable, y, and the independent variable, x, have been collected using simple random sampling: (i) Construct a scatter plot for these data. Based on the scatter plot, how would you describe the relationship between the two variables? (ii) Compute the correlation coefficient.	16	–	–

x	10	14	16	12	20	18	16	14	16	18
y	120	130	170	150	200	180	190	150	160	200

OR

8 An article in the Journal of Sound and Vibration (Vol. 151, 1991, pp. 383–394) described a study investigating the relationship between noise exposure and hypertension. The following data are representative of those reported in the article. — 16 – –

y	1	0	1	2	5	1	4	6	2	3
x	60	63	65	70	70	70	80	90	80	80
y	5	4	6	8	4	5	7	9	7	6
x	85	89	90	90	90	90	94	100	100	100

(i) Draw a scatter diagram of y (blood pressure rise in millimeters of mercury) versus x (sound pressure level in decibel). Does a simple linear regression model seem reasonable in this situation?
(ii) Fit the simple linear regression model using least squares. Find an estimate of σ^2.
(iii) Find the predicted mean rise in blood pressure level associated with a sound pressure level of 85 decibels.

9 a Explain the general procedure for Hypothesis Tests. — 08 – –

b The mean water temperature downstream from a power plant cooling tower discharge pipe should be no more than 100°F. Past experience has indicated that the standard deviation of temperature is 2°F. The water temperature is measured on nine randomly chosen days, and the average temperature is found to be 98°F. — 08 – –

(i) Should the water temperature be judged acceptable with $\alpha = 0.05$?
(ii) What is the P-value for this test?
(iii) What is the probability of accepting the null hypothesis at $\alpha = 0.05$ if the water has a true mean temperature of 104°F?

OR

10 Grades in a statistics course and an operations research course taken simultaneously were as follows for a group of students. — 16 – –

Statistics Grade	Operations Research Grade			
	A	B	C	Other
A	25	6	17	13
B	17	16	15	6
C	18	4	18	10
Other	10	8	11	20

Are the grades in statistics and operations research related? Use $\alpha = 0.01$ in reaching your conclusion. What is the P-value for this test?

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E. Examinations Oct/Nov - 2022

Industrial Engineering and Management

ENGINEERING STATISTICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Give an example where median is a more appropriate measure of central tendency as compared to the mean.	01
	1.2	What are the variables which are plotted on the X axis and Y xis of a Histogram?	01
	1.3	Give the classical definition of probability.	01
	1.4	Give the statement of the total probability theorem.	01
	1.5	Sketch the shape of discrete uniform probability mass function.	01
	1.6	Define a geometric random variable.	01
	1.7	Write down the mathematical form of a Normal distributed random variable with mean μ and standard deviation σ.	01
	1.8	Give an example for the applicability of a Weibull probability distribution.	01
	1.9	What is the meaning of bias in estimation process?	01
	1.10	Write down the expression for a Maximum Likelihood function.	01
	1.11	State the central limit theorem.	01
	1.12	Which one of the interval will be wider a 99% confidence interval or a 95% confidence interval for the mean of a population when standard deviation is known?	01
	1.13	Write down the formula for the sample correlation coefficient.	01
	1.14	What is the meaning of the term Empirical models?	01
	1.15	Mention any one of the common abuses of regression.	01
	1.16	Diagrammatically sketch the simple linear regression model with 95% confidence interval included.	01
	1.17	Give any of the guideline for specifying the alternate hypothesis for statistical inferencing.	01
	1.18	What is the concept of a pooled variance?	01
	1.19	Which test statistic is used for testing the Goodness of fit of a hypothesized distribution?	01
	1.20	What is the meaning of Test of association?	01

PART B

2	a	The following data represent the yield on 34 consecutive batches of ceramic substrate to which a metal coating has been applied by a vapor – deposition process. 87.1 89.6 83.9 88.5 93.7 87.9 85.1 88.2 80.2 89.2 89.8 86.0 83.4 84.7 89.8 96.6 89.7 94.3 86.3 86.7 90.6 81.2 89.8 94.8 88.2 79.2 86.4 96.8 85.5 87.3 93.9 85.0 83.6 85.4 i) Construct a frequency distribution for the yield data. Use six bins take start class intervals as 76 and bin width as 4. ii) Construct histogram and comment on important features that you noticed in the plot of the histogram.	08
	b	A digital scale is used that provides weights to the nearest grams. Define the sample space for this experiment? Let A denote the event that a weight exceeds 11 grams, let b denote the event that a weight is less than or equal to 15 grams, and let C denote the event that weight is greater than or equal to 8 grams and less than 12 grams. Describe the following events. $i)\ A \cup B$ $ii)\ A \cap B$ $iii)\ A'$ $iv)\ A \cup B \cup C$	08
3	a	The probability is 0.80 that a randomly selected technician will finish his or her project successfully. Let x be the number of technicians among a randomly selected group of 10 technicians who will finish their projects successfully. i. Find the probability distribution of the random variable x. ii. Find the probabilities that at least three technicians will finish their project successfully. iii. Find the probability that at most five technicians will finish their project successfully. iv. Find the probability that between four and six (inclusive) technicians will finish their project successfully.	08
	b	A total of 75 parts are received from the suppliers. We are informed that 8 defective parts were shipped by mistake, and 5 parts have already been installed on machines. i. What is the probability that exactly 1 defective part was installed on a machine. ii. What is the probability of finding les than 2 defective parts?	08
		OR	
4	a	The speed of a file transfer from a server on campus to a personal computer at a student's home on a weekday evening is normally distributed with a mean of 60 kilobits per second and a standard deviation of 4 kilobits per second. i. What is the probability that the file will transfer at a speed of 70 kilobits per second or more? ii. What is the probability that the file will transfer at a speed of less than 58 kilobits per second? iii. If the file is 1 megabyte, what is the average time it will take to transfer the file? (Assume eight bits per byte)	08

	b	The life of automobile voltage regulators has an exponential distribution with a mean life of six years. You purchase an automobile that is six years old, with a a working voltage regulator, and plan to own it for six years. i. What is the probability that the voltage regulator fails during your ownership? ii. If your regulator fails after you own the automobile for three years and it is replaced, what is the mean time until the next failure.	08
5	a	Suppose that n=100 random samples of water from a fresh water lake were taken and the calcium concentration (milligrams per liter) measured. A 95% CI on the mean calcium concentration is $0.49 \leq \mu \leq 0.82$. i. Would a 99% CI calculated from the same sample data been wider or narrower? ii. Consider the following statement: There is a 95% chance that μ is between 0.49 and 0.82. Is this statement correct? Explain your answer. iii. Consider the following statement: If n = 100 random samples of water from the lake were taken and the 95% CI on μ computed, and this process was repeated 1000 times, 950 of the CI's will contain the true value of μ. *I*s this statement correct? Explain your answer.	08
	b	A post mix beverage machine is adjusted to release a certain amount of syrup into a chamber where it is mixed with carbonated water. A random sample of 25 beverages was found to have mean syrup content of x = 120. Fluid ounces and a standard deviation of CI's = 0.015fluid ounces. Find a 95% on the mean volume of syrup dispensed. X=1.10 fluid ounces.	08
		OR	
6	a	A confidence interval estimate is desired for the gain in a circuit on a semiconductor device. Assume that gain is normally distributed with standard deviation $\sigma = 20$. Find a 99% CI for μ when n = 10 and xbar = 1000. Also a confidence interval estimate is desired for the gain in a circuit on a semiconductor device. Assume that gain is normally distributed with standard deviation $\sigma = 20$. Find a 99% CI for μ when n = 25 and xbar = 1000.	08
	b	Following are two confidence interval estimates of the mean μ *of the cyles* to failures of an automotive doors latch mechanism. $3124.9 \leq \mu \leq 3215.7$ $3110.5 \leq \mu \leq 3230.1$ i) What is the value of the sample mean cycles to failure? ii) The confidence level for one of these CI's is 95% and the confidence level for the other is 99%. Both CI's are calculated from the same sample data. Which is the 95% CI's? Explain why.	08

7 a	The distance Y necessary for stopping a vehicle is a function of the speed of travel of the vehicles X. Suppose the following set of data were observed for 12 vehicles travelling at different speeds as shown in the table below:	

$Car\ No$	1	2	3	4	5	6	7	8	9	10	11
$Speed\ (kph)$	40	9	100	50	15	65	25	60	9	65	30
$Stopping\ distance\ (m)$	15	2	40	15	4	25	5	25	30	24	8

i. Plot the stopping distance versus the speed of the travel.

ii. Assume that the stopping distance is linear function of the speed, i.e $E(Y|x) = \alpha + \beta x$. Estimate the regression coefficients, $\alpha, \beta,$ and the conditional standard deviation $S_{y/x}$.

iii. Also, determine the correlation coefficient between Y and X. 10

b Scraps of iron were selected on the basis of their densities, xi, and their iron contents, yi were measured. The results were as follows:

Xi	2.8	2.9	3	3.1	3.2	3.2	3.2	3.3	3.4
yi	27	23	30	28	30	32	34	33	30

Find the regression equation of y on x by the method of least squares. 06

8 a	A textile fiber manufacturer is investigating a new drapery yarn, which the company claims has a mean thread elongation of 12 kilograms with a standard derivation of 0.5 kilograms. The company wishes to test hypothesis $H_0: \mu = 12\ against\ H_1: \mu < 12\ using$ a random sample of four specimens.	

i) What is the type I error probability if the critical region is defined as $x < 11.5\ kilograms$?

ii) Find β for the case where the true mean elongation is 11.25 kilograms.

iii) Find the boundary of the critical region, if the type I error probability is specified to be $\alpha = 0.01$.

iv) Find the boundary of the critical region, if the type I error probability is specified to be $\alpha = 0.05$. 10

b In each of the following situations, state whether it is a correctly stated hypothesis testing problem and why.

i. $H_0: \mu = 25, H_1: \mu \neq 25$

ii. $H_0: \sigma > 10, H_1: \sigma = 10$

iii. $H_0: x = 50, H_1: x \neq 50$ 06

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IV Semester B. E.Grade Improvement ExaminationsOct-2021

Industrial Engineering and Management

ENGINEERING STATISTICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer any FIVE full questions out of TEN. Each carries 20 marks
2. Use of statistical tables along with statistical formula handbooks and probability distribution formulae is permitted.

1	1.1	An experiment fields 3 mutually exclusive events $A, B$and .If $P(A) = 2$ and $P(B) = 3$, then $P(C) =$_____.	01
	1.2	If a car agency sells 50% of its inventory of a certain foreign car equipped with side airbags, find a formula for the probability distribution of the number of cars with side airbags among the next4 cars sold by the agency.	02
	1.3	A random variable x has a $pdf\ f(x)$ where $f(x) = e^{-x}, 0 \leq x \leq \infty$. Find the probability of $0 \leq x \leq 2$.	02
	1.4	The probability that a person, living in a certain city, owns a dog is estimated to be 0.3. Find the probability that the tenth person randomly interviewed in that city is the fifth one to own a dog.	02
	1.5	Three people toss a fair coin and the odd man pays for coffee. If the coins all turn up the same, they are tosses again. Find the probability that fewer than 4 tosses are needed.	02
	1.6	If a random variable X has the gamma distribution with $a = 2$ and $\beta = 1$, find $P(1.8 < X < 2.4)$.	02
	1.7	Then length of life X in hours of an item in the machine shop has a Weibull distribution with $a = 0.01$ and $\beta = 2$. What is the probability that it fails before eight hours of usage?	02
	1.8	List any two methods of point estimation.	02
	1.9	The average monthly electricity consumption for a sample of 100 families is 1250units. Assuming the standard deviation of electric consumption of all families is 150units, construct a 95% confidence interval estimate of the actual mean electric consumption.	02
	1.10	A multiple linear regression analysis considering “Sugars” and “Fat” as the explanatory variables and “Rating” as the response variable shows the following output in Minitab. Predictor Coef StDev T P Constant 61.089 1.953 31.28 0.000 Fat -3.066 1.036 -2.96 0.004 Sugars -2.2128 0.2347 -9.43 0.000 S = 8.755 R-Sq = 62.2% R-Sq(adj) = 61.2% Write the multiple regression model for this.	
	1.11	State the assumptions governing significance test in correlation.	01
	1.12	Differentiate between type-I and type-II errors in hypothesis testing.	02

2	a	A university bought 45%, 25% and 30% of computers from *HCL, WIPRO* and *IBM* respectively, and 2%, 3% and 1% of these were found to be defective. Find the probability of a computer selected at random is found to be defective.	10
	b	The following table shows the number of inches of (melted) precipitation, yearly, in New york city, (1869 – 1957). 43.6 37.8 49.2 40.3 45.5 44.2 38.6 40.6 38.7 46.0 37.1 34.7 35.0 43.0 34.4 49.7 33.5 38.3 41.7 51.0 54.4 43.7 37.6 34.1 46.6 39.3 33.7 40.1 42.4 46.2 36.8 39.4 47.0 50.3 55.5 39.5 35.5 39.4 43.8 39.4 39.9 32.7 46.5 44.2 56.1 38.5 43.1 36.7 39.6 36.9 50.8 53.2 37.8 44.7 40.6 41.7 41.4 47.8 56.1 45.6 40.4 39.0 36.1 43.9 53.5 49.8 33.8 49.8 53.0 48.5 38.6 45.1 39.0 48.5 36.7 45.0 45.0 38.4 40.8 46.9 36.2 36.9 44.4 41.5 45.2 35.6 39.9 36.2 36.5 For the above data construct i) Stem and leaf plot ii) Histogram Draw suitable conclusion.	10
3	a	The following table summarizes the analysis of samples of galvanized steel for coating weight and surface roughness: *Coating weight* / *High* / *Low* *Surface Roughness* / *High* / 12 / 16 *Surface Roughness* / *Low* / 88 / 34 i) If the coating weight of a sample is high, what is the probability that surface roughness is high? ii) If the surface roughness of sample is high, what is the probability that the coating weight is high? iii) If the surface roughness of the sample is low, what is the probability that the coating weight is low?	10
	b	The investigation in medical research involves collecting resting and exercise pulse rates from all medical students in the institution. The data is as follows: Resting rates :- 68, 76, 84, 80, 76, 72, 60, 68, 68, 80, 68, 80, 64, 64, 72, 76, 72, 68, 56, 88, 80, 76, 68, 56, 64, 60, 92, 72, 84, 72 Exercise rate :- 148, 136, 157, 151, 121, 139, 137, 129, 127, 129, 155, 141, 133, 153, 161, 153, 127, 135, 144, 146, 136, 131, 133, 159, 127, 142, 133, 150, 164,161 Construct a box plot compare the above parameters on a common scale.	10
4	a	A restaurant chef prepares a tossed salad containing, on average, 5 vegetables. Find the probability that the salad contains more than 5 vegetables. i) On a given day. ii) On 3 of the next 4 days. iii) For the first time in April on April 5.	10
	b	A lawyer commutes daily from his suburban home to his mid town office. The average time for a one-way trip is 24 minutes, with a standard deviation of 3.8 minutes. Assume the distribution of trip times to be normally distributed.	

		i) What is the probability that a trip will take at least ½ hour?	10

		ii) If the office opens at $9.00AM$ and leaves his office at $8.45AM$ daily, what percentage of the time is he late for work? iii) If he leaves the house at $8.35AM$ and coffee is served at the office from $8.50AM$until $9.00\ AM$, what is the probability that he misses coffee? iv) Find the length of time above which we find the slowest 15% of the trips. v) Find the probability that 2 of the next 3 trips will take at least ½ hour.	
5	a	In airport luggage screening it is known that 3%of people screened have questionable objects in their luggage. What is the probability that a string of 15 people pass through successfully before an individual is caught with a questionable object? What is the expected number in a row that pass through before an individual is stopped. ?	06
	b	Suppose that the time, in hours, taken to repair a heat pump is a random variable X having a gamma distribution with parameters$\alpha = 2$ and $\beta = 1/2$. What is the probability that the next service call will require. i) At most 1 hour to repair the heat pump? ii) At least 2 hours to repair the heat pump?	06
	c	The length of time for one individual to be served at a cafeteria is a random variable having an exponential distribution with a mean 4 minutes. What is the probability that a person is served in less than 3 minutes on at least 4 of the next 6 days?	08
6	a	An electronics company manufactures resistors that have a mean resistance of 100ohms and a standard deviation of 10ohms. The distribution of resistance is normal. Find the probability that a random sample of $n = 25$ resistors will have an average resistance less than 95ohms.	08
	b	The posted speed limit on a particular segment of an interstate highway is 65mph. From past experience , the standard deviation of vehicle speed is 10 mph. To estimate the average speed of vehicles, the speed of 50 vehicles are recorded and the average speed is estimated to be $67mph$. i) Determine the 95% confidence interval for the mean vehicle speed. ii) Determine the 95% lower and upper confidence limits for the mean vehicle speed.	08
	c	PVCpipe is manufactured with a mean diameter of 1.01 inch and a standard deviation of 0.003 inch. Find the probability that a random sample of $n = 9$ sections of pipe will have a sample mean diameter greater than 1.009 inch and less than 1.012 inch.	04
7	a	Speed data collected on an urban roadway yielded a standard deviation in speeds of ± 4.8 mile/hr.If an engineer wishes to estimate the average speed on the roadway at a 95% confidence level so that the estimate is within $\pm 2mi/hr$ of the true average, how many spot speeds should be collected?If the estimate of the average must be within $\pm 1mi/hr$, what should the sample size be?	08
	b	A sample of 10 concrete blocks manufactured by a certain process had a mean compressive strength of $\bar{X} = 1312MPa$, with a standard deviation $= 25MPa$. Find a 95% prediction interval for the strength of a block whose strength has yet to be measured.	04

	c	The department of transportation wishes to survey state residents to determine what proportion of the population would like to increase state-	08

		wide highway speed limits to 75mph from 65mph. How many residents do they need to survey if they want to be atleast99% confident that the sample proportion is within 0.05 of the true proportion?	
8	a	The success of a shopping center can be represented as a function of the distance(in miles) from the center of the population and the number of clients(in hundreds of people) who will visit. The data is given in the table below: $Number\ of\ customers(x)$: 8, 7, 6, 4, 2, 1 $Distance\ (y)$: 15, 19, 25, 23, 34, 40 i) Calculate the correlation coefficient using covariance. ii) If the mall is located 2 miles from the centre of the population, how many customers should the shopping centre expect? iii) To receive 500 customers, at what distance from the centre of the population should be shopping centre be located.	12
	b	Briefly explain the different types of regression with examples.	08
9		An article is the journal of Environmental Engineering (vol.115, No. 3, 1989, pp. $608-619$) reported the results of a study on the occurrence of sodium and chloride in surface streams in central Rhode island. The following data are chloride concentration Y (in milligrams per liter) and roadway area in the watershed X (in percentage). $Y \rightarrow 4.4, 6.6, 9.7, 10.6, 10.8, 10.9, 11.8, 12.1, 14.3, 14.7, 15.0, 17.3, 19.2, 23.1, 27.4, 27.7, 31.8, 39.5$ $X \rightarrow 0.19, 0.15, 1.57, 0.70, 0.67, 0.63, 0.47, 0.7, 0.6, 0.78, 0.81, 0.78, 0.69, 1.30, 1.05, 1.06, 1.74, 1.62$ i) Draw a scatter diagram of the data. Does a simple linear regression model seem appropriate here ii) Fit the simple linear regression model using the method of least squares. Find an estimate of σ^2. iii) Estimate the mean chloride concentration for a watershed that has 1% roadway area. iv) Find the fitted value corresponding to $x = 0.47$ and the associated residual.	20
10	a	An inventor has developed a new, energy –efficient lawn mower engine. He claims that the engine will run continuously for 5 hours (300munites) on a single gallon of regular gasoline. From his stock of 2000 engines, the inventor selects a simple random sample of 50 engines for testing. The engines run for an average of 295 minutes, with a std deviation of 20minutes. Test the null hypothesis that the mean run time is 300 minutes against the alternative hypothesis that the mean run time is not 300 minutes. Use a 0.05 level of significance. (assume that run times for the population of engines are normally distributed)	08
	b	Four identical six-sided dice, each with faces marked 1 to 6 are rolled 200 times. At each rolling, a record is made of the number of dice whose score on the uppermost face is even. The results are as follows: $Number\ of\ even\ scores$: 0, 1, 2, 3, 4 $Frequency$: 10, 41, 70, 57, 22 Conduct Chi-square goodness – of –fit using Binomial distribution.	12

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IV Semester B. E. Grade Improvement Examinations Nov-2020

Industrial Engineering and Management

ENGINEERING STATISTICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Use of statistical tables along with statistical formula handbooks giving probability distribution formulae is permitted.

PART A

1	1.1	Which plot can quickly identify outliers in a given data set?	01
	1.2	Give an example for finite sample space. Define the random experiment and write down the sample space.	01
	1.3	State the necessary condition for the conditional probability theorem.	01
	1.4	If X is a random variable denoting the number of darts hitting the bull's eye on the dashboard. State whether X is discrete or continuous random variable.	01
	1.5	Write down the probability distribution of a geometric random variable.	01
	1.6	Give the correct definition of a Hypergeometric random variable.	01
	1.7	If the power outage in an area can occur with the same probability anytime during the 24hr interval, what is the most appropriate continuous probability distribution which is applicable to this situation?	01
	1.8	Give an example for the random variable which could be modeled using exponential distribution.	01
	1.9	Mention the advantage of an interval estimate over a point estimate.	01
	1.10	Write down the principle behind the maximum likelihood estimator.	01
	1.11	If you can construct an interval estimate based on the sample information for the population mean, given the sample information such as $n, \bar{x}, \sigma$ given $\alpha = 0.05$ and you are required to repeat the same with $\alpha = 0.01$.Which of the two intervals will be wider?	01
	1.12	What is the formula for estimating the sample size n, required for a given α, margin of error E and sampling information?	01
	1.13	Illustratively sketch to demonstrate the principle of least squares for a simple linear regression model.	01
	1.14	Why do you need to assume that the error distribution is normal for developing least square estimates of the parameters of a simple linear regression model?	01
	1.15	What is the theoretical range for the correlation coefficient?	01
	1.16	Write down the formula for prediction of new observations based on least square estimates?	01

	1.17	Failing to reject the null hypothesis when it is false. Which type of error are we committing in this case?	01
	1.18	Write down the test statistic for testing for goodness of fit?	01
	1.19	What is significance of the P value with respect to hypothesis testing?	01
	1.20	What is the difference between the statistical significance and practical significance with respect to hypothesis tests?	01

PART B

2	a	The following data represents the length of life in years, measured to the nearest tenth of 30 similar fuel pumps 2.0 3.0 0.3 3.3 1.3 0.4 0.2 6.0 5.5 6.5 0.2 2.3 1.5 4.0 5.9 1.8 4.7 0.7 4.5 0.3 1.5 0.5 2.5 5.0 1.0 6.0 5.6 6.0 1.2 0.2 Construct the box and whisker plot. Comment on the outliers in the data.	08
	b	The probability is 1% that an electrical connecter that is kept dry fails during the warranty period of a portable computer. If the connector is ever wet, the probability of a failure during the warranty period is 5%. If 90% of the connectors are kept dry and 10% are wet, what proportion of connectors fail during warranty period.	04
	c	Samples of emulsion from three suppliers are classified for conformance to air quality specifications. The results from 100 samples are summarized as follows: $Conforms$ / $Supplier$: 1: YES 22, NO 8; 2: YES 25, NO 5; 3: YES 30, NO 10 Let A denote the event that a sample is from supplier 1 and Let B denote the event that a sample conforms to specifications: i) Are events A and B independent ii) Determine $p(B/A)$	04
3	a	Let X denote the number of bits received in error in a digital communication channel and assume X is a binomial random variable with $p = 0.001$. If 1000 bits are transmitted determine the following: i) $p(X = 1)$ ii) $p(X \geq 1)$ iii) $P(X \leq 2)$ iv) Mean and variance of X	06
	b	The number of messages sent to a computer bulletin board is a poisson random variable with a mean of 5 messages per hour: i) What is the probability that 5 messages are received in 1 hour? ii) What is the probability that 10 messages are received in 1.5 hours? iii) What is the probability that less than two messages are received in one-half hour?	08
	c	Which of the discrete probability distribution has the mean and variance which are equal?	02
		OR	

4	a	The compressive strength of samples of cement can be modeled by a normal distribution with a mean of 6000 kilograms per square centimeter and a standard deviation of 100 kilograms per square centimeter. i) What is the probability that a sample strength is less than $6250\ kg/cm^2$? ii) What is the probability that a sample strength is in between $5800\ kg/cm^2$ and $5900\ kg/cm^2$? iii) What strength is exceeded by 95% of the samples?	08
	b	The time to failure (in hours) of fans in a personal computer can be modeled by an exponential distribution with $\lambda = 0.0003$ i) What proportion of fans will last at least 10,000 hours? ii) What proportion of the fans will last at most 7,000 hours?	04
	c	Suppose that X has a weibull distribution with $\beta = 0.2$ and $\delta = 100$ hours. Determine the following: i) $p(X < 10{,}000)$ ii) $p(X > 5000)$	04
5	a	Suppose that $\widehat{\theta_1}$ and $\widehat{\theta_2}$ are unbiased estimates of the parameter θ. We know that $V(\widehat{\theta_1}) = 10$ and $V(\widehat{\theta_2}) = 4$. Which estimate is best and in what sense is it best?	04
	b	Consider the probability density function $f(x) = c(1 + \theta x) \ \ -1 \le x \le 1$ i) Find the value of constant C ii) What is the moment estimate for θ? iii) Show that $\hat{\theta} = 3\bar{x}$ is an unbiased estimator for θ.	08
	c	Write briefly on the Method of Moments.	04
		OR	
6	a	A post m in beverage machine is adjusted to release a certain amount of syrup into a chamber where it is mixed with carbonated water. A random sample of 25 beverages was found to have a mean syrup content of $\bar{X} = 1.10$ fluid ounces and a standard deviation of $S = 0.015$ fluid ounces. Find a 95% confidence interval on the mean volume of syrup dispensed.	06
	b	An izod impact test was performed on 20 specimens of PVC pipe. The sample mean is $\bar{X} = 1.25$ and the sample standard deviation is $S = 0.25$. Find a 99% lower confidence band on izod impact strength.	04
	c	The percentage of titanium n an alloy used in aerospace castings is measured in 51 randomly selected parts. The sample standard deviation is $S = 0.37$**.** Construct a 95% two sided confidence interval for σ. Do the same for 99% confidence interval. Which is wider?	06

7 a The following data were collected to determine the relationship between pressure and corresponding scale reading for the purpose of calibration

$pressure, x\ (lb/sq.in.)$	$scale\ reading, y$
10	13
10	18
10	16
10	15
10	20
50	86
50	90
50	88
50	88
50	92

i) Plot the scatter diagram
ii) Find the equation of the regression line
iii) The purpose of calibration in this application is to estimate pressure from an observed scale reading. Estimate the pressure for a observed scale reading of 54. 10

b Compute the correlation coefficient and interpret the same for the following grades of 6 students selected at random:

$Mathematics\ grade$	70	92	80	74	65	83
$English\ grade$	74	84	63	87	78	90

If two more pairs of valves are made available to you later $(80, 92)$ and $(70, 74)$, recompute the correlation coefficient and interpret the same. 06

8 a Let X denote the number of flaws observed on a large coil of galvanized steel. Seventy five coils are inspected and the following data were observed for the values of X.

$value$	1	2	3	4	5	6	7	8
$observed\ frequency$	1	11	8	13	11	12	10	9

Does the assumption of the poisson distribution seem appropriate as a probability model for the data? Use $\propto = 0.01$ 08

b An engineer who is studying the tensile strength of steel alloy intended for use in golf club shafts knows that the tensile strength is approximately normally distributed with $\sigma = 60psi$. A random sample of 12 specimens has a mean tensile strength of $\bar{X} = 3250psi$.

i) Test the hypothesis that the mean strength is $3500psi$ use $\propto = 0.01$
ii) Explain how you would answer the above question with a two sided confidence interval on mean tensile strength.
iii) What is the smallest level of significance at which you would be willing to reject the null hypothesis. 08

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E. Fast Track Examinations Oct-2020

Industrial Engineering and Management

ENGINEERING STATISTICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Use of statistical tables and handbook is permitted

PART A

1	1.1	The modal class in the table below showing the number of products sold in each class of sizes is _____	

$Class$	$21" - 30"$	$31" - 40"$	$41" - 50"$	$51" - 60"$	$61" - 70"$
$No\ of\ occurrences$	12	16	22	20	36

	1.2	For the above data in Q.No 1, the 40th percentile location would be at the location _____.	01
	1.3	Differentiate between radar chart and stem and leaf plot?	02
	1.4	If random variables X and Y represent the number on a die, when it is thrown, $X + 2Y$ will take on the values _____.	02
	1.5	List the four conditions of applicability which must be checked for the suitability of a binomial distribution in a random experiment.	02
	1.6	The life of a certain type of automobile tyre is normally distributed with mean 40,000 miles and standard deviation 5000 miles. How many tires would you expect to last less than 40000 miles?	02
	1.7	For some constant, C, the random variable x has the probability density function $f(x) = \begin{cases} Cx^4, & 0 < x < 2 \\ 0, & otherwise \end{cases}$ Determine 'C'	02
	1.8	In a linear regression model $y = \beta_0 a + \beta_1 x + e$, the error component is distributed as _____ distribution (include parameters also)	01
	1.9	In simple linear regression, the term linear means _____.	01
	1.10	The regression line derived from 10 observed pairs (x, y) is: $y = 32 + 4.6x$. The computed values of S_{YY} and S_{XY} are 204 and 37 respectively. The proportion of variability explained by the model is _____	02
	1.11	Accepting a null hypothesis when it is false is a _____ error.	01
	1.12	Based on a sample size of 9 with sample variance as 20.25 and sample mean 121, the rejection region corresponding to $H_1: \mu > 100$ is _____ at a significance level $\alpha = 1\%$	02
	1.13	To improve the precision of a confidence interval at a given significance level, _____ has to be increased.	01

PART B

2	a	The following distribution is with regard to weight (in grams) of apples of a given variety. If an apple is less than 122 grams it is considered unsuitable for export $\begin{array}{\|c\|c\|}\hline Weight\ (in\ grams) & No.\ of\ apples \\ \hline 100-110 & 10 \\ \hline 110-120 & 20 \\ \hline 120-130 & 48 \\ \hline 130-140 & 38 \\ \hline 140-150 & 25 \\ \hline 150-160 & 15 \\ \hline 160-170 & 5 \\ \hline \end{array}$ Draw a less than type ogive and deduce how many apples will be suitable for export.	08
	b	It is known that screws produced by a certain company will be defective with probability 0.01 independently of each other. The company sells the screws in package of 10 and offers a replacement guarantee on the package if more than 1 screw is defective in a package. What proportion of the packages sold must the company replace?	08
3	a	Suppose the response time X at a certain on-line computer terminal (the elapsed time between the end of a users inquiry and the beginning of the systems response to that of an inquiry) has an exponential distribution with expected response time equal to 5sec. Compute: i) $E(x)$ ii) The probability that the response time is atmost 10 sec iii) The probability that the response time is between 5 and 10sec. iv) Give one more instance of the exponential distribution frequently used.	08
	b	X denotes the number of creatures of a particular type captured in a trap during a given month. X has a poisson distribution with an average of 4.5 creatures trapped per month. Evaluate: i) The probability that exactly 5 creatures are trapped in a month. ii) The probability that a month has at most 5 creatures trapped. iii) The probability that 6 creatures are trapped in a 2 month period. iv) Probability that no creatures are trapped in a fortnight period.	08
		OR	
4	a	15% of parts fail on being subjected to a qualification test. The parts fail independently of other parts. 300 parts are tested every day. Compute the foll: i) Probability that between 35 and 45 parts fail in a day (exceeding 35 and 45) ii) Probability that there are atmost 25 part failures in a day. iii) Of 10 consecutive days when parts are tested, what is the probability that there were exactly 3 days with 25 or less part failures?	10
	b	Customers arrive according to a poisson process at a counter with a mean rate of a customer for every 2 minutes. i) What are the mean & variance of the inter arrival times? ii) Compute the probability that inter arrival time between two customers is atmost 2 mins.	06

5	a	10 values collected at random from a poisson distribution with mean μ are: $10, 11, 15, 17, 9, 13, 18, 11, 15, 16$. Estimate the mean parameter μ using the method of moments. What significance does the mean square error hold for an estimator? How is it used to evaluate the estimator?	10
	b	Briefly explain the following properties of estimators and how they are computed? i) Bias ii) Mean square error.	06
		OR	
6	a	A small-business website contains 100 pages and 60%, 30% and 10% of the pages contain low, moderate and high graphic content respectively. A sample of four pages is selected without replacement, and X and Y denote the number of pages with moderate and high graphics output in the sample. Determine: i) $fxy(x, y)$ ii) $fx(x)$ iii) $E(x)$ iv) $E(Y/X = 3)$	10
	b	The random variable X represents the number of defective parts in a lot of 90 parts. A sample of 25 lots give $x_i, i = 1,2 - - - 25$, where each $x_i =$ number of defective parts in i^{th} lot. Derive the method of moments estimator for the proportion of a part being defective.	06
7	a	What are the assumptions made in linear regression analysis?	04
	b	A study was made to model the relation between weekly advertising expenditures and sales: The following data were recorded: $Ad.Cost$: 20, 25, 30, 35, 40, 45, 50, 55, 60 $Weekly\ sales$: 400, 420, 405, 480, 475, 490, 525, 560, 515 i) Plot a scatter diagram and recommend suitable model. ii) Find the equation of regression line to predict weekly sales from advertising expenditure. iii) Estimate the weekly sales when ad costs are 35 iv) Compute coefficient of determination R^2r interpret about model.	12
8	a	The time in hours to install or commission a machine is as follows for a sample of 10 machines chosen at random: $192, 201, 190, 183, 185, 176, 191, 214, 220, 205$ If the above data is assumed to come from a normal population, can you conclude that the population mean is greater than 205 hours at 5% significance level?	10
	b	For a random sample of 1000 families owning TV sets in Bangalore, it was found that 240 families had TV of particular brand i) Find 95% CI for the proportion of families in the city owning that brand. ii) How large a sample is required if it is desired to be 95% confident that our estimate of binomial parameter p is within ± 0.02 of the true value.	06

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Appendix A Statistical Tables and Charts

Table I Summary of Common Probability Distributions

Name	Probability Distribution	Mean	Variance	Section in Book
Discrete				
Uniform	$\frac{1}{n}, a \le b$	$\frac{(b+a)}{2}$	$\frac{(b-a+1)^2-1}{12}$	3-5
Binomial	$\binom{n}{x}p^x(1-p)^{n-x}$, $x = 0, 1, \ldots, n, 0 \le p \le 1$	np	$np(1-p)$	3-6
Geometric	$(1-p)^{x-1}p$, $x = 1, 2, \ldots, 0 \le p \le 1$	$1/p$	$(1-p)/p^2$	3-7.1
Negative binomial	$\binom{x-1}{r-1}(1-p)^{x-r}p^r$ $x = r, r+1, r+2, \ldots, 0 \le p \le 1$	r/p	$r(1-p)/p^2$	3-7.2
Hypergeometric	$\frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$ $x = \max(0, n-N+K), 1, \ldots \min(K, n), K \le N, n \le N$	np, where $p = \frac{K}{N}$	$np(1-p)\left(\frac{N-n}{N-1}\right)$	3-8
Poisson	$\frac{e^{-\lambda}\lambda^x}{x!}, x = 0, 1, 2, \ldots, 0 < \lambda$	λ	λ	3-9
Continuous				
Uniform	$\frac{1}{b-a}, a \le x \le b$	$\frac{(b+a)}{2}$	$\frac{(b-a)^2}{12}$	4-5
Normal	$\frac{1}{\sigma\sqrt{2\pi}}e^{-1/2\left(\frac{x-\mu}{\sigma}\right)^2}$ $-\infty < x < \infty, -\infty < \mu < \infty, 0 < \sigma$	μ	σ^2	4-6
Exponential	$\lambda e^{-\lambda x}, 0 \le x, 0 < \lambda$	$1/\lambda$	$1/\lambda^2$	4-8
Erlang	$\frac{\lambda^r x^{r-1}e^{-\lambda x}}{(r-1)!}, 0 < x, r = 1, 2, \ldots$	r/λ	r/λ^2	4-9.1
Gamma	$\frac{\lambda x^{r-1}e^{-\lambda x}}{\Gamma(r)}, 0 < x, 0 < r, 0 < \lambda$	r/λ	r/λ^2	4-9.2
Weibull	$\frac{\beta}{\delta}\left(\frac{x}{\delta}\right)^{\beta-1}e^{-(x/\delta)^\beta}$, $0 < x, 0 < \beta, 0 < \delta$	$\delta\Gamma\left(1+\frac{1}{\beta}\right)$	$\delta^2\Gamma\left(1+\frac{2}{\beta}\right) - \delta^2\left[\Gamma\left(1+\frac{1}{\beta}\right)\right]^2$	4-10
Lognormal	$\frac{1}{x\sigma\sqrt{2\pi}}\exp\left(\frac{-[\ln(x)-\theta]^2}{2\omega^2}\right)$	$e^{\theta+\omega^2/2}$	$e^{2\theta+\omega^2}(e^{\omega^2}-1)$	4-11

$$\Phi(z) = P(Z \le z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2}\, du$$

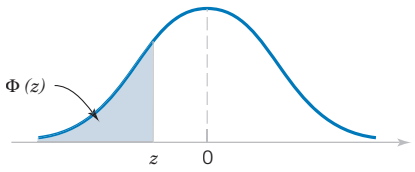


Table II Cumulative Standard Normal Distribution

z	−0.09	−0.08	−0.07	−0.06	−0.05	−0.04	−0.03	−0.02	−0.01	−0.00
−3.9	0.000033	0.000034	0.000036	0.000037	0.000039	0.000041	0.000042	0.000044	0.000046	0.000048
−3.8	0.000050	0.000052	0.000054	0.000057	0.000059	0.000062	0.000064	0.000067	0.000069	0.000072
−3.7	0.000075	0.000078	0.000082	0.000085	0.000088	0.000092	0.000096	0.000100	0.000104	0.000108
−3.6	0.000112	0.000117	0.000121	0.000126	0.000131	0.000136	0.000142	0.000147	0.000153	0.000159
−3.5	0.000165	0.000172	0.000179	0.000185	0.000193	0.000200	0.000208	0.000216	0.000224	0.000233
−3.4	0.000242	0.000251	0.000260	0.000270	0.000280	0.000291	0.000302	0.000313	0.000325	0.000337
−3.3	0.000350	0.000362	0.000376	0.000390	0.000404	0.000419	0.000434	0.000450	0.000467	0.000483
−3.2	0.000501	0.000519	0.000538	0.000557	0.000577	0.000598	0.000619	0.000641	0.000664	0.000687
−3.1	0.000711	0.000736	0.000762	0.000789	0.000816	0.000845	0.000874	0.000904	0.000935	0.000968
−3.0	0.001001	0.001035	0.001070	0.001107	0.001144	0.001183	0.001223	0.001264	0.001306	0.001350
−2.9	0.001395	0.001441	0.001489	0.001538	0.001589	0.001641	0.001695	0.001750	0.001807	0.001866
−2.8	0.001926	0.001988	0.002052	0.002118	0.002186	0.002256	0.002327	0.002401	0.002477	0.002555
−2.7	0.002635	0.002718	0.002803	0.002890	0.002980	0.003072	0.003167	0.003264	0.003364	0.003467
−2.6	0.003573	0.003681	0.003793	0.003907	0.004025	0.004145	0.004269	0.004396	0.004527	0.004661
−2.5	0.004799	0.004940	0.005085	0.005234	0.005386	0.005543	0.005703	0.005868	0.006037	0.006210
−2.4	0.006387	0.006569	0.006756	0.006947	0.007143	0.007344	0.007549	0.007760	0.007976	0.008198
−2.3	0.008424	0.008656	0.008894	0.009137	0.009387	0.009642	0.009903	0.010170	0.010444	0.010724
−2.2	0.011011	0.011304	0.011604	0.011911	0.012224	0.012545	0.012874	0.013209	0.013553	0.013903
−2.1	0.014262	0.014629	0.015003	0.015386	0.015778	0.016177	0.016586	0.017003	0.017429	0.017864
−2.0	0.018309	0.018763	0.019226	0.019699	0.020182	0.020675	0.021178	0.021692	0.022216	0.022750
−1.9	0.023295	0.023852	0.024419	0.024998	0.025588	0.026190	0.026803	0.027429	0.028067	0.028717
−1.8	0.029379	0.030054	0.030742	0.031443	0.032157	0.032884	0.033625	0.034379	0.035148	0.035930
−1.7	0.036727	0.037538	0.038364	0.039204	0.040059	0.040929	0.041815	0.042716	0.043633	0.044565
−1.6	0.045514	0.046479	0.047460	0.048457	0.049471	0.050503	0.051551	0.052616	0.053699	0.054799
−1.5	0.055917	0.057053	0.058208	0.059380	0.060571	0.061780	0.063008	0.064256	0.065522	0.066807
−1.4	0.068112	0.069437	0.070781	0.072145	0.073529	0.074934	0.076359	0.077804	0.079270	0.080757
−1.3	0.082264	0.083793	0.085343	0.086915	0.088508	0.090123	0.091759	0.093418	0.095098	0.096801
−1.2	0.098525	0.100273	0.102042	0.103835	0.105650	0.107488	0.109349	0.111233	0.113140	0.115070
−1.1	0.117023	0.119000	0.121001	0.123024	0.125072	0.127143	0.129238	0.131357	0.133500	0.135666
−1.0	0.137857	0.140071	0.142310	0.144572	0.146859	0.149170	0.151505	0.153864	0.156248	0.158655
−0.9	0.161087	0.163543	0.166023	0.168528	0.171056	0.173609	0.176185	0.178786	0.181411	0.184060
−0.8	0.186733	0.189430	0.192150	0.194894	0.197662	0.200454	0.203269	0.206108	0.208970	0.211855
−0.7	0.214764	0.217695	0.220650	0.223627	0.226627	0.229650	0.232695	0.235762	0.238852	0.241964
−0.6	0.245097	0.248252	0.251429	0.254627	0.257846	0.261086	0.264347	0.267629	0.270931	0.274253
−0.5	0.277595	0.280957	0.284339	0.287740	0.291160	0.294599	0.298056	0.301532	0.305026	0.308538
−0.4	0.312067	0.315614	0.319178	0.322758	0.326355	0.329969	0.333598	0.337243	0.340903	0.344578
−0.3	0.348268	0.351973	0.355691	0.359424	0.363169	0.366928	0.370700	0.374484	0.378281	0.382089
−0.2	0.385908	0.389739	0.393580	0.397432	0.401294	0.405165	0.409046	0.412936	0.416834	0.420740
−0.1	0.424655	0.428576	0.432505	0.436441	0.440382	0.444330	0.448283	0.452242	0.456205	0.460172
0.0	0.464144	0.468119	0.472097	0.476078	0.480061	0.484047	0.488033	0.492022	0.496011	0.500000

$$\Phi(z) = P(Z \le z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2}\, du$$

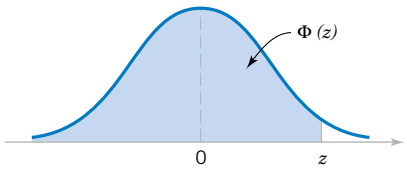


Table II Cumulative Standard Normal Distribution (*continued*)

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.500000	0.503989	0.507978	0.511967	0.515953	0.519939	0.532922	0.527903	0.531881	0.535856
0.1	0.539828	0.543795	0.547758	0.551717	0.555760	0.559618	0.563559	0.567495	0.571424	0.575345
0.2	0.579260	0.583166	0.587064	0.590954	0.594835	0.598706	0.602568	0.606420	0.610261	0.614092
0.3	0.617911	0.621719	0.625516	0.629300	0.633072	0.636831	0.640576	0.644309	0.648027	0.651732
0.4	0.655422	0.659097	0.662757	0.666402	0.670031	0.673645	0.677242	0.680822	0.684386	0.687933
0.5	0.691462	0.694974	0.698468	0.701944	0.705401	0.708840	0.712260	0.715661	0.719043	0.722405
0.6	0.725747	0.729069	0.732371	0.735653	0.738914	0.742154	0.745373	0.748571	0.751748	0.754903
0.7	0.758036	0.761148	0.764238	0.767305	0.770350	0.773373	0.776373	0.779350	0.782305	0.785236
0.8	0.788145	0.791030	0.793892	0.796731	0.799546	0.802338	0.805106	0.807850	0.810570	0.813267
0.9	0.815940	0.818589	0.821214	0.823815	0.826391	0.828944	0.831472	0.833977	0.836457	0.838913
1.0	0.841345	0.843752	0.846136	0.848495	0.850830	0.853141	0.855428	0.857690	0.859929	0.862143
1.1	0.864334	0.866500	0.868643	0.870762	0.872857	0.874928	0.876976	0.878999	0.881000	0.882977
1.2	0.884930	0.886860	0.888767	0.890651	0.892512	0.894350	0.896165	0.897958	0.899727	0.901475
1.3	0.903199	0.904902	0.906582	0.908241	0.909877	0.911492	0.913085	0.914657	0.916207	0.917736
1.4	0.919243	0.920730	0.922196	0.923641	0.925066	0.926471	0.927855	0.929219	0.930563	0.931888
1.5	0.933193	0.934478	0.935744	0.936992	0.938220	0.939429	0.940620	0.941792	0.942947	0.944083
1.6	0.945201	0.946301	0.947384	0.948449	0.949497	0.950529	0.951543	0.952540	0.953521	0.954486
1.7	0.955435	0.956367	0.957284	0.958185	0.959071	0.959941	0.960796	0.961636	0.962462	0.963273
1.8	0.964070	0.964852	0.965621	0.966375	0.967116	0.967843	0.968557	0.969258	0.969946	0.970621
1.9	0.971283	0.971933	0.972571	0.973197	0.973810	0.974412	0.975002	0.975581	0.976148	0.976705
2.0	0.977250	0.977784	0.978308	0.978822	0.979325	0.979818	0.980301	0.980774	0.981237	0.981691
2.1	0.982136	0.982571	0.982997	0.983414	0.983823	0.984222	0.984614	0.984997	0.985371	0.985738
2.2	0.986097	0.986447	0.986791	0.987126	0.987455	0.987776	0.988089	0.988396	0.988696	0.988989
2.3	0.989276	0.989556	0.989830	0.990097	0.990358	0.990613	0.990863	0.991106	0.991344	0.991576
2.4	0.991802	0.992024	0.992240	0.992451	0.992656	0.992857	0.993053	0.993244	0.993431	0.993613
2.5	0.993790	0.993963	0.994132	0.994297	0.994457	0.994614	0.994766	0.994915	0.995060	0.995201
2.6	0.995339	0.995473	0.995604	0.995731	0.995855	0.995975	0.996093	0.996207	0.996319	0.996427
2.7	0.996533	0.996636	0.996736	0.996833	0.996928	0.997020	0.997110	0.997197	0.997282	0.997365
2.8	0.997445	0.997523	0.997599	0.997673	0.997744	0.997814	0.997882	0.997948	0.998012	0.998074
2.9	0.998134	0.998193	0.998250	0.998305	0.998359	0.998411	0.998462	0.998511	0.998559	0.998605
3.0	0.998650	0.998694	0.998736	0.998777	0.998817	0.998856	0.998893	0.998930	0.998965	0.998999
3.1	0.999032	0.999065	0.999096	0.999126	0.999155	0.999184	0.999211	0.999238	0.999264	0.999289
3.2	0.999313	0.999336	0.999359	0.999381	0.999402	0.999423	0.999443	0.999462	0.999481	0.999499
3.3	0.999517	0.999533	0.999550	0.999566	0.999581	0.999596	0.999610	0.999624	0.999638	0.999650
3.4	0.999663	0.999675	0.999687	0.999698	0.999709	0.999720	0.999730	0.999740	0.999749	0.999758
3.5	0.999767	0.999776	0.999784	0.999792	0.999800	0.999807	0.999815	0.999821	0.999828	0.999835
3.6	0.999841	0.999847	0.999853	0.999858	0.999864	0.999869	0.999874	0.999879	0.999883	0.999888
3.7	0.999892	0.999896	0.999900	0.999904	0.999908	0.999912	0.999915	0.999918	0.999922	0.999925
3.8	0.999928	0.999931	0.999933	0.999936	0.999938	0.999941	0.999943	0.999946	0.999948	0.999950
3.9	0.999952	0.999954	0.999956	0.999958	0.999959	0.999961	0.999963	0.999964	0.999966	0.999967

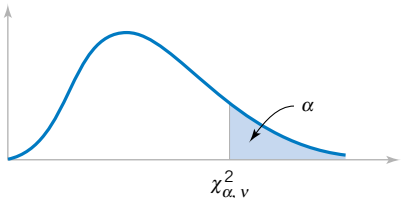


Table III Percentage Points $\chi^2_{\alpha,\nu}$ of the Chi-Squared Distribution

ν \ α	.995	.990	.975	.950	.900	.500	.100	.050	.025	.010	.005
1	.00+	.00+	.00+	.00+	.02	.45	2.71	3.84	5.02	6.63	7.88
2	.01	.02	.05	.10	.21	1.39	4.61	5.99	7.38	9.21	10.60
3	.07	.11	.22	.35	.58	2.37	6.25	7.81	9.35	11.34	12.84
4	.21	.30	.48	.71	1.06	3.36	7.78	9.49	11.14	13.28	14.86
5	.41	.55	.83	1.15	1.61	4.35	9.24	11.07	12.83	15.09	16.75
6	.68	.87	1.24	1.64	2.20	5.35	10.65	12.59	14.45	16.81	18.55
7	.99	1.24	1.69	2.17	2.83	6.35	12.02	14.07	16.01	18.48	20.28
8	1.34	1.65	2.18	2.73	3.49	7.34	13.36	15.51	17.53	20.09	21.96
9	1.73	2.09	2.70	3.33	4.17	8.34	14.68	16.92	19.02	21.67	23.59
10	2.16	2.56	3.25	3.94	4.87	9.34	15.99	18.31	20.48	23.21	25.19
11	2.60	3.05	3.82	4.57	5.58	10.34	17.28	19.68	21.92	24.72	26.76
12	3.07	3.57	4.40	5.23	6.30	11.34	18.55	21.03	23.34	26.22	28.30
13	3.57	4.11	5.01	5.89	7.04	12.34	19.81	22.36	24.74	27.69	29.82
14	4.07	4.66	5.63	6.57	7.79	13.34	21.06	23.68	26.12	29.14	31.32
15	4.60	5.23	6.27	7.26	8.55	14.34	22.31	25.00	27.49	30.58	32.80
16	5.14	5.81	6.91	7.96	9.31	15.34	23.54	26.30	28.85	32.00	34.27
17	5.70	6.41	7.56	8.67	10.09	16.34	24.77	27.59	30.19	33.41	35.72
18	6.26	7.01	8.23	9.39	10.87	17.34	25.99	28.87	31.53	34.81	37.16
19	6.84	7.63	8.91	10.12	11.65	18.34	27.20	30.14	32.85	36.19	38.58
20	7.43	8.26	9.59	10.85	12.44	19.34	28.41	31.41	34.17	37.57	40.00
21	8.03	8.90	10.28	11.59	13.24	20.34	29.62	32.67	35.48	38.93	41.40
22	8.64	9.54	10.98	12.34	14.04	21.34	30.81	33.92	36.78	40.29	42.80
23	9.26	10.20	11.69	13.09	14.85	22.34	32.01	35.17	38.08	41.64	44.18
24	9.89	10.86	12.40	13.85	15.66	23.34	33.20	36.42	39.36	42.98	45.56
25	10.52	11.52	13.12	14.61	16.47	24.34	34.28	37.65	40.65	44.31	46.93
26	11.16	12.20	13.84	15.38	17.29	25.34	35.56	38.89	41.92	45.64	48.29
27	11.81	12.88	14.57	16.15	18.11	26.34	36.74	40.11	43.19	46.96	49.65
28	12.46	13.57	15.31	16.93	18.94	27.34	37.92	41.34	44.46	48.28	50.99
29	13.12	14.26	16.05	17.71	19.77	28.34	39.09	42.56	45.72	49.59	52.34
30	13.79	14.95	16.79	18.49	20.60	29.34	40.26	43.77	46.98	50.89	53.67
40	20.71	22.16	24.43	26.51	29.05	39.34	51.81	55.76	59.34	63.69	66.77
50	27.99	29.71	32.36	34.76	37.69	49.33	63.17	67.50	71.42	76.15	79.49
60	35.53	37.48	40.48	43.19	46.46	59.33	74.40	79.08	83.30	88.38	91.95
70	43.28	45.44	48.76	51.74	55.33	69.33	85.53	90.53	95.02	100.42	104.22
80	51.17	53.54	57.15	60.39	64.28	79.33	96.58	101.88	106.63	112.33	116.32
90	59.20	61.75	65.65	69.13	73.29	89.33	107.57	113.14	118.14	124.12	128.30
100	67.33	70.06	74.22	77.93	82.36	99.33	118.50	124.34	129.56	135.81	140.17

ν = degrees of freedom.

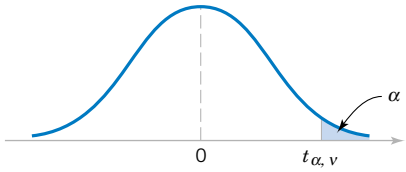


Table IV Percentage Points $t_{\alpha,\nu}$ of the t-Distribution

ν \ α	.40	.25	.10	.05	.025	.01	.005	.0025	.001	.0005
1	.325	1.000	3.078	6.314	12.706	31.821	63.657	127.32	318.31	636.62
2	.289	.816	1.886	2.920	4.303	6.965	9.925	14.089	23.326	31.598
3	.277	.765	1.638	2.353	3.182	4.541	5.841	7.453	10.213	12.924
4	.271	.741	1.533	2.132	2.776	3.747	4.604	5.598	7.173	8.610
5	.267	.727	1.476	2.015	2.571	3.365	4.032	4.773	5.893	6.869
6	.265	.718	1.440	1.943	2.447	3.143	3.707	4.317	5.208	5.959
7	.263	.711	1.415	1.895	2.365	2.998	3.499	4.029	4.785	5.408
8	.262	.706	1.397	1.860	2.306	2.896	3.355	3.833	4.501	5.041
9	.261	.703	1.383	1.833	2.262	2.821	3.250	3.690	4.297	4.781
10	.260	.700	1.372	1.812	2.228	2.764	3.169	3.581	4.144	4.587
11	.260	.697	1.363	1.796	2.201	2.718	3.106	3.497	4.025	4.437
12	.259	.695	1.356	1.782	2.179	2.681	3.055	3.428	3.930	4.318
13	.259	.694	1.350	1.771	2.160	2.650	3.012	3.372	3.852	4.221
14	.258	.692	1.345	1.761	2.145	2.624	2.977	3.326	3.787	4.140
15	.258	.691	1.341	1.753	2.131	2.602	2.947	3.286	3.733	4.073
16	.258	.690	1.337	1.746	2.120	2.583	2.921	3.252	3.686	4.015
17	.257	.689	1.333	1.740	2.110	2.567	2.898	3.222	3.646	3.965
18	.257	.688	1.330	1.734	2.101	2.552	2.878	3.197	3.610	3.922
19	.257	.688	1.328	1.729	2.093	2.539	2.861	3.174	3.579	3.883
20	.257	.687	1.325	1.725	2.086	2.528	2.845	3.153	3.552	3.850
21	.257	.686	1.323	1.721	2.080	2.518	2.831	3.135	3.527	3.819
22	.256	.686	1.321	1.717	2.074	2.508	2.819	3.119	3.505	3.792
23	.256	.685	1.319	1.714	2.069	2.500	2.807	3.104	3.485	3.767
24	.256	.685	1.318	1.711	2.064	2.492	2.797	3.091	3.467	3.745
25	.256	.684	1.316	1.708	2.060	2.485	2.787	3.078	3.450	3.725
26	.256	.684	1.315	1.706	2.056	2.479	2.779	3.067	3.435	3.707
27	.256	.684	1.314	1.703	2.052	2.473	2.771	3.057	3.421	3.690
28	.256	.683	1.313	1.701	2.048	2.467	2.763	3.047	3.408	3.674
29	.256	.683	1.311	1.699	2.045	2.462	2.756	3.038	3.396	3.659
30	.256	.683	1.310	1.697	2.042	2.457	2.750	3.030	3.385	3.646
40	.255	.681	1.303	1.684	2.021	2.423	2.704	2.971	3.307	3.551
60	.254	.679	1.296	1.671	2.000	2.390	2.660	2.915	3.232	3.460
120	.254	.677	1.289	1.658	1.980	2.358	2.617	2.860	3.160	3.373
∞	.253	.674	1.282	1.645	1.960	2.326	2.576	2.807	3.090	3.291

ν = degrees of freedom.

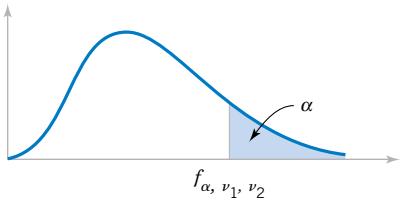


Table V Percentage Points f_{α,ν_1,ν_2} of the F-Distribution

$f_{0.25,\nu_1,\nu_2}$

ν_2 \ ν_1	Degrees of freedom for the numerator (ν_1)																		
	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
1	5.83	7.50	8.20	8.58	8.82	8.98	9.10	9.19	9.26	9.32	9.41	9.49	9.58	9.63	9.67	9.71	9.76	9.80	9.85
2	2.57	3.00	3.15	3.23	3.28	3.31	3.34	3.35	3.37	3.38	3.39	3.41	3.43	3.43	3.44	3.45	3.46	3.47	3.48
3	2.02	2.28	2.36	2.39	2.41	2.42	2.43	2.44	2.44	2.44	2.45	2.46	2.46	2.46	2.47	2.47	2.47	2.47	2.47
4	1.81	2.00	2.05	2.06	2.07	2.08	2.08	2.08	2.08	2.08	2.08	2.08	2.08	2.08	2.08	2.08	2.08	2.08	2.08
5	1.69	1.85	1.88	1.89	1.89	1.89	1.89	1.89	1.89	1.89	1.89	1.89	1.88	1.88	1.88	1.88	1.87	1.87	1.87
6	1.62	1.76	1.78	1.79	1.79	1.78	1.78	1.78	1.77	1.77	1.77	1.76	1.76	1.75	1.75	1.75	1.74	1.74	1.74
7	1.57	1.70	1.72	1.72	1.71	1.71	1.70	1.70	1.70	1.69	1.68	1.68	1.67	1.67	1.66	1.66	1.65	1.65	1.65
8	1.54	1.66	1.67	1.66	1.66	1.65	1.64	1.64	1.63	1.63	1.62	1.62	1.61	1.60	1.60	1.59	1.59	1.58	1.58
9	1.51	1.62	1.63	1.63	1.62	1.61	1.60	1.60	1.59	1.59	1.58	1.57	1.56	1.56	1.55	1.54	1.54	1.53	1.53
10	1.49	1.60	1.60	1.59	1.59	1.58	1.57	1.56	1.56	1.55	1.54	1.53	1.52	1.52	1.51	1.51	1.50	1.49	1.48
11	1.47	1.58	1.58	1.57	1.56	1.55	1.54	1.53	1.53	1.52	1.51	1.50	1.49	1.49	1.48	1.47	1.47	1.46	1.45
12	1.46	1.56	1.56	1.55	1.54	1.53	1.52	1.51	1.51	1.50	1.49	1.48	1.47	1.46	1.45	1.45	1.44	1.43	1.42
13	1.45	1.55	1.55	1.53	1.52	1.51	1.50	1.49	1.49	1.48	1.47	1.46	1.45	1.44	1.43	1.42	1.42	1.41	1.40
14	1.44	1.53	1.53	1.52	1.51	1.50	1.49	1.48	1.47	1.46	1.45	1.44	1.43	1.42	1.41	1.41	1.40	1.39	1.38
15	1.43	1.52	1.52	1.51	1.49	1.48	1.47	1.46	1.46	1.45	1.44	1.43	1.41	1.41	1.40	1.39	1.38	1.37	1.36
16	1.42	1.51	1.51	1.50	1.48	1.47	1.46	1.45	1.44	1.44	1.43	1.41	1.40	1.39	1.38	1.37	1.36	1.35	1.34
17	1.42	1.51	1.50	1.49	1.47	1.46	1.45	1.44	1.43	1.43	1.41	1.40	1.39	1.38	1.37	1.36	1.35	1.34	1.33
18	1.41	1.50	1.49	1.48	1.46	1.45	1.44	1.43	1.42	1.42	1.40	1.39	1.38	1.37	1.36	1.35	1.34	1.33	1.32
19	1.41	1.49	1.49	1.47	1.46	1.44	1.43	1.42	1.41	1.41	1.40	1.38	1.37	1.36	1.35	1.34	1.33	1.32	1.30
20	1.40	1.49	1.48	1.47	1.45	1.44	1.43	1.42	1.41	1.40	1.39	1.37	1.36	1.35	1.34	1.33	1.32	1.31	1.29
21	1.40	1.48	1.48	1.46	1.44	1.43	1.42	1.41	1.40	1.39	1.38	1.37	1.35	1.34	1.33	1.32	1.31	1.30	1.28
22	1.40	1.48	1.47	1.45	1.44	1.42	1.41	1.40	1.39	1.39	1.37	1.36	1.34	1.33	1.32	1.31	1.30	1.29	1.28
23	1.39	1.47	1.47	1.45	1.43	1.42	1.41	1.40	1.39	1.38	1.37	1.35	1.34	1.33	1.32	1.31	1.30	1.28	1.27
24	1.39	1.47	1.46	1.44	1.43	1.41	1.40	1.39	1.38	1.38	1.36	1.35	1.33	1.32	1.31	1.30	1.29	1.28	1.26
25	1.39	1.47	1.46	1.44	1.42	1.41	1.40	1.39	1.38	1.37	1.36	1.34	1.33	1.32	1.31	1.29	1.28	1.27	1.25
26	1.38	1.46	1.45	1.44	1.42	1.41	1.39	1.38	1.37	1.37	1.35	1.34	1.32	1.31	1.30	1.29	1.28	1.26	1.25
27	1.38	1.46	1.45	1.43	1.42	1.40	1.39	1.38	1.37	1.36	1.35	1.33	1.32	1.31	1.30	1.28	1.27	1.26	1.24
28	1.38	1.46	1.45	1.43	1.41	1.40	1.39	1.38	1.37	1.36	1.34	1.33	1.31	1.30	1.29	1.28	1.27	1.25	1.24
29	1.38	1.45	1.45	1.43	1.41	1.40	1.38	1.37	1.36	1.35	1.34	1.32	1.31	1.30	1.29	1.27	1.26	1.25	1.23
30	1.38	1.45	1.44	1.42	1.41	1.39	1.38	1.37	1.36	1.35	1.34	1.32	1.30	1.29	1.28	1.27	1.26	1.24	1.23
40	1.36	1.44	1.42	1.40	1.39	1.37	1.36	1.35	1.34	1.33	1.31	1.30	1.28	1.26	1.25	1.24	1.22	1.21	1.19
60	1.35	1.42	1.41	1.38	1.37	1.35	1.33	1.32	1.31	1.30	1.29	1.27	1.25	1.24	1.22	1.21	1.19	1.17	1.15
120	1.34	1.40	1.39	1.37	1.35	1.33	1.31	1.30	1.29	1.28	1.26	1.24	1.22	1.21	1.19	1.18	1.16	1.13	1.10
∞	1.32	1.39	1.37	1.35	1.33	1.31	1.29	1.28	1.27	1.25	1.24	1.22	1.19	1.18	1.16	1.14	1.12	1.08	1.00

Degrees of freedom for the denominator (ν_2)

Table V Percentage Points of the *F*-Distribution (*continued*)

$f_{0.10,v_1,v_2}$

v_2 \ v_1	Degrees of freedom for the numerator (v_1)																		
	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
1	39.86	49.50	53.59	55.83	57.24	58.20	58.91	59.44	59.86	60.19	60.71	61.22	61.74	62.00	62.26	62.53	62.79	63.06	63.33
2	8.53	9.00	9.16	9.24	9.29	9.33	9.35	9.37	9.38	9.39	9.41	9.42	9.44	9.45	9.46	9.47	9.47	9.48	9.49
3	5.54	5.46	5.39	5.34	5.31	5.28	5.27	5.25	5.24	5.23	5.22	5.20	5.18	5.18	5.17	5.16	5.15	5.14	5.13
4	4.54	4.32	4.19	4.11	4.05	4.01	3.98	3.95	3.94	3.92	3.90	3.87	3.84	3.83	3.82	3.80	3.79	3.78	3.76
5	4.06	3.78	3.62	3.52	3.45	3.40	3.37	3.34	3.32	3.30	3.27	3.24	3.21	3.19	3.17	3.16	3.14	3.12	3.10
6	3.78	3.46	3.29	3.18	3.11	3.05	3.01	2.98	2.96	2.94	2.90	2.87	2.84	2.82	2.80	2.78	2.76	2.74	2.72
7	3.59	3.26	3.07	2.96	2.88	2.83	2.78	2.75	2.72	2.70	2.67	2.63	2.59	2.58	2.56	2.54	2.51	2.49	2.47
8	3.46	3.11	2.92	2.81	2.73	2.67	2.62	2.59	2.56	2.54	2.50	2.46	2.42	2.40	2.38	2.36	2.34	2.32	2.29
9	3.36	3.01	2.81	2.69	2.61	2.55	2.51	2.47	2.44	2.42	2.38	2.34	2.30	2.28	2.25	2.23	2.21	2.18	2.16
10	3.29	2.92	2.73	2.61	2.52	2.46	2.41	2.38	2.35	2.32	2.28	2.24	2.20	2.18	2.16	2.13	2.11	2.08	2.06
11	3.23	2.86	2.66	2.54	2.45	2.39	2.34	2.30	2.27	2.25	2.21	2.17	2.12	2.10	2.08	2.05	2.03	2.00	1.97
12	3.18	2.81	2.61	2.48	2.39	2.33	2.28	2.24	2.21	2.19	2.15	2.10	2.06	2.04	2.01	1.99	1.96	1.93	1.90
13	3.14	2.76	2.56	2.43	2.35	2.28	2.23	2.20	2.16	2.14	2.10	2.05	2.01	1.98	1.96	1.93	1.90	1.88	1.85
14	3.10	2.73	2.52	2.39	2.31	2.24	2.19	2.15	2.12	2.10	2.05	2.01	1.96	1.94	1.91	1.89	1.86	1.83	1.80
15	3.07	2.70	2.49	2.36	2.27	2.21	2.16	2.12	2.09	2.06	2.02	1.97	1.92	1.90	1.87	1.85	1.82	1.79	1.76
16	3.05	2.67	2.46	2.33	2.24	2.18	2.13	2.09	2.06	2.03	1.99	1.94	1.89	1.87	1.84	1.81	1.78	1.75	1.72
17	3.03	2.64	2.44	2.31	2.22	2.15	2.10	2.06	2.03	2.00	1.96	1.91	1.86	1.84	1.81	1.78	1.75	1.72	1.69
18	3.01	2.62	2.42	2.29	2.20	2.13	2.08	2.04	2.00	1.98	1.93	1.89	1.84	1.81	1.78	1.75	1.72	1.69	1.66
19	2.99	2.61	2.40	2.27	2.18	2.11	2.06	2.02	1.98	1.96	1.91	1.86	1.81	1.79	1.76	1.73	1.70	1.67	1.63
20	2.97	2.59	2.38	2.25	2.16	2.09	2.04	2.00	1.96	1.94	1.89	1.84	1.79	1.77	1.74	1.71	1.68	1.64	1.61
21	2.96	2.57	2.36	2.23	2.14	2.08	2.02	1.98	1.95	1.92	1.87	1.83	1.78	1.75	1.72	1.69	1.66	1.62	1.59
22	2.95	2.56	2.35	2.22	2.13	2.06	2.01	1.97	1.93	1.90	1.86	1.81	1.76	1.73	1.70	1.67	1.64	1.60	1.57
23	2.94	2.55	2.34	2.21	2.11	2.05	1.99	1.95	1.92	1.89	1.84	1.80	1.74	1.72	1.69	1.66	1.62	1.59	1.55
24	2.93	2.54	2.33	2.19	2.10	2.04	1.98	1.94	1.91	1.88	1.83	1.78	1.73	1.70	1.67	1.64	1.61	1.57	1.53
25	2.92	2.53	2.32	2.18	2.09	2.02	1.97	1.93	1.89	1.87	1.82	1.77	1.72	1.69	1.66	1.63	1.59	1.56	1.52
26	2.91	2.52	2.31	2.17	2.08	2.01	1.96	1.92	1.88	1.86	1.81	1.76	1.71	1.68	1.65	1.61	1.58	1.54	1.50
27	2.90	2.51	2.30	2.17	2.07	2.00	1.95	1.91	1.87	1.85	1.80	1.75	1.70	1.67	1.64	1.60	1.57	1.53	1.49
28	2.89	2.50	2.29	2.16	2.06	2.00	1.94	1.90	1.87	1.84	1.79	1.74	1.69	1.66	1.63	1.59	1.56	1.52	1.48
29	2.89	2.50	2.28	2.15	2.06	1.99	1.93	1.89	1.86	1.83	1.78	1.73	1.68	1.65	1.62	1.58	1.55	1.51	1.47
30	2.88	2.49	2.28	2.14	2.03	1.98	1.93	1.88	1.85	1.82	1.77	1.72	1.67	1.64	1.61	1.57	1.54	1.50	1.46
40	2.84	2.44	2.23	2.09	2.00	1.93	1.87	1.83	1.79	1.76	1.71	1.66	1.61	1.57	1.54	1.51	1.47	1.42	1.38
60	2.79	2.39	2.18	2.04	1.95	1.87	1.82	1.77	1.74	1.71	1.66	1.60	1.54	1.51	1.48	1.44	1.40	1.35	1.29
120	2.75	2.35	2.13	1.99	1.90	1.82	1.77	1.72	1.68	1.65	1.60	1.55	1.48	1.45	1.41	1.37	1.32	1.26	1.19
∞	2.71	2.30	2.08	1.94	1.85	1.77	1.72	1.67	1.63	1.60	1.55	1.49	1.42	1.38	1.34	1.30	1.24	1.17	1.00

Degrees of freedom for the denominator (v_2)

Table V Percentage Points of the *F*-Distribution (*continued*)

$f_{0.05,v_1,v_2}$

v_2 \ v_1	Degrees of freedom for the numerator (v_1)																		
Degrees of freedom for the denominator (v_2)	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	243.9	245.9	248.0	249.1	250.1	251.1	252.2	253.3	254.3
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.36
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.17	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.55	1.43	1.35	1.25
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

Table V Percentage Points of the *F*-Distribution (*continued*)

$f_{0.025,v_1,v_2}$

v_2 \ v_1	Degrees of freedom for the numerator (v_1)																		
Degrees of freedom for the denominator (v_2)	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
1	647.8	799.5	864.2	899.6	921.8	937.1	948.2	956.7	963.3	968.6	976.7	984.9	993.1	997.2	1001	1006	1010	1014	1018
2	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39	39.40	39.41	39.43	39.45	39.46	39.46	39.47	39.48	39.49	39.50
3	17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47	14.42	14.34	14.25	14.17	14.12	14.08	14.04	13.99	13.95	13.90
4	12.22	10.65	9.98	9.60	9.36	9.20	9.07	8.98	8.90	8.84	8.75	8.66	8.56	8.51	8.46	8.41	8.36	8.31	8.26
5	10.01	8.43	7.76	7.39	7.15	6.98	6.85	6.76	6.68	6.62	6.52	6.43	6.33	6.28	6.23	6.18	6.12	6.07	6.02
6	8.81	7.26	6.60	6.23	5.99	5.82	5.70	5.60	5.52	5.46	5.37	5.27	5.17	5.12	5.07	5.01	4.96	4.90	4.85
7	8.07	6.54	5.89	5.52	5.29	5.12	4.99	4.90	4.82	4.76	4.67	4.57	4.47	4.42	4.36	4.31	4.25	4.20	4.14
8	7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43	4.36	4.30	4.20	4.10	4.00	3.95	3.89	3.84	3.78	3.73	3.67
9	7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10	4.03	3.96	3.87	3.77	3.67	3.61	3.56	3.51	3.45	3.39	3.33
10	6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.85	3.78	3.72	3.62	3.52	3.42	3.37	3.31	3.26	3.20	3.14	3.08
11	6.72	5.26	4.63	4.28	4.04	3.88	3.76	3.66	3.59	3.53	3.43	3.33	3.23	3.17	3.12	3.06	3.00	2.94	2.88
12	6.55	5.10	4.47	4.12	3.89	3.73	3.61	3.51	3.44	3.37	3.28	3.18	3.07	3.02	2.96	2.91	2.85	2.79	2.72
13	6.41	4.97	4.35	4.00	3.77	3.60	3.48	3.39	3.31	3.25	3.15	3.05	2.95	2.89	2.84	2.78	2.72	2.66	2.60
14	6.30	4.86	4.24	3.89	3.66	3.50	3.38	3.29	3.21	3.15	3.05	2.95	2.84	2.79	2.73	2.67	2.61	2.55	2.49
15	6.20	4.77	4.15	3.80	3.58	3.41	3.29	3.20	3.12	3.06	2.96	2.86	2.76	2.70	2.64	2.59	2.52	2.46	2.40
16	6.12	4.69	4.08	3.73	3.50	3.34	3.22	3.12	3.05	2.99	2.89	2.79	2.68	2.63	2.57	2.51	2.45	2.38	2.32
17	6.04	4.62	4.01	3.66	3.44	3.28	3.16	3.06	2.98	2.92	2.82	2.72	2.62	2.56	2.50	2.44	2.38	2.32	2.25
18	5.98	4.56	3.95	3.61	3.38	3.22	3.10	3.01	2.93	2.87	2.77	2.67	2.56	2.50	2.44	2.38	2.32	2.26	2.19
19	5.92	4.51	3.90	3.56	3.33	3.17	3.05	2.96	2.88	2.82	2.72	2.62	2.51	2.45	2.39	2.33	2.27	2.20	2.13
20	5.87	4.46	3.86	3.51	3.29	3.13	3.01	2.91	2.84	2.77	2.68	2.57	2.46	2.41	2.35	2.29	2.22	2.16	2.09
21	5.83	4.42	3.82	3.48	3.25	3.09	2.97	2.87	2.80	2.73	2.64	2.53	2.42	2.37	2.31	2.25	2.18	2.11	2.04
22	5.79	4.38	3.78	3.44	3.22	3.05	2.93	2.84	2.76	2.70	2.60	2.50	2.39	2.33	2.27	2.21	2.14	2.08	2.00
23	5.75	4.35	3.75	3.41	3.18	3.02	2.90	2.81	2.73	2.67	2.57	2.47	2.36	2.30	2.24	2.18	2.11	2.04	1.97
24	5.72	4.32	3.72	3.38	3.15	2.99	2.87	2.78	2.70	2.64	2.54	2.44	2.33	2.27	2.21	2.15	2.08	2.01	1.94
25	5.69	4.29	3.69	3.35	3.13	2.97	2.85	2.75	2.68	2.61	2.51	2.41	2.30	2.24	2.18	2.12	2.05	1.98	1.91
26	5.66	4.27	3.67	3.33	3.10	2.94	2.82	2.73	2.65	2.59	2.49	2.39	2.28	2.22	2.16	2.09	2.03	1.95	1.88
27	5.63	4.24	3.65	3.31	3.08	2.92	2.80	2.71	2.63	2.57	2.47	2.36	2.25	2.19	2.13	2.07	2.00	1.93	1.85
28	5.61	4.22	3.63	3.29	3.06	2.90	2.78	2.69	2.61	2.55	2.45	2.34	2.23	2.17	2.11	2.05	1.98	1.91	1.83
29	5.59	4.20	3.61	3.27	3.04	2.88	2.76	2.67	2.59	2.53	2.43	2.32	2.21	2.15	2.09	2.03	1.96	1.89	1.81
30	5.57	4.18	3.59	3.25	3.03	2.87	2.75	2.65	2.57	2.51	2.41	2.31	2.20	2.14	2.07	2.01	1.94	1.87	1.79
40	5.42	4.05	3.46	3.13	2.90	2.74	2.62	2.53	2.45	2.39	2.29	2.18	2.07	2.01	1.94	1.88	1.80	1.72	1.64
60	5.29	3.93	3.34	3.01	2.79	2.63	2.51	2.41	2.33	2.27	2.17	2.06	1.94	1.88	1.82	1.74	1.67	1.58	1.48
120	5.15	3.80	3.23	2.89	2.67	2.52	2.39	2.30	2.22	2.16	2.05	1.94	1.82	1.76	1.69	1.61	1.53	1.43	1.31
∞	5.02	3.69	3.12	2.79	2.57	2.41	2.29	2.19	2.11	2.05	1.94	1.83	1.71	1.64	1.57	1.48	1.39	1.27	1.00

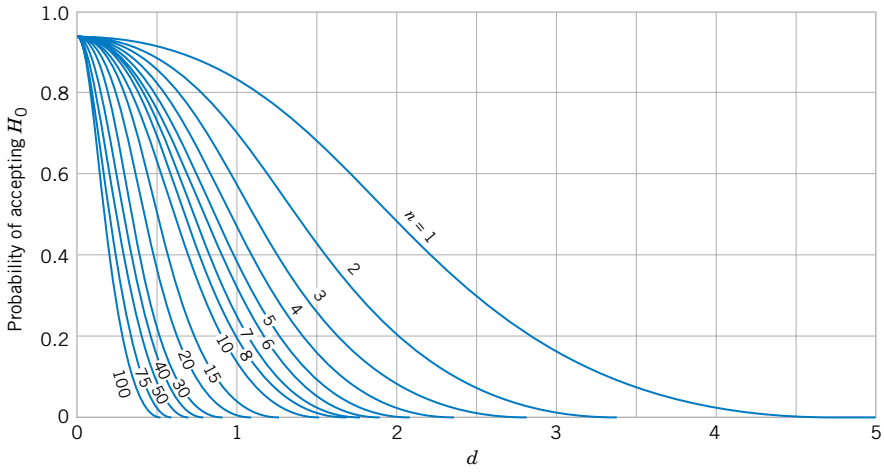
Table V Percentage Points of the *F*-Distribution (*continued*)

$f_{0.01,v_1,v_2}$

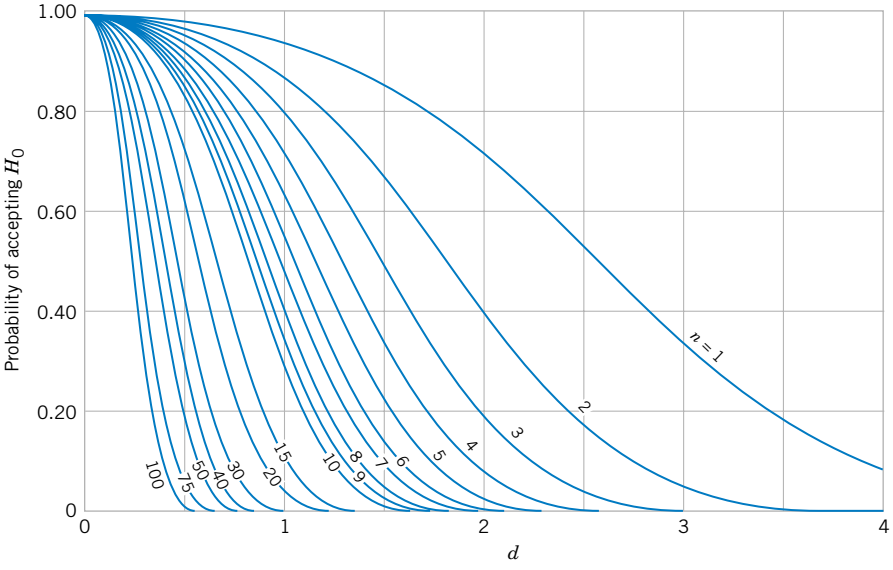
v_2 \ v_1	Degrees of freedom for the numerator (v_1)																		
	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
1	4052	4999.5	5403	5625	5764	5859	5928	5982	6022	6056	6106	6157	6209	6235	6261	6287	6313	6339	6366
2	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39	99.40	99.42	99.43	99.45	99.46	99.47	99.47	99.48	99.49	99.50
3	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35	27.23	27.05	26.87	26.69	26.00	26.50	26.41	26.32	26.22	26.13
4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55	14.37	14.20	14.02	13.93	13.84	13.75	13.65	13.56	13.46
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05	9.89	9.72	9.55	9.47	9.38	9.29	9.20	9.11	9.02
6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.72	7.56	7.40	7.31	7.23	7.14	7.06	6.97	6.88
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62	6.47	6.31	6.16	6.07	5.99	5.91	5.82	5.74	5.65
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81	5.67	5.52	5.36	5.28	5.20	5.12	5.03	4.95	4.46
9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26	5.11	4.96	4.81	4.73	4.65	4.57	4.48	4.40	4.31
10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85	4.71	4.56	4.41	4.33	4.25	4.17	4.08	4.00	3.91
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54	4.40	4.25	4.10	4.02	3.94	3.86	3.78	3.69	3.60
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30	4.16	4.01	3.86	3.78	3.70	3.62	3.54	3.45	3.36
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10	3.96	3.82	3.66	3.59	3.51	3.43	3.34	3.25	3.17
14	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14	4.03	3.94	3.80	3.66	3.51	3.43	3.35	3.27	3.18	3.09	3.00
15	8.68	6.36	5.42	4.89	4.36	4.32	4.14	4.00	3.89	3.80	3.67	3.52	3.37	3.29	3.21	3.13	3.05	2.96	2.87
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69	3.55	3.41	3.26	3.18	3.10	3.02	2.93	2.84	2.75
17	8.40	6.11	5.18	4.67	4.34	4.10	3.93	3.79	3.68	3.59	3.46	3.31	3.16	3.08	3.00	2.92	2.83	2.75	2.65
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51	3.37	3.23	3.08	3.00	2.92	2.84	2.75	2.66	2.57
19	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43	3.30	3.15	3.00	2.92	2.84	2.76	2.67	2.58	2.59
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37	3.23	3.09	2.94	2.86	2.78	2.69	2.61	2.52	2.42
21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40	3.31	3.17	3.03	2.88	2.80	2.72	2.64	2.55	2.46	2.36
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26	3.12	2.98	2.83	2.75	2.67	2.58	2.50	2.40	2.31
23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30	3.21	3.07	2.93	2.78	2.70	2.62	2.54	2.45	2.35	2.26
24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17	3.03	2.89	2.74	2.66	2.58	2.49	2.40	2.31	2.21
25	7.77	5.57	4.68	4.18	3.85	3.63	3.46	3.32	3.22	3.13	2.99	2.85	2.70	2.62	2.54	2.45	2.36	2.27	2.17
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18	3.09	2.96	2.81	2.66	2.58	2.50	2.42	2.33	2.23	2.13
27	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.26	3.15	3.06	2.93	2.78	2.63	2.55	2.47	2.38	2.29	2.20	2.10
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12	3.03	2.90	2.75	2.60	2.52	2.44	2.35	2.26	2.17	2.06
29	7.60	5.42	4.54	4.04	3.73	3.50	3.33	3.20	3.09	3.00	2.87	2.73	2.57	2.49	2.41	2.33	2.23	2.14	2.03
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98	2.84	2.70	2.55	2.47	2.39	2.30	2.21	2.11	2.01
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80	2.66	2.52	2.37	2.29	2.20	2.11	2.02	1.92	1.80
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63	2.50	2.35	2.20	2.12	2.03	1.94	1.84	1.73	1.60
120	6.85	4.79	3.95	3.48	3.17	2.96	2.79	2.66	2.56	2.47	2.34	2.19	2.03	1.95	1.86	1.76	1.66	1.53	1.38
∞	6.63	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41	2.32	2.18	2.04	1.88	1.79	1.70	1.59	1.47	1.32	1.00

Degrees of freedom for the denominator (v_2)

Chart VI Operating Characteristic Curves



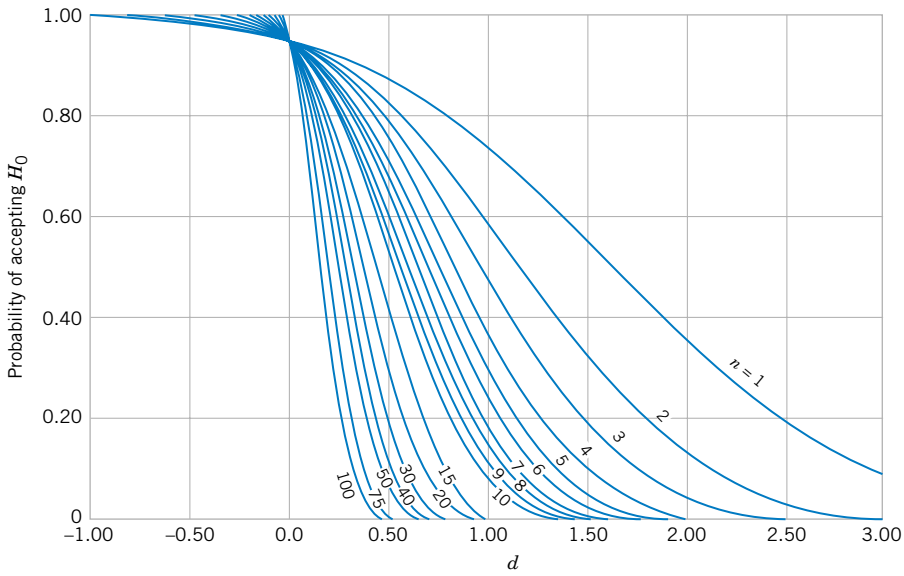
(*a*) *O.C.* curves for different values of n for the two-sided normal test for a level of significance $\alpha = 0.05$.



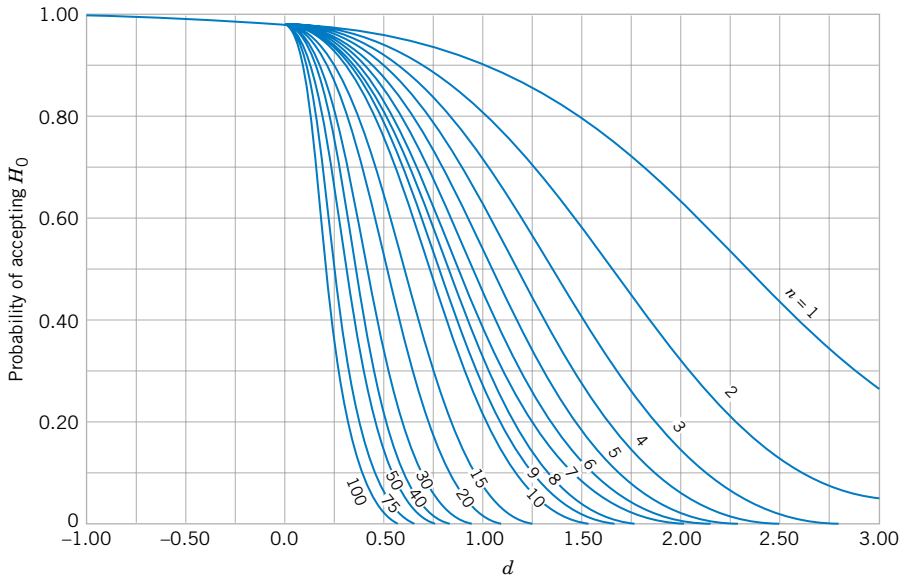
(*b*) *O.C.* curves for different values of n for the two-sided normal test for a level of significance $\alpha = 0.01$.

Source: Charts VI*a, e, f, k, m,* and *q* are reproduced with permission from "Operating Characteristics for the Common Statistical Tests of Significance," by C. L. Ferris, F. E. Grubbs, and C. L. Weaver, *Annals of Mathematical Statistics,* June 1946.
Charts VI*b, c, d, g, h, i, j, l, n, o, p,* and *r* are reproduced with permission from *Engineering Statistics*, 2nd Edition, by A. H. Bowker and G. J. Lieberman, Prentice-Hall, 1972.

Chart VI Operating Characteristic Curves (*continued*)

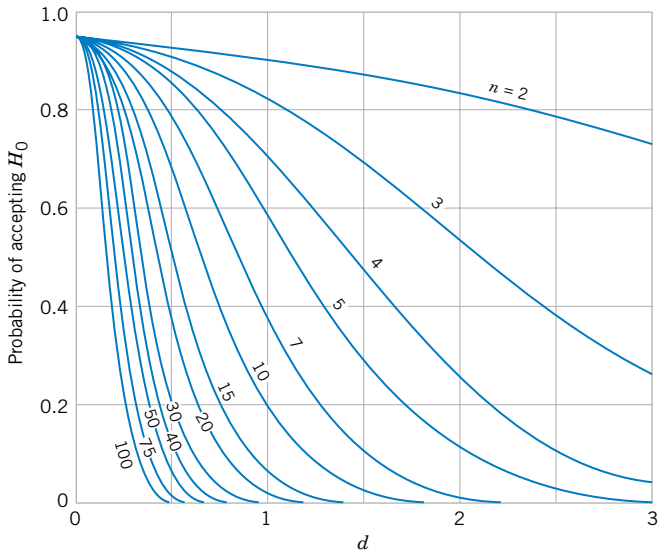


(*c*) *O.C.* curves for different values of n for the one-sided normal test for a level of significance $\alpha = 0.05$.

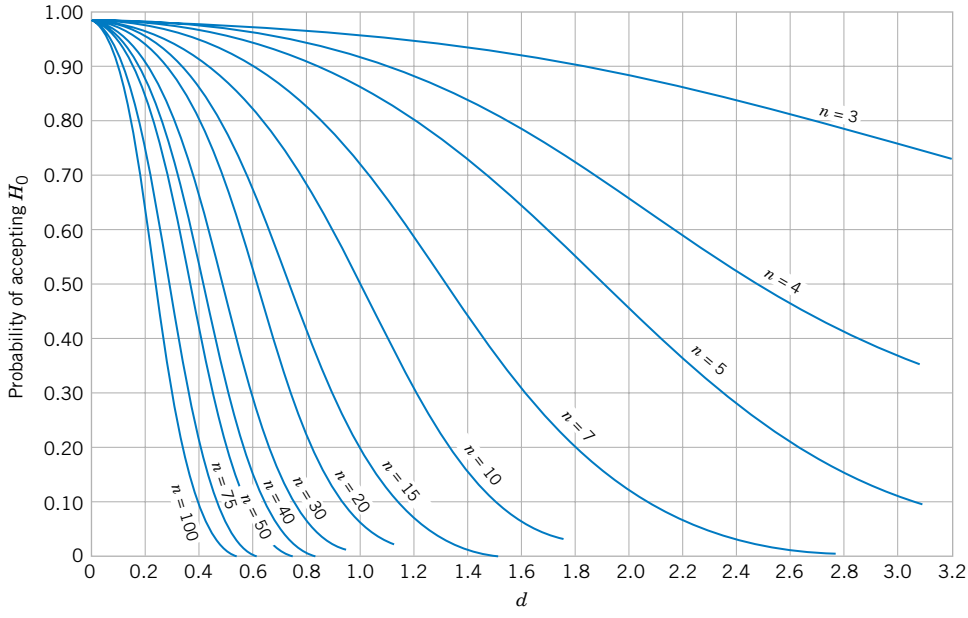


(*d*) *O.C.* curves for different values of n for the one-sided normal test for a level of significance $\alpha = 0.01$.

Chart VI Operating Characteristic Curves (*continued*)

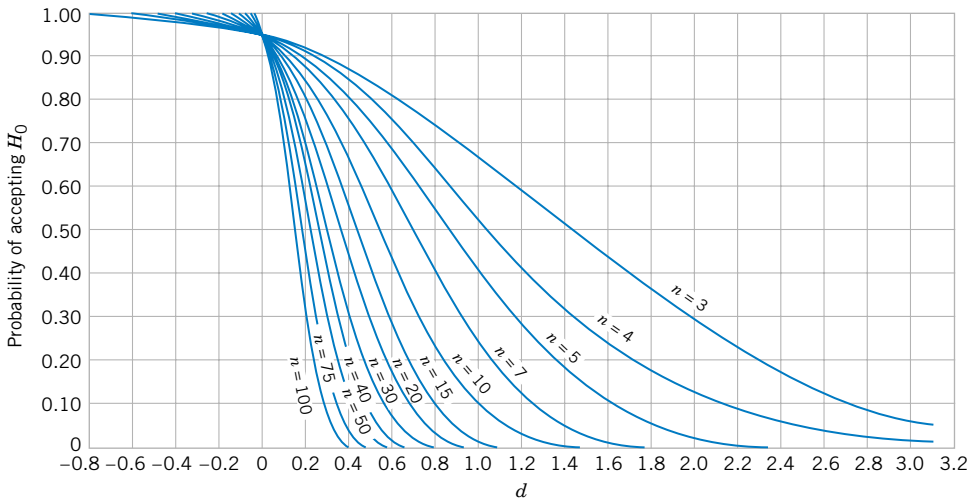


(*e*) *O.C.* curves for different values of n for the two-sided t-test for a level of significance $\alpha = 0.05$.

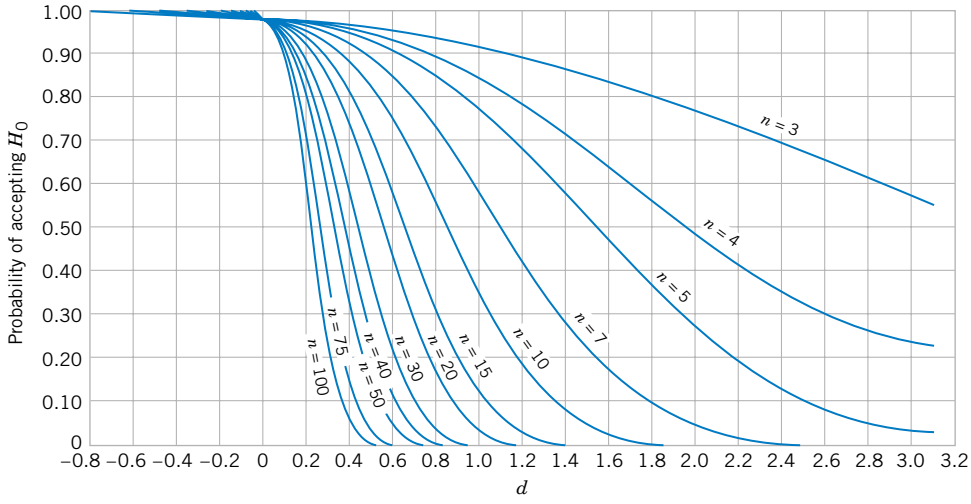


(*f*) *O.C.* curves for different values of n for the two-sided t-test for a level of significance $\alpha = 0.01$.

Chart VI Operating Characteristic Curves (*continued*)

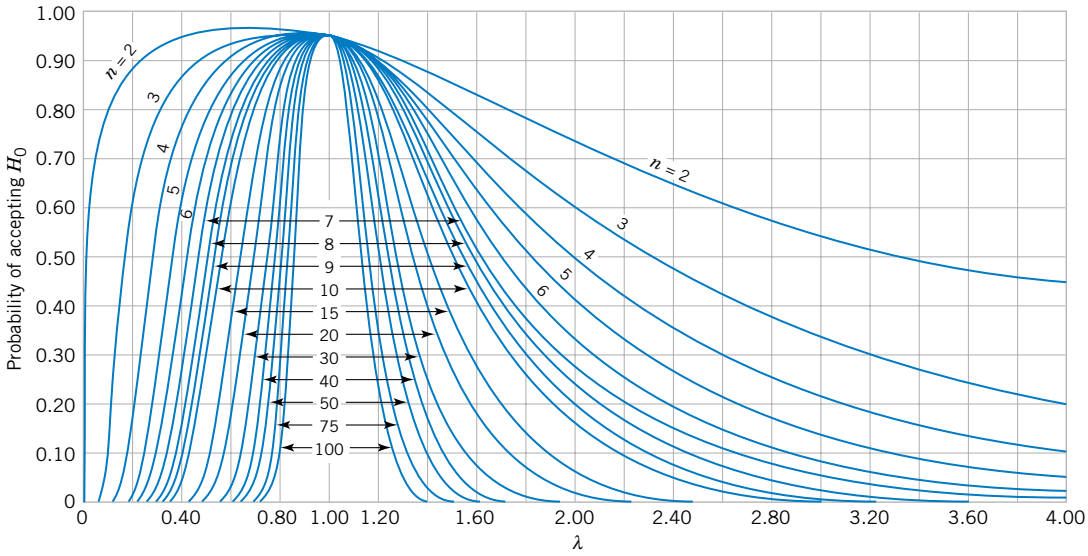


(*g*) *O.C.* curves for different values of n for the one-sided t-test for a level of significance $\alpha = 0.05$.

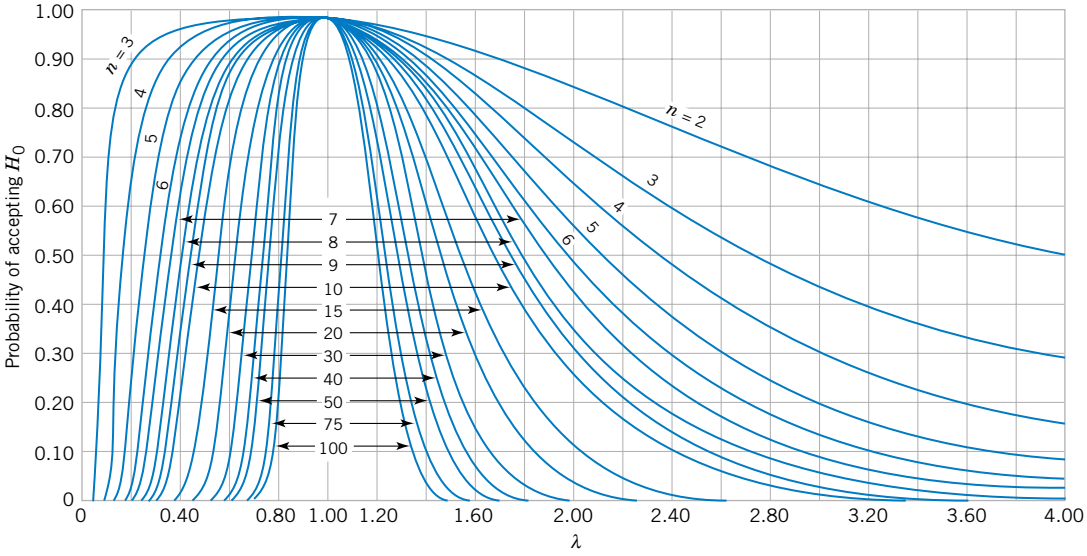


(*h*) *O.C.* curves for different values of n for the one-sided t-test for a level of significance $\alpha = 0.01$.

Chart VI Operating Characteristic Curves (*continued*)

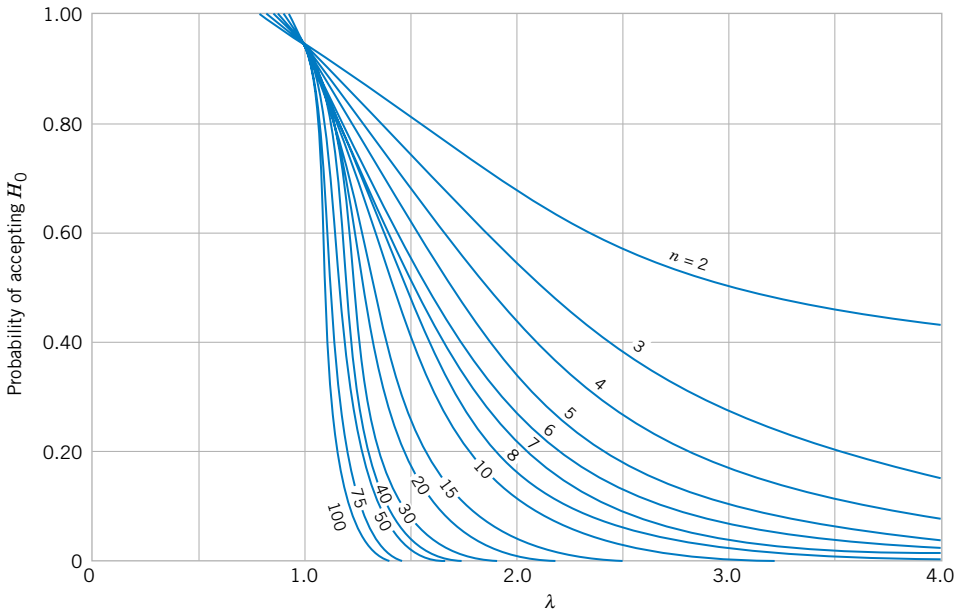


(*i*) *O.C.* curves for different values of n for the two-sided chi-square test for a level of significance $\alpha = 0.05$.

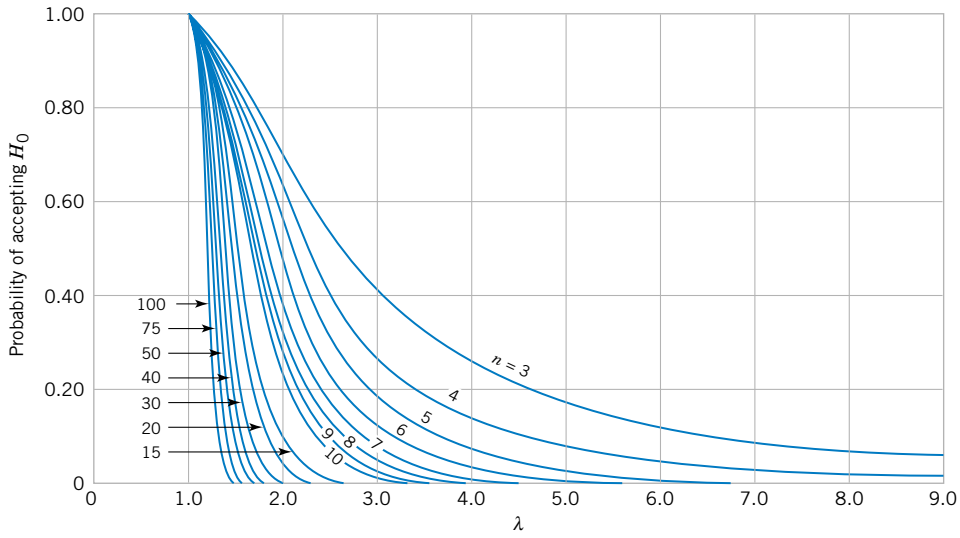


(*j*) *O.C.* curves for different values of n for the two-sided chi-square test for a level of significance $\alpha = 0.01$.

Chart VI Operating Characteristic Curves (*continued*)

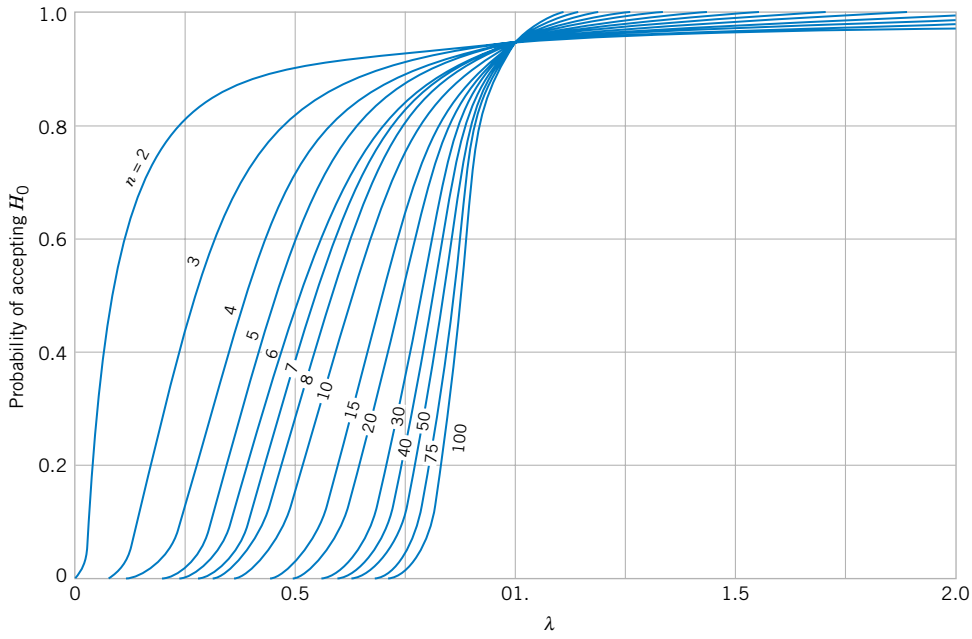


(*k*) *O.C.* curves for different values of *n* for the one-sided (upper tail) chi-square test for a level of significance $\alpha = 0.05$.

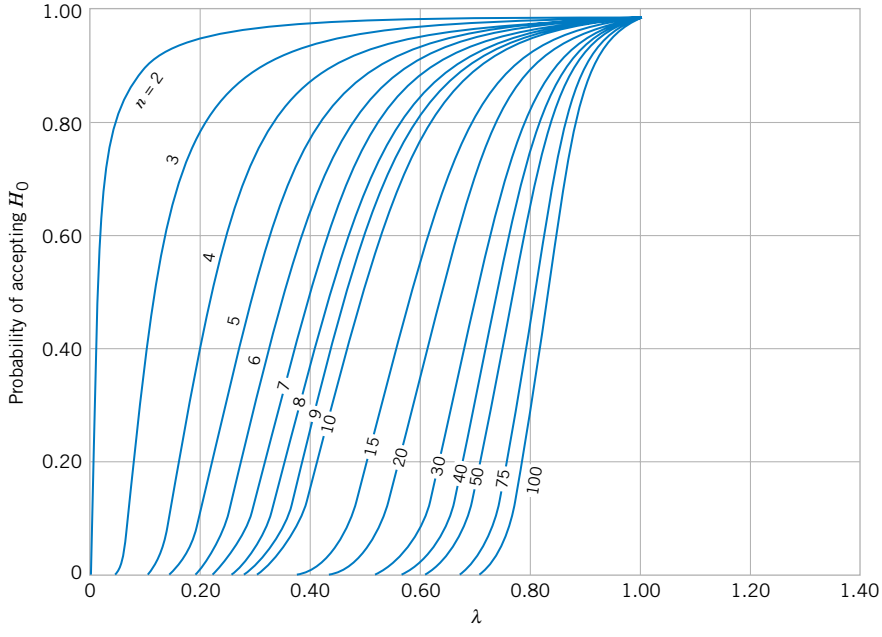


(*l*) *O.C.* curves for different values of *n* for the one-sided (upper tail) chi-square test for a level of significance $\alpha = 0.01$.

Chart VI Operating Characteristic Curves (*continued*)

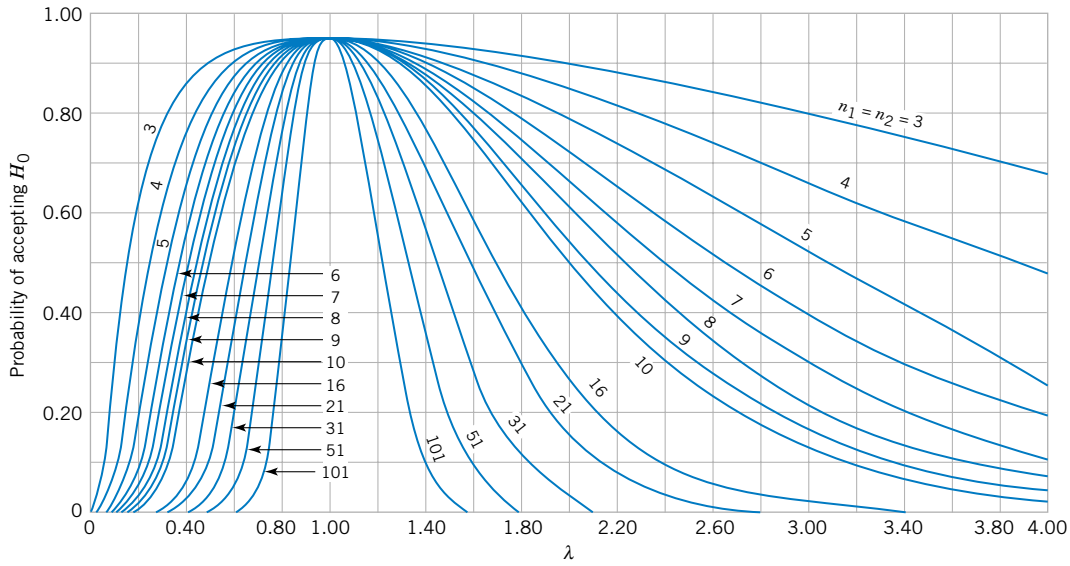


(*m*) *O.C.* curves for different values of *n* for the one-sided (lower tail) chi-square test for a level of significance $\alpha = 0.05$.

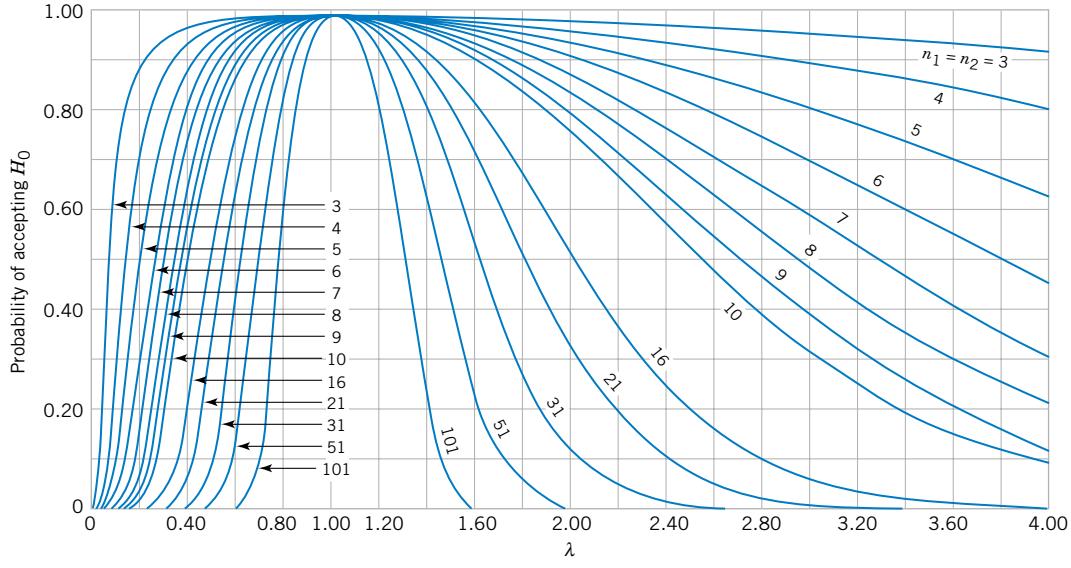


(*n*) *O.C.* curves for different values of *n* for the one-sided (lower tail) chi-square test for a level of significance $\alpha = 0.01$.

Chart VI Operating Characteristic Curves (*continued*)

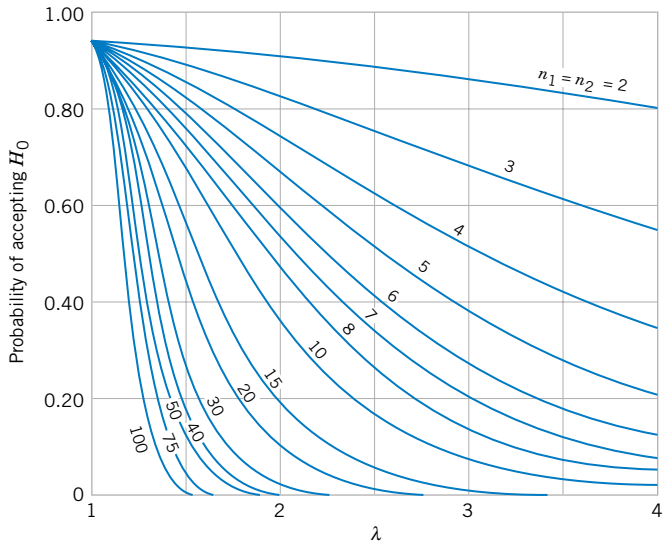


(*o*) *O.C.* curves for different values of n for the two-sided F-test for a level of significance $\alpha = 0.05$.

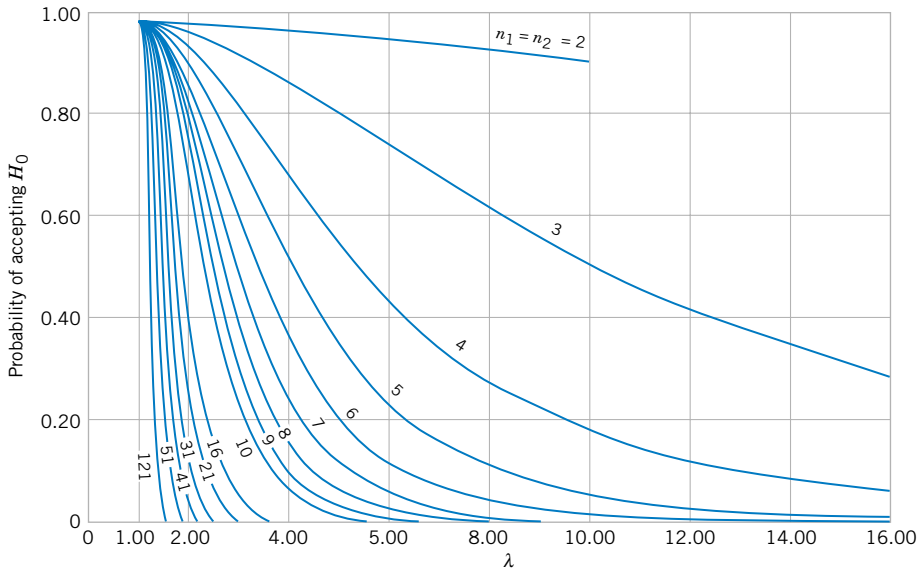


(*p*) O.C. curves for different values of n for the two-sided F-test for a level of significance $\alpha = 0.01$.

Chart VI Operating Characteristic Curves (*continued*)



(*q*) *O.C.* curves for different values of *n* for the one-sided *F*-test for a level of significance $\alpha = 0.05$.



(*r*) *O.C.* curves for different values of *n* for the one-sided *F*-test for a level of significance $\alpha = 0.01$.

Table VII Critical Values for the Sign Test

r^*_α

n \ α	0.10	0.05	0.01	Two-sided tests	n \ α	0.10	0.05	0.01	Two-sided tests
	0.05	0.025	0.005	One-sided tests		0.05	0.025	0.005	One-sided tests
5	0				23	7	6	4	
6	0	0			24	7	6	5	
7	0	0			25	7	7	5	
8	1	0	0		26	8	7	6	
9	1	1	0		27	8	7	6	
10	1	1	0		28	9	8	6	
11	2	1	0		29	9	8	7	
12	2	2	1		30	10	9	7	
13	3	2	1		31	10	9	7	
14	3	2	1		32	10	9	8	
15	3	3	2		33	11	10	8	
16	4	3	2		34	11	10	9	
17	4	4	2		35	12	11	9	
18	5	4	3		36	12	11	9	
19	5	4	3		37	13	12	10	
20	5	5	3		38	13	12	10	
21	6	5	4		39	13	12	11	
22	6	5	4		40	14	13	11	

Table VIII Critical Values for the Wilcoxon Signed-Rank Test

w^*_α

n^* \ α	0.10	0.05	0.02	0.01	Two-sided tests
	0.05	0.025	0.01	0.005	One-sided tests
4					
5	0				
6	2	0			
7	3	2	0		
8	5	3	1	0	
9	8	5	3	1	
10	10	8	5	3	
11	13	10	7	5	
12	17	13	9	7	
13	21	17	12	9	
14	25	21	15	12	
15	30	25	19	15	
16	35	29	23	19	
17	41	34	27	23	
18	47	40	32	27	
19	53	46	37	32	
20	60	52	43	37	
21	67	58	49	42	
22	75	65	55	48	
23	83	73	62	54	
24	91	81	69	61	
25	100	89	76	68	

* If $n > 25$, W^+ (or W^-) is approximately normally distributed with mean $n(n + 1)/4$ and variance $n(n + 1)(2n + 1)/24$.

Table IX Critical Values for the Wilcoxon Rank-Sum Test

$w_{0.05}$

n_2 \ n_1*	4	5	6	7	8	9	10	11	12	13	14	15
4	10											
5	11	17										
6	12	18	26									
7	13	20	27	36								
8	14	21	29	38	49							
9	15	22	31	40	51	63						
10	15	23	32	42	53	65	78					
11	16	24	34	44	55	68	81	96				
12	17	26	35	46	58	71	85	99	115			
13	18	27	37	48	60	73	88	103	119	137		
14	19	28	38	50	63	76	91	106	123	141	160	
15	20	29	40	52	65	79	94	110	127	145	164	185
16	21	31	42	54	67	82	97	114	131	150	169	
17	21	32	43	56	70	84	100	117	135	154		
18	22	33	45	58	72	87	103	121	139			
19	23	34	46	60	74	90	107	124				
20	24	35	48	62	77	93	110					
21	25	37	50	64	79	95						
22	26	38	51	66	82							
23	27	39	53	68								
24	28	40	55									
25	28	42										
26	29											
27												
28												

*For n_1 and $n_2 > 8$, W_1 is approximately normally distributed with mean $\frac{1}{2}n_1(n_1 + n_2 + 1)$ and variance $n_1n_2(n_1 + n_2 + 1)/12$.

Table IX Critical Values for the Wilcoxon Rank-Sum Test (*continued*)

$w_{0.01}$

n_2 \ n_1	4	5	6	7	8	9	10	11	12	13	14	15
5		15										
6	10	16	23									
7	10	17	24	32								
8	11	17	25	34	43							
9	11	18	26	35	45	56						
10	12	19	27	37	47	58	71					
11	12	20	28	38	49	61	74	87				
12	13	21	30	40	51	63	76	90	106			
13	14	22	31	41	53	65	79	93	109	125		
14	14	22	32	43	54	67	81	96	112	129	147	
15	15	23	33	44	56	70	84	99	115	133	151	171
16	15	24	34	46	58	72	86	102	119	137	155	
17	16	25	36	47	60	74	89	105	122	140		
18	16	26	37	49	62	76	92	108	125			
19	17	27	38	50	64	78	94	111				
20	18	28	39	52	66	81	97					
21	18	29	40	53	68	83						
22	19	29	42	55	70							
23	19	30	43	57								
24	20	31	44									
25	20	32										
26	21											
27												
28												

Table X Factors for Constructing Variables Control Charts

	Factor for Control Limits						
	$\overline{X}$ Chart			R Chart		S Chart	
n*	A_1	A_2	d_2	D_3	D_4	c_4	n
2	3.760	1.880	1.128	0	3.267	0.7979	2
3	2.394	1.023	1.693	0	2.575	0.8862	3
4	1.880	.729	2.059	0	2.282	0.9213	4
5	1.596	.577	2.326	0	2.115	0.9400	5
6	1.410	.483	2.534	0	2.004	0.9515	6
7	1.277	.419	2.704	.076	1.924	0.9594	7
8	1.175	.373	2.847	.136	1.864	0.9650	8
9	1.094	.337	2.970	.184	1.816	0.9693	9
10	1.028	.308	3.078	.223	1.777	0.9727	10
11	.973	.285	3.173	.256	1.744	0.9754	11
12	.925	.266	3.258	.284	1.716	0.9776	12
13	.884	.249	3.336	.308	1.692	0.9794	13
14	.848	.235	3.407	.329	1.671	0.9810	14
15	.816	.223	3.472	.348	1.652	0.9823	15
16	.788	.212	3.532	.364	1.636	0.9835	16
17	.762	.203	3.588	.379	1.621	0.9845	17
18	.738	.194	3.640	.392	1.608	0.9854	18
19	.717	.187	3.689	.404	1.596	0.9862	19
20	.697	.180	3.735	.414	1.586	0.9869	20
21	.679	.173	3.778	.425	1.575	0.9876	21
22	.662	.167	3.819	.434	1.566	0.9882	22
23	.647	.162	3.858	.443	1.557	0.9887	23
24	.632	.157	3.895	.452	1.548	0.9892	24
25	.619	.153	3.931	.459	1.541	0.9896	25

*$n > 25$: $A_1 = 3/\sqrt{n}$ where $n =$ number of observations in sample.

Table XI Factors for Tolerance Intervals

Values of k for Two-Sided Intervals									
Confidence Level	0.90			0.95			0.99		
Percent Coverage	0.90	0.95	0.99	0.90	0.95	0.99	0.90	0.95	0.99
2	15.978	18.800	24.167	32.019	37.674	48.430	160.193	188.491	242.300
3	5.847	6.919	8.974	8.380	9.916	12.861	18.930	22.401	29.055
4	4.166	4.943	6.440	5.369	6.370	8.299	9.398	11.150	14.527
5	3.949	4.152	5.423	4.275	5.079	6.634	6.612	7.855	10.260
6	3.131	3.723	4.870	3.712	4.414	5.775	5.337	6.345	8.301
7	2.902	3.452	4.521	3.369	4.007	5.248	4.613	5.488	7.187
8	2.743	3.264	4.278	3.136	3.732	4.891	4.147	4.936	6.468
9	2.626	3.125	4.098	2.967	3.532	4.631	3.822	4.550	5.966
10	2.535	3.018	3.959	2.839	3.379	4.433	3.582	4.265	5.594
11	2.463	2.933	3.849	2.737	3.259	4.277	3.397	4.045	5.308
12	2.404	2.863	3.758	2.655	3.162	4.150	3.250	3.870	5.079
13	2.355	2.805	3.682	2.587	3.081	4.044	3.130	3.727	4.893
14	2.314	2.756	3.618	2.529	3.012	3.955	3.029	3.608	4.737
15	2.278	2.713	3.562	2.480	2.954	3.878	2.945	3.507	4.605
16	2.246	2.676	3.514	2.437	2.903	3.812	2.872	3.421	4.492
17	2.219	2.643	3.471	2.400	2.858	3.754	2.808	3.345	4.393
18	2.194	2.614	3.433	2.366	2.819	3.702	2.753	3.279	4.307
19	2.172	2.588	3.399	2.337	2.784	3.656	2.703	3.221	4.230
20	2.152	2.564	3.368	2.310	2.752	3.615	2.659	3.168	4.161
21	2.135	2.543	3.340	2.286	2.723	3.577	2.620	3.121	4.100
22	2.118	2.524	3.315	2.264	2.697	3.543	2.584	3.078	4.044
23	2.103	2.506	3.292	2.244	2.673	3.512	2.551	3.040	3.993
24	2.089	2.489	3.270	2.225	2.651	3.483	2.522	3.004	3.947
25	2.077	2.474	3.251	2.208	2.631	3.457	2.494	2.972	3.904
30	2.025	2.413	3.170	2.140	2.529	3.350	2.385	2.841	3.733
40	1.959	2.334	3.066	2.052	2.445	3.213	2.247	2.677	3.518
50	1.916	2.284	3.001	1.996	2.379	3.126	2.162	2.576	3.385
60	1.887	2.248	2.955	1.958	2.333	3.066	2.103	2.506	3.293
70	1.865	2.222	2.920	1.929	2.299	3.021	2.060	2.454	3.225
80	1.848	2.202	2.894	1.907	2.272	2.986	2.026	2.414	3.173
90	1.834	2.185	2.872	1.889	2.251	2.958	1.999	2.382	3.130
100	1.822	2.172	2.854	1.874	2.233	2.934	1.977	2.355	3.096

Table XI Factors for Tolerance Intervals (*continued*)

Values of k for One-Sided Intervals									
Confidence Level	0.90			0.95			0.99		
Percent Coverage	0.90	0.95	0.99	0.90	0.95	0.99	0.90	0.95	0.99
2	10.253	13.090	18.500	20.581	26.260	37.094	103.029	131.426	185.617
3	4.258	5.311	7.340	6.155	7.656	10.553	13.995	17.370	23.896
4	3.188	3.957	5.438	4.162	5.144	7.042	7.380	9.083	12.387
5	2.742	3.400	4.666	3.407	4.203	5.741	5.362	6.578	8.939
6	2.494	3.092	4.243	3.006	3.708	5.062	4.411	5.406	7.335
7	2.333	2.894	3.972	2.755	3.399	4.642	3.859	4.728	6.412
8	2.219	2.754	3.783	2.582	3.187	4.354	3.497	4.285	5.812
9	2.133	2.650	3.641	2.454	3.031	4.143	3.240	3.972	5.389
10	2.066	2.568	3.532	2.355	2.911	3.981	3.048	3.738	5.074
11	2.011	2.503	3.443	2.275	2.815	3.852	2.898	3.556	4.829
12	1.966	2.448	3.371	2.210	2.736	3.747	2.777	3.410	4.633
13	1.928	2.402	3.309	2.155	2.671	3.659	2.677	3.290	4.472
14	1.895	2.363	3.257	2.109	2.614	3.585	2.593	3.189	4.337
15	1.867	2.329	3.212	2.068	2.566	3.520	2.521	3.102	4.222
16	1.842	2.299	3.172	2.033	2.524	3.464	2.459	3.028	4.123
17	1.819	2.272	3.137	2.002	2.486	3.414	2.405	2.963	4.037
18	1.800	2.249	3.105	1.974	2.453	3.370	2.357	2.905	3.960
19	1.782	2.227	3.077	1.949	2.423	3.331	2.314	2.854	3.892
20	1.765	2.028	3.052	1.926	2.396	3.295	2.276	2.808	3.832
21	1.750	2.190	3.028	1.905	2.371	3.263	2.241	2.766	3.777
22	1.737	2.174	3.007	1.886	2.349	3.233	2.209	2.729	3.727
23	1.724	2.159	2.987	1.869	2.328	3.206	2.180	2.694	3.681
24	1.712	2.145	2.969	1.853	2.309	3.181	2.154	2.662	3.640
25	1.702	2.132	2.952	1.838	2.292	3.158	2.129	2.633	3.601
30	1.657	2.080	2.884	1.777	2.220	3.064	2.030	2.515	3.447
40	1.598	2.010	2.793	1.697	2.125	2.941	1.902	2.364	3.249
50	1.559	1.965	2.735	1.646	2.065	2.862	1.821	2.269	3.125
60	1.532	1.933	2.694	1.609	2.022	2.807	1.764	2.202	3.038
70	1.511	1.909	2.662	1.581	1.990	2.765	1.722	2.153	2.974
80	1.495	1.890	2.638	1.559	1.964	2.733	1.688	2.114	2.924
90	1.481	1.874	2.618	1.542	1.944	2.706	1.661	2.082	2.883
100	1.470	1.861	2.601	1.527	1.927	2.684	1.639	2.056	2.850

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RV COLLEGE OF ENGINEERING®

(Autonomous Institution affiliated to VTU)

IV Semester B. E. Fast Track Examinations July-19

Industrial Engineering and Management

ENGINEERING STATISTICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of statistical tables and probability distribution formulae permitted.

PART A

1	1.1	The graphical tool for identifying outlier in a given dataset is _____.	01
	1.2	Give any one area of application for the radar diagram (illustrate the graphical plot of radar diagram).	01
	1.3	Suppose that you are required to apply the definition of probability to estimate the number of occupational hazards in a shop floor environment due to lack of safe practices, which of the following would you choose? i. Classical definition ii. Limit frequency approach to probability iii. Expert judgement based probability and why?	02
	1.4	Give the probability distribution function for a hyper geometric distribution with the usual notation.	01
	1.5	If X is defined as the number of electrons hitting a screen in a given interval of time, which distribution is most appropriate for the random variable X.	01
	1.6	Given X is a random variable representing the number of times failures due to CPU faults is poisson distributed and Y is a random variable representing the time between failures is _____ distributed.	01
	1.7	Suppose $f(x) = \frac{x}{8}$ $for\ 3 < x < 5$. Determine the following probability $p(x < 4)$	01
	1.8	Define the property of unbiasedness of estimator.	01
	1.9	For small samples, the t distribution is more appropriate to a normal distribution (z-distribution). Show through sketch the comparison between a normal distribution and the students t distribution.	01
	1.10	Suppose that $\hat{\theta}_1$ and $\hat{\theta}_2$are unbiased estimates of the parameter θ. We know that $V(\hat{\theta}_1) = 10; V(\hat{\theta}_2) = 04$. Which estimator is the best and in what sense it is best?	02
	1.11	Mention any one of the common abuses of regression.	01
	1.12	Given any one reason as to why the scatter plot (diagram) between the dependent variable (say purity level) and the independent variable (say hydrocarbon level) is the first tool used for data visualization.	01

	1.13	If the residuals from a regression model is defined as $e_i = y_i - \hat{y}_i$, i=1,2……n, where y_i is an actual observation and $\hat{y}_i$ is the fitted value from the regression model. Show the pattern for residual plot which can be termed as satisfactory.	01
	1.14	Suppose the effective life of a cutting tool depends on the cutting speed and the tool angle. Propose a multiple regression model that might be used to describe the relationship.	01
	1.15	Distinguish between point estimate and an interval estimate (list any two distinctions).	02
	1.16	Mention the test statistic for carrying out goodness of fit test for a hypothesized distribution, with usual notation.	01
	1.17	Suppose H_o-Null hypothesis is given as $\mu = H_o$ (single sample mean test).List the three possible alternative ways of defining the alternate hypothesis. (use the usually accepted convention to define alternate hypothesis).	01

PART B

2	a	The following are the weight of $50kg$ cement bags measured at the terminal inspection stage after packing (weight in kilograms) 50.4 | 50.6 | 49.8 | 50.3 | 50.5 | 50.3 | 50.2 | 49.9 i. Calculate the mean, median, variance and standard deviation. ii. Construct the boxplot for the data.	08
	b	The probability is 1% that an electric connector that is kept dry fails during warranty period of a portable computer. If the connector is ever wet, the probability of failure during warranty period is 5%. If 90% of the connectors are kept dry and 10% are wet, what proportion of connectors fail during the warranty period.	06
	c	State Baye's law.	02

3	a	Let 'x' denote the number of bits received in error in a digital communication channel and assume that 'X' is a binomial random variable with $p = 0.001. If$ 1000 bits are transmitted determine the following i. $P(x = 1)$ ii. $P(x \geq 1)$ iii. $P(X \leq 2)$ iv. $Mean\ and\ variance\ of\ X.$	06
	b	The number of flows in both of cloth in textile manufacturing is assumed to be poisson distributed with a mean of 0.1 flaw per square meter. i. What is the probability that there are two flaws in 1 square meter of cloth? ii. What is the probability that there is one flow in 10 square meters of cloth? iii. What is the probability that here are atleast two flaws in 10 square meters of cloth?	06
	c	Define a Geometric random variable. Give an illustrative example.	04
		OR	

4	a	The compressive strength of samples of cement can be modeled by a normal distribution with a mean of 6000 kilograms per square centimeter and a standard deviation of 100 kilograms per square centimeter. i. What is the probability that a sample strength is less than 6250 kg/cm^2? ii. What is the probability that a sample strength is between 5800 and5900 kg/cm^2? iii. What strength is exceeded by 95% of the samples?	06
	b	The time to failure (in hours) of fans in a personal computer can be modeled by an exponential distribution with $\lambda = 0.003$ fialures/hr. i. What proportion of fans will last atleast10,000 $hours$? ii. What proportion of fans will last atmost7,000 $hours$?	06
	c	The thickness of conductive coating has density function of $600x^{-2}$for $100\mu m < x < 120\mu m$. i. Determine mean and variance of coating thickness. ii. If the coating cost \$0.50 per micrometer of thickness on each part, what is the average cost of the coating per part?	04

5	a	Consider the Poisson distribution $f(x) = \frac{e^{-\lambda}\lambda^x}{\angle x} \qquad x = 0,1,2 \ldots.$ Find the maximum likelihood estimator of λ, based on a random sample of size n.	06
	b	Consider the probability density function $f(x) = c(1 + \theta x) \qquad -1 \le x \le 1$ i. Find the value of the constant C ii. What is the moment estimator for θ? iii. Show that $\hat{\theta} = 3\bar{x}$ is an unbiased estimator for θ. iv. Find the maximum likelihood estimator for θ.	10
		OR	
6	a	PVC pipe is manufactured with a mean diameter of 1.01 $inch$ and a standard deviation of0.003 $inch$.Find the probability that a random sample of $n = 9$ sections of pipe will have a sample mean diameter greater than 1.009 inch and less than 1.012$inch$.	06
	b	Define Minimum variance unbiased estimator. Give an example.	04
	c	A normal population has a mean 100 and variance 25. How large must the random sample be if we want the standard error of the sample average to be 1.5?If the variance is reduced to 16 keep the mean at the same level what should be the size of the sample, assuming standard error of the sample average to be at the same level of 1.5?	06

7	a	Cost accountants often estimate overhead based on level of production. At the M/S Standard Knitting company, data has been collected on overhead expenses and units produced at different plants and is as under	

overhead	191	170	172	155	280	173	234	116	153	178
units	40	42	53	35	56	39	48	30	37	40

i. Draw the scatter plot for the data.
ii. Develop the regression line based on the data. Plot the line on the scatter plot.
iii. Predict the overhead when 50 units are produced.
iv. Calculate the standard error of estimate. 12

	b	What are the common abuses of regression models? Illustratively depict these on a graphical plot of a hypothetical data set.	04

8	a	The life in hours of battery is known to be approximately normally distributed with standard deviation $\sigma = 1.25 hours$. A random sample of 10 batteries has a mean life of $\overline{X} = 40.5\ hours$. i. Is there evidence to support the claim that the battery life exceeds $40 hrs$?use $\propto = 0.05$. ii. What is the p-value for the test? iii. What is the $\beta - error$ for the test in part (i) above if the true mean life is 42hours?.	08
	b	Write down the following i. Test statistic for testing the goodness of fit with respect to hypothesized distribution. ii. Criteria for the acceptance of a proposed null hypothesis for the testing of goodness of fit iii. Usefulness of the goodness of fit test.	04
	c	In each of the following situations state whether it is a correctly stated hypothesis testing problem and why. i. $H_o: \mu = 25$ $H_1 \mu \neq 25$ ii. $H_o: \sigma > 10$ $H_1 \sigma = 25$ iii. $H_o: p = 0.1$ $H_1 p = 0.5$ iv. $H_o: S = 30$ $H_1 S > 30$	04

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU

IV Semester B. E. Examinations April/May-19

Industrial Engineering and Management

ENGINEERING STATISTICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of statistical tables and probability distribution formulae permitted.

PART A

1	1.1	Suppose that we have a fuse box containing 20 fuses, of which 5 are defective. If 2 fuses are selected at random and removed from the box in succession without replacing the first, what is the probability that both fuses are defective?	02
	1.2	If a sample space contains a finite number of possibilities or an unending sequence with as many elements as there are whole numbers, it is called a _____ sample space.	01
	1.3	A shipment of 8 similar microcomputers to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the probability distribution for the number of defectives.	02
	1.4	What is a Poisson experiment?	01
	1.5	Physical measurements in areas such as meteorological experiment, rainfall studies, and measurements of manufactured parts are often more than adequately explained with _____ distribution.	01
	1.6	The normal distribution if often a good approximation to a discrete distribution when the latter takes on a _____.	01
	1.7	The special gamma distribution for which $a = 1$ is called the _____ distribution.	01
	1.8	An industrial engineer may be interested in modeling the time T between arrivals at a congested intersection during rush hour in a large city. An arrival represents the _____ event. (State the name of appropriate probability distribution)	01
	1.9	List any two probability distributions that are applied to reliability and life testing problems such as the time to failure.	02
	1.10	For most populations, if the sample size is greater than _____, the Central Limit theorem approximation is good.	01
	1.11	The _____ is an adjustment, made when approximating a discrete distribution with a continuous one, that can improve the accuracy of approximation.	01
	1.12	What is confidence level?	01
	1.13	The sample mean and standard deviation for the cold cranking amperages of 100 batteries are 185.5 and 5.0 respectively. Find the standard error that is used in interval estimation.	01

	1.14	The hypothesis test on population proportion requires that the sample proportion be approximately _____ distributed.	01
	1.15	State any two assumptions of errors of a linear regression model.	02
	1.16	When the vertical spread in a scatter plot doesn't vary too much, the scatter plot is said to be _____.	01

PART B

2	a	A university warehouse has received a shipment of 25 printers, of which 10 are laser printers and 15 are inkjet models. If 6 of these 25 are selected at random to be checked by a particular technician, what is the probability that exactly 3 of those selected are laser printers (so that the other 3 are inkjets)? What is the probability that exactly 4 of the printers selected are inkjet models?	06
	b	The tragedy that befell the space shuttle Challenger and its astronauts in 1986 led to a number of studies to investigate the reasons for mission failure. Attention quickly focused on the behavior of the rocket engine's O-rings. Here is data consisting of observations on $x = O$-ring temperature(°F) for each test firing or actual launch of shuttle rocket engine (Presidential Commission on the Space Shuttle Challenger Accident, $Vol.1,1986: 129 - 131$). 84 49 61 40 83 67 45 66 70 69 80 58 68 60 67 72 73 70 57 63 70 78 52 67 53 67 75 61 70 81 76 79 75 76 58 31 For the above data, construct stem and leaf plot and histogram. Draw conclusions.	10
3	a	Consider writing onto a computer disk and then sending it through a certifier that counts the number of missing pulses. Suppose this number X has a Poisson distribution with parameter $\lambda = 0.2$. i) What is the probability that a disk has exactly one missing pulse? ii) What is the probability that a disk has at least two missing pulses? iii) If two disks are independently selected, what is the probability that neither contain a missing pulse?	08
	b	Suppose the force acting on a column that helps to support a building is normally distributed with mean 15.0 kips and standard deviation 1.25 kips. what is the probability that the force i) Is at most 18 kips? ii) Is between 10 and 12 kips? iii) Differs from 15.0 kips by at most 2 standard deviations?	08
		OR	
4	a	Suppose that 20% of all copies of a particular textbook fail a certain binding strength test. Let X denote the number among 15 randomly selected copies that fail the test. i) What is the probability that at most 8 fail the test? ii) What is the probability that exactly 8 fail? iii) What is the probability that failure is between 4 and 7, inclusive?	06
	b	Suppose the survival time X in weeks of a randomly selected male mouse exposed to 240 rads of a gamma radiation has a gamma distribution with $\alpha = 8$ and $\beta = 15$. What is the probability that a mouse survives between 60 and 120 weeks?	06
	c	State the properties and applications of Weibull distribution.	04

5	a	An electrical firm manufactures light bulbs that have a length of life that is approximately normally distributed, with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a random sample of 16 bulbs will have an average life of less than 775 hours.	06
	b	An important manufacturing process produces cylindrical component parts for the automotive industry. It is important that the process produces parts having a mean of 5.0 millimeters. The engineer involved conjectures that, the population mean is 5.0 millimeters. An experiment is conducted in which 100 parts produced by the process are selected randomly and the diameter measured on each. It is known that the population standard deviation $a = 0.1$. The experiment indicated a sample average diameter $x = 5.027$ millimeters. Does this sample information appear to support or refute the engineer's conjecture?	06
	c	What are the properties of good estimator? Explain.	04
		OR	
6	a	The average life of a bread-making machine is 7 years, with a standard deviation of 1 year. Assuming that the lives of these machines follow approximately a normal distribution, find i) The probability that the mean life of a random sample of 9 such machines falls between 6.4 and 7.2 years. ii) The value of a; to the right of which 15% of the means computed from random samples of size 9 would fall.	06
	b	If the standard deviation of the mean for the sampling distribution of random samples of size 36 from a large or infinite population is 2, how large must the size of the sample become if the standard deviation is to be reduced to 1.2?	04
	c	The heights of 1000 students are approximately normally distributed with a mean of 174.5 centimeters and standard deviation of 6.9 centimeters. If 200 random samples of size 25 are drawn from this population and the means recorded to the nearest tenth of a centimeter, determine: i) The mean and standard deviation of the sampling distribution of X. ii) The number of sample means that fall between 172.5 and 175.8 centimeters inclusive.	06
7	a	A mathematics placement test is given to all entering freshmen at a small college. A student who receives a grade below 35 is denied admission to the regular mathematics course and placed in a remedial class. The placement test scores and the final grades for 20 students who took regular course were recorded as follows:	

Placement test	*Course grade*	*Placement test*	*Course grade*
50	53	90	54
35	41	80	91
35	61	60	48
40	56	60	71
55	68	60	71
65	36	40	47
35	11	55	53
60	70	50	68
90	79	65	57
35	59	50	79

Find the equation of regression line to predict course grades from placement test scores. If 60 is the minimum passing grade, below which placement test score should students in the future be denied admission to this course? 08

b

It is important that scientific researchers in the area of forest products be able to study correlation among the anatomy and mechanical properties of trees. According to the study conducted by the Department of Forestry and Forest products at the Virginia Polytechnic Institute and State University, an experiment in which 29 loblolly pines were randomly selected for the investigation yielded the data shown below, on the specific gravity in $grams/cm^3$ and the modulus of rupture in kilopascals (kPa). Compute and interpret the sample correlation coefficient and coefficient determination.

$Specific\ gravity, x(g/cm^3)$	$Modulus\ of\ Rupture, y\ (kPa)$	$Specific\ gravity, x(g/cm^3)$	$Modulus\ of\ Rupture, y\ (kPa)$
0.414	29,189	0.581	85,156
0.383	29,266	0.557	69,571
0.399	26,215	0.550	84,160
0.402	30,162	0.531	73,466
0.442	38,867	0.550	78,610
0.422	37,831	0.556	97,657
0.466	44,576	0.523	74,017
0.500	46,097	0.602	87,291
0.514	59,698	0.569	86,836
0.530	67,705	0.544	82,540
0.569	66,088	0.557	81,699
0.558	78,486	0.530	82,096
0.577	89,869	0.547	75,657
0.572	77,369	0.585	80,490
0.548	67,095		

08

8 a

The Edison Electric Institute has published figures on the annual number of kilowatt hours expended by various home appliances. It is claimed that a vacuum cleaner expends an average of 46 kilowatt hours per year. If a random sample of 12 homes included in a planned study indicated that vacuum cleaners expend an average 42 kilowatt hours per year with a standard deviation of 11.9 kilowatt hours, does this suggest at the 0.05 level of significance that vacuum cleaners expend, on the average , less than 46 kilowatt hours annually? Assume the population of kilowatt hours to be normal.

06

b

The number of days 'f' in a 50 days period during which x automobile accidents occurred in a city. Identify the appropriate probability and test the goodness of fit.

$Number\ of\ Accidents, x$	0	1	2	3	4
$Number\ of\ days\ , f$	21	18	7	3	1

10

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Fast Track Examinations July-18

Industrial Engineering and Management

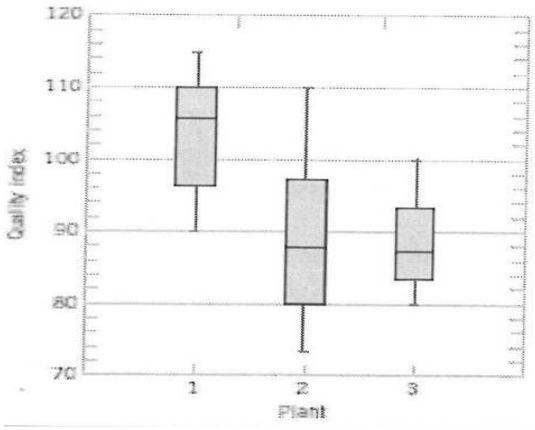
ENGINEERING STATISTICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

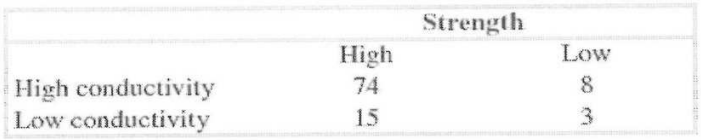
6. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
7. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
8. Use of statistical tables permitted.

PART A

1	1.1	A subject that can result in different outcomes, even though it is repeated in the same manner every time, is called _____.	01
	1.2	Differentiate between discrete and continuous sample spaces.	02
	1.3	Samples of polycarbonate plastic are analyzed for scratch and shock resistance. The results from 50 samples are summarized as follows: shock resistance: high / low scratch resistance: high — 40, 4; low — 1, 5 Let A denote the event that a sample has high shock resistance, and let B denote the event that a sample has high resistance. Determine the number of samples in $A \cup B$.	01
	1.4	Form the box plot given below, state which plant has more variability.  	02
	1.5	Give the expression for the probability of probability distribution function.	01
	1.6	State any two applications of Erlang distribution.	02
	1.7	State the central limit theorem.	01
	1.8	State the conditions for Normal approximation to the binomial distribution.	02
	1.9	What are critical values in inference theory?	01

	1.10	What is power of a test? How do you compute it?	02
	1.11	Define p-value in hypothesis tests.	01
	1.12	What is Coefficient of Determination (R^2)? State its usefulness.	02
	1.13	Write the normal equations that are used in solving multiple linear regression analysis.	02

PART B

2	a	A batch of 140 semiconductor chips is inspected by choosing a sample of 5 chips. Assume 10 of the chips do not confirm to customer requirements. i) How many different samples are possible? ii) How many samples of five contain exactly one nonconforming chip? iii) How many samples of five contain at least one nonconforming chip?	08
	b	Strands of copper wire from a manufacturer are analyzed for strength and conductivity. The results from 100 strands are as follows:  i) If a strand is randomly selected, what is the probability that its conductivity is high and its strength is high? ii) If a strand is randomly selected, what is the probability that its conductivity is low or its strength is low? iii) Consider the event that a strand has low conductivity and the event that the strand has low strength. Are these two events mutually exclusive?	08
3	a	Batches that consist of 50 coil springs from a production process are checked for conformance to customer requirements. The mean number of nonconforming coil springs in a batch is 5. Assume that the number of nonconforming springs in a batch, denoted as X, is a binomial random variable. i) What are n and p? ii) What is $P(X >= 2)$? iii) What is $P(X > 49)$?	08
	b	The probability of a successful optical alignment in the assembly of an optical data storage product is 0.8. Assume the trials are independent. i) What is the probability that the first successful alignment requires exactly four trials? ii) What is the probability that the first successful alignment requires at most four trials? iii) What is the probability that the first successful alignment requires at least four trials?	08
		OR	
4	a	The sick-leave time of employees in a firm in a month is normally distributed with a mean of 100 hours and a standard deviation of 20 hours. i) What is the probability that the sick-leave time for next month will be between 50 and 80 hours? ii) How much time should be budgeted for sick-leave if budgeted amount should be exceeded with a probability of only 10%?	08

	Strength	
	High	Low
High conductivity	74	8
Low conductivity	15	3

	b	Assume the life of a packaged magnetic disk exposed to corrosive gases has a Weibull distribution with $\beta = 0.5$ and the mean life is 600 hours. i) Determine the probability that a packaged disk lasts at least 500 hours. ii) Determine the probability that a packaged disk fails before 400 hours.	08
5	a	PVC pipe is manufactured with a mean diameter of 1.01 inch and a standard deviation of 0.003 inch. Find the probability that a random of $n = 9$ sections of pipe will have a sample mean diameter greater than 1.009 inch and less than 1.012 inch.	04
	b	The amount of time that a customer spends waiting at an airport check-in counter is a random variable with mean 8.2 minutes and standard deviation 1.5 minutes. Suppose that a random sample of $n = 49$ customers is observed. Find the probability that the average time waiting in line for these customers is i) Less than 10 minutes. ii) Between 5 and 10 minutes. iii) Less than 6 minutes.	08
	c	A normal population has mean 100 and variance 25. How large must the random sample be if we want the standard error of the sample average to be 1.5?	04
		OR	
6	a	The compressive strength of concrete is normally distributed with $\mu = 2500$ psi and $\sigma = 50$ psi. Find the probability that a random sample of $n = 5$ specimens will have a sample mean diameter that falls in the interval from 2499 psi to 2510 psi.	06
	b	Enumerate the properties of maximum likelihood estimator.	04
	c	Data on pull-off force (pounds) for connectors used in an automobile engine are as follows: 79.3, 75.1, 78.2, 74.1, 73.9, 75.0, 77.6, 77.3, 73.8, 74.6, 75.5, 74.0, 74.7, 75.9, 72.9, 73.8, 74.2, 78.1, 75.4, 76.3, 75.3, 76.2, 74.9, 78.0, 75.1, 76.8. i) Calculate a point estimate of the mean pull-off force of all connectors in the population. State which estimator is used and why? ii) Calculate a point estimate of the pull-off force value that separates the weakest 50% of the connectors in the population from the strongest 50%. iii) Calculate point estimates of the population variance and the population standard deviation.	06
7	a	The number of pounds of steam used per month by a chemical plant is thought to be related to the average ambient temperature (in degree F) for that month. The past year's usage and temperature are shown in the following table.	

Month	Temp.	Usage/1000	Month	Temp.	Usage/1000
Jan.	21	185.79	July	68	621.55
Feb.	24	214.47	Aug.	74	675.06
Mar.	32	288.03	Sept.	62	562.03
Apr.	47	424.84	Oct.	50	452.93
May	50	454.58	Nov.	41	369.95
June	59	539.03	Dec.	30	273.98

		i) Assuming that a simple linear regression model is appropriate, fit the regression model relating steam usage (y) to the average temperature (x). What is the estimate of σ^2. ii) What is the estimate of expected steam usage when the average temperature is 55 degree F? iii) What change in mean steam usage is expected when the monthly average temperature changes by 1 degree F?	10
	b	The weight and systolic blood pressure of 26 randomly selected males in the age group 25 to 30 are shown in the following table. Assume that weight and blood pressure are joint jointly normally distributed.	

Subject	Weight	Systolic BP	Subject	Weight	Systolic BP
1	165	130	14	172	153
2	167	133	15	159	128
3	180	150	16	168	132
4	155	128	17	174	149
5	212	151	18	183	158
6	175	146	19	215	150
7	190	150	20	195	163
8	210	140	21	180	156
9	200	148	22	143	124
10	149	125	23	240	170
11	158	133	24	235	165
12	169	135	25	192	160
13	170	150	26	187	159

		Estimate the correlation coefficient.	06
8	a	Compute a 99% tolerance interval on the impact strength of PVC pipe that has confidence level 90%. Compare the length of the tolerance interval with the length of the 99% CI on the population mean. Which interval is shorter? Discuss the difference in interpretation of these two intervals.	05
	b	Medical researchers have developed a new artificial heart constructed primarily of titanium and plastic. The heart will last and operate almost indefinitely once it is implanted in the patient's body, but the battery pack needs to be recharged about every four hours. A random sample of 50 battery packs is selected and subjected to a life test. The average life of these batteries is 4.05 hours. Assume that battery life is normally distributed with standard deviation of 0.2 hour. i) Is there evidence to support the claim that the mean battery life exceeds 4 hours? ii) Compute the power of the test if the mean battery life is 4.5 hours.	11

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Examinations April/May-18

Industrial Engineering and Management

ENGINEERING STATISTICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

4. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
5. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
6. Use of statistical tables and probability distribution formulae permitted.

PART A

1	1.1	Distinguish between probability mass function and probability density function.	02
	1.2	Define conditional probability.	02
	1.3	State the assumptions of Binomial distribution.	02
	1.4	Write down the density function of normal distribution.	02
	1.5	For the following set of data which is the best measure of central tendency and why? $15000, 16000, 45000, 100000, 45000, 45000.$	02
	1.6	Compare the one-tailed test and two-tailed test.	02
	1.7	Mention the different methods used in correlation.	02
	1.8	Define type I and type II error.	02
	1.9	Differentiate between estimator and estimate.	02
	1.10	What is the purpose of level of significance?	02

PART B

2	a	The energy commission initiates a study in a certain geographical area. The Survey covered 40 domestic consumers. Observations related to the energy consumption with measurements of solar energy usage source in the domestic usage matrix. Construct the stem and leaf diagram for the following data. And also determine the mean and mode using the diagram. Data Pertaining to daily solar energy usage in consumption mix (in kW/hr) 0.52,0.62,0.61,0.65,0.48,0.54,0.53,0.62,0.68,0.66 0.49,0.52,0.51,0.57,0.59,0.64,0.66,0.69,0.67,0.61 0.46,0.48,0.54,0.56,0.62,0.63,0.62,0.72,0.56,0.46 0.66,0.65,0.66,0.63,0.61,0.68,0.62,0.74,0.72,0.63	08
	b	The sample space of a random experiment is $\{a, b, c, d, e, f\}$ and each outcome is equally likely. A random variable is defined as follows: $Outcome$ | a | b | c | d | e | f X | 0 | 0 | 1.5 | 1.5 | 2 | 3 Determine the probability mass function and cumulative distribution of X.	08

3	a	A multiple choice test contains 25 questions, each with four answers. Assume a student just guesses on each question. i) What is the probability that the student answers more than 20 questions correctly? ii) What is the probability that the student answers less than 5 questions correctly?	08
	b	The number of messages sent to a computer bulletin board is a poisson random variable with a mean of 5 messages per hour. i) What is the probability that 5 messages are received in 1 hour? ii) What is the probability that 10 messages are received in 1.5 hours? iii) What is the probability that less than two messages are received in one-half hours?	08
		OR	
4	a	The time between arrivals of taxis at a busy intersection is exponentially distributed with a mean of 10 minutes. i) What is the probability that you wait longer than one hour for a taxi? ii) Suppose you have already been waiting for one hour for a taxi, what is the probability that one arrives within next 10 minutes?	08
	b	Assume that the life of a roller bearing follows a weibull distribution with parameters $\beta = 2$ and $\delta = 10{,}000$ hours. i) Determine the probability that a bearing lasts at least 8000 hours. ii) Determine the mean time until failure of a bearing. iii) If 10 bearings are in use and failure occurs independently, what is the probability that all 10 bearings last at least 8000 hours?	08
5	a	In a large quantity, out of a sample of 100 school boys 20 were found to have defective eye sight. Out of 50 girls, 18 were found to have defective eye sight. Estimate 85% confidence between two proportions.	10
	b	Explain what is estimation? List the properties of good estimator? Also differentiate between point estimate and interval estimate.	06
		OR	
6	a	Suppose that X has a discrete uniform distribution $f(x) = \begin{cases} 1/3, & x = 1,2,3 \\ 0, & Otherwise \end{cases}$ A random sample of $n = 36$ is selected from this population. Find the probability that the sample mean is greater than 2.1 but less than 2.5, assuming that the sample mean would be measured to the nearest tenth.	10
	b	PVC pipe is manufactured with a mean diameter of 1.01 inch and a standard deviation of 0.003 inch. Find the probability that a random sample of $n = 9$ sections of pipe will have a sample mean diameter greater than 1.009 inch and less than 1.012 inch.	06
7	a	The heights of a sample of 10 fathers and their eldest sons are given below (to the nearest cm): Height of father (X) 170,167,162,163,167,166,169,171,164,165 Height of son (Y) 168,167,166,166,168,165,168,170,165,168 i) Compute the correlation coefficient Y ii) Find the regression of Y on X. iii) Compute the coefficient of determination and give your comments.	12

	b	Mention the different types of regression.	04
8	a	The heights of 10 children selected at random from a given locality had a mean $63.2cms$ and variance $6.25cms$. Test at 5% level of significance the hypothesis that the children of the given locality are on the average less than 65cms in all. Given for 9 degrees of freedom $P(t > 1.83) = 0.05$	08
	b	Explain the steps involved in testing of Hypothesis.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Fast Track Examinations July-17

Industrial Engineering and Management

STATISTICS FOR DECISION MAKING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
2. Answer FIVE full questions from Part B.
3. Use of statistical tables is permitted.

PART-A

1	1.1	The population has a mean of 14 and a standard deviation of 3. The sample size of your sampling distribution is $N = 10$. What is the mean of the sampling distribution of the mean?	01
	1.2	What is the power of test in hypothesis testing?	01
	1.3	In correlation analysis, when the points scatter widely about the regression line, this means that the correlation is _____.	01
	1.4	The standard deviation of one set of numbers $3, 4, 5$ and another set $1003, 1004, 1005$ is _____.	01
	1.5	The population has a mean of 30 and a standard deviation of 6. The sample size of your sampling distribution is $N = 9$. What is the variance of the sampling distribution of the mean?	01
	1.6	Number of molecules in a cubic centimeter of air is an example of _____ distribution.	01
	1.7	A function that assigns a real number to each outcome in the sample space of a random experiment is known as _____.	01
	1.8	The probability of intersection of two or more events is known as _____ rule.	01
	1.9	In _____ study, a sample from a population is used to make inference to a future population and needs stability.	01
	1.10	The population has a mean of 120 and a standard deviation of 12. The sample size of your sampling distribution is $N = 16$. What is the standard error of the mean?	01
	1.11	The interval can be narrowed without suffering a loss of confidence by increasing the _____.	01
	1.12	The standard error will_____ with increase in sample size.	01
	1.13	An _____ is the numerical result of the estimator.	01
	1.14	An estimator is unbiased if the _____ of the sampling distribution equals the corresponding parameter.	01
	1.15	An estimate is consistent if, as n increases, the value of the statistic approaches the _____.	01
	1.16	In a class of 100, the mean score on a certain exam was 50, the standard deviation was 0. What can you conclude?	01
	1.17	The sample standard deviation is a measure of central tendency around the _____.	01
	1.18	Does correlation imply causation?	01

	1.19	Selecting a random sample from a box containing blue and red balls is an example of _____ distribution.	01
	1.20	What type of decision is reached when the calculated value of any test statistic falls in the critical region when a false null hypothesis is being tested?	01

PART-B

2	a	Draw a box and whisker plot for the numbers given below: $13, 72, 73, 85, 86, 87, 89.$	05
	b	*History Test Scores* *Steam* | *Leaf* 6 | 1 1 4 6 7 8 7 | 2 3 5 7 9 8 | 1 3 5 6 6 7 7 8 9 9 | 0 0 3 4 6 8 9 9 10 | 0 0 i) What is the best test score? ii) How many students took the test? iii) How many students scored 90? iv) What is the lowest score? v) Find the range for the scores.	05
	c	For a group of 150 observations mean and standard deviation were quoted as 120 and 15 respectively. However, it was discovered that one of the observations was taken as 15 instead of 105. Find the correct values of mean and standard deviation.	06
		OR	
3	a	The compressive strength in kg of 45 specimens of a new alloy undergoing evaluation as a possible material for aircraft structural elements are given below: 190 218 208 181 228 176 165 157 201 193 154 245 163 142 146 178 131 229 110 97 76 149 183 156 168 105 207 186 169 237 200 196 194 151 87 180 134 123 221 121 199 160 141 174 143 i) Construct frequency distribution. ii) Determine the mean, median, mode. iii) Determine the sample variance and sample standard deviation.	08
	b	In an exam, two reasoning problems, 1 and 2 are asked. 35% students solved problem 1 and 15% students solved both the problems. How many students who solved the first problem will also solve the second one?	04
	c	Find the mean and variance of the following discreet probability distribution x: -10, -20, 30, 75, 80 $P(X=x)$: $\frac{1}{5}$, $\frac{3}{20}$, $\frac{1}{2}$, $\frac{1}{10}$, $\frac{1}{20}$	04
4	a	In a Binomial distribution, mean = 6 and variance = 1.5, find i) $P(X=2)$ ii) $P(X \leq 2)$	08

	b	To avoid detection at customs office 6 narcotic tablets are placed in a bottle having 9 vitamin pills that are similar in appearance. If the custom official selects 3 tablets for analysis, what is the probability that the traveler is arrested for illegal transport of narcotic drugs?	04
	c	The lifetime of a mechanical assembly in a vibration test is exponentially distributed with a mean of 400 hours. i) What is the probability that an assembly on test fails in less than 100 hours? ii) What is the probability that an assembly operates for more than 500 hours before failure?	04
		OR	
5	a	The thickness measurements of the coating process are made to the nearest 100th of a mm. These measurements are uniformly distributed with values $0.15, 0.16, 0.17, 0.18$ and 0.19. Find the mean and variance.	04
	b	How is Erlang distribution related to Gamma distribution?	04
	c	The length of an injection – molded plastic case that holds magnetic tape is normally distributed with a length of $90.2\ mm$ and a standard deviation of $0.1\ mm$. i) What is the probability that a part is longer that $90.3\ mm$ or shorter than $89.7\ mm$? ii) What should the process mean be set at to obtain the greatest number of parts between 89.7 and $90.3\ mm$? iii) If parts that are not between 89.7 and $90.3\ mm$ are scrapped, what is the yield for the process mean that you selected in part ii?	08
6	a	Let X and Y be two jointly continuous random variables with joint PDF $f_{XY}(x,y) = \begin{cases} x + cy^2 & 0 \le x \le 1, \quad 0 \le y \le 1 \\ 0 & otherwise \end{cases}$ Find the constant c.	08
	b	Suppose the weights of randomly selected American female college students are normally distributed with unknown mean μ and standard deviation σ. A random sample of 10 American female college students yielded the following weights (in pounds). 115 122 130 127 149 160 152 138 149 180 Based on the definitions given above, identify the likelihood function and the maximum likelihood estimator of μ, the mean weight of all American female college students. Using the given sample, find a maximum likelihood estimate of μ as well.	08
		OR	
7	a	State Central limit theorem and give its applications.	04
	b	Provide a Poisson example for maximum likelihood estimator.	06
	c	The scores of 12 students in their mathematics and physics classes are: $Mathematics$: 2 3 4 4 5 6 6 7 7 8 10 10 $Physics$: 1 3 2 4 4 4 6 4 6 7 9 10 Find the covariance of the distribution.	06

8	a	The data on compressive strength x and intrinsic permeability y of various concrete mixes and cures is presented below. Assume that the two variables are related according to the simple linear regression model. $n = 14, \sum yi = 72, \sum yi^2 = 23530, \sum xi = 43, \sum xi^2 = 157.42, \sum xi\, yi = 1697.80$ i) Calculate the least squares estimate of the slope and intercept. ii) Use the equation of the fit line to predict what permeability would be observed when the compressive strength is 4.3 iii) Give a point estimate of the mean permeability when compressive strength is 3.7 iv) Suppose that the observed value of permeability at $x = 3.7$ is $y = 46.1$, determine the value of the corresponding residual.	10
	b	State the assumptions of the Simple Linear regression Model.	06
		OR	
9	a	Given the lines of regression as $8x - 10y + 66 = 0$ and $40x - 18y = 214$ and variance of x is 9, determine i) Mean of x-series. ii) Mean of y-series.	10
	b	Write a note on the prediction of future observations providing suitable examples.	06
10	a	For a random sample of 1000 families owning *TV* sets in Bengaluru, it was found that 240 families had a *TV* of a particular brand. i) Find 95% confidence interval for the proportion of families in the city owning that brand. ii) How large a sample is required if it is desired to be 95% confident that our estimate of binomial parameter p is within ± 0.02 of the true value.	10
	b	200 out patients are diagnosed for major depression. Patients were assigned to two groups. One group went to clinic A and the other went to B. After 8 weeks, 19 out of 'A' patients showed improvement, whereas 27 of 'B' patients showed improvement. Is there any reason to believe that B is effective in treating major depression?	06
		OR	
11	a	For the data given below, conduct goodness – of –fit test using Poisson distribution at $\alpha = 5\%$ *Value* | *Observed Frequency* 0 | 24 1 | 30 2 | 31 3 | 11 4 | 4	10
	b	Suppose an editor of a publishing company claims that the mean time to write a textbook is at most 15 months. A sample of 16 textbook authors is randomly selected and it is found that the mean time taken by them to write a textbook was 12.5. Assume also that the standard deviation is known to be 3.6 months. Assuming the time to write a textbook is normally distributed and using a 0.025 level of significance, would you conclude the editor's claim is true?	06

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Examinations April/May-17

Industrial Engineering and Management

STATISTICS FOR DECISION MAKING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
2. Answer FIVE full questions from Part B.
3. Use of statistical tables is permitted.

PART A

1	1.1	Rejection of the null hypothesis when it should be accepted is known as _____.	01
	1.2	What do you mean by level of significance?	02
	1.3	Hospital records show that of the patients suffering from a certain disease, 75 % die of it. What is the probability that of 6 randomly selected patients, 4 will die?	02
	1.4	Vehicles pass through a junction on a busy road at an average rate of 300 per hour. Find the probability that none passes in a given minute.	02
	1.5	List any two properties of a normal distribution.	02
	1.6	Define parameters and statistic.	02
	1.7	Differentiate between an Estimator and an Estimate.	02
	1.8	What is standard Normal probability distribution?	01
	1.9	State the assumptions of Binomial distribution.	02
	1.10	A random sample of 100 students gave a mean weight of $58\,kg$ with std deviation of $4\,kg$. Set a null and alternate hypothesis to test the mean weight in the population is $60\,kg$.	02
	1.11	Bring out the key differences between Descriptive statistics and Inferential statistics.	02

PART B

2	a	The $A.C.$ Neilson company publishes data on the television viewing habits of Americans. A sample of 15 people's television weekly viewing times, in hours, is shown below. Based on the data, move the minimum and maximum data points to complete a box plot for the television-viewing data? 5,7,11,15,18,21,22,24,25,26,28,30,31,31.	08
	b	Swifty Markets company's prices charged for identical items in all of its food stores. Here are the prices charged by each store for a pound of bacon last week: 1.08 0.98 1.09 1.24 1.33 1.14 1.55 1.08 1.22 1.05 i) Calculate the median price per pound ii) Calculate the mean price per pound iii) Which value is the better measure of the central tendency or this set of data?	08

OR

3	a	Friendly's department store has been the target of many shoplifters during the past month, but owing to increased security precautions, 250 shoplifters have been caught. Each shoplifter's sex is noted; also noted is whether the perpetrator was a first-time or repeat offender. The data are summarized in the table. Sex | $First-time\ offender$ | $Repeat\ offender$ $Male$ | 60 | 70 $Female$ | 44 | 76 | 104 | 146 Assuming that an apprehender shoplifter is chosen at random, find i) The probability that the shop lifter is male. ii) The probability that the shop lifter is a first time offender, given that the shop lifter is male. iii) The probability that the shop lifter is female, given that the shoplifter is a repeat offender. iv) The probability that the shop lifter is female, given that the shoplifter is a first-time offender.	08
	b	A doctor has decided to prescribe two new drugs to 200 heart patients as follows: 50 get drug A, 50 get drug B and 100 get both. The 200 patients were chosen so that each had an 80 % chance of having heart attack if given neither drug. Drug A reduces the probability of a heart attack by 35 %, drug B reduces the probability by 20 % and the two drugs when taken together work independently. If a randomly selected patient in the program has a heart attack, what is the probability that the patient was given both drugs?	08
4	a	Let X denote that no.of bits received in error in a digital communication channel and assume that X is a binomial random variable with $P = 0.001$. If 1000 bits are transmitted, determine the following: i) $P(X = 1)$ ii) $P(X \geq 1)$ iii) $P(X \leq 2)$ iv) Mean and variance of X	08
	b	Let the random variable X has a discrete uniform distribution on the integers $1 \leq X \leq 3$. Determine the mean and variance of X.	03
	c	Thickness measurements of a coating process are made to the nearest hundredth of a millimeter. The thickness measurements are uniformly distributed with values $0.15, 0.16, 0.17, 0.18$ and 0.19. Determine the mean and variance of the coating thickness for this process.	05
		OR	
5	a	Suppose the time taken for a data collection operater to fill out an electronic form for a database is uniformly between 1.5 and 2.2 minutes. i) What is the mean and variance of the time it takes an operator to fill out the form? ii) What is the probability that it will take less than two minutes to fill out the form? iii) Determine cumulative distribution function of the time it takes to fill out the form.	08

	b	Dennis Hogan is the supervisor for the Cordwingo hydroelectric Dam. Mr. Hogan knows that the dam's turbines generate electricity at the peak rate only when at least 1,000,000 gallons of water pass through dam each day. He also knows, from his experience, that the daily flow is normally distributed, with the mean equal to previous day's flow and a standard deviation of 200,000 gallons. Yesterday 850,000 gallons flowed through the dam. What is the probability that the turbines will generate peak rate today?	08
6	a	A small- business website contains 100 pages and 60 %, 30 % and 10 % of the pages contain low, moderate and high graphic content, respectively. A sample of four pages is selected without replacement and X and Y denote the number of pages with moderate and high graphics output in the sample. Determine: i) $f_{xy}(x,y)$ ii) $f_x(x)$ iii) $E(x)$ iv) $f_{y/3}(y)$ v) $E(y/x=3)$.	10
	b	State the properties of a joint probability density function. Give a typical mathematical function for a joint probability density function.	06
		OR	
7	a	A PVC pipe is manufactured with a mean diameter of 1.01 inch and a standard deviation of 0.003 inch. Find the probability that a random sample of $n = 9$ sections of pipe will have a sample mean diameter greater than 1.009 inch and less than 1.012 inch.	08
	b	A synthetic fiber used in manufacturing carpet has tensile strength that is normally distributed with mean 75.5 psi and standard deviation 3.5 psi. Find the probability that random sample of $n = 6$ fiber specimens will have sample mean tensile strength that exceeds 75.75 psi.	08
8		The final test and exam averages for 20 randomly selected students taking a course in engineering statistics and a course in operations research follow. Assume that the final averages are jointly normally distributed. $Statistics$ 86 75 69 75 90 94 83 86 71 65 84 71 62 90 83 75 71 76 84 97 OR 80 81 75 81 92 95 80 81 76 72 85 72 65 93 81 70 73 72 80 98 a) Find the regression line relating the statistics final average to the OR final average. b) Test for significance of regression using $\alpha = 0.05$ c) Estimate the correlation coefficient d) Test the hypothesis that $\rho = 0$ using $\alpha = 0.05$ e) Test the hypothesis that $\rho = 0.5$ using $\alpha = 0.05$ f) Construct a 95 % confidence interval for correlation coefficient.	16
		OR	

9		Campus stores has been selling the Believe it or not: wonders of statistics study guide for 12 semesters and would like to estimate the relationship between sales and number of sections of elementary statistics taught each semester. The following data have been collected: $Sales\ (units)$: 83, 38, 24, 61, 52, 45, 65, 82, 29, 63, 50, 79 $Number\ of\ sections$: 3, 7, 6, 6, 10, 12, 12, 13, 12, 13, 14, 15 a) Develop the estimating equation that best fits the data. b) Calculate the sample coefficient of determination and the sample coefficient of correlation.	16
10	a	In each of the following situations state whether it is a correctly stated hypothesis testing problem and why. i) $H_O: \mu = 25, H_1: \mu \neq 25$ ii) $H_O: \theta > 10, H_1: \theta = 10$ iii) $H_O: \bar{X} = 50,\ H_1: \bar{X} \neq 50$ iv) $H_O: p = 0.1\ , H_1:\ p = 0.5$ v) $H_O: S = 30\ , H_1: S > 30$	08
	b	A stenographer claims that she can type at the rate of 120 words per min. Can we reject her claim on the basis of 100 trials in which she demonstrates a mean of 116 words with a standard deviation of 15 words? Use of 5 % level significance.	08
		OR	
11	a	State the general procedure for hypothesis test.	08
	b	A random sample of 1000 workers from south India was that their mean wages are $Rs.47$ per week with a standard deviation of $Rs.28$. A random sample of 1500 from north India gives a mean wage of $Rs.49$ per week with standard deviation of $Rs.40$. Is there any significant difference between their mean level of wages?	08

12IM41 / 10IM43 / 07IM42

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IVSemester B. E. Fast Track Examinations July-16

Industrial Engineering and Management

STATISTICS FOR DECISION MAKING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
2. Answer FIVE full questions from Part B.
3. Use of statistical tables is permitted.

PART-A

1	1.1	In a throw of a 6-sided fair die, given that a prime number is on top, what is the probability of a number at most 3 being on top?	01
	1.2	For mutually exclusive events A, B & C, the probability $P[(A \cup B \cap C)']$ is given by _____.	01
	1.3	When quartiles are available for a data set, its inter-quartile range is computed as _____.	01
	1.4	The expectation of a continuous random variable X with probability density function $f(x)$ is _____.	01
	1.5	X is normally distributed with mean and variance 9 and 8 respectively. The standardized value for $X = 24$ is _____.	01
	1.6	In a location, earthquakes occur as a Poisson process with a mean of 3 quakes in a decade. P (at least one earth quake in the next decade) = _____.	01
	1.7	A factory producing 500 parts in a day has 8% of its production defective. Then the mean and variance of number of defective parts in a day is _____ and _____ respectively.	01
	1.8	Historically, 2% of stray dogs are unlicensed in a city. On a given day if 7 dogs are picked up, what is the probability that at least one is unlicensed?	01
	1.9	X, the number of flaws per km of a cable is Poisson distributed with a mean of 250 flaws per km. Then Y, the distance between flaws in km has a variance of _____.	01
	1.10	A normally distributed characteristic has _____ % of all the values within ± 3 standard deviations away from its mean.	01
	1.11	_____ is the extent to which observed value and predicted value differ in simple linear regression.	01
	1.12	The primary assumption about error terms in simple linear regression is _____.	01
	1.13	Type II error is committed when the null hypothesis is _____ and the conclusion is _____ the null hypothesis.	01
	1.14	The test statistic used in a chi-squared goodness of fit test is given as _____.	01
	1.15	The property of an estimator which combines the effect of its bias $b$and variance V is _____ and it is computed as _____.	02
	1.16	The median of the ungrouped data set: $86, 52, 44, 43, 35, 31, 30, 11$ is _____.	02

	1.17	A binomially distributed variable X has its mean and variance as 0.6 and 0.48 respectively. X can take on values _____.	02

PART-B

2	a	A random variable X has the following cumulative distribution: $F(x) = 0, -\infty < x < 1$ $0.4, 1 \leq x \leq 3$ $0.5, 3 \leq x \leq 8$ $1.0, 8 \leq x \leq \infty$ Determine the following: i) $P[X\ being\ at\ most\ 3]$ ii) $P[X\ between\ 1\ \&2, both\ inclusive]$ iii) $P[X\ being\ at\ least\ 2]$ iv) $Median\ of\ X.$	08
	b	X is discrete in the range $-5, -4, \ldots\ldots 0, 1, \ldots\ldots 10$ with mass function $f(x) = K$. Compute $K, E(X)$ & $var\ (X)$.	04
	c	25 boxes containing parts were inspected and the number of defective parts in each box was recorded as shown below. Draw a box plot and comment on the distribution. $10, 12, 19, 22, 15, 19, 6, 12, 18, 8, 9, 11, 7, 21, 17, 15, 13, 14, 16, 8, 9, 11, 2, 18, 15.$	04
		OR	
3	a	Three events A, B, C occur with probabilities 0.35, 0.45 & 0.2 respectively. When $A, B\ or\ C$ has already occurred, the probabilities of another related event Y occurring are 0.8, 0.65 & 0.3 respectively. Find the probabilities of A, B, C occurring when it is known that Y has occurred.	06
	b	A card is drawn from a pack of 52 playing cards. Define the following events: $A =$ black card drawn, $B =$ Red card is drawn, $C =$ even numbered card drawn, $D =$ card with a number is drawn. Identify or compute: i) Mutually exclusive events ii) Complementary events iii) $P(A\|C)$ iv) $P(A \cap D')$ v) $P(C\|A)$.	10
4	a	A certain company producing screws has 1% of the production as defectives. The company sells screws in packages of 10 screws each. It offers a replacement guarantee on the package if more than 1 screw is found defective in a package. What is the propagation of solid packages? What is the proportion of sold packages being replaced by the company?	06
	b	The time between calls to a plumbing supply business is exponentially distributed with a mean time of 15 minutes between calls. i) What is the probability that there are no calls within a 30 minute period? ii) What is the probability of at least one call arriving within a 10-minute interval? iii) What is the probability of the first call arriving within 5 and 10 minutes after opening? iv) Find the probability of at least one call in the interval is 0.9.	10
		OR	

5	a	Life time of the CPU of a personal computer is exponentially distributed with a mean of 6 years. i) What is the probability that the CPU fails in next 3 years, given that it has not failed for the past 3 years? ii) If there are 10 such $CPU's$ in the company and all of them fail or function independently, what is the probability that at least one fails within the next 3 years?	08
	b	In a digital communication channel 50 bits are transmitted as a packet every micro second, and the probability that a bit is received in error is 0.1 bits are independent. Compute the probability that at most 2 bits are received in error for a packet. Compare the exact answer to the approximate answer with continuity correction. Comment on the difference.	08
6	a	A joint probability mass function with 2 variables x&y is defined as $f_{xy}(x,y) = C(x+y)$, for $x = 1,2,3$ & $y = 1,2,3$. Determine the value of C and the correlation coefficient. Are x & y independent?	10
	b	Suppose $x_1, x_2, \ldots x_{20}$ is a random sample from a normal population with mean and variance parameters as μ & σ^2. Derive the method of moments estimators for μ & σ^2.	06
		OR	
7	a	X & Y are processing times measured in seconds in two sequential processes, and are jointly distributed as $f_{xy}(x,y) = 2 \times 10^{-6} e^{-0.001x - 0.002y}, x \geq 0, y \geq 0$. Are x & y independent? Determine $P(x > 1000, y < 1000)$.	10
	b	From a random sample, $x_1, x_2, \ldots \ldots x_{10}$ of size 10 drawn from a population, the population mean μ is estimated using two estimators: $\hat{\mu}_1 = \frac{1}{5}(x_1 + x_3 + x_5 + x_7 + x_9)$ & $\hat{\mu}_2 = \frac{1}{10}\sum_{i=1}^{10} x_i$. Which is the better estimator? Why?	06
8	a	Compressive strength, x and permeability y of cured concrete mixes were studied and the summary quantities are :$\sum x_i = 43$, $\sum x_i^2 = 157.42, \sum y_i = 572, \sum y_i^2 = 23530, \sum x_i y_i = 1697.80$. Assume a linear relation between x & y. Number of observations (x_i, y_i) is 14. i) Compute the least square estimates of slope and intercept. ii) Estimate the permeability when compressive strength is $x = 3.7$. If the observed value $y = 46.1$ when $x = 3.7$. If the observed value $y = 46.1$ when $x = 3.7$, what is the corresponding residual? iii) What is the estimate of underlying variance of the error terms?	12
	b	What kind of misuses one has to guard against when dealing with simple linear regression?	04
		OR	
9		In finance, the overall market return x and a particular stock's return y are observed at 10 different instances as below:	

$x(\%)$	11	15	3	18	10	12	6	7	18	13
$y(\%)$	10	12	8	15	9	11	8	10	13	11

		i) Plot a scatter diagram of $y\ vs\ x$. What does the figure imply? ii) Compute slope and intercept coefficients. iii) Compute residuals at $x = 3, 7$ & 11 iv) With what assumption are the regression coefficients computed using least squares estimation? How will you validate the assumption using residual plots? v) Compute the variance of the error terms. vi) What is the proportion of variability explained?	16
10	a	Number of defects in a part were observed for 3420 parts in a month as given below. Do these values indicate that the data is from a Poisson distribution? Test at 5% significance. $Number\ of\ defects, i$ | 0 | 1 | 2 | 3 | 4 + $Number\ of\ parts\ with\ i\ defects$ | 2962 | 382 | 47 | 25 | 4	08
	b	The life of an electronic sensor is normally distributed. A random sample of 20 sensors has a mean of 1000 hours and a standard deviation of 100 hours. i) Construct a 99% 2 sided confidence interval for the mean life of sensors, if population variance is unknown. ii) If the population standard deviation is known to be 105 hours, construct a 99% 2-sided interval for the mean life. iii) Comment on the difference in answers to (i) and (ii).	08
		OR	
11	a	A *TV* documentary claimed that Americans are about 10 pounds overweight on average. To test this claim, 18 randomly selected individuals were examined and their average excess weight was 12.4 pounds, with their standard deviation 2.7 pounds. At 1% significance, can we doubt the validity of the claim?	08
	b	From a sample of 42 petrol bunks in a state, the average price of a fuel additive was found to be $Rs.1.12$ per liter with sample standard deviation $Rs.0.04$. Construct a 99% confidence interval on the mean price of the additive.	03
	c	From a population with known standard deviation of 1.65 and a sample mean of 34.8 from a sample of 32 values: i) Find the standard error of the mean ii) Compute an interval estimate that will include the population with a probability of 99.7%	05

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Industrial Engineering and Management

STATISTICS FOR DECISION MAKING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
2. Answer FIVE full questions from Part B.
3. Use of statistical tables is permitted.

PART-A

1	1.1	A _____ is a graph that present the median, first and third quartiles, and any outliers that are present in a sample.	01
	1.2	A _____ is an approximation to a probability density function.	01
	1.3	The events A and B are said to be _____ if they have no outcomes in common.	01
	1.4	The _____ of a random variable X is a description of the probabilities associated with the possible values of X.	01
	1.5	If the range of X is the set $\{0, 1, 2, 3, 4\}$ and $P(X = x) = 0.2$, then the mean of the random variable is _____.	01
	1.6	If a random variable X equals the number of trials that result in a success out of 'n' trial is called _____ random variable.	01
	1.7	A _____ random variable represents the number of occurrences of an event of interest in a unit interval specified.	01
	1.8	An _____ random variable represents the length of an interval from a certain point until the next success in a Poisson process.	01
	1.9	A _____ random variable represents the length of an interval until the next success for various types of success rate.	01
	1.10	Statistical _____ is concerned with making decisions about a population based on the information contained in a random sample from that population.	01
	1.11	The general idea behind the _____ is to equate population moments, defined in terms of expected values to the corresponding sample moments.	01
	1.12	The probability distribution of a statistic is a _____ distribution.	01
	1.13	The approach used to estimate the regression coefficients is the method of _____.	01
	1.14	The _____ describes the error in the fit of the model to the ith observation y_i.	01
	1.15	The correlation is a measure of the _____ of association between the two variables.	01

	1.16	The _____ is interpreted as the proportion of variability in the observed response variable that is explained by the linear regression model.	01
	1.17	A $100(1-\alpha)$ percent upper confidence interval on the mean of a normal distribution, variance unknown but $n > 30$ is mathematically expressed as _____.	01
	1.18	For a normal population with known variance σ^2 the confidence level for the interval $\bar{x} - 2.49\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + 2.49\frac{\sigma}{\sqrt{n}}$ is _____ %.	01
	1.19	Failing to reject the null hypothesis when it is false is defined as a _____ error.	01
	1.20	The Chi-square random variable is non-negative and the probability distribution is skewed to the _____.	01

PART-B

2	a	The data on the motor fuel octane ratings of several blends of gasoline is given in following table: 93 90.2 91 92.4 90.5 88.7 89.8 89.5 89.8 84.6 91.0 87.3 92.1 94.8 90.2 89.8 88.8 96.7 88.7 91.3 85.9 85.6 84.7 89.3 89.0 89.7 92.6 87.7 92.3 92.3 86.5 89.5 87.3 88.7 94.1 87.2 85.4 90.7 Construct a Stem and Leaf Diagram and highlight the important feature that can be noticed from histogram.	05
	b	A manufacturing firm employs three analytical plans for the design and development of a particular product. For cost reasons, all three are used at varying times. In fact, plans $1, 2$ and 3 are used for $30\%, 20\%$ and 50% of the products respectively. The defect rate is different for the three procedures. The defect rate for plan, 1,2 and 3 are $0.01, 0.03$ and 0.02. If a random product was observed and found to be defective, which plan was more likely to be used and thus responsible?	07
	c	The probability distribution of the number of imperfections per 10 meters of a synthetic fabric in continuous rolls of uniform width, is given by x | 0 | 1 | 2 | 3 | 4 $f(x)$ | 0.41 | 0.37 | 0.16 | 0.05 | 0.01 Determine mean and variance of x.	04
		OR	
3	a	The data on the joint temperatures of the O-rings (°F)for each test firing or actual launch of the space shuttle rocket motor is given below: 84 49 61 40 83 67 45 66 70 69 80 58 68 60 67 72 73 70 78 52 67 53 67 75 61 70 81 76 79 75 76 58 31 Construct a box plot the data and comment on the important features and the possible presence of outliers.	08

	b	The probability that an automobile being filled with gasoline will also need an oil change is 0.25, the probability that it needs a new oil filter is 0.40; and the probability that both the oil and filter need changing is 0.14. i) If the oil had to be changed, what is the probability that a new oil filter is needed? ii) If a new oil filter is needed, what is the probability that the oil has to be changed?	04
	c	The proportion of the budget for a certain type of industrial company that is allotted to environmental and pollution control is coming under scrutiny. A data collection project determines that the distribution of these proportions is given by $f(x) = \begin{cases} 5(1-x)^4 & 0 \le x \le 1 \\ 0, & elsewhere \end{cases}$ i) Verify that the above is a valid density function. ii) What is the probability that a company chosen at random expends less than 10% of its budget on environmental and pollution controls? iii) What is the probability that a company selected at random spends more than 50% of its budget on environmental and pollution controls?	04
4	a	A manufacturing process has 100 customer orders to fill. Each order requires one component part that is purchased from a supplier. However, typically, 2% of the components are identified as defective, and the components can be assumed to be independent. i) If the manufacturer stocks 100 components, what is the probability that the 100 orders can be filled without reordering components? ii) If the manufacturer stocks 102 components, what is the probability that the 100 orders can be filled without reordering components?	04
	b	The time between calls to a plumbing supply business is exponentially distributed with a mean time between calls of 15 minutes. i) What is the probability that there are no calls within 30 minute interval? ii) What is the probability that the first call arrives within 5 and 10 minutes after opening? iii) Determine the length of an interval of time such that the probability of at least one call in the interval is 0.90.	06
	c	Assume that the life of a roller bearing follows a Weibull distribution with parameters $\beta = 2$ and $\delta = 10000$ hours. i) Determine the probability that a bearing lasts at least 8000 hours. ii) Determine the mean time until failure of a bearing. iii) If 10 bearings are in use and failures occur independently, what is the probability that all 10 bearings last at least 8000 hours.	06

OR

5 a The number of surface laws in plastic panels used in the interior of automobiles has a distribution with a mean of 0.05 flaws per square foot of plastic panel. Assume that an automobile interior contains 10 square feet of plastic panel.

i) What is the probability that there are no surface flaws in an auto's interior?

ii) If 10 cars are sold to a rental company, what is the probability that at most 1 car has any surface flaws? 06

b The thickness of a flange on an aircraft component is uniformly distributed between 0.95 and 1.05mm

i) Determine the cumulative distribution function of flange thickness.

ii) Determine the proportion of flanges that exceeds 1.02mm.

iii) What thickness is exceeded by 90% of the flanges? 06

c The reaction time of a driver to a visual stimulus is normally distributed with a mean of 0.4 seconds and a standard deviation of 0.05 seconds.

i) What is the probability that a reaction requires more than 0.5 seconds?

ii) What is the reaction time that is exceeded 90% of time? 04

6 a Determine the covariance and correlation for the following joint probability distribution:

X	1	1	2	4
Y	3	4	5	6
$f_{x,y}(x,y)$	1/8	1/4	1/2	1/8

08

b Consider the probability density function $f(x) = c(1+\theta x), 1 \le x \le 1$

i) Find the value of the constant c

ii) What is the moment estimator for θ?

iii) Show that $\hat{\theta} = 3X$ is an unbiased estimator for θ. 08

OR

7 a Let X denotes the number of times a certain numerical control machine will malfunction: 1,2, or 3 times on any given day. Let Y denote the number of times a technician is called on an emergency call. Their joint probability distribution is given as

$f(x,y)$		x		
		1	2	3
y	1	0.5	0.05	0.1
	2	0.05	0.1	0.35
	3	0.0	0.2	0.1

i) Evaluate the marginal distribution of X

ii) Evaluate the marginal distribution of Y

iii) Compute $P(Y = 3 \mid X = 2)$. 06

	b	A consumer electronics company is comparing the brightness of two different types of picture tubes for use in its television sets. Tube type A has mean brightness of 100 and standard deviation of 16, while tube type B has unknown mean brightness, but the standard deviation is assumed to be identical to that for type A. A random sample of $n = 25$ tubes of each type is selected, and $\overline{X_B} - \overline{X_A}$ is computed. If μ_B equals or exceeds μ_A, the manufacturer would like to adopt type B for use. The observed difference is $\overline{X_B} - \overline{X_A} = 3.5$. What decision would you make, and why?	06
	c	Determine the value of c that makes the function $f(x,y) = c(x+y)$ a joint probability density function over the range $0 \le x \le 3$ and $0 \le y \le x$ and the Marginal probability distribution of X.	04
8	a	What are the assumptions made in linear regression analysis?	04
	b	A study was made to model the relation between weekly advertising expenditures and sales. During the study following data were recorded:	

Advertising cost	20	25	30	35	40	45	50	55	60
Weekly Sales	400	420	405	480	475	490	525	560	515

i) Plot a scatter diagram and recommend suitable model.
ii) Find the equation of the regression line to predict weekly sales from advertising expenditures.
iii) Estimate the weekly sales when advertising costs are 35.
iv) Compute coefficient of determination R^2 and interpret about model. 12

OR

9	a	Discuss the abuses to regression analysis.	04
	b	The following data gives the hourly numbers of units spoiled and the hourly number of items produced for 10 press operators:	

Operators	1	2	3	4	5	6	7	8	9	10
Units produced/hour (X)	94	98	106	114	107	93	98	88	104	95
Units spoiled/hour (Y)	4	5	6	7	6	5	6	4	7	5

i) Plot the scatter diagram and comment on the nature of correlation.
ii) Determine the strength of relationship between correlation units produced and spoiled and comment on strength of relationship.
iii) Estimate the underlying value of variance σ^2 08

	c	Briefly explain the methods used to check the model of adequacy.	04
10	a	A manufacturer of sprinkler systems used for fire protection in office buildings claims that the true average system – activation temperature is 130℃. A sample of $n = 9$ systems, when tested, yields a sample average activation temperature of 131.08℃. If the distribution of activation times is normal with standard deviation 1.5℃, does the data contradict the manufacturer's claim at significance level $\alpha = 0.05$? Also answer the question by constructing appropriate confidence interval.	08

	b	The number of defects in printed circuit boards is hypothesized to follow a Poisson distribution. A random sample of $n = 60$ printed boards has been collected and the following number of defects observed. $Number\ of\ defects$ | 0 | 1 | 2 | 3 $Observed\ frequency$ | 32 | 15 | 9 | 4 Does the assumption of the Poisson distribution seem appropriate as a probability model for this data? Use $\alpha = 0.05$	08
		OR	
11	a	A research engineer for a tyre manufacturer is investigating tyre life for a new rubber compound and has built 16 tyres and tested them to end-of-life in a road test. The sample mean and standard deviation are 60,139.7 and 3645.94 kilometers. i) Can you conclude, using $\alpha = 0.05$, that the standard deviation of tyre life exceeds 200 kilometers? State any necessary assumptions about the underlying distribution of the data. ii) Answer the question (i) by constructing appropriate confidence interval.	08
	b	The production of radar component is checked by sample of four. The following table shows the number of defective found in 300 samples. $Number\ of\ defectives$ | 0 | 1 | 2 | 3 | 4 $Number\ of\ samples$ | 90 | 130 | 58 | 20 | 2 Fit the binomial distribution for the data and test the goodness of the fit at level of significance of 0.05.	08

CAD/CAM, Robotics and CIM

IV Semester SEE Question Papers

Course codes: IM343AI, 21IM43, 16IM44, 18IM44, 16IM44 / 12IM63

- Jun/Jul 2025 (IM343AI)
- Sep/Oct 2024 (IM343AI)
- Oct 2023 (21IM43)
- Oct/Nov 2022 (16IM44)
- Oct/Nov 2022 (18IM44)
- Grade Improvement Nov 2020 (18IM44) - Set 1 [DSpace: GRADE IMPROVEMENT APR-21 MAY-21]
- Grade Improvement Nov 2020 (18IM44) - Set 2 [DSpace: UG APR-20 MAY-20 GRADE IMPROVEMEN
- Fast Track Oct 2020 (16IM44)
- Fast Track Jul 2019 (16IM44)
- Apr/May 2019 (16IM44)
- Fast Track Jul 2018 (16IM44 / 12IM63)
- Apr/May 2018 (16IM44)

IM343AI

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CAD/CAM & ROBOTICS

Time: 03 Hours ***Maximum Marks: 100***

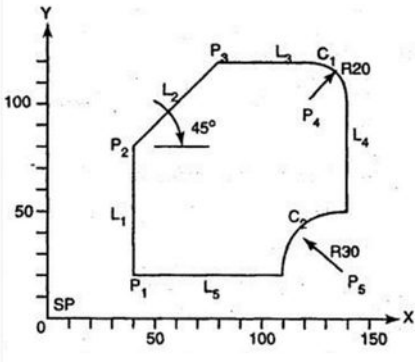
Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	**M**	**BT**	**CO**
1	1.1	Why is conceptual design an important phase in the Shigley model?	02	1	1
	1.2	What is the purpose of a line drawing algorithm?	02	1	1
	1.3	What is a part program in NC machining?	02	2	2
	1.4	What is tool offset in CNC machining?	02	2	2
	1.5	What is mechanization?	02	2	2
	1.6	List two motivating factors for robot usage in industries.	02	1	3
	1.7	The power source of a robot can be electric, hydraulic, or _____.	01	1	3
	1.8	What is meant by payload in robotics?	02	2	4
	1.9	Give two examples of exteroceptive sensors.	02	1	4
	1.10	What is lead-through programming?	02	1	4
	1.11	A _____ is a device at the end of a robot arm, designed to interact with the environment.	01	1	4

PART-B

2	a	Use Bresenham's Line Algorithm to rasterize a line from (10, 15) to (20, 25). Show all intermediate steps.	10	4	1
	b	Discuss the various applications of CAD in different industries such as mechanical, civil, and electrical engineering.	06	3	1
3	a	Analyze the effect of incorrect tool offset values in CNC milling programs.	06	1	2
	b	Compare APT programming with manual NC programming in terms of flexibility, accuracy, and complexity.	10	3	2
		OR			

4	a	Prepare a program for milling the given component by milling process using APT. Post Processor: Machine Mill, 1 Feed rate: 60 mm/min Spindle speed: 500 rpm Cutter diameter: 5 mm Tolerance: ± 0.02 mm Thickness: 10 mm 	16	4	2
5	a	Explain the social issues associated with automation.	08	1	2
	b	Explain reasons for adopting automation in industries.	08	3	3
		OR			
6	a	Explain how automation contributes to sustainable manufacturing.	08	4	3
	b	Explain the different types of automation with examples.	08	2	2
7	a	Discuss the selection criteria for industrial robots in manufacturing applications.	08	2	3
	b	Explain the robot wrist and end-of-arm tooling (EOAT) with examples.	08	2	3
		OR			
8	a	A family-owned metal fabrication workshop is evaluating whether to automate its manual cutting and welding operations. However, the owner is hesitant. Analyze the reasons why automation might not be suitable for such a workshop. Provide alternatives.	10	4	4
	b	Explain the classification of robots based on configuration.	06	1	3
9	a	Describe in detail the role of sensors in robotics and how they improve robot performance.	08	3	4
	b	How do exteroceptive sensors support robotic adaptability in dynamic environments?	08	2	4
		OR			
10	a	A bakery needs a robot to pick and place soft pastries into boxes. Recommend an end effector and justify it based on hygiene, delicacy, and cycle time.	10	5	4
	b	Explain on-line robot programming and its advantages.	06	1	4

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IV Semester B.E. Examinations Sept/Oct – 2024

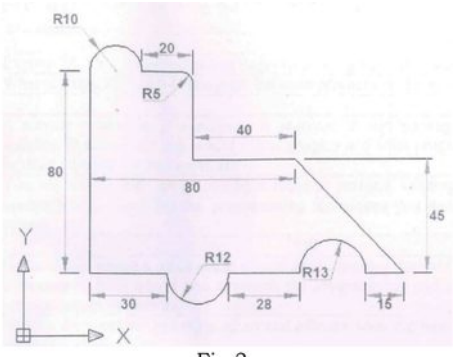
Industrial Engineering and Management

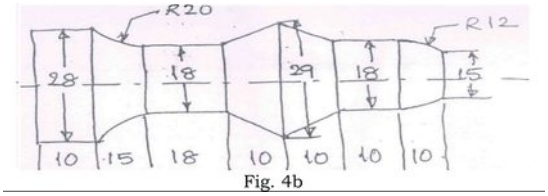
CAD/CAM AND ROBOTICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	What is rendering in computer graphics?	02	2	1
	1.2	What is the Z-buffer algorithm used for?	02	2	1
	1.3	List the main elements of an NC system.	02	2	2
	1.4	Identify two key components of a DNC system.	02	1	2
	1.5	How does mechanization differ from automation?	02	1	2
	1.6	What are the primary goals of automation?	02	2	1
	1.7	Define zeroth law of Robotics.	02	1	1
	1.8	List any two features of fixed automation.	02	1	2
	1.9	Define pitch, yaw and roll?	02	2	2
	1.10	What is meant by work envelope?	02	1	1
		PART-B			
2	a	Consider a scenario where you are designing a complex mechanical part using CAD. Discuss how you would utilize the different stages of the Shigley Model to ensure a successful design.	08	2	1
	b	Express a line from (5,5) to (13,9) on a raster screen using Bresenham's Line Drawing Algorithm. Show the result in a Cartesian graph.	08	3	2
3	a	Figure 3a below shows the final profile to be generated on a MS by using a CNC milling center. Write CNC program and show the tool path.  Fig 3a	12	3	3
	b	What key factors should be considered during the Problem Definition stage of the Shigley Model when designing a mechanical part?	04	2	2
		OR			
4	a	Explain the APT statements: (i) GOTO and GODLTA (ii) INTOL and OUTTOL (iii) TANTO and GOBLACK.	06	3	2

	b	Write a manual part program for the component shown below.  Fig. 4b	10	3	2
5	a	Explain the concept of Low-Cost Automation (LCA) and its significance in small and medium enterprises (SMEs).	08	2	2
	b	Imagine you are managing a small manufacturing unit. How would you implement Low-Cost Automation to improve productivity without incurring high expenses? Provide specific example.	08	4	3
		OR			
6	a	A factory is considering the shift from manual assembly to automated assembly lines. Discuss the issues they might face during this transition and how these challenges can be mitigated.	08	4	4
	b	How would you design a strategy for automation in a factory that currently relies heavily on manual labor? Outline the key steps involved.	08	3	3
7	a	Briefly explain the various components of an industrial robot. State the main function of each of the components.	08	2	3
	b	What do you understand by degree of freedom (DOF)? How many DOFs are required to position an end effector at any point in 3-D space?	08	2	2
		OR			
8	a	Explain SCARA robot indicating its joints and degree of freedom with the applications.	08	2	3
	b	What is a robotic joint? Distinguish between prismatic and revolute joints.	08	2	3
9	a	A robotic system in a warehouse is required to sort packages of varying sizes and weights. Discuss how you would select grippers and what programming approach would be most suitable for this application.	08	3	3
	b	You are tasked with programming a robot to perform welding tasks in an automotive assembly line. Describe the programming techniques you would use and justify your choice.	08	3	4
		OR			
10	a	In a scenario where a robot needs to perform a complex assembly task involving multiple components, how would you approach the programming and integration of the robot's sensors and end effectors?	08	4	4
	b	Explain the working principle of an end effector with the help of a sketch and give its important applications.	08	2	3

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RV COLLEGE OF ENGINEERING®

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IV Semester B.E. Examinations October – 2023

Industrial Engineering and Management

CAD/CAM AND ROBOTICS

Time: 03 Hours ***Maximum Marks: 100***

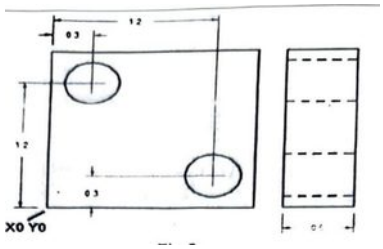
Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Define Robot Programming.	02	–	–
	1.2	List any two advantages of suction grippers.	02	–	–
	1.3	Identify any two objectives of an Industrial Robot.	02	–	–
	1.4	If given an opportunity to build a robot to be used in the department of IEM at RVCE, what problem would the robot solve and what would be its functionality?	02	–	–
	1.5	_____ and _____ are special purpose tool which are used to facilitate production (machining, assembling and inspection operations) when work piece is based on the concept of interchangeability according to which every part will be produced within an established tolerance.	02	–	–
	1.6	Students who carry out productivity studies at industries as part of major project sometimes tend to conclude that one or set of robots can be used to automate the entire process and enhance productivity by multifold and cost of commissioning it can be recovered in a very short period by compensating the salaries of the employee who can be removed from the process. Review such inference from the standpoint of India as a country.	02	–	–
	1.7	Identify two disadvantages of automation.	02	–	–
	1.8	Identify two applications of robots in industrial sector and service sector.	02	–	–
	1.9	Tactile sensors are also called as _____ and are sensitive to _____.	02	–	–
	1.10	State the Zeroth Law of robotics.	02	–	–

PART-B

2	a	Explain the Shigley model with a block diagram.	10	–	–
	b	Enumerate any six benefits of CAD in automation industry.	06	–	–
3	a	Write a CNC peck drilling program for the profile given in Fig 3a. The cutting tool used is slot drill cutter and the depth of cut is 5 mm.  Fig 3a	10	–	–
	b	Discuss the various elements of a typical Numerical Control System.	06	–	–

		OR			
4	a	Explain with the help of an example (i) Linear interpolation (ii) Circular interpolation.	06	–	–
	b	Explain any five types of Distributed Numerical Control Systems.	10	–	–
5	a	Identify major socio-economic considerations favoring automation.	08	–	–
	b	With example, state the difference between mechanization and automation.	08	–	–
		OR			
6	a	Explain how low-cost automation is achieved in manufacturing industry.	10	–	–
	b	Describe six major issues for Automation in Factory Operations.	06	–	–
7	a	The robot has to do the following functions. Recommend the types of sensors it should be fitted with to carry out the operations successfully. Also mention one specification of the recommended sensor. (i) Functionality 1: A binary output signal when it comes in contact with an object. (ii) Functionality 2: To sense and indicate presence of any object without physical contact. (iii) Functionality 3: To identify the distance of sensor from the object. (iv) Functionality 4: To estimate the speed with which a manipulator is moved.	08	–	–
	b	Explain the advantages and disadvantages of vacuum/suction grippers.	08	–	–
		OR			
8	a	Explain the characteristics of an industrial robot.	10	–	–
	b	How are the robots classified? Explain.	06	–	–
9	a	Assume that the robot must pick up objects from a chute and place them in successive boxes. Choose the sequence of robot activity for which robot programming will have to be written at a later stage.	08	–	–
	b	Explain any two robot programming techniques.	08	–	–
		OR			
10	a	Explain the factors considered for selection of gripper.	10	–	–
	b	Write short notes on types of sensors in robots.	06	–	–

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RV COLLEGE OF ENGINEERING®

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IV Semester B. E. Examinations Oct/Nov - 2022

Industrial Engineering and Management

COMPUTER INTEGRATED MANUFACTURING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	What does Interactive Computer Graphics (ICG) denotes in CAD environment?	02
	1.2	What is meant by super sampling?	02
	1.3	Mention the method of predicting the behavior of a real structure under specified load and displacement conditions.	02
	1.4	What is meant by Iso-parametric element?	02
	1.5	Distinguish between two methods for specifying zero point in NC environment namely Fixed zero and Floating zero.	02
	1.6	Identify the two main units of machine control unit.	02
	1.7	List the three types of automation.	02
	1.8	What does the term 'automated assembly' refer to?	02
	1.9	What are the two broad classes of process planning in computer aided environment?	02
	1.10	Interpret the term 'Flexibility' in Flexible manufacturing system.	02

PART B

2	a	Examine the four major reasons for implementing CAD.	04
	b	Examine various steps for the design process as outlined by Shigley.	06
	c	Review the four types of techniques used for scan conversion.	06
3	a	Describe the six steps involved in Finite Element Analysis.	12
	b	Summarize the four phases in solution derival for a typical finite element problem.	04
		OR	
4	a	Examine the three stages involved in applying finite element modeling to engineering problems.	12
	b	Mention and write a couple of sentences about a minimum of four requirements for making a valid mesh in FEA.	04

5	a	Describe the steps to be accomplished in utilizing numerical control in manufacturing.	10
	b	Examine the characteristics of NC machine tool systems according to the type of machine.	06
		OR	
6	a	DNC consists of four basin components. With an appropriate explanation, illustrate the configuration of the basic DNC system.	10
	b	The computer's job in computer assisted part programming consists of four steps. Examine the activities performed in each of the steps.	06
7	a	Analyze the features of general methods of transporting work pieces in automated flowlines.	09
	b	The hardware system that delivers components to the work head in an automated assembly system consists of several elements. Review the five elements of the part delivery system.	07
8	a	What is meant by computer aided process planning? Illustrate with the help of a figure, the typical computer aided process planning system.	06
	b	Write short notes on the following: i) Benefits of Flexible Manufacturing systems. ii) Applications of Flexible Manufacturing systems.	10

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RV COLLEGE OF ENGINEERING®

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IV Semester B. E. Examinations Oct/Nov-2022

Industrial Engineering and Management

COMPUTER AIDED DESIGN AND MANUFACTURING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

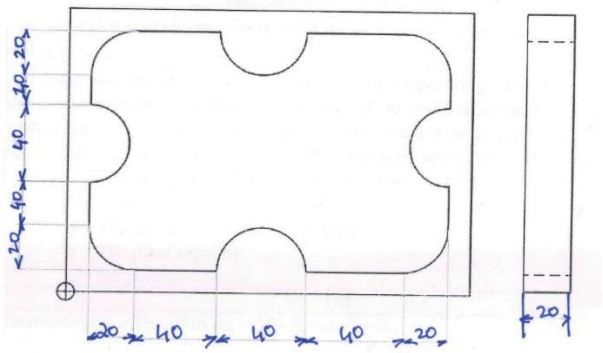
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	DDA stands for _____.	01
	1.2	The process of repositioning an object along a circular path is called _____.	01
	1.3	What happens if the values of scaling factors s_x and s_y are less than 1 (i.e., $s_x < 1$ and $s_y < 1$)?	01
	1.4	_____ algorithm is more accurate than any other circle drawing algorithms as it avoids the use of round off function.	01
	1.5	Each pixel has some intensity value which is represented in memory of the computer. This memory part if known as the _____.	01
	1.6	During the execution of a CNC part program block $NO20\ GO2\ X45.0\ Y25.0\ R5.0$, the type of tool motion will be _____.	01
	1.7	A _____ automated system is one that can produce a variety of products (or parts) with virtually no time lost for changeovers from one product to the next.	01
	1.8	The information-processing cycle, represented by the outer ring can be described as consisting of four functions, they are Business functions, Product design, Manufacturing planning and _____.	01
	1.9	When coordinates going into a program are specified relative to the program zero point, it is called _____.	01
	1.10	The finite element method formulation of the problem result in a system of _____.	01
	1.11	To solve an FEM problem, it subdivides a large problem results in a system of _____.	01
	1.12	Two or more processing (or assembly) operations are being performed _____ on the same work part, thus reducing total processing time.	01
	1.13	_____ is a technology concerned with the application of mechanical, electronic and computer-based systems to operate and control production.	01
	1.14	_____ is the detailed planning process that tracks manufacturing output and matches this against customer orders that have been placed.	01
	1.15	Components of the operation time is $T_o = T_m + T_h + T_{th}$. Here T_{th} stands for _____.	01

	1.16	_____ is the interface between product development and manufacturing system development.	01
	1.17	The capability of the machines to a wide range of products of operations and part cycles is known as _____.	01
	1.18	How do storage buffer affect transfer line downtime?	02
	1.19	What kind of automation would you recommend for cement manufacturing?	01

PART B

2	a	The process of designing something is characterized by Shigley as an interactive procedure which consists of six identifiable steps. Elaborate the steps with block diagram.	10
	b	Differentiate between DDA and Bresenham's line algorithm.	06
3	a	What is numerical control? What are the three basic components of an NC system?	06
	b	Write a manual part program for the component shown in Fig 3b. Assume missing dimensions.  Fig 3b	10
		OR	
4	a	List and explain the different types of control system in CNC machines with neat sketches.	06
	b	Write a complete APT program to perform drilling operations of the job given in Fig 4b below. Assume the missing data.	

Post processor	Machine Drill, 1
Feed rate	30mm/min
Spindle speed	1500rpm
Cutter diameter	20mm
Tolerance	$\pm 0.05mm$
Thickness	15mm

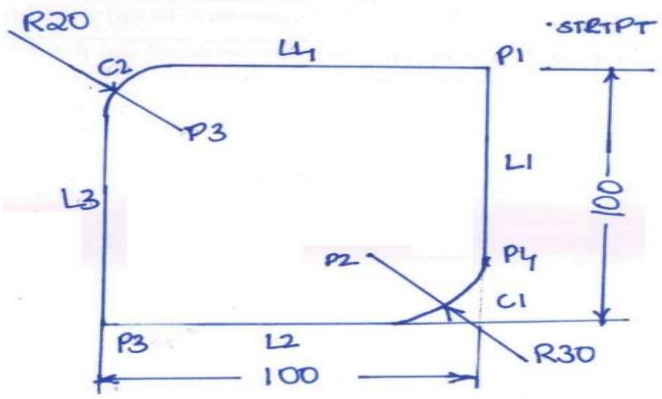


Fig 4b

10

5	a	Modern film projector uses mechanisms which convert a linear motion into a rotational motion which is made easy for rotating the film. Discuss the transfer mechanism used and its process.	08
	b	A typical part produced in a certain batch manufacturing plant must be processed on an average through 8 machines. There are 25 new batches of parts launched each week. Average operation time is 6min, average setup time is 5hours, average batch size is 25 parts, average non-operational time per batch is 10hours. There are 18 machines in the plant. The plant operates on an average of 70 production hours per week. Scrap is negligible. Determine the following. i. Manufacturing lead time for an average part ii. The plant capacity iii. The plant utilization.	08
		OR	
6	a	Enumerate the effect of using buffer storage in any manufacturing industry.	06
	b	Evaluate the various automation strategies in detail.	10
7	a	Elaborate the steps involved in finite element analysis.	08
	b	List out the advantages and disadvantages of FEM.	08
8	a	Explain with block diagram, variant CAPP system of automated process planning.	06
	b	What is FMS? Explain the components of FMS.	10

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IV Semester B. E.Grade Improvement ExaminationsNov-2020

Industrial Engineering and Management

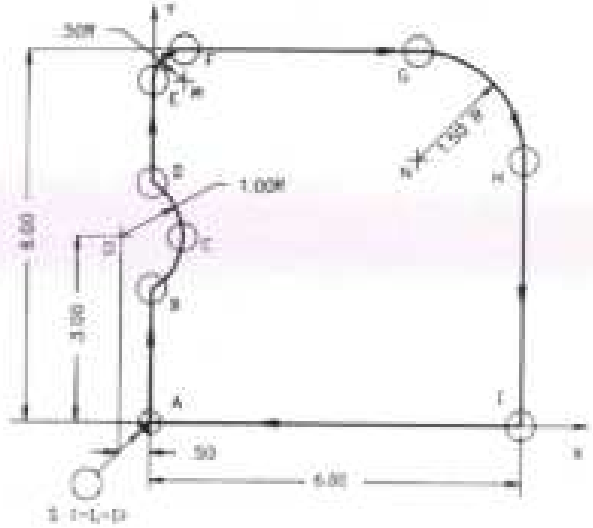
COMPUTER AIDED DESIGN MANUFACTURING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

Answer any FIVE full questions out of TEN. Each carries 20 marks

1	1.1	CAD/CAM is the relationship between ____ and ____.	01
	1.2	The use of computers to control the operation of the production process is known as ____.	01
	1.3	In _____ system, the coordinates are mentioned in the program with respect to one reference point.	01
	1.4	The matrix representation for translation in homogeneous coordinates is _____.	01
	1.5	A triangular plane stress element has _____ degrees of freedom.	01
	1.6	If the structure is more complex in order to simplify the model, we need to subdivide the structure into substructures. These substructures are termed as _____.	01
	1.7	For truss analysis _____ type of element is used.	01
	1.8	For 1-D bar elements if the structure is having 3 nodes then the stiffness matrix formed is having an order of _____.	01
	1.9	A combination of equipment and controls which handles, stores and retrieves materials with precision, accuracy and speed under a defined degree of automation is known as _____.	01
	1.10	_____ is an independent and self-operated vehicle which moves on defined guide ways.	01
	1.11	List different types of Automated Guided Vehicle(AGV).	02
	1.12	List the applications of Automated Guided Vehicle(AGV).	02
	1.13	Ina CNC program block, N02 GO2 G91 X40 Z40……., GO2 and G91 refer to _____.	02
	1.14	In an NC machining operation, the tool has to be moved from point (5,4) to point (7,2) along a circular path with centre at (5,2). Before starting the operations , the tool is at (5,4). The G and N codes for this motion are _____.	02
	1.15	During the execution of a CNC part program block NO20 GO2 X45.0 Y25.0 R5.0 the type of tool motion will be _____.	02
2	a	"CAD system can beneficially be used in four of the design phases". Substantiate your answer with Shigley design process.	08
	b	Given the centre point coordinates(0,0) and radius as 8, generate all the points to form a circle using Bresenham's Circle Drawing Algorithm.	12
3	a	Discuss the importance of implementing the computer aided drawing systems.	10
	b	Using Digital Differential analyzer (DDA) Line Drawing Algorithm draw line segments from point (5,6) to (8,12).	10

4	a	With the help of a neat sketch, explain the following motion control system: i. Point to point system ii. Contouring path system.	10
	b	Enumerate the importance of using direct numerical control system.	10
5	a	Name the function of miscellaneous function words for the following: i. m 00 ii. m 03 iii. m 04 iv. m 05 v. m 06 vi. m 07 vii. m 09 viii. m 45	08
	b	Write a manual part program to mill the component as shown in Fig 5.b. Make suitable assumptions.  Fig 5.b	12
6	a	Describe Automation and discuss the different types of automation.	10
	b	A production machine is operated 65h/week at full capacity. Its production rate is 20units/hr. During a certain week, the machine produced 1000 good parts and was idle the remaining time. i. Determine the production capacity of the machine. ii. What was the utilization of the machine during the week under consideration?	10
7	a	Enumerate the importance of automation strategies.	10
	b	A single station robotic assembly system performs a series of five assembly elements, each of which adds a different component to a base part. Each element takes 4.5 sec. In addition , the handling time needed to move the base part into and out of position is 4sec. The fraction defect rate is 0.005 for all components, and the probability of a jam by a defective component is 0.7. Average downtime per occurrence=2.5min. Determine i. Production rate ii. Yield of good product in the output iii. Uptime efficiency iv. Proportion of the output that contains a defective type 3 component.	10
8	a	State the purpose of finite element analysis in Computer Aided Design. List and explain the steps involved in finite element analysis.	08
	b	Differentiate between structured and unstructured mesh generation techniques in finite element analysis.	12
9	a	List the requirements on mesh generation in a finite element method.	12
	b	Differentiate between local and global coordinate system in FEM?	08

10	a	Illustrate with a neat block diagram the retrieval computer aided process planning system.	08
	b	Discuss the flexibility meaning for the following approaches in a flexible manufacturing system. i. Manufacturing ii. Operational iii. Customer iv. Strategic.	12

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RV COLLEGE OF ENGINEERING®

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IV Semester B. E. Grade Improvement Examinations Nov-2020

Industrial Engineering and Management

COMPUTER AIDED DESIGN MANUFACTURING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

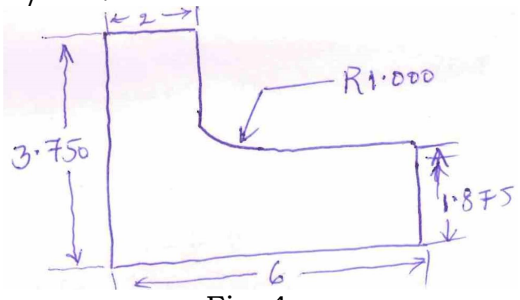
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART A

1	1.1	Modern CAD systems are based on Interactive Computer Graphics (ICG). What does ICG denote?	02
	1.2	There are number of variations in the characteristics of computer graphics that may be classified into three categories. What are they?	02
	1.3	What is Numerical Control in production systems?	02
	1.4	List the basic components of DNC System.	02
	1.5	Categorize the types of automated assembly systems based on the type of work transfer system.	02
	1.6	Categorize the types of automated assemble systems based on the type of work transfer system.	02
	1.7	In a typical finite element problem, the solution is usually desired in four separate phases. What are they?	02
	1.8	What is meant by ISO-parametric element?	02
	1.9	Bring out the primary differentiating feature between retrieval type and generative type of process planning.	02
	1.10	What is the meaning of 'flexibility' in Flexible Manufacturing System?	02

PART B

2	a	Explain the various steps for design process as outlined by Shigley. Depict through a block diagram.	08
	b	Review the principles involved in the four techniques used in scan conversion.	08
3	a	Enumerate the steps involved in utilizing Numerical Control in manufacturing.	08
	b	Illustrate the generic configuration of computer numerical control system. Also review the functions of CNC.	08
		OR	

4	a	Write an APT program to cut the profile as shown in Fig. 4.a below. Use the cutter diameter as 0.256 inches and the cutting speed of 900 rpm and feed rate of 3 in/min.  Fig. 4.a	08
	b	Review the classification of NC machine tools according to the type of control system.	08
5	a	Illustrate and explain the relationship between physical activities and information processing activities in an organization. Explain the role of automation and computer integrated manufacturing philosophies in integrating both in organizations.	08
	b	Review the two forms of workflow configurations in automated flow lines.	08
		OR	
6	a	Classify the general methods of transferring / transporting work pieces on flow lines. Explain any one method in detail.	08
	b	Describe the elements of a typical part delivery system.	08
7	a	Enumerate the six steps involved in finite element analysis works.	08
	b	Identify the requirements that make a mesh valid in mesh generation. Explain in detail any four.	08
8	a	Mention the two functions of variant (retrieval) $CAPP$ systems. Also with an appropriate illustration, describe the procedure involved in Retrieval type $CAPP$ system.	08
	b	Justify the purpose of factory data collection system in shop floor control. Also review the Factory Data Collection systems.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E. Fast Track Examinations Oct-2020

Industrial Engineering and Management

COMPUTER INTEGRATED MANUFACTURING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART A

1	1.1	List any two reasons for using a computer-aided design system to support the engineering design function	02
	1.2	Distinguish between the terms 'rendering' and 'shading'	02
	1.3	List the three stages of finite element analysis	02
	1.4	What is meant by 'Numerical control'?	02
	1.5	Mention the four types of statements in the *APT* language	02
	1.6	List any two advantages of computer numerical control over conventional numerical control	02
	1.7	List any two features of flexible automation that are distinguishable from programmable automation	02
	1.8	Illustrate the configuration of a casual assembly system	02
	1.9	Identify the typical material handling systems used for 'In-line' type layout configuration of flexible manufacturing system	02
	1.10	The order scheduling module in shop floor control is intended to solve two problems in production control. Identify them	02

PART B

2	a	With a suitable illustration, explain the design process as advocated by Shigley.	08
	b	Provide a review of graphic primitives; along with appropriate illustrations.	08
3	a	Explain in detail the general procedure for finite element analysis.	12
	b	Write a brief note on "Mesh generation techniques".	04
		OR	
4	a	Enumerate the *CAD* applications to finite element method.	08
	b	What is an Isoparametric element? Give a brief account of the steps involved in isoparametric formulation of the bar element.	08

5	a	Explain in brief the basic components of Numerical control along with an illustration.	08
	b	Analyze the three types of motion control used in Numerical control.	08
		OR	
6	a	Illustrate with suitable explanation, the general configuration and components of Direct Numerical Control(DNC)	08
	b	Provide a general configuration of CNC system. Also list atleast five features of CNC system.	08
7	a	Deliberate on any eight reasons for automation.	08
	b	List different types of automated assembly systems based on the type of work transfer system used. Explain in detail any two of them.	08
8	a	Provide a brief note on the approaches followed in computer aided process planning systems.	08
	b	Write short notes on: i) Factory data collection system ii) Bar code technology.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E. Fast Track Examinations July-19

Industrial Engineering and Management

COMPUTER INTEGRATED MANUFACTURING

Time: 03 Hours ***Maximum Marks: 100***

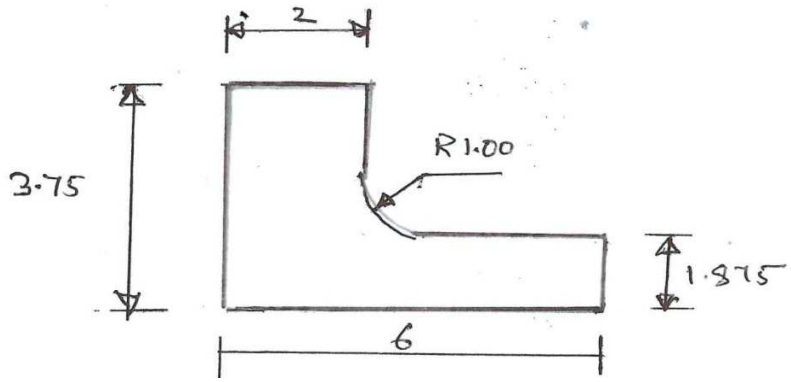
Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART A

1	1.1	In computer aided design _____ is concerned with computer compatible mathematical description of the geometry of an object.	01
	1.2	_____ may involve the use of stress-strain calculation, heat transfer computation, or use differential equation to describe the dynamic behavior of system being designed.	01
	1.3	Two commonly used incremental shading techniques are _____ shading and _____ shading.	02
	1.4	In _____ phase of solving finite element problem, the physical region of interest is divided into elements and element type and appropriate functions are selected.	01
	1.5	Elements whose shape (or geometry) and field variables are described by the same interpolation functions of the same order are known as _____ elements.	01
	1.6	The _____ stage of applying finite element modeling to engineering problem helps in visualization and interpretation of result.	01
	1.7	The tool of an NC machine has to move along a circular arc from $(5,5)$ to $(10,10)$ while performing an operation. The centre of the arc is at $(10,5)$. Write the NC tool path command to perform the above mentioned operation	02
	1.8	Rotation about X-axis is called as _____.	01
	1.9	_____ M code is used for calling a sub-program.	01
	1.10	Finite element method is also called as _____.	01
	1.11	List the three stages involved in applying finite element method to engineering problem.	02
	1.12	_____ G code is used for metric input	01
	1.13	The _____ module of shop floor control provides the documentation needed to process a production order.	01
	1.14	_____ element of the part delivery system physically places the components in the correct location at the workstation for the assembly operation by the work head.	01
	1.15	Withrespect to flexible manufacturing system _____ performs machining operations on families of parts.	01
	1.16	Retrieval-type of computer-aided process planning system is also called as _____ $CAPP$ system.	01
	1.17	_____ auxiliary statement is used in APT program for end of program.	01

PART B

2	a	The process of designing something is characterized by Shigley as an interactive procedure which consists of six identifiable steps. Elaborate the steps with a block diagram.	08
	b	Briefly explain the four techniques used for scan conversion in computer graphics.	08
3	a	Explain the three stages involved in applying the FEM to engineering problem.	10
	b	List any eight facilities that should be provided by a CAD system for automatic mesh generation and presentation of data in FEM.	06
		OR	
4	a	Explain with help of an example the four phases for deriving solution to a finite element problem.	08
	b	Give details of various types of element shapes usually employed for modeling components.	08
5	a	Differentiate between point-to-point and continuous path control in a motion control system.	06
	b	Explain with help of a block diagram, the components of direct numerical control.	10
		OR	
6	a	Write an APT program to cut the profile as shown in the fig 6a. With the cutter of diameter 0.256 inches, cutting speed of 900 rpm and feed rate of 3 in/min.  Fig 6a	10
	b	Explain the basic components of a numerical control system.	06
7	a	Elucidate the ten strategies for automation and process improvement.	10
	b	Explain with the help of suitable example, linear transfer mechanism.	06
8	a	Explain with help of a block diagram, the three phases in shop floor control system.	08
	b	With a suitable flow chart, explain the Retrieval type computer aided process planning system.	08

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RV COLLEGE OF ENGINEERING®

(An autonomous Institution affiliated to VTU)

IV Semester B. E. Examinations April/May-19

Industrial Engineering and Management

COMPUTER INTEGRATED MANUFACTURING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

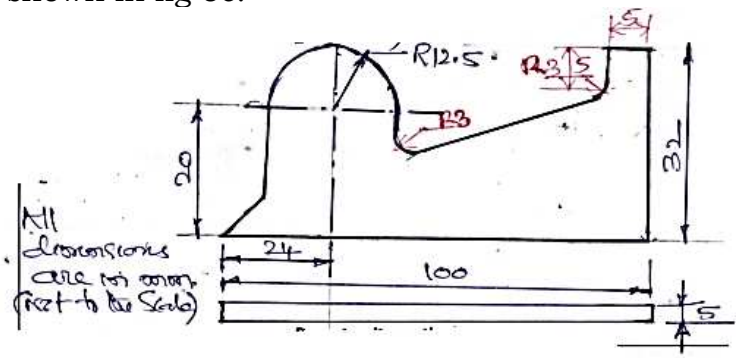
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART A

1	1.1	List the reasons for implementing CAD.	02
	1.2	What is the meaning of the terms "rendering" and "shading"?	02
	1.3	List the three stages of finite element analysis.	02
	1.4	Distinguish between CNC and DNC machine tools.	02
	1.5	Write the four types of statements used in APT.	02
	1.6	Distinguish between fixed zero and floating zero.	02
	1.7	List any two automation strategies.	02
	1.8	List the components of parts feeding device.	02
	1.9	What is the importance of computer aided process planning?	02
	1.10	List the elements of a flexible manufacturing system.	02

PART B

2	a	With a block diagram, explain the design process (Shigley's model).	08
	b	With suitable sketches, explain the graphic primitives for solid modeling.	08
3	a	With suitable example, explain the general procedure for finite element analysis.	08
	b	Explain the following terms: i) Automatic mesh generation ii) Natural coordinates of one dimensional element.	08
		OR	
4	a	With suitable example, explain iso-parametric elements.	08
	b	Explain the CAD application to finite element method.	08
5	a	Explain the NC procedure.	06
	b	List the various functions and features of CNC machine tools.	04
	c	With a block diagram explain DNC system.	06
		OR	

6	a	With suitable G codes and M codes explain the steps involved in manual part programming.	06
	b	Differentiate between absolute and incremental positioning.	02
	c	With suitable assumptions, write the complete APT program to machine the component as shown in fig 6c.  Fig 6c	08
7	a	With suitable example, explain the following types of automation: i) Fixed Automation ii) Programmable Automation iii) Flexible Automation.	08
	b	Explain with the help of neat sketches, walking beam linear transfer mechanism.	08
8	a	With a block diagram explain the generative type computer aided process planning.	08
	b	Explain the following terms related to shop floor control: i) Factory data collection system ii) Bar code technology.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV / VI Semester B. E. Fast Track Examinations July-18

Industrial Engineering and Management

COMPUTER INTEGRATED MANUFACTURING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

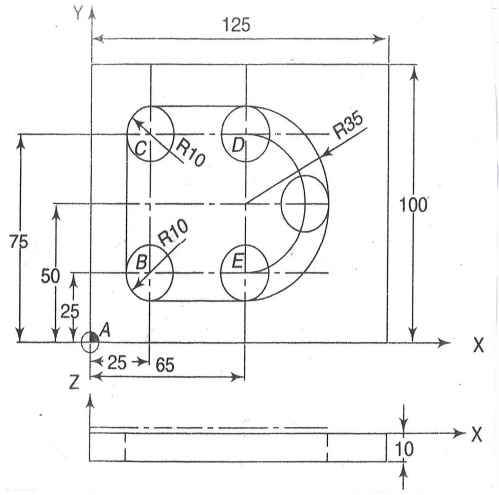
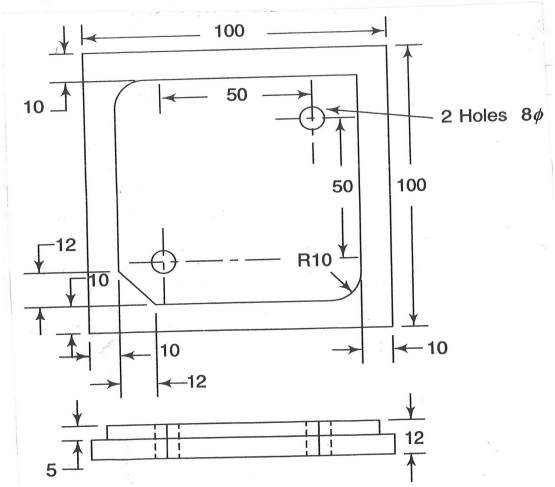
9. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
10. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART A

1	1.1	List any four reasons for implementing CAD.	02
	1.2	Write the matrix for rotation of a line.	01
	1.3	Write the basic equation for the FEA technique.	01
	1.4	Write the steps involved in the finite element analysis.	02
	1.5	Write the block diagram of a single NC system components.	02
	1.6	Name the components of a DNC system.	02
	1.7	Name the different statements in a computer assisted part programming.	02
	1.8	What are the achievements of a CAD/CAM in industrial environment?	02
	1.9	Name any four work transport mechanisms/devices.	02
	1.10	Write the final equation for the analysis of a stator assembly machine.	02
	1.11	Give the classification for material handling devices.	01
	1.12	What is the use of a bar code technology?	01

PART B

2	a	With a block diagram, explain the applications of computers in design process.	10
	b	With suitable sketches, explain the important graphic primitives.	06
3	a	Explain the various steps in the finite element analysis.	10
	b	Explain the following terms with examples: i) Natural coordinates. ii) Isoperimetric elements.	06
		OR	
4	a	Explain the three dimensional shape functions and mesh generation.	08
	b	Explain the general structure of a finite element procedure.	08
5	a	With block diagrams, explain the NC coordinate system for turning, drilling and milling operations.	06

	b	The component to be machined is as shown in figure 5b. The outer profile needs to be maintained using a slot drill of $\phi 16\, mm$. With suitable assumptions of machining condition, write a complete machined part programming using a word address format. All dimensions are in mm.  Figure 5b	10
		OR	
6	a	Explain the functions and features of a CNC machine tools for industrial use.	06
	b	Write a complete manual part programming to machine a component as shown in the figure 6b. Make suitable assumptions and machining conditions while preparing the part program. All dimensions are in mm.  Figure 6b	10
7	a	Explain any four automation strategies to improve the productivity in manufacturing of a complex shapes.	06
	b	With a sketch explain any one type of a work part transport machine.	06
	c	Draw a neat sketch of a part feeding device (any two).	04
8	a	With a block diagram, explain the generative type computer aided process planning.	06
	b	Write a note on applications and benefits of a flexible manufacturing system (FMS).	06
	c	Explain any two methods of a factory data collection system.	04

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Examinations April/May-18

Industrial Engineering and Management

COMPUTER INTEGRATED MANUFACTURING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

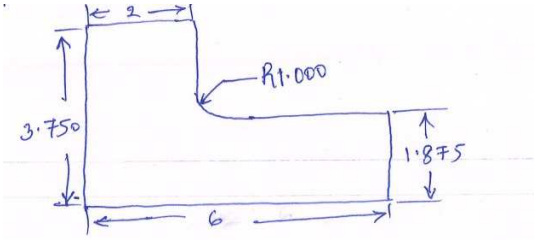
7. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
8. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART A

1	1.1	Identify the reasons for implementing CAD in organizations.	02
	1.2	What is meant by the term primitives in CAD environment?	02
	1.3	Classify the three categories of computer graphics based on the variations in their characteristics.	02
	1.4	Automatic mesh generation reduces the cost involved in Finite Element Analysis-Justify.	02
	1.5	Surfaces are designated for shell meshing in three ways. List the three ways.	02
	1.6	Distinguish between Fixed zero and Floating zero concepts in NC(Numerical Control) systems.	02
	1.7	List the codes to represent the following preparatory functions: a) Thread cutting b) Pattern Repeating.	02
	1.8	List the two categories of mass production.	02
	1.9	Illustrate the configuration of a typical carousel assembly system.	02
	1.10	What are the three basic components of a flexible manufacturing system?	02

PART B

2	a	What is meant by scan conversion? Explain different techniques of scan conversion.	08
	b	What is Rendering? Explain the different stages of rendering an image.	08
3	a	Explain the three stages involved in applying finite element modeling to engineering problems.	08
	b	What are isoparametric elements? Determine the shape functions for Hexahedron elements.	08
		OR	
4	a	Explain the basic requirements behind making a mesh valid, thereby producing correct FEA results.	08

	b	In a typical finite element problem, the solution is generally derived in four separate phases in a particular order.Explain with suitable illustrations.	08
5	a	Explain steps to be accomplished for utilizing numerical control in manufacturing.	08
	b	Provide a classification of NC systems based on the type of machine and also the type of programming method used.	08
		OR	
6	a	Illustrate the configuration of the basic DNC system along with a brief explanation of its basic components.	08
	b	Write an APT program to cut the profile shown below. Assume the following parameters: -cutter diameter: 0.256 inches -cutting speed: 900 rpm -Feed rate: 3 inches/min. 	08
7	a	Provide a brief overview of different automation strategies and their probable effects on different production parameters.	10
	b	Review different types of automated assembly systems.	06
8	a	Illustrate the general procedure for using computer aided process planning system based on the principles of group technology and parts classification and coding.	08
	b	Elaborate on the eight categories of functions performed by the FMS computer control system.	08

Operations Research / Decision Sciences

IV Semester SEE Question Papers

Course codes: IM244AI, 18IM45, 16IM46

- Jun/Jul 2025 (IM244AI)
- Sep/Oct 2024 (IM244AI)
- Nov 2022 (18IM45)
- Grade Improvement Oct 2021 (18IM45)
- Grade Improvement Nov 2020 (18IM45)
- Fast Track Oct 2020 (16IM46)
- Fast Track Jul 2019 (16IM46)
- Apr/May 2019 (16IM46)
- Apr/May 2018 (16IM46)

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B.E. Regular/Supplementary Examinations June/July – 2025

Industrial Engineering and Management

OPERATIONS RESEARCH

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of normal distribution table is permitted.

PART-A

			M	BT	CO
1	1.1	Operations research is a _____ approach to problem solving for executives.	01	2	1
	1.2	Any two isoprofit or isocost lines for a given linear programming problem are _____ to each other.	01	1	2
	1.3	The constraints may be in the form of _____ or _____.	02	1	2
	1.4	Define sensitivity analysis in linear programming.	02	1	1
	1.5	What does the shadow price in duality represent?	02	2	2
	1.6	What is the goal in a transportation problem?	01	1	2
	1.7	What is the cost matrix in assignment problems?	01	2	2
	1.8	What are the steps of the Hungarian method?	02	2	3
	1.9	What is the full form of CPM?	01	1	1
	1.10	What is a critical path?	01	2	2
	1.11	What are the advantages of crashing a network?	02	2	3
	1.12	Define saddle point in a game matrix.	02	1	2
	1.13	When is a game solved using the arithmetic method?	02	2	4

PART-B

2	a	With an example, explain a work-scheduling problem in linear programming.	06	3	2
	b	Explain the concept of feasible region and optimal solution in a linear programming problem using an example.	06	3	2
	c	Define Operations Research and mention any four of its applications.	04	1	1
3	a	Use penalty (Big-M) method to solve the following LP problem. Minimize Z = $5x_1 + 3x_2$ subject to the constraints (i) $2x_1 + 4x_2 \leq 12$, (ii) $2x_1 + 2x_2 = 10$, (iii) $5x_1 + 2x_2 \geq 10$ and $x_1, x_2 \geq 0$.	08	3	2
	b	Formulate the dual of the following problem and interpret the dual variables economically: Maximize Z = 6x + 8y s.t. $2x + 3y \leq 30$, $x + 2y \leq 20$, $x, y \geq 0$	08	3	2
		OR			
4	a	Write the standard format of simplex method for the following LP problem. Maximize Z = $3x_1 + 5x_2 + 4x_3$ subject to the constraints (i) $2x_1 + 3x_2 \leq 8$, (ii) $2x_2 + 5x_3 \leq 10$, (iii) $3x_1 + 2x_2 + 4x_3 \leq 15$ and $x_1, x_2, x_3 \geq 0$	06	4	2

	b	Conduct sensitivity analysis for the objective coefficients of the LP: Maximize $Z = 2x + 3y$ s.t. $x + y \leq 4$, $x + 2y \leq 6$, $x, y \geq 0$ after obtaining optimal solution.	10	4	2
5	a	A company wants to assign 4 salesmen (S1, S2, S3, S4) to 4 different sales regions (R1, R2, R3, R4). The expected profits (in ₹'000s) from each salesman in each region are given in the table below. Assign the salesmen to the regions such that the total profit is maximized.	10	3	3

	R1	R2	R3	R4
S1	25	40	35	20
S2	30	45	40	25
S3	20	35	30	50
S4	10	25	20	30

	b	Compare the three methods of finding initial basic feasible solution: North West Corner, Least Cost, Vogel's Method.	06	4	2
		OR			
6	a	Discuss degeneracy in a transportation problem. How is it identified and resolved?	06	2	3
	b	Explain the general structure and assumptions of a Transportation Problem.	06	2	1
	c	Explain any two methods to find the initial basic feasible solution.	04	2	2
7	a	A project consists of the following activities with the duration in days:	10	4	2

Activity	A	B	C	D	E	F	G	H	I
Precedence	–	A	A	B, C	A	D, E	C	F, G	H
Duration	10	4	12	10	9	8	7	9	6

(i) Draw the network of the above project.
(ii) Identify the critical path and determine the project duration.
(iii) Calculate independent float, free float for all the activities.

	b	An assembly is to be made from two parts X and Y. Both parts must be turned on a lathe. Y must be polished whereas X need not be polished. The sequence of activities, together with their predecessors, is given below.	06	4	3

Activity	Description	Predecessor Activity
A	Open work order	–
B	Get material for X	A
C	Get material for Y	A
D	Turn X on lathe	B
E	Turn Y on lathe	B, C
F	Polish Y	E
G	Assemble X and Y	D, F
H	Pack	G

Draw a network diagram of activities for the project.

		OR			
8	a	'PERT takes care of uncertain durations.' How far is this statement correct? Explain with reasons.	08	4	3
	b	Explain the following in the context of project management: (i) Activity variance (ii) Project variance	08	2	3
9	a	Explain: Minimax and Maximin principle used in the theory of games.	06	2	4

<table>
<tr><td></td><td>b</td><td>By applying dominance rules, reduce the given payoff matrix and identify the optimal pure strategies and value of the game.
<table><tr><td></td><td>B1</td><td>B2</td><td>B3</td><td>B4</td></tr><tr><td>A1</td><td>1</td><td>2</td><td>1</td><td>3</td></tr><tr><td>A2</td><td>0</td><td>−1</td><td>0</td><td>2</td></tr><tr><td>A3</td><td>2</td><td>3</td><td>1</td><td>4</td></tr></table></td><td>10</td><td>4</td><td>4</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td>10</td><td>a</td><td>Obtain the optimal strategies for both persons and the value of the game for two-person zero-sum game whose payoff matrix is as follows:
<table><tr><td>Player A</td><td>B_1</td><td>B_2</td></tr><tr><td>A_1</td><td>1</td><td>−3</td></tr><tr><td>A_2</td><td>3</td><td>5</td></tr><tr><td>A_3</td><td>−1</td><td>6</td></tr><tr><td>A_4</td><td>4</td><td>1</td></tr><tr><td>A_5</td><td>2</td><td>2</td></tr><tr><td>A_6</td><td>−5</td><td>0</td></tr></table></td><td>10</td><td>3</td><td>4</td></tr>
<tr><td></td><td>b</td><td>Game theory provides a systematic quantitative approach for analysing competitive situations in which the competitors make use of logical processes and techniques in order to determine an optimal strategy for winning. Comment.</td><td>06</td><td>5</td><td>4</td></tr>
</table>

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B.E. Regular Examinations Sep/Oct – 2024

Industrial Engineering and Management

OPERATIONS RESEARCH

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	What is the difference between a descriptive and a prescriptive models in operations research?	02	2	2
	1.2	List the major assumptions of linear programming.	02	2	2
	1.3	In simplex method, what is the purpose of the pivot operation?	02	3	3
	1.4	When can the dual Simplex method be applied?	02	3	3
	1.5	List the methods used to arrive at an initial basic feasible solution in transportation model.	02	2	2
	1.6	How does a travelling salesman problem differ from a routine assignment model?	02	3	3
	1.7	If an activity has its optimistic, most likely and pessimistic times as 2, 4 and 9 respectively, then what is its expected time and variance?	02	3	3
	1.8	Which project management tool introduces the estimates for optimistic time and pessimistic time?	02	2	2
	1.9	Define pay-off as applied to game theory.	02	2	2
	1.10	What is a two person zero sum game?	02	2	2

PART-B

2	a	List the seven step model building process with an example for each step.	04	2	2
	b	An Electric Company produces two products P_1 and P_2. Products are produced and sold on a weekly basis. The weekly production cannot exceed 25 for product P_1 and 35 for product P_2 because of limited available facilities. The company employs total of 60 workers. Product P_1 requires 2 man-weeks of labor, while P_2 requires one man-week of labour. Profit margin on P_1 is Rs. 60 and P_2 is Rs. 40. Formulate this problem as an LP problem and solve that using graphical method.	12	4	4
3	a	Discuss the following with respect to linear program: (i) Slack variable (ii) Surplus variable.	04	2	2
	b	Solve the following linear programming problem by simplex method: Maximize Z = $3x_1 + 5x_2 + 4x_3$ subject to the constraints $2x_1 + 3x_2 \leq 8$, $2x_2 + 5x_3 \leq 10$, $3x_1 + 2x_2 + 4x_3 \leq 15$ and $x_1, x_2, x_3 \geq 0$.	12	3	3
		OR			
4	a	Enumerate important characteristics of Duality.	06	2	2
	b	Solve by dual simplex method the following linear programming problem: Minimize Z = $6x_1 + 7x_2 + 3x_3 + 5x_4$ subject to $5x_1 + 6x_2 - 3x_3 + 4x_4 \geq 12$, $x_2 + 5x_3 - 6x_4 \geq 10$, $2x_1 + 5x_2 + x_3 + x_4 \geq 8$ and $x_1, x_2, x_3, x_4 \geq 0$.	10	3	3

5	a	Solve the initial basic feasible solution of transportation problem by least cost entry method: <table><tr><td>Source</td><td>Kanpur</td><td>Pune</td><td>Delhi</td><td>Supply</td></tr><tr><td>Jaipur</td><td>4</td><td>5</td><td>1</td><td>40</td></tr><tr><td>Udaipur</td><td>3</td><td>4</td><td>3</td><td>60</td></tr><tr><td>Mumbai</td><td>6</td><td>2</td><td>8</td><td>70</td></tr><tr><td>Demand</td><td>70</td><td>40</td><td>60</td><td></td></tr></table>	04	2	2
	b	Find the optimum solution to the following transportation problem in which the cells contain the transportation cost in rupees. <table><tr><td></td><td>W_1</td><td>W_2</td><td>W_3</td><td>W_4</td><td>W_5</td><td>Available</td></tr><tr><td>F_1</td><td>7</td><td>6</td><td>4</td><td>5</td><td>9</td><td>40</td></tr><tr><td>F_2</td><td>8</td><td>5</td><td>6</td><td>7</td><td>8</td><td>30</td></tr><tr><td>F_3</td><td>6</td><td>8</td><td>9</td><td>6</td><td>5</td><td>20</td></tr><tr><td>F_4</td><td>5</td><td>7</td><td>7</td><td>8</td><td>6</td><td>10</td></tr><tr><td>Required</td><td>30</td><td>30</td><td>15</td><td>20</td><td>5</td><td></td></tr></table>	12	3	3
		OR			
6	a	Four different jobs can be done at four different machines. The set up and take down time costs are assumed to be prohibitively high for changeovers. The matrix below gives the cost in rupees of producing job i on machine j. <table><tr><td></td><td>M_1</td><td>M_2</td><td>M_3</td><td>M_4</td></tr><tr><td>J_1</td><td>5</td><td>7</td><td>11</td><td>6</td></tr><tr><td>J_2</td><td>8</td><td>5</td><td>9</td><td>6</td></tr><tr><td>J_3</td><td>4</td><td>7</td><td>10</td><td>7</td></tr><tr><td>J_4</td><td>10</td><td>4</td><td>8</td><td>3</td></tr></table> How should the jobs be assigned to the various machines so that the total cost is minimized.	08	3	3
	b	ABC Ice cream company has a distribution depot in Greater Kailash Part 1 for distributing ice-cream in South Delhi. There are four vendors located in different parts of South Delhi (call them A, B, C and D) who have to be supplied ice-cream every day. The following matrix displays the distances (in kilometers) between the depot and the four vendors: <table><tr><td>From \ To</td><td>Depot</td><td>Vendor A</td><td>Vendor B</td><td>Vendor C</td><td>Vendor D</td></tr><tr><td>Depot</td><td>∞</td><td>3.5</td><td>3</td><td>4</td><td>2</td></tr><tr><td>Vendor A</td><td>3.5</td><td>∞</td><td>4</td><td>2.5</td><td>3</td></tr><tr><td>Vendor B</td><td>3</td><td>4</td><td>∞</td><td>4.5</td><td>3.5</td></tr><tr><td>Vendor C</td><td>4</td><td>2.5</td><td>4.5</td><td>∞</td><td>4</td></tr><tr><td>Vendor D</td><td>2</td><td>3</td><td>3.5</td><td>4</td><td>∞</td></tr></table> What route should the company van follow so that the total distance travelled is minimized?	08	3	3
7	a	List the steps for drawing a network.	08	2	2

	b	A project schedule has the following characteristics: (i) Construct the network. (ii) Compute E_i and L_j values for each network. (iii) Find the critical path and total duration.	08	3	3

Activity	Time (weeks)	Activity	Time (weeks)
1–2	4	5–6	4
1–3	1	5–7	8
2–4	1	6–8	1
3–4	1	7–8	2
3–5	6	8–10	5
4–9	5	9–10	7

OR

8	a	What is float? What are the different types of floats?	04	2	2
	b	A project schedule has the following characteristics: (i) Calculate the expected time for each activity. (ii) Draw the network diagram. (iii) Find the critical path and duration. (iv) Calculate total float for each activity.	12	3	3

Activity	10–20	20–30	20–40	20–50	30–60	40–60	50–70	60–70
Optimistic	1	8	3	3	4	7	6	1
Most likely	2	10	4	5	5	8	8	4
Pessimistic	3	12	6	7	6	9	10	7

9	a	A company management and the labor union are negotiating a new three-year settlement. Each of these has 4 strategies: (i) Hard and aggressive bargaining, (ii) Reasoning and logical approach, (iii) Legalistic strategy, (iv) Conciliatory approach. The costs to the company are given for every pair of strategy choice. Find the saddle point and value of the game.	06	3	3

Union \ Company	I	II	III	IV
I	20	15	12	35
II	25	14	8	10
III	40	2	10	5
IV	-5	4	11	0

	b	Use the graphical method for solving the following game and find the value of the game.	10	3	3

Player A \ Player B	B_1	B_2	B_3	B_4
A_1	2	2	3	-2
A_2	4	3	2	6

OR

10	a	List the assumption of a game theory.	08	2	2
	b	Reduce the following game by dominance and find the optimal strategies and value of the game.	08	3	3

Player A \ Player B	I	II	III	IV
I	3	2	4	0
II	3	4	2	4
III	4	2	4	0
IV	0	4	0	8

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E. Examinations November - 2022

Industrial Engineering and Management

DECISION SCIENCES-DETERMINISTIC MODELS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Three main components of a linear programming problem are _____, _____, and _____.	01
	1.2	Graphical method is applicable when the number of decision variables are _____.	01
	1.3	The property that the line joining any two points in the feasible region entirely lies inside the feasible region is known as _____.	01
	1.4	In an LPP with 4 variables and 6 constraints (all '≤' constraints), the number of basic and non basic variables are _____ and _____ respectively.	01
	1.5	A degenerate solution contains _____ in an LPP.	01
	1.6	In maximization LPP with objective function $z = 2x_1+3x_2-4x_3$, there are two '≥' type constraints. The modified objective in Big M method is: z = _____	01
	1.7	Change in the objective function value for a unit change in availability of resources is known as _____ of the resource.	01
	1.8	An LPP is given as: maximize $z = 2x_1+3x_2-4x_3$, subject to $x_1-x_2 \geq 5$ and $x_1+x_2 \geq 8$. The objective in Phase I of the two phase method is _____.	01
	1.9	In a graphical solution of a 2 variable LPP, the constraint line shifts parallelly if _____ is changed.	01
	1.10	The problem of allotting the right job for a right person can be solved using _____ algorithm.	01
	1.11	A transportation problem has 5 sources and 7 destinations, and is balanced, A degenerate solution has _____ allocations.	01
	1.12	Which method of obtaining an initial solution to a transportation problem is generally considered to be the most efficient?	01
	1.13	Can a travelling salesman problem be solved like an assignment problem? Why or Why not?	01
	1.14	In a project network diagram, the critical path is a path from start to end with _____ duration.	01
	1.15	Earliest & Latest start & finish times of activities in a project network diagram are computed using _____ and _____.	01
	1.16	The requirements: B precedes D, and B& C precede E can be shown in a project network diagram as :	01

1.17	In a 2 person 0 sum game, a row strategy A_1 is dominated by another row strategy A_2 if _____.	01
1.18	Games with no saddle point have _____ strategies.	01
1.19	Is the game with pay off matrix $\begin{pmatrix} 2 & 4 \\ -1 & 0 \end{pmatrix}$ biased? Why or Why not? (Row strategies are A_1, A_2 and column strategies are B_1, B_2)	02

PART B

2	a	A paint manufacturer is contracted to produce two types of paint: premium and regular, involving usage of additives, blending using machines and packing. The additive quantity required, packing and blending rates, revenue and cost information ae given in the table:	

	Premium	Regular	Availability	Cost
Additive used (Kg/lit)	0.3	0.15	600 tons	Rs.1.2/Kg
Blending rate (Kilo lit/shift)	2	3	100 shifts	Rs.500/shift
Packing rate (Kilo lit/shift)	12	10	80 shifts	Rs.300/shift
Sales Revenue (Rs/Kilo lit)	6,000	4,000		
Demand (Kilo lit/month)	100	400		

		Formulate the problem as an LPP (Do not solve)	12
	b	Solve graphically: maximize $z = 5x_1+4x_2$, subject to $6x_1+4x_2 \le 24$; $x_1+2x_2 \le 6$; $x_2 \le 2$; $x_1, x_2 \ge 0$ show feasible region, objective function and identity binding constraints. Compute the slacks if any.	04
3		The graphical solution for the LPP: Minimize $z = 5x_1+8x_2$ subject to $x_1 \le 4$; $x_2 \ge 2$; $x_1+x_2 \ge 5$; $x_1, x_2 \ge 0$ is given in Fig 3:	

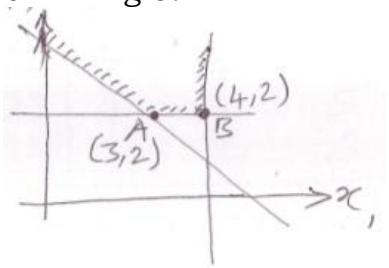


Fig: 3

i) Find the optimal solution
ii) Compute the dual prices of the constraints
iii) Compute feasibility ranges for the constraints. 16

OR

4	a	For the LPP of Question 3, find the optimality ranges of the profit coefficients.	06
	b	Using two-phase method, derive the starting table of Phase II for the LPP: Minimize $z = x_1+x_2$, subject to $2x_1+x_2 \ge 4$; $x_1+7x_2 \ge 7$; $x_1, x_2 \ge 0$. Identify the pivot element.	10

5 a

A company having its manufacturing facilities at locations A, B, C is contracted to supply to destinations P, Q, R. The unit costs of production and transportation are as given in a Table 5a. The destinations P,Q,R levy a penalty of non supply at the rate of Rs.5,3 and 2 per unit respectively. Find the optimum allocation starting from the least-cost method.

Table 5 a: (Costs, demand & Supplies)

	P	Q	R	Supplies
A	5	1	7	10
B	6	4	6	80
C	3	2	5	15
Demand	75	20	50	

12

b

A job shop needs to assign jobs J_1, J_2, J_3, J_4 to W_1, W_2, W_3, W_4. The associated costs are given in Table 5b. Additionally, W_1 cannot be assigned J_3 and W_3 cannot be assigned J_4. Use Hungarian method to find optimal assignment.

Table 5 b: Assignment Costs (Rs)

	J_1	J_2	J_3	J_4
W_1	5	5	-	2
W_2	7	4	2	3
W_3	9	3	5	-
W_4	7	2	6	7

04

OR

6 a

In the transportation problem shown in table 6a, if a unit from source S_i is not shipped out, a storage cost is incurred at the unit rate of Rs.5,4,3 respectively for i=1,2,3. Additionally, all the supply at source S_2 must be shipped out completely. Determine optimum shipping schedule starting from Vogel's solution.

Table 6a: Unit costs of shipping, supplies and demand

	D_1	D_2	D_3	Supplies
S_1	1	2	1	20
S_2	3	4	5	40
S_3	2	3	3	30
Demand	30	20	20	

10

b

A job shop has 4 jobs J_1, J_2, J_3, J_4 to be assigned to 4 out of 5 works W_1, W_2, W_3, W_4, W_5. W_1 cannot be assigned J_3 and W_3 cannot be assigned J_4. Find optimal assignment.

Table 6 b: Cost of Assignment (Rs)

	J_1	J_2	J_3	J_4
W_1	50	50	-	20
W_2	70	40	20	30
W_3	90	30	50	-
W_4	70	20	60	70
W_5	60	45	30	80

06

7 A project consisting of 6 activities has the precedence relations as well as the normal and crashed duration and costs as given in the table 7:

Table 7: Precedence, duration and costs

Activities	A	B	C	D	E	F
Predecessors	-	-	B	C	D, F	B
Normal Duration (days)	120	20	40	30	50	60
Crashed duration (days)	100	15	30	20	40	45
Normal cost (Rs in thousands)	12	1.8	16	1.4	3.6	13.5
Crashed cost (Rs in thousands)	14	2.8	22	2	4.8	18

i) Draw an 'Activity on Arrow' Network diagram and find the critical path and total project duration.

ii) Compute the free float and total float for all the non-critical activities, using forward and backward pass.

iii) Decide the activity should be crashed and determine if the critical path changes when it is crashed.

iv) In general, what further steps are followed to complete the crashing analysis, with the given information? 16

8 a A biker who loves to drive very fast has two options to reach another city from his current location. But the routes (R_1, R_2) are patrolled by the local police which has a limited manpower. If the full police force is allocated either R_1 alone or R_2 alone the biker definitely will get a speeding ticket of Rs.100/-. If the police force is split 50-50 between the two routes, the biker has 50% chance of being fined Rs.100/- on the route R_1, and a 30% chance of being fined Rs.100/- on route R_2. It is known that R_1 is a broad, fast highway and R_2 is a narrow and long winding road.

i) Develop the pay-off matrix from the biker's point of view (Hint: 2 strategies for biker and 3 for police force)

ii) Solve the game and find optimal strategies and the value of the game. Is it a fair game? 12

b State the dominance rules for a game with pay off values x_{ij}, i=1,2…m for player A and j= 1,2,3…n for player B. x_{ij} represents A's gain and B's loss in a 2-person 0 sum game. 04

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IV Semester B. E.Grade Improvement ExaminationsOctober-2021

Industrial Engineering and Management

DECISION SCIENCES – DETERMINISTICMODELS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

Answer any FIVE full questions out of TEN. Each carries 20 marks.

1	1.1	Mention any one of the application of OR methodology in the area of finance.	01
	1.2	List any one limitation of the OR methodology.	01
	1.3	What is the meaning of a unique optimal solution?	01
	1.4	List any one feature of LINDO/LINGO software platform.	01
	1.5	Give any one reason as to why there may be a need to conduct post optimality analysis or sensitivity analysis on a LPP model.	01
	1.6	Mathematically express the condition for optimality using the simplex method.	01
	1.7	How do you diagnose an infeasible solution while applying the two phase simple method?	01
	1.8	List any one benefit of using an Excel solver for solving a linear or non-linear programing problem.	01
	1.9	What is the condition for degeneracy in a transportation model?	01
	1.10	How you diagnose multiple optimal solutions while solving a transportation problem?	01
	1.11	Why should the assignment model always begin with a square matrix?	01
	1.12	Which method of determining the IBFS gives a better initial solution?	01
	1.13	Which organization first demonstrated the application of the PERT methodology?	01
	1.14	Define free float.	01
	1.15	Mention any one utility of Gantt chart from the project control point of view.	01
	1.16	Define critical path based on the idea of floats.	01
	1.17	What is a two person zero sum game?	01
	1.18	What is the practical meaning of a saddle point?	01
	1.19	State the rule for row dominance or column dominance.	01
	1.20	Graphically illustrate a max-min point of $2 * n$ Game.	01
2	a	"Operations Research includes a variety of techniques used in modeling business applications for both better understanding the system in question and making best decisions". Critically analyze this statement. List out any five of the OR techniques used for modeling business applications. Elaborate on any one of the techniques.	06
	b	A company makes two types of products A and B. The company makes a profit of $Rs.4$ on type A and profit of $Rs.5$ on type B products. Both A and B can be produced by the machines X and Y. The processing times of products A and B on machines X and Y is given as follows.	

		<table><tr><th>Products</th><th>Machine X</th><th>Machine Y</th></tr><tr><td>A</td><td>2 minutes</td><td>3 minutes</td></tr><tr><td>B</td><td>2 minutes</td><td>2 minutes</td></tr></table> It is known that Machine X is available for maximum 5 *hours* & 30 *minutes* during any working day and Machine Y is available for maximum 8*hours* during any working day. Formulate the problem as LPP and solve graphically.	14
3	a	A company manufactures two types of products: '*XYZ*' and '*MNS*'. Each product part of '*MNS*' takes twice as long as to produce one '*XYZ*', the company has the time to make maximum of 2000 products per day. The supply of raw material is sufficient to produce a total of 1500 products/day. (Both products require same raw material). '*MNS*'require a special subassembly, of which only 600 pieces/day are available. The company makes a profit of $Rs\,30$ and $Rs\,50$ per product respectively on '*XYZ*' and '*MNS*' i) Formulate the problem as *LPP* ii) How should the manufacturer schedule production in order to maximize the profit? Solve graphically.	12
	b	Explain with suitable example: i) Slack variable ii) Surplus variable iii) Artificial variable.	08
4	a	Solve the following *LPP* using simplex method. Maximize $Z = 6x_1 + 3x_2 + x_3$ Subject to $6x_1 - 3x_2 + x_3 \geq 2$ $3x_1 + 4x_2 - 2x_3 \geq 5$ $x_1, x_2, x_3 \geq 0$	10
	b	Solve the following *LPP* using two phase simplex method. Minimize $Z = 6x_1 + 3x_2 + x_3$ Subject to $6x_1 - 3x_2 + x_3 \geq 2$ $3x_1 + 4x_2 - x_3 \geq 5$ $x_1, x_2, x_3 \geq 0$	10
5	a	Consider the following LPP. Maximize $Z = 3x_1 + x_2 + 4x_3$ Subjected to $6x_1 + 3x_2 + 5x_3 \leq 25$ $3x_1 + 4x_2 + 5x_3 \leq 20$ $x_1, x_2, x_3 \geq 0$ The optimum basic variables and the associated inverse matrix are given below. $(x_1, x_2)\begin{pmatrix} \frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{5} & \frac{2}{5} \end{pmatrix}$ i) Suppose the objective function is changes to $2x_1 + 4x_2 + x_3$, find the new optimum solution. ii) If the coefficient of x_2 in the two constraints are changed to (8,10), determine the new optimal solution. iii) If an additional constraint namely $2x_1 + 3x_2 + 4x_3 \leq 15$ is added, does the optimum solution change? If it does, find the new optimum solution.	

		iv) If the RHS of the constraints are changes from $\begin{pmatrix}25\\20\end{pmatrix}$ to $\begin{pmatrix}15\\10\end{pmatrix}$,find the new optimum solution	14
	b	Briefly explain the concept of Degeneracy with respect to LPP problems being solved using the Simplex method with the necessary supporting logic.	06
6	a	Four factories A, B, C and D produce sugar and the capacity of each factory is given below. Factory A produces 10tons of sugar, Factory B produces 8tons of sugar, Factory C produces 5tons of sugar and that of D is 6tons of sugar. The sugar has demand in three markets X,Y and Z. The demand for sugar in market X is 7tons, that of market Y is 12 tons and the demand of market Z is 4 tons. The following matrix gives the transportation cost of 1ton of sugar from each factory to the destination. Determine the IBFS using the north-west corner rule or Vogel's approximation method. What is the optimum shipping schedule?	14
	b	"An assignment problem can be considered a highly degenerate version of the Transportation problem". Explain this with clarity.	06
7	a	The coach of an age group swim team needs to assign swimmers to a 200-yard medley relay team to send to the Junior Olympics. Since most of his best swimmers are very fast in more than one stroke, it is not clear which swimmer should be assigned to each of the four strokes. The five fastest swimmers and the best times (in seconds) they have achieved in each of the strokes (for 50yards) are	
		The coach wishes to determine how to assign four swimmers to the four different strokes to minimize the sum of the corresponding best times. Obtain an optimal solution. Which swimmer will not have an assignment?	14
	b	Write down the typical matrix structure of a traveling salesman problem.	06
8	a	Christine Phillips is in charge of planning and coordinating next spring's sales management training program for her company. Christine has listed the following activity information for this project.	

Factories	Cost per ton (× 100)			Availability in tons
	X	Y	Z	
A	4	3	2	10
B	5	6	1	8
C	6	4	3	5
D	3	5	4	6
Requirements in tons	7	12	4	

Stroke	Carl	Chris	David	Tony	Ken
Back stroke	37.7	32.9	33.8	37.0	35.4
Breast stroke	43.4	33.1	42.2	34.7	41.8
Butterfly	33.3	28.5	38.9	30.4	33.6
Freestyle	29.2	26.4	29.6	28.5	31.1

Activity	Activity description	Predecessors	Duration
A	Select location	-	2 weeks
B	Obtain speakers	-	3 weeks
C	Make speaker travel plans	A,B	2 weeks
D	Prepare and mail brochure	A,B	2 weeks
E	Take reservations	D	3 weeks

		Construct the project network for this project. i) Find all the paths and path lengths through this project network. ii) Which of these paths is a critical path? iii) Find the earliest start time and earliest finish time for each activity. iv) Find the latest start time and latest finish time for each activity. v) Find the slack for each activity. Which of the paths is a critical path?	14
	b	PERT technique treats activity times which are distributed according to Beta distribution. Graphically illustrate this concept. Write down the formulae for the expected activity times and project duration. Also write down the formula for computing variance of the activity times and the project as a whole.	06
9	a	Differentiate between $PERT$and CPM.	06
	b	For the data given below find the following: i) Earliest expected and latest occurrence time of each event. ii) The critical path iii) The probability of completing the project in 4 weeks more than expected completion time.	14
10	a	Consider the game having the following payoff table.	14
	b	Clearly explain the following terms. i) Two person zero sum games ii) Pure strategy iii) Mixed strategy.	06

$Duration\ in\ Weeks$	$Task$	A	B	C	D	E	F	G	H	I	J	K	L
	t_o	3	1	2	2	8	0	5	6	1	3	8	2
	t_m	5	2	4	8	12	0	7	9	2	6	15	4
	t_p	7	3	6	14	16	0	9	12	3	9	22	6

Strategy		Player 2			
		1	2	3	4
Player 1	1	2	-3	-1	1
	2	-1	1	-2	2
	3	-1	2	-1	3

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DECISION SCIENCES-DETERMINISTIC MODELS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of normal distribution tables permitted.

PART A

1	1.1	State the basic four assumptions necessary for all linear programming models.	02
	1.2	Distinguish between unbounded solution and infeasible solution.	02
	1.3	Let M be the number of units to make and B be the number of units to buy. If it costs $Rs2$ to make a unit and $Rs\ 3$ to buy a unit and 4000 units are needed, the objective function is _____.	01
	1.4	Differentiate between convex and non convex solution spaces.	02
	1.5	What is the shadow price of a resource in an LPP?	01
	1.6	Degeneracy in a $m \times n$ transportation model occurs when the number of occupied cells is _____.	01
	1.7	The number of nonbasic variables in the balanced transportation problem with 'm' rows and 'n' columns is _____.	01
	1.8	In a travelling salesman problem, no assignment should be made along the _____ of the cost matrix.	01
	1.9	A major constraint in the use of an assignment technique is that the number of tasks must be _____ to the number of facilities.	01
	1.10	If the earliest starting time for an activity is 8 weeks., the latest finish time is 37 weeks and the duration time of the activity is 11 weeks, then the total float is equal to _____.	01
	1.11	A $PERT$ network has 3 activities on critical path with mean time 3, 8 and 6 and standard deviation 1,3 and 3 respectively. The probability that the project will be completed in 20 days is _____.	02
	1.12	What is the cost slope with respect to network?	01
	1.13	The _____ in a payoff matrix is one which is smallest value in its row and the largest value in its column.	01
	1.14	When 2 or more moves are considered by a player for optimal strategy pay- offs then the game has a _____ strategy.	01

	1.15	Find the ranges of values of 'P' and 'q' which will render the entry (2,2) a saddle point for the game player B $Player\ A$ / $Player\ B$: 1: 2, 4, 5; 2: 10, 7, q; 3: 4, P, 6	02

PART B

2	a	A company manufactures two types of products: 'XYZ' and 'MNS'. Each product part of 'MNS' takes twice as long as to produce one 'XYZ', the company has the time to make maximum of 2000 products per day. The supply of raw material is sufficient to produce a total of 1500 products/day. (Both products require same raw material). 'MNS' require a special subassembly, of which only 600 pieces/day are available. The company makes a profit of $Rs\ 30$ and $Rs\ 50$ per product respectively on 'XYZ' and 'MNS' i) Formulate the problem as LPP ii) How should the manufacturer schedule production in order to maximize the profit? Solve graphically.	10
	b	Explain with suitable example: i) Slack variable ii) Surplus variable iii) Artificial variable.	06
3	a	Solve the following LPP using BigM method Maximize $Z = 3\ x_1 + 2x_2 + 3x_3$ Subject to $2x_1 + x_2 + x_3 \leq 2$ $3\ x_1 + 4x_2 + 2x_3 \geq 8$	08
	b	Use two phase simplex method to solve the LPP Minimize $Z = 4\ x_1 + x_2$ Subject to $x_1 + 2x_2 \leq 3$ $4x_1 + 3x_2 \geq 6$ $3x_1 + x_2 = 3$ $x_1, x_2 \geq 0$	08
		OR	
4	a	Solve the following LPP by simplex method. Maximize $Z = 2x_1 + 5\ x_2$ Subject to $x_1 + 2x_2 \leq 8$ $x_1 \leq 4$ $x_2 \leq 3$ $x_2 \geq 0$, x_1 is unrestricted in sign.	10
	b	Assuming suitable example explain sensitivity analysis as applied to LPP [any 1 case]	06

5 A car rental agency has 08 outlets in a city. The agency found from experience that, it is best to begin each day with all their cars evenly distributed among the 08 outlets, assuming that all 80 cars are present. Suppose that at the end of a particular day the number of cars at the outlet is as follows:

$Outlet:$	A	B	C	D	E	F	G	H
$Number\ of\ Cars:$	1	11	3	15	20	5	7	18

The agency wish to transport the cars from the scrplus outlets to the deficit outlets to attain an even distribution. The best set of movement is that which has the smallest total distance travelled. The distance in km between the outlets are given below. Solve the problem by transportation algorithm.

		$To\ Outlet$							
		A	B	C	D	E	F	G	H
$From\ outlet$	A	–	13	3	11	12	5	8	7
	B	–	–	11	11	25	16	23	9
	C	–	–	–	19	11	16	7	15
	D	–	–	–	–	31	26	16	3
	E	–	–	–	–	–	4	9	12
	F	–	–	–	–	–	–	8	9
	G	–	–	–	–	–	–	–	14
	H								

16

OR

6 a Give the mathematical formulation of the assignment problem. 06

b Solve the travelling salesman problem given by the following data: $C_{12} = 20,\ C_{13} = 4,\ C_{14} = 10, C_{23} = 5,\ \ C_{34} = 6,\ C_{25} = 10,\ C_{35} = 6,\ C_{45} = 20$ where $Cij{=}Cji,$ there is no route between cities i & j, if the values are not shown. 10

7 a Differentiate between $PERT$ and CPM. 04

b For the data given below find the following:

i) Earliest expected and latest occurrence time of each event.

ii) The critical path

iii) The probability of completing the project in 4 weeks more than expected completion time.

	$Task$	A	B	C	D	E	F	G	H	I	J	K	L
$Duration\ in\ Weeks$	t_o	3	1	2	2	8	0	5	6	1	3	8	2
	t_m	5	2	4	8	12	0	7	9	2	6	15	4
	t_p	7	3	6	14	16	0	9	12	3	9	22	6

12

8 a In a game of matching coins with 2 players, suppose A wins 1 unit of value when there are two heads, wins nothing when there are two tails and loses half a unit of value when there is one head and one tail. Determine the pay off matrix and the best strategies for each of the player and the value of the game. 08

b	Solve the following game					
				$Player\ B$		
				I	II	III
		$Player\ A$	I	-4	6	3
			II	-3	-3	4
			III	2	-3	4
						08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E. Fast Track Examinations Oct-2020

Industrial Engineering and Management

OPERATIONS RESEARCH

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of statistical tables permitted.

PART A

1	1.1	State the 3 points to be borne in mind while formulating a problem as an LPP?	02
	1.2	Illustrate graphically infeasible solution.	02
	1.3	What is duality in linear programming?	02
	1.4	In simplex method, how will you identify multiple optimal solutions?	02
	1.5	Given any two differences between transportation and assignment problem.	02
	1.6	Determine the value of p and q that will render the cell entry (2,2) a saddle point.	02
	1.7	Define i) Saddle point ii) Pay-off matrix with respect to theory of game.	02
	1.8	In $PERT$, it is assumed that activity declaration follows _____ distribution and project completion follows _____ distribution.	02
	1.9	Define float and list the different types of floats.	02
	1.10	Enlist Kendal's and Lee notation for queuing theory analysis.	02

	B_1	B_2	B_3
A_1	2	4	5
A_2	10	7	q
A_3	4	p	8

PART B

2	a	State and explain the main phases of OR study.	06
	b	Products P & Q which serve as fodder for cattle contain different amounts of nutritive ingredients N_1 and N_2 which are to be provided to cattle in certain specified quantity. The products also contain an ingredient N_3 which is harmful if present in excessive quantity. The following table gives the necessary data. If the prices of P & Q per unit are $Rs.20$ and $Rs.40$ respectively, determine the optimum product mix of the fodder with prescribed ingredient contents using the graphical method.	

		Ingredient	*Amount present in product* P / Q	*Minimum or maximum amount needed in units*
		N_1	9 / 3	45 min
		N_2	1 / 4	16 min
		N_3	2 / 2	20 max
				10
3	a	With respect to LPP, explain slack, surplus and unrestricted variables.		06
	b	Solve the following LPP by using the simplex method. Maximize $z = 3x_1 + 2x_2$ Subject to $-2x_1 + x_2 \leq 1$ $x_1 \leq 2$ $x_1 + x_2 \leq 3$ $x_1 \geq 0$ and $x_2 \geq 0$		10
		OR		
4		Solve the following LPP by using the Big-M method. Maximize $z = 3x_1 - x_2$ Subject to $2x_1 + x_2 \geq 2$ $x_1 + 3x_2 \leq 3$ $x_2 \leq 4$ $x_1, x_2 \geq 0$		16
5	a	What is degeneracy in transportation problem? How to resolve it?		06
	b	A manufacturer must produce a product in sufficient quantity to meet contractual sales in next 4 *months*. The production capacity and unit cost of production vary from month to month. The products produced in one month may be held for sale in later months at an estimated storage cost of $Rs.1$ per unit per month. No storage cost is incurred for its goods sold in the same month in which they are produced. There is no opening inventory and none is desired at its end of $4th\ month$. The necessary details are given in the table below. Find BFS by LCM. How should the manufacturer produce each month to minimize cost?		
		Month	*Sales*	*Maximum production* / *Unit cost of production*
		1	20	40 / 14
		2	30	50 / 16
		3	50	30 / 15
		4	40	50 / 17
		Solve using the transportation model structure.		10
		OR		
6	a	Write a note on travelling salesman problem.		04
	b	The cost of assigning 4 operators to 4 existing machines is given in the table. Note that operator 1 can't be assigned to machine 4. A new machine is available whose respective assignment costs are 2,1,2,8. Determine whether it is economical to replace an existing machine with a new one. If the answer is yes, which machine should be replaced? What would be the savings?		

				Machines			
				1	*2*	*3*	*4*
	Operators		*1*	**5**	**5**	**_**	**2**
			2	**7**	**4**	**2**	**3**
			3	**9**	**3**	**5**	**_**
			4	**9**	**2**	**6**	**7**

12

7 a Identify any four differences between CPM and $PERT$. 04

b A marketing manager wants to launch a new project. The activities of network are given below with their N_d, N_c, C_d, C_c. Indirect cost is $Rs.90/week$.

$Activity$	N_d	N_c	C_d	C_c
$1-2$	3	50	2	100
$1-3$	2	75	1	150
$1-4$	6	140	4	260
$2-4$	5	100	3	180
$3-4$	2	80	2	80
$4-5$	4	100	2	240
$2-5$	7	115	6	145

If the manager wants to launch the project in $7\,weeks$, what will be your advice to him?
Also determine the optimum duration and cost and also minimum project duration and cost. 12

8 a Two competitors of a consumer product are likely to introduce their product through advertising media. One of them has found out the pay-off matrix through market survey at different strategies.

		B		
		I	II	III
A	I	3	2	3
	II	2	3	4
	III	5	4	2

Find out best strategy of each company and the value of the game.
If the pay-off of company A is increased by $5\,units$ for strategy II against all moves of company B, what is the value of the game? 08

b In a barber shop customers arrive according to Poisson distribution with a mean arrival rate of $5/hr$. The hair cutting time is exponentially distributed with a haircut time of $10\,minutes$ on average. Assuming that the customers are always willing to wait, determine

i) Average number of customers in the shop
ii) Average waiting time of a customer
iii) Probability that customer wait for more than $10\,minutes$ in queue. 08

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RV COLLEGE OF ENGINEERING®

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IV Semester Fast Track Examinations July-19

Industrial Engineering and Management

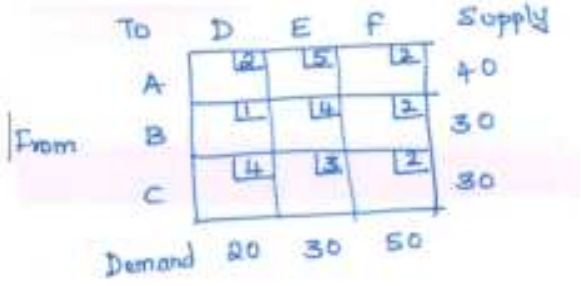
OPERATIONS RESEARCH

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of statistical tables permitted.

PART A

1	1.1	_____ entails an attempt to translate the problem definition into Mathematical relationships.	01
	1.2	List the application of operations research model in the field of Finance and Marketing.	02
	1.3	Consider the objective function, Z $(x_1,x_2)=3x_1+9x_2$ and the constraints $x_1+x_2 \leq 8$, $x_1+2x_2 \leq 4$, x_1 & $x_2 \geq 0$. What is the maximum value of the objective function?	02
	1.4	State all assumptions that form the basis for an LPP model.	02
	1.5	What is sensitivity analysis?	02
	1.6	In a 6 X 6 assignment problem, the number of non – basic cells are _____	01
	1.7	When is, a dummy destination added to a transportation problem?	01
	1.8	Consider the following transportation problem and find the cost of initial solution using northwest corner method 	02
	1.9	What is cost slope with respect to networks?	01
	1.10	_____ is a physically identifiable part of the project that consumes time and resources and is represented by an arrow.	01
	1.11	_____ is the time by which the completion of an activity can be delayed beyond the earliest start of a subsequent activity.	01
	1.12	An arriving customer may take a decision not to join the queue due to long waiting time is known as _____	01
	1.13	List any 4 characteristics of queuing model.	02
	1.14	A _____ is an element of pay off matrix which is the smallest element in its row and the largest in its Column.	01

PART B

2	a	Enumerate on the different methods of solving operations research problems?	06
	b	A diet for a sick person must contain atleast 4000 units of vitamins, 50 units of minerals and 1400 calories. Two foods A and B are available at a cost of Rs 4 and Rs 3 per unit respectively, In one unit of A contains 200 units of vitamins, one unit of mineral and 40 calories and one unit of food B contains 100 units of vitamins, 2 units of minerals and 40 calories. Determine the least cost food combination (food mix) using the graphical method. Show clearly the iso cost line in the graph and clearly mark the feasible solution space.	10

3	a	Solve the following LP problem by simplex method. $Max\ z = 6x_1 - 2x_2$ $Subjectto\ 2x_1 - x_2 \leq 2$ $X1 \leq 4$ $\&x_1, x_2 \geq 0$	08
	b	Use two phase method to solve: $Maximize\ Z = 3x_1 + 2x_2 + 3x_3$ $S.t \quad 2x_1 + x_2 + x_3 \leq 2$ $3x_1 + 4x_2 + 2x_3 \geq 8$ $x_1, x_2, x_3 \geq 0$	08
		OR	
4	a	Solve the LPP using Big M method $Maximize\ Z = 2x_1 + 5x_2 + 3x_3$ $S.t \quad 3x_1 - 6x_2 \geq 30$ $6x_1 + 12x_2 + 3x_3 = 75$ $x_1, x_2, x_3 \geq 0$	10
	b	Explain with suitable example: i) SLACK variable ii) SURPLUS variable iii) ARTIFICIAL variable	06

5	a	A company produces small components for an industrial product and distributes it to 5 wholesalers at a fixed delivery price of Rs. 250 per unit. Sales forecasts indicate that monthly deliveries will be 300, 300, 100, 500 and 400 units to wholesalers 1,2,3,4 & 5 respectively. The direct cost of production of each unit are Rs. 100, Rs. 90 and Rs. 80at plants 1,2& 3 respectively. Transportation costs of shipping a unit from plants to wholesalers are as follows:	

		Wholesalers				
		1	2	3	4	5
Plants	1	5	7	10	15	15
	2	8	6	9	12	14
	3	10	9	8	10	15

		Find how many components each plant must supply to each wholesalers to maximize the profit? What is the maximum total profit? Take the monthly production capacities of plants 1,2& 3 as 500, 100 and 1250 units respectively.	12
	b	Differentiate between a transportationand an assignment problem.	04
		OR	

6	a	Solve the following travelling salesman problem: Cities | | A | B | C | D | E | |---|---|---|---|---|---| | A | ∞ | 2 | 5 | 7 | 1 | | B | 6 | ∞ | 3 | 8 | 2 | | C | 8 | 7 | ∞ | 4 | 7 | | D | 12 | 4 | 6 | ∞ | 5 | | E | 1 | 3 | 2 | 8 | ∞ |	06
	b	Solve the following assignment problem: | | 1 | 2 | 3 | 4 | 5 | 6 | |---|---|---|---|---|---|---| | A | 12 | 10 | 15 | 22 | 18 | 8 | | B | 10 | 18 | 25 | 15 | 16 | 12 | | C | 11 | 10 | 3 | 8 | 5 | 9 | | D | 6 | 14 | 10 | 13 | 13 | 12 | | E | 8 | 12 | 11 | 7 | 13 | 10 |	10

7	a	A project consists of following activities with the duration in days. a) Draw the network diagram of above project. b) Identify the critical path and determine project duration. c) Calculate Early Start Time, Early Finish Time, Late Start Time, Late Finish Time, Total float, Free float and Independent float for each activity | Activity | Precedence | Duration in days | |---|---|---| | A | - | 10 | | B | A | 8 | | C | A | 12 | | D | B | 9 | | E | B | 7 | | F | B, C | 12 | | G | B, C | 5 | | H | D, F | 8 |	12
	b	List the basic differences in PERT and CPM.	04

8	a	In a railway yard, goods train at a rate of 30 trains per day. Assume that the inter arrival time follows an exponential distribution and the service time is also exponential with an average 36 minutes. Calculate the following: i) The average number of trains in the system. ii) The probability that the queue size exceeds 10. iii) The changes in a, b if the input of trains increases to an average of 33 per day.	08
	b	Solve the following game by principle of dominance. | | | Player B | | | | |---|---|---|---|---|---| | Player A | | B_1 | B_2 | B_3 | B_4 | | | A_1 | 3 | 2 | 4 | 0 | | | A_2 | 3 | 4 | 2 | 4 | | | A_3 | 4 | 2 | 4 | 0 | | | A_4 | 0 | 4 | 0 | 8 |	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E. Examinations April/May-19

Industrial Engineering and Management

OPERATIONS RESEARCH

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Normal distribution table is permitted.

PART A

1	1.1	What is Mathematical model or symbolic Model? State example.	02
	1.2	The objective function for a $L.P$ model is $3x_1 + 2x_2$, if $x_1 = 20$, and $x_2 = 30$ what is the value of the objective function?	02
	1.3	Give reason for choosing the penalty method rather than simplex method.	02
	1.4	State the condition for applying initial basic feasible solution to any transportation problem.	02
	1.5	When does degeneracy happen in transportation problem?	02
	1.6	State the basic assumption of the Assignment problem?	02
	1.7	Using the following data, what is the total project variance?	02
	1.8	List the basic elements of queuing system.	02
	1.9	Distinguish between transient and steady state queuing system.	02
	1.10	List the characteristics of a competitive game.	02

$Activity$	a	m	b	$Predecessors$
A	6	10	14	–
B	8	9	13	A
C	4	12	20	A
D	6	8	10	B, C

PART B

2	a	Explain any two assumptions of Linear Programming Problem with an illustration.	04
	b	A manufacturing company manufactures two types of teaching aids A & B of Operations Research for a class of IV Sem. Each type of A requires 9 labour hours of fabricating and 1 hour for finishing. Each type of B requires 12 labour hours of fabricating and 3 hours for finishing. For fabricating and finishing, the maximum labour hours available per week are 180 and 30 respectively. The company makes a profit of $Rs.80$ on each type of A and $Rs.120$ on each type of B. How many pieces of types of $A \& B$ are manufactured per week to get a maximum profit? Formulate it as an Linear Programming problem and solve by graphical method.	12

3 a What is sensitivity analysis and why do we use it. 06

b Solve the following Linear Programming problem by simplex method:
Maximize $Z = 2x_1 + 4x_2$ subject to
$2x_1 + 3x_2 \leq 48$
$x_1 + 3x_2 \leq 42$
$x_1 + x_2 \leq 21$
x_1, x_2 all ≥ 0 10

OR

4 a Solve the following Linear Programming problem by two phase simplex method: Minimize $Z = 3x_1 + 8x_2$ Subject to
$x_1 + x_2 = 200$
$x_1 \leq 80$
$x_2 \geq 60$
$x_1, x_2\ all\ \geq 0$ 16

5 a Find the initial basic feasible solution to the following Transportation Problem using North-West Corner Rule:

	1	2	3	4	*Supply*
1	10	30	25	15	14
2	20	15	20	10	10
3	10	30	20	20	15
4	30	40	35	45	13
Demand	10	15	12	15	

06

b The following table shows all the necessary information on the available supply to each warehouse, the requirement of each market and the unit transportation cost from each warehouse to each market.

		Market				*Supply*
		I	*II*	*III*	*IV*	
	A	5	2	4	3	22
Warehouse	*B*	4	8	1	6	15
	C	4	6	7	5	8
Requirement		7	12	17	9	

The shipping clerk has worked out the following schedule from experience: 12 units from A to II, 1 unit from A to III, 9 units from A to IV, 15 units from B to III, 7 units from C to I and 1 unit from C to III.

i) Check that the clerk has the optimal schedule.

ii) Find the optimum schedule and minimum shipping cost. 10

OR

6 a A head department has four lecturers to assign to Pure Maths(1), Mechanics(2), Statistics(3) and Quantitative Techniques(4). All of the teachers have taught the courses in the past and have been evaluated with a score from 0 to 100. The scores are shown in the table below:

		Courses			
		1	2	3	4
Teachers	*A*	80	55	45	45
	B	58	35	70	50
	C	70	50	80	65
	D	90	70	40	80

b The head of the department wishes to know the optimal assignment of teachers to courses that will maximize the overall score. Use the Hungarian algorithm to solve the problem. 08

b A travelling salesman plans to visit five cities I, II, III, IV and V. The travel time (in hours) between these cities is given below. How should he schedule his touring plan in order to minimize the total travel time, solve by Travelling Salesman problem.

	I	II	III	IV	V
I	∞	2	5	7	1
II	6	∞	3	8	2
III	8	7	∞	4	7
IV	12	4	6	∞	5
V	1	3	2	8	∞

08

7 a Draw the network diagram and determine the early start and late start, free float in respect of all node points and identify critical path for the following activities and duration.(week)

Activity	A	B	C	D	E	F	G	H	I	J	K	L	M
Predecessor	–	–	–	A	B	C	D	D	F	E, H, I	F	G	K
Duration	10	8	9	8	16	7	6	7	7	12	5	13	15

08

b The below table shows the jobs of a network along with their time estimates (days).

Jobs	$1-2$	$1-6$	$2-3$	$2-4$	$3-5$	$4-5$	$5-8$	$6-7$	$7-8$
Most likely	3	2	6	2	5	3	1	3	4
Optimistic time	6	5	12	5	11	6	4	9	19
Pessimistic time	15	14	30	8	17	15	7	27	27

i) Draw the project network.
ii) Find the critical path.
iii) Find the probability that the project is completed in 31 days. 08

8 a Customers arrive at a one window drive in a bank according to Poisson distribution with mean 10 per hour. Service time per customer is exponential with mean 5 minutes. The space in front of the window including that for the serviced car can accommodate a maximum of 3 cars. Others can wait outside this space.

i) What is the probability that an arriving customer can drive directly to the space in front of the window?
ii) What is the probability that an arriving customer will have to wait outside the indicated space?
iii) How long is an arriving customer expected to wait before starting service. 08

b Obtain the optimal strategies for both persons and the value the game by graphical method, whose payoff matrix is as follows:

		Player B	
		B_1	B_2
Player A	A_1	1	-3
	A_2	3	5
	A_3	-1	6
	A_4	4	1
	A_5	2	2
	A_6	-5	0

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Examinations April/May-18

Industrial Engineering and Management

OPERATIONS RESEARCH

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

11. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
12. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
13. Use of statistical tables permitted.

PART A

1	1.1	Define OR in two statements.	02
	1.2	Mention any four applications of OR.	02
	1.3	Graphically explain infeasible solution.	02
	1.4	List the features of canonical form of LPP.	02
	1.5	What is degeneracy in LPP? How to resolve it?	02
	1.6	Mention any two differences between transportation and assignment problems.	02
	1.7	Explain any two rules to be followed while drawing project network.	02
	1.8	What is the necessity of three-time estimates in $PERT$ analysis?	02
	1.9	Explain two-person zero sum game.	02
	1.10	State the characteristics of queuing theory.	02

PART B

2	a	List and explain briefly the different phases of OR.	06
	b	A company manufactures washing machines and driers. Details are given in the following table:	

$Sl.No$	$Department$	$Monthly\ capacity$	
		$Washing$	$driers$
1	$Stamping$	10000 or	10000
2	$Motors\ and\ transmission$	16000 or	7000
3	$Assembly\ for\ washing$	9000	-
4	$Assembly\ for\ driers$	-	5000

The net contribution to profit for each washing machine is $Rs.900/-$ and that for driers is $Rs.1000/-$. Assuming that any combination of products can be sold in the market, determine the optimal product mix. What is the corresponding maximum profit? 10

3	a	With respect to LPP, give the meaning of basic variable, slack variable and surplus variables.	06

	b	Solve the following LP model using Simplex method. Maximize, $z = 4x_1 + 3x_2 + 6x_3$ s.f $2x_1 + 3x_2 + 2x_3 \leq 440$ $4x_1 + 3x_3 \leq 470$ $2x_1 + 5x_2 \leq 430$ $x_1, x_2, x_3 \geq 0$	10
		OR	
4	a	Solve the following by Big-M method Maximize, $z = 3x_1 + 2x_2 + x_3$ s.f $-3x_1 + 2x_2 + 2x_3 = 8$ $-3x_1 + 4x_2 + x_3 = 7$ $x_1, x_2, x_3 \geq 0.$	16
5	a	What is a transportation problem? Give its applications. Express a general transportation model in matrix form with all notations.	06
	b	Determine the optimal transportation cost for the following problem.	

Source	*Unit transportation cost*				*Supply*
	Destination				
	D_1	D_2	D_3	D_4	
S_1	21	16	25	13	11
S_2	17	18	14	23	13
S_3	32	27	18	41	19
Demand	6	10	12	15	

10

OR

6	a	Distinguish clearly between assignment and transportation problems. Give the applications of assignment problem.	06
	b	A company has 4 territories and 4 salesmen. It is estimated that looking under the same conditions, salesman yearly sales would be as follows:	

Territory	*I*	*II*	*III*	*IV*
Annual sales(Rs)	60000	50000	40000	30000

Salesman	*A*	*B*	*C*	*D*
Proportion	7	5	5	4

Assign the best salesman to the richest territory so as to maximize the total expected sales. Also indicate the sales. 10

7	a	Draw the network for the following project and number the events using Fulkerson's rule.	

Activity	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*I*	*J*	*K*
Preceeded by	-	-	*A*	*B*	*A*	*B*	*C,D*	*G,F*	*E*	*H,I*	*J*

06

	b	A small project is given below:	

Activity	$1-2$	$1-3$	$1-4$	$2-5$	$3-5$	$4-6$	$5-6$
t_o	1	1	2	1	2	2	3
t_m	1	4	2	1	5	5	6
t_p	7	7	8	1	14	8	15

Find the expected completion time of each activity. Draw the network and find the critical path. Find the probability of completing the project 4 days earlier and 4 days later than the expected time. 10

8	a	Solve the following game graphically: $Player\ B$ $Player\ A$ A_1: B_1 2, B_2 -1, B_3 5, B_4 -2, B_5 6 A_2: -2, 4, -3, 1, 0	08
	b	Workers in a machine shop receive tools required by them for completing the work assigned to them. The average time between successive arrivals of workers is 60 sec and its arrivals are assumed to be Poisson distribution. The average time of service per worker is 40 secs. Determine the average queue length, average number of workers in the system, average waiting time of a worker in a queue, idle time cost of the server per day and the waiting cost of the workers per day. The worker is paid $Rs.4/hr$ and the server cost is $Rs.0.75/hr$. working hours/day $= 8$.	08

Marketing Management

IV Semester SEE Question Papers

Course codes: IM345AT, 21IM44

- Jun/Jul 2025 (IM345AT)
- Sep/Oct 2024 (IM345AT)
- Oct/Nov 2023 (21IM44)

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B.E. Regular/Supplementary Examinations June/July – 2025

Industrial Engineering and Management

MARKETING MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	Differentiate between push and pull marketing.	02	2	2
	1.2	Mention any four characteristics of a good content.	02	1	1
	1.3	What is the primary goal of social media marketing?	02	1	1
	1.4	What is the primary objective of Search Engine Optimization (SEO)?	02	2	1
	1.5	How bounce rate affects website performance assessment.	02	2	2
	1.6	Why is segmentation important in e-mail marketing campaigns?	02	2	2
	1.7	Why is it important to integrate SEO in website development?	02	2	4
	1.8	What is the role of geo-targeting in mobile marketing?	02	2	2
	1.9	Why is the 'Satisfaction' component important in the AIDAS model?	02	1	2
	1.10	What is meant by the "end-to-end customer experience" in digital analytics?	02	2	3

PART-B

2	a	Discuss how website analysis contributes to enhanced customer experience and conversion optimization.	08	3	2
	b	Explain how content research tools help marketers discover new opportunities and gain a competitive edge.	08	3	2
3	a	Analyse how businesses can utilize Facebook and Instagram for different stages of a marketing funnel.	08	3	2
	b	Develop an SEO content plan for an e-commerce startup targeting eco-friendly household products.	08	3	2
		OR			
4	a	Illustrate how social media analytics contribute to creating audience profiles.	08	3	3
	b	Construct a targeted marketing plan using SEO tools like Google Analytics and SEMrush for a B2B company.	08	4	3
5	a	Design a simple marketing plan for an e-learning platform using insights gathered from web analytics tools.	08	4	3
	b	Analyze a failed e-mail marketing campaign and identify areas where marketing research could have improved outcomes.	08	3	3
		OR			
6	a	Evaluate the effectiveness of responsive email design in maintaining a competitive edge in digital marketing.	08	4	4
	b	Critically evaluate the contribution of Google Tag Manager in optimizing marketing channels for global companies.	08	3	4

7	a	Demonstrate how tools like Google Analytics and Hotjar can be used to develop insights into user behavior and improve marketing decisions.	08	3	3
	b	Formulate a mobile-based marketing plan for a local food delivery startup to reach a targeted student segment.	08	3	3
OR					
8	a	Evaluate the effectiveness of mobile marketing in facing market changes due to rapid technological advancements.	08	4	2
	b	Create a strategy to use new digital channels like mobile apps, chatbots, and social media integration to strengthen a brand's global online presence.	08	4	4
9	a	What are the key performance indicators (KPIs) used to evaluate website optimization, and how do they contribute to increased conversions?	08	3	4
	b	Analyze how market segmentation can be refined using digital analytics data.	08	3	3
OR					
10	a	Design a basic digital dashboard that reflects holistic marketing performance.	08	4	2
	b	Analyze how a company can use the AIDAS model to structure content on its product landing page.	08	3	3

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

IV Semester B. E. Examinations Sept/Oct – 2024

Industrial Engineering and Management

MARKETING MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Comment on the importance of demographic segmentation.	02	2	1
	1.2	Identify the 4 P's of marketing.	02	1	1
	1.3	Assume you are sitting around in the dining area in casual attire during your vacation. Your uncle who is a marketing manager himself wants to test your marketing skill and gives a pen (as a product) to sell right at that moment. How would you respond?	02	3	1
	1.4	Identify two types of research data.	02	1	1
	1.5	Provide your own Psychographic and Demographic profile when you are being assessed by any marketing personnel.	02	1	2
	1.6	Highlight one advantage and one limitation of psychographic segmentation.	02	1	2
	1.7	You want to test an app you have developed which addresses some needs of engineering college students. How can you use digital marketing to get initial quantitative feedback?	02	3	3
	1.8	List any 4 online marketing tools that can be useful for beginners in the field of marketing.	02	1	3
	1.9	Identify two key benefits of A/B testing in optimizing email marketing conversions.	02	1	4
	1.10	What is the primary difference between mobile advertising and mobile marketing?	02	2	4

PART-B

2	a	Explain any 4 key principles of digital marketing.	08	2	1
	b	Explain any 4 digital marketing channels to spread awareness about new product.	08	2	1
3	a	Develop a social media marketing campaign strategy for a local bakery looking to increase online sales.	08	4	1
	b	Explain the difference between on-site and off-site SEO.	08	2	1
		OR			
4	a	Discuss the importance of SEO in digital marketing strategies.	08	3	1
	b	Discuss the benefits and challenges of using social media for business marketing.	08	2	1

5	a	Explain the key features and functionalities of Google Analytics which can be used to track website traffic, user behaviour and conversions.	10	3	2
	b	Discuss the key elements of an effective email marketing campaign.	06	2	2
		OR			
6	a	Discuss the importance of web analytics.	10	3	2
	b	Write short notes on any two from the following: i. Bounce rate ii. Monthly unique visitors.	06	2	2
7	a	Outline the steps involved in publishing basic website for a small business.	08	2	3
	b	Discuss any 4 important website metrics.	08	2	3
		OR			
8	a	Discuss the challenges of using mobile marketing for sales and promotion.	08	3	3
	b	Elucidate the importance of understanding User centered design concepts while developing digital marketing strategies.	08	3	3
9	a	Explain the AIDAS model and its role in conversion optimization.	08	3	4
	b	Discuss the purpose of website optimization.	08	2	4
		OR			
10	a	Explain how advancements in technology, data collection methods and analytical tools have transformed the way businesses understand and optimize their online presence.	08	3	4
	b	Analyze the evolution of Digital Analytics in the last decade and predict the changes we might see in the next decade from marketing perspective.	08	4	4

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B.E. Examinations Oct/Nov – 2023

Industrial Engineering and Management

MARKETING MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	**M**	**BT**	**CO**
1	1.1	List any 2 principles of digital marketing.	02	–	–
	1.2	Differentiate between market segmentation and market mix.	02	–	–
	1.3	Define Hashtag. List the importance of Hashtag in social media marketing.	02	–	–
	1.4	What is keyword research in SEO?	02	–	–
	1.5	Define "Dimensions" and "Metrics" in Google analytics.	02	–	–
	1.6	List any two key components of a successful email marketing campaign.	02	–	–
	1.7	List any two aspects of branding and consistency in web design.	02	–	–
	1.8	List any two differences between Mobile Web marketing and Mobile App marketing.	02	–	–
	1.9	What is the role of AIDAS framework in the conversion optimization?	02	–	–
	1.10	What is the impact of user behavior analysis on website optimization?	02	–	–
		PART-B			
2	a	What is Digital marketing? Explain the differences between Traditional Marketing and Digital Marketing. Highlight any 5 key differences.	08	–	–
	b	List and explain various pricing strategies used in marketing management with suitable examples.	08	–	–
3	a	Discuss the key metrics used to measure the success of a social media marketing campaign and explain the importance of it for evaluating performance.	08	–	–
	b	Enumerate briefly on the best practices for creating compelling and shareable content on social media platforms to increase brand visibility and engagement.	08	–	–
		OR			
4	a	Explain the concept of Search Engine optimization and its significance in digital marketing.	08	–	–
	b	Describe the role of content in SEO. How can business create high quality SEO friendly content?	08	–	–
5	a	Discuss the concept of "Acquisition channels" in Google Analytics. How can business use this information to optimize their marketing efforts.	08	–	–
	b	Explain the significance of A/B testing and conversion rate optimization in Web Analytics. How can these practices help improve website performance.	08	–	–
		OR			

6	a	Describe the difference between transactional emails and promotional emails. Provide examples of each and explain when to use them.	08	–	–
	b	List and explain the various reports that can be generated using google analytics.	08	–	–
7	a	What is the role of user Interface and user Experience designs in web design for marketing? How can businesses ensure a positive user Journey on their websites?	08	–	–
	b	Describe the importance of testing and optimization in web design for Marketing. What tools and methods can be used to analyze and improve Website performance?	08	–	–
		OR			
8	a	Discuss briefly on Push Notification in Mobile Marketing. Explain how can businesses use them to engage and retain Mobile App users? Provide examples of effective push notification strategies.	08	–	–
	b	Explain the significance of mobile analytics in mobile marketing campaigns. What key metrics should marketers track to measure the success of their mobile strategies?	08	–	–
9	a	Describe the conversion funnel in digital marketing. What are the key stages and how can business identify and address drop offs in the funnel?	08	–	–
	b	Explain the concept of social proof and its impact on conversion rates. Provide examples of how business can use social proof effectively in marketing.	08	–	–
		OR			
10	a	Define Digital Analytics. Describe the key metrics and KPIs used in digital analytics. How can these metrics help business measure the success of their online campaigns?	10	–	–
	b	Describe the process of setting up and tracking goals in google analytics. How can businesses use goal tracking to measure website performance.	06	–	–

Universal Human Values

IV Semester SEE Question Papers

Course codes: HS248AT, 21HSU48

- Jun/Jul 2025 (HS248AT)
- Sep/Oct 2024 (HS248AT)
- Oct 2023 (21HSU48)

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
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R V College of Engineering

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

IV Semester B.E. Regular/Supplementary Examinations June/July-2025.
Common to all programs
Course : Universal Human Values-HS248AT

Time : 2 Hours **Maximum Marks : 50**

Instructions to the students

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer THREE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4 & 5 and 6.

Part A

Question No	Question	M	CO	BT
1.1	How does ensuring justice in relationships contribute to society? a) It leads to fearlessness and co-existence b) It leads to prosperity and right understanding c) It promotes self-regulation and health d) It ensures better exchange and storage	01	3	1
1.2	Imagination is continuous with A. Time B. Soul C. Body D. None	01	1	1
1.3	Identify the solution which helps human being to transform from animal consciousness to human consciousness. (a) Right understanding (b) Realization (c) Value education (d) Physical facilities.	01	1	1
1.4	Which of the following is a fundamental human aspiration related to relationships? a) Physical comfort. b) Financial security. c) Mutual understanding and fulfillment. d) Accumulating wealth.	01	2	2
1.5	Which of the following capacity leads to desires A. Power B. Expectation C. Realization D. Thoughts	01	2	1

1.6	Which of the following is considered the foundation of a strong human relationship? (a) Intelligence (b) Trust (c) Wealth (d) Appearance	01	1	2
1.7	Respect in a relationship means: a) Agreeing to everything b) Controlling the other person's actions c) Valuing the other person's opinions and boundaries d) Always trying to win argument.	01	3	1
1.8	The participation of the human being in ensuring the role of physical facility to help and preserve its utility is called its -------- (a) Utility value (b) Artistic value (c) Activity (d) Energy	01	3	2
1.9	A harmonious world is created by values at 4 levels. These are: (a) Home, family, society, country (b) Individual, family, society, universe (c) School, home, office, temple (d) None of these	01	1	1
1.10	When nature is submerged in space it is known as a) Conformance b) Acceptance c) Mixing d) Co-existence	01	3	2

Part B

Question No	Question	M	CO	BT
2a	What do you understand by the value of an entity? What is the value of a human being?	08	1	2
2b	For human being physical facility is necessary, but relationship is also necessary: Justify with an example comparing with animal order.	06	2	4
3a	"Discuss how right understanding in relationships contributes to harmony in the family. Analyze the role of trust, respect, and responsibility in maintaining family unity, and provide examples to support your answer."	07	4	4
3b	Articulate the four goals of human beings living in a society. Critically examine the state of the society today in context with the fulfilment of a comprehensive human goal.	06	3	3
	OR			
4a	How do justice and preservation contribute to a harmonious society? Analyze the shortcomings of focusing solely on punishment in cases of injustice, and explain how true preservation ensures prosperity and coexistence with nature.	06	3	4
4b	Explain the concept of harmony in the family. Why is it important, and how can it be achieved through right understanding and mutual relationships?	07	3	2
5a	What is sanskaar? Explain its effects or the conformance of the human order.	06	4	4
5b	What is the svabhava (natural characteristic) of a unit? Mention the natural characteristics of the material, pranic, animal and human orders. Give examples	07	4	3
	OR			
6a	What do you mean by 'conformance'? Explain the conformance in the four orders.	07	3	4
6b	Explain the meaning of co-existence.	06	3	3

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RV COLLEGE OF ENGINEERING®

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IV Semester B. E. Regular Examinations Sept/Oct – 2024

Common to All branches

UNIVERSAL HUMAN VALUES – I

Time: 02 Hours ***Maximum Marks: 50***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7.

PART-A

			M	BT	CO
1	1.1	The word "society" is primarily used in the context of human_____ relationship. a) Human b) Nature c) Both d) None	01	1	1
	1.2	Prosperity deals with a) Right understanding in the self b) Fulfilment in relationship c) Ensuring more than required physical facility d) None	01	1	1
	1.3	_____ is the feeling of responsibility towards body of my relative. a) Care b) Guidance c) Respect d) Affection	01	1	2
	1.4	_____ means harmony within myself. a) Pleasure b) Happiness c) Excitement d) Fearlessness	01	1	2
	1.5	The forth order of nature is _____. a) Bio order b) Animal order c) Human order d) Material order	01	1	3
	1.6	The continuity of a plant species is maintained in nature by _____ method. a) Constitution conformance b) Speed conformance c) Right value / sanskaar d) Breed conformance	01	1	4
	1.7	There is _____ among the four orders of nature. a) Inherent struggle b) No dependency c) Mutual fulfillment d) Opposition	01	1	4

	1.8	Our natural acceptance is to live in _____ with nature. a) Harmony b) Struggle for survival c) Opposition d) None of the above	01	1	2
	1.9	Production and work for physical facilities leads to _____ in family and _____ with nature. a) Prosperity, existence b) Happiness, existence c) Happiness, co-existence d) Prosperity, co-existence	01	1	1
	1.10	Ensuring justice in relationship, on the basis of values leads to _____ in society.	01	1	3

PART-B

2	a	"To be in a state of harmony is happiness". Explain this statement and illustrate with two examples from your day to day life.	06	2	1
	b	What do you mean by values? How do they differ from skills? How are values and skills complementary?	06	1	1
3	a	What is the role of value system in family harmony?	05	2	2
	b	"Discrimination leads to acrimony in relationships". Explain. What problems are created when we discriminate?	05	2	3
	c	Define love. Or How can you say that love is the complete value?	04	1	2
		OR			
4	a	What do you understand by trust? Differentiate between intention and competence with examples.	05	1	3
	b	Realization and understanding are essential for happiness and harmony. Explain.	05	1	2
	c	Describe the concept of an undivided society and the universal order and explain how both these can help to create a world family.	05	2	3
5	a	Existence is co-existence of mutually interacting units in all-pervasive space. Explain.	05	2	3
	b	Define harmony in nature and why is it important. Explain with examples.	05	1	2
	c	What do you mean by co-existence?	04	1	4
		OR			
6	a	What is the natural characteristics (swabhava) of human order? Explain.	05	1	4
	b	What are the four orders of nature? Briefly explain them.	05	2	4
	c	How can we say that 'nature is Self-Organized and in space Self-Organization Is Available.'	04	2	3

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

IV Semester B. E. Examinations Oct-2023

COMMON TO ALL PROGRAMS

UNIVERSAL HUMAN VALUES AND PROFESSIONAL ETHICS

Time: 02 Hours ***Maximum Marks: 50***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer THREE full questions from Part B. In Part B question number 2 is compulsory.

PART-A

1	1.1	There are _____ comprehensive human goals.	01
	1.2	Human goal at the level of individual is _____.	01
	1.3	Human-human interaction is called as _____.	01
	1.4	Human values are essential for _____.	01
	1.5	_____ is the central in relationship.	01
	1.6	What do you understand by complete value?	01
	1.7	Mutual happiness in human-human interaction leads to _____.	01
	1.8	The value or participation of different orders in existence is also referred to as _____.	01
	1.9	Sah-astitva means _____.	01
	1.10	Which activity of self indicates that “there is a complemntarity in nature and no opposition”	01

PART-B

2	a	Explain the process of self-exploration with a diagram.	06
	b	What is value education? Describe the need of value education with its guidelines.	06
3	a	Explain the problems faced due to differentiation in relationship.	05
	b	What is ‘justice’? What are its four elements? Is it a continuous or a temporary need?	05
	c	What do you understand by trust? Differentiate between intention and competence with examples.	04
		OR	
4	a	Distinguish between the needs of the Self and the needs of the Body.	05
	b	Feeling of love by lay down the basis of undivided society. Explain.	05
	c	Bring out the differences between Respect and differentiation.	04

5	a	Describe briefly the nature of coexistence.	05
	b	What are the four orders of nature? Briefly explain them.	05
	c	What do you understand by holistic technology? Explain briefly.	04
		OR	
6	a	Define harmony in nature. Why is it important? Illustrate with examples.	05
	b	What is ethical human conduct? Explain in terms of values, policies and character with appropriate examples.	05
	c	Describe briefly the nature of universal human order.	04

Engineering Economy / Financial Accounting

IV Semester SEE Question Papers

Course codes: 21IM45, 18IM43, 12IM44, 12IM44 / 07IM36

- Oct 2023 (21IM45)
- Oct/Nov 2022 (18IM43)
- Grade Improvement Oct 2021 (18IM43)
- Grade Improvement Nov 2020 (18IM43)
- Fast Track Jul 2019 (12IM44)
- Fast Track Jul 2018 (12IM44)
- Fast Track Jul 2017 (12IM44)
- Apr/May 2017 (12IM44)
- Fast Track Jul 2016 (12IM44 / 07IM36)
- Apr/May 2016 (12IM44)

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

IV Semester B. E. Examinations Oct-2023

Industrial Engineering and Management

FINANCIAL ACCOUNTING AND MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

1	1.1	Mention the three broad categories of financial decision making.	02
	1.2	What is meant by Interest Rate in the context of Financial market returns?	02
	1.3	What is the role of 'Financial Intermediaries' in the context of Financial system?	02
	1.4	Distinguish between the terms – 'Debit balance' and 'Credit balance' in ledger accounts.	02
	1.5	List any two reasons for balancing a 'Ledger'.	02
	1.6	What is meant by $GAAP$ in the context of Financial accounting?	02
	1.7	List the four different rights of equity share holders.	02
	1.8	'Preference capital is a hybrid form of financing' why?	02
	1.9	Distinguish between the terms – IPO & FPO.	02
	1.10	Distinguish between 'Gross working capital' and 'Net working capital'.	02

PART-B

2	a	Review the characteristics of any four different types of business organizations.	10
	b	Th price of a stock at the beginning of the year is $Rs.60/-$. The dividend paid at the end of the year is $Rs.2.40/-$. The stock price at the end of the year is quoted as $Rs.69/-$. Determine the Rate of Return on the stock. Also compute the current yield and capital gain or loss yield as the case may be.	06
3	a	Provide a brief note on the three Golden rules of accounting.	06
	b	A person started a business with $Rs.1{,}00{,}000/-$ on April 1st 2022. The other transactions involved for the month are as follows:	

Date	*Particulars*
2/4/22	*Purchased furniture for cash of* $Rs.7000/-$
8/4/22	*Purchased goods for cash of* $Rs.2000/-$ *and for credit of* $Rs.1000/-$ *from Akshara's store*
14/4/22	*Sold goods to Aditya & Bro's* $Rs.12000/-$ *and cash sales of* $Rs.5000/-$
18/4/22	*Owner withdrew of worth* $Rs.2000/-$ *for personal work*
22/4/22	*Paid Akshara's store* $Rs.500/-$
26/4/22	*Received* $Rs.10000/-$ *from Aditya & Bro's*
30/4/22	*Paid salaries expense* $Rs.2000/-$

Journalize the above transactions. 10

OR

4 a Write a brief note on Double Entry Book keeping. 04

b The transactions for the month of *Jan'*2012 of an organization are passed in the following Journal. Post the entries to appropriate general ledger accounts.

General Journal

Date	*A/c Title & Explanation*	*Amount*	
		Debit	*Credit*
2/1/2012	*cash Computer*	50,000	
	owner's equity – MS. Malini	20,500	
	(started biz with cash & computers)		70,500
4/1/2012	*Office supplies*	4,000	
	Accounts payable		4,000
	(office supplies purchased on credit)		
10/1/2012	*Cash*	20,000	
	Bank loan (bank loan obtained)		20,000
12/1/2012	*Utility bills*	2,750	
	Cash (utility bill paid)		2,750
15/1/2012	*Account payable*	3,000	
	Cah (paid partial A/C payable)		3,000
24/1/2012	*A/C's receivable – Lakshmi Flowers*	18,300	
	Advertising services		18,300
	(bill to consumer for services rendered)		
27/1/2012	*Cash*	5,500	
	A/C's receivable – Lakshmi flowers		5,500
	(Received cash from customer billed previously)		
30/1/2012	*Drawing*	6,000	
	cash (withdrew of cash by owner)		6,000
	Total	130050	130050

12

5 a Ascertain the gross profit of the organization for the year ending 31*st* March 2023, from the following details:

Purchases – Rs. 3,00,000 *Wages – Rs.* 30,000
Sales – Rs. 5,00,000 *Carriage inwards – Rs.* 4,000
Opening stock – Rs. 40,000 *Sales returns – Rs.* 2,500
Carriage outwards – Rs. 3,000 *Freight and clearing charges – Rs.* 5,000
Stock as on 31*st March* 2023 *– Rs.* 42,000

06

b Prepare a Trading and Profit-Loss Account for the year ending 31*st March* 2023 considering the following details:

Stock as on 1*st April* 2022 *– Rs.* 50,000
Sales – Rs. 2,80,000 *Purchases – Rs.* 2,43,000
Sales returns – Rs. 2,000 *Purchase returns – Rs.* 3000
Freight inwards – Rs. 4,000 *Printing – Rs.* 7,000
Carriage outwards – Rs. 6,000 *Discount received – Rs.* 900
Salaries & wages – Rs. 30,000 *Audit fees – Rs.* 3,000
Discount allowed – Rs. 600 *Trade expenses – Rs.* 600
Stock as on 31*st March* 2023 *– Rs.* 70,000

10

OR

6	a	Ascertain the net profit of the organization from the following details: $Gross\ profit\ -\ Rs.82{,}000$ $Wages - Rs.8{,}000$ $Salaries - Rs.18{,}000$ $Printing\ /\ stationary - Rs.2{,}000$ $Rent, Rates\ /\ Taxes\ -\ Rs.5000$ $Commission\ received - Rs.1000$ $Depreciation\ Machinery - Rs.8000$ $Advertisement - Rs.3000$	06
	b	Prepare a balance sheet of $M/S\ XYZ$ for the year ending $31st$ March 2022. $Machinery - Rs.20{,}000$ $Furniture - Rs.1{,}600$ $Depreciation - Rs.2{,}000$ $Closing\ stock - Rs.16800$ $Capital - Rs.35{,}000$ $Interest\ on\ capital - Rs.1750$ $Net\ profit - Rs.13{,}428$ $Debtors - Rs.9880$ $Creditors - Rs.3{,}903$ $Prepaid\ Rent - Rs.201$ $Outstanding\ salaries - Rs.400$ $Cash\ at\ Bank - Rs.8000$	10
7	a	Four forms are used in the procedure towards Rights issue. Elaborate on each of them.	06
	b	Enumerate the steps involved in raising long term finance through term loans.	10
		OR	
8	a	There are three types of stock market indices. What are they? Review the features of each type.	06
	b	Several participants are involved in the successful transactions of securities in the market. In the above context list the participants and explain their role.	10
9	a	Explain the three strategies followed by organizations in financing their current assets requirements.	06
	b	Enumerate the factors influencing the working capital requirements in organizations.	10
		OR	
10	a	Explain the concepts of 'operating cycle' and 'cash cycle' with the aid of a cash flow line.	06
	b	The financial information for a manufacturing unit is given below: *Particulars* | *Balance sheet data* *P&L A/C data* | *Year Beginning* | *Year end* *Sales* 800 | *Inventory* 96 102 *Cost of goods sold* 720 | *A/C Receivable* 86 90 | *A/C payable* 56 60 Determine the operating cycle and cash cycle for the manufacturing unit.	10

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E. Examinations Oct/Nov-2022

Industrial Engineering and Management

ENGINEERING ECONOMY AND COSTING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of compound Interest tables permitted.

PART A

1	1.1	Draw the cash flow diagrams based on borrower's point of view and Lender's point of view.	02
	1.2	Write the formula and the cash flow diagram for uniform series sinking fund factor.	02
	1.3	For a deposit of Rs.10,000/- made today, what equal amounts can be withdrawn at the end of each quarter for the next 4 years, if the nominal interest rate is 10%.	02
	1.4	The annual interest rate is 6% and the interest is compounded quarterly. What is the quarterly nominal interest rate? What is the effective annual interest rate if compounded quarterly?	02
	1.5	Mention any two conditions used in comparing the present worth of alternatives.	02
	1.6	Write the situations for equivalent annual worth comparisons.	02
	1.7	Differentiate between external rate of return and Internal rate return.	02
	1.8	Bring out the relationship between economic life and the equivalent annual cost and equivalent annual worth.	02
	1.9	Give one example each for the elements of cost.	02
	1.10	What is inflation? What are the causes of inflation?	02

PART B

2	a	Mention any six reasons to bring out the importance of engineering economy in decision making	06
	b	Determine the effective interest rate for a nominal rate of 8% that is compounded i) Semiannually ii) Quarterly iii) Monthly iv) Weekly v) Daily	10

3 a Draw the cashflow diagram and determine its present worth and future worth of the following cash flow for an interest rate of 10%

Year	0	1	2	3	4	5
Cash flow (Rs)	-1000	+2500 & -2000	+2500 & -2000	+2500 & -2000	+2500 & -2000	+3000 & -2400

08

b A loan of Rs.1000 is made today under an agreement that Rs.1400 will be received sometime in future. When should be Rs.1400 received, if the loan is to earn interest at a rate of 10% compounded quarterly. 08

OR

4 a A film star is at the height of his career. He wants to invest Rs.10 Lakhs from the end of this year & follow it up with 9 lakhs, 8 lakhs and so on for the next five years, when his income would go on diminishing. Find the maturity amount 6 years later if a film producer agrees to pay him 15% rate of interest compounded annually. 08

b A professor has 10 years of service before his retirement. He now plans to deposit Rs.25000/- at the end of first year & thereafter an annual increase of Rs.5000/- for the remaining years. If he can expect a return of 8%, find the future amount at the time of professor's retirement. 08

5 a Two models of small machines perform the same function. Type 1 m/c has low initial cost of Rs.9500, relatively high operating cost of Rs.1900/yr more than that of type 2 m/c and a short life of 4yrs.The more expensive Type 2 m/c costs Rs.25100 and can be kept in service economically for 8 years. The scrap value from either m/c at the end of the service life will barely cover its removal cost. Which is preferred when the interest rate is 8%? Use EAW approach. 08

b The following table details the two alternatives for investment. Each alternative has a life of five years with no salvage value. The MARR is 12%

Alternatives	Initial Investment (Rs)	Annual Income (Rs)
I	6 Lakhs	1.7 Lakhs
II	8 Lakhs	2.8 Lakhs

i) Find the best alternative

ii) Find the RoR at which both alternatives are equally good. 08

OR

6 a Highlight the IRR misconceptions with suitable examples. 08

b Following alternatives can perform the same function. At an annual interest rate of 12%, rank the alternatives as per the equivalent annual cost.

Alternative	Initial cost (Rs)	Life (years)	Salvage value (Rs)	Annual maintenance cost (Rs)
A	12000	6	4000	1600
B	6000	3	2000	2000
C	4000	3	1000	2400

08

7	a	The following details pertain to the manufacture of 2000 electric stoves. Direct material cost is Rs. 5,36,000/- Direct labour rate is Rs.2/hr Direct labour hours are Rs.32,000/- Factory overheads is Rs.3/direct labour hour. Administrative overheads are Rs.50,000/- If the manufacturer desires a profit of 20% of selling price, find the selling price of each electric stove.	08
	b	What are the elements of cost? Explain with suitable examples.	08
8	a	Calculate material variances from the following details. Standard quantity of material for producing 1 unit of finished product 'P' is 5kg.The standard price is Rs.6/kg. During a particular period, 500 units of 'P' were produced. Actual material consumed was 2700 kg at a cost of Rs.16,200.	08
	b	Draw the chart which shows the relationship between various labour variances and also write the formulas to calculate various labour variances.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E. Grade Improvement Examinations October-2021

Industrial Engineering and Management

ENGINEERING ECONOMY AND COSTING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

Answer any FIVE full questions out of TEN. Each carries 20 marks.

1	1.1	If you deposit \$1000 over three years at 9% annual interest (meaning that you will make one lump sum withdrawal at the end of three years), what is the interest earned in the 2nd year?	01
	1.2	What amount must be paid in two years to settle a current debt of \$1000 if the interest rate is 6% ?	02
	1.3	Depict a cash flow diagram of the following. i. Equal amounts of A occurring at the end of n years. ii. End of year amounts increasing by an equal annual gradient G.	02
	1.4	What is the present worth of the following income strings at an interest rate of 10% ? n | Net cash flow 1 | \$11,200 2 | \$12,100 3 | \$0 4 | \$14,641	01
	1.5	Why the payback method is often considered inferior to discounted cash flow in capital investment appraisal?	01
	1.6	You purchased a CNC machine for \$18,000. It is expected to have a useful life of 10 *years* and a salvage value of \$3,000. At $i = 15\%$, what is the annual capital cost of this machine?	01
	1.7	If the interest rate is 10% and cash flows are \$1000 at the end of year one and \$2000 at the end of year two, then the present value of these cash flows is _____.	01
	1.8	What is the present worth at 5% interest of a series of \$20,000 payments every 25 *years* (forever)? The first payment occurs 25 *years* from now.	01
	1.9	You have built a product which solves a particular problem. Out of 100 people you have interviewed, 78 people have shown interest. You get an insight that they have lot of desire to buy products. Should you consider this number as a demand and start producing? Comment.	02
	1.10	State any four conditions for computing EAW.	02
	1.11	List any four principles of engineering economy.	02
	1.12	Differentiate between simple interest and compound interest.	02
	1.13	Showcase your understanding of "Marginal Cost" taking an example of your choice.	02
2	a	A person deposits a sum of $Rs.20{,}000$ at the interest rate 18% compounded annually for 10 *years*. Find the maturity value after 10 *years*.	04

	b	A person who is now 35 years of old is planning for his retired life. He plans to invest an equal sum of $Rs.10{,}000$ at the end of every year for the next25 years starting from the end of next year. The bank gives 20% interest rate, compounded annually. Find the maturity value of his account when he is60 years old.	08
	c	Cash flow diagram is very helpful in comparing between alternatives and showcase it to stakeholders who sometimes may be naïve but might be interested in investing in novel ideas. Choose a very simple idea of your choice, Pitch two financial alternatives for the naïve stakeholder. Showcase your understanding of cash flow diagram	08
3	a	Illustrate through calculation, how soon the money will double if it is invested at 8% interest?	04
	b	Below is a representative table showcasing inflation rate of India from past four years. Discuss how inflation affects depreciation at the Macro level (Country level).	08
	c	A loan of $Rs.2000$ if the interest rate is $110\%\ per\ year$. If interest had not been paid each year but had been allowed to compound, how much interest would be due to the lender as a lump sum at the end of six years?	08
4	a	Given the following information, suggest which technology should be selected based on present worth method, assuming 15%interest rate compounded annually.	12
	b	List and explain the conditions for the present worth comparisons.	08
5	a	List any 2 advantages and disadvantages of payback period method.	08
	b	Alpha industry is planning to expand its production operation. It has identified three different technologies for meeting the goal. The initial outlay and annual revenues with respect to each of the technologies are summarized in Table. Suggest the best technology which is to be implemented based on the present worth method of comparison assuming 20% interest rate, compounded annually.	12
6	a	Consider a machine that costs Rs.40,000 and a 10year useful life. At the end of 10year, it can be sold for Rs.5,000 after tax adjustment. If the firm could earn an after-tax revenue of Rs.10,000 per year with this machine, should it be purchased at an interest rate of 15%, compounded annually?	10

Characteristic	Inflation rate compared to previous year
2020	6.2%
2019	4.76%
2018	3.43%
2017	3.6%

Technology	$Initial\ Cost\ (Rs)$	$Service\ life(years)$	$Annual\ O\ \&\ M\ costs\ (Rs.)$
A	4,00,000	15	25,000
B	5,00,000	15	29,000

	Initial outlay (Rs)	Annual revenue (Rs)	Life (years)
Technology 1	1200000	400000	10
Technology 2	2000000	600000	10
Technology 3	1800000	500000	10

	b	A certain individual firm desires an economic analysis to determine which of the two machines is attractive in a given interval of time. The minimum attractive rate of return for the firm is 15%. The following data are to be used in the analysis. Which machine would you choose? Base your answer on annual equivalence cost.	10

	Machine X	Machine Y
First cost	Rs.1,50,000	Rs.2,40,000
Estimated life	12 years	12 years
Salvage value	Rs.0	Rs.0
Annual maintenance	Rs.0	Rs.4,500

7	a	A firm has identified three mutually exclusive investment proposals whose details are given below. The life of all the three alternatives is estimated to be five years with negligible salvage value. The minimum attractive rate of return for the firm is 12%.	12
	b	Write short notes on the following i) Minimum Acceptance Rate of Return ($MARR$) ii) Internal Rate of Return (IRR)	08
8	a	Discuss the various elements of cost.	06
	b	In a manufacturing unit, raw material passes through four processes, I, II, III and IV and the output of each process is the input for the subsequent process. The losses in the four processes are $25\%, 20\%, 20\%$ and $16\frac{2}{3}$ % respectively for I, II, III and IV processes of the input. If the end product at the end of the IV process is $40{,}000\ kg$, what is the quantity of raw material required to be fed at the beginning of process I and the cost of the same at $Rs.5$ per kg?	14
9	a	Enumerate the need for costing.	06
	b	Distinguish between job costing and process costing.	08
	c	Identify any 5 advantages and disadvantages of activity based costing.	06
10	a	100 workers are working in a factory at a standard wage of $Rs.4.80\ per\ hour$. During a month, there are four weeks of $40\ hours$ each. The standard performance is set at $360\ units\ per\ hour$. The following is the summary of the wages paid during the month. 91 workers were paid @ $Rs.4.80\ per\ hour$ 5 workers were paid @ $Rs.5.00\ per\ hour$ The remaining were paid @ $Rs.4.60\ per\ hour$ Power failure stopped production for $2\ hours$. Actual production is $57{,}960\ units$. Calculate labour variances.	10

	Alternatives		
	A1	A2	A3
Investment	Rs.1,50,000	Rs.2,10,000	Rs.2,55,000
Annual net income	Rs.45,570	Rs.58,260	Rs.69,000

	b	A manufacturing concern which had adopted standard costing furnishes the following information. Standard: Material for 70kgs, finished products 100kgs, price of material Rs.1 per kg. Actual: Calculate i. Material usage variance ii. Material price variance iii. Material cost variance Also justify your answers obtained.	10

Output	2,10,000kgs
Materials used	2,80,000kgs
Cost of materials	Rs.2,52,000

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RV COLLEGE OF ENGINEERING®

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IV Semester B. E. Grade Improvement Examinations Nov-2020

Industrial Engineering and Management

ENGINEERING ECONOMY AND COSTING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Depict a cash flow diagram from a borrower's and lender's point of view	02
	1.2	The annual interest rate is 6% and the interest is compounded quarterly. What is the quarterly nominal interest rate? what is the effective annual interest rate if compounded quarterly	02
	1.3	A project costs 2 lakhs and yields cash inflows of $Rs\,50{,}000/year$. In this case, the payback period is _____.	02
	1.4	The present value of 2000 to be collected in 3 years using a rate of 11% is _____.	02
	1.5	Bring out the relationship between economic life and equivalent annual cost and equivalent annual worth	02
	1.6	Distinguish between internal rate of return and external rate of return	02
	1.7	Give one example each for three elements of cost	02
	1.8	Differentiate between job costing and process costing	02
	1.9	List the components of standard cost	02
	1.10	List the four ways of devise standards	02

PART-B

2	a	Determine the effective rate of interest for a nominal annual rate of 8% that is compounded: i) Semiannually ii) quarterly iii) monthly iv) weekly.	08
	b	List all the compound interest factors. Deduce an equation for single payment compound amount factor	08
3	a	A professor has 10 years of service before his retirement. He now plans to deposit $Rs.25{,}000/-$ at the end of first year and thereafter an annual increase of $Rs.5000/-$ for the remaining years. If he can expect a return of 8%, find the future amount at the time of professor's retirement	08

b Novel Investment Ltd accepts $Rs.10{,}000/-$ at the end of every year for 20 years and pays the investor Rs 8 lakhs at the end of 20th year. Innovative investment Ltd accepts $Rs.10{,}000/-$ at the end of every year for 20 years and pays the investor Rs 15 Lakhs at the end of 25th year. Which is the best investment alternative? Use present worth base with $i = 12\%$. 08

OR

4 a Draw cash flow diagram and determine its present worth and future worth for the following cash flow and consider an interest rate of 10%.

Year	0	1	2	3	4	5
Cash flows in Rs	-1000	$+2500$ & -2000	$+2500$ & -2000	$+2500$ & -2000	$+2500$ & -2000	$+3000$ & -2400

08

b An automobile company recently advertised its car for a down payment of Rs 1,50,000. Alternatively the car can be taken by customers without making any payment but they have to pay an equal yearly amount of $Rs.25000/-$ for 15 years at an interest rate of 18% compounded annually. Suggest the best alternative based on present worth of customer. 08

5 a Following alternatives can perform the same function.

Alternative	*Initial Cost (Rs)*	*Life(yr)*	*Salvage (rs)*	*Annual Maintenance Cost (Rs)*
A	1200	6	4000	1600
B	6000	3	2000	2000
C	4000	3	1000	2400

At an annual interest rate of 12%, rank the alternatives as per the equivalent annual cost. 08

b The following table details the 2 alternatives for investment. Each alternative has a life of 5 years with no salvage value. The *MARR* is 12%.

Alternative	*Initial Investment (Rs)*	*Annual Income(Rs)*
1	6,00,000	1,70,000
2	8,00,000	2,80,000

i) Find the best alternative

ii) Find the $R \circ R$ at which both alternatives are equally good. 08

OR

6 a An asset was purchased five years ago for $Rs.5{,}20{,}000$. It was expected to have an economic life of 8 years at which time its salvage value would be $Rs.40{,}000$. If the function that the asset was serving is no longer needed, for what price must it be sold now to recover the invested capital when $i = 12\%$? 08

b A Company is trying to diversify its business in a new product line. The line of the project is 10 years with no salvage value at the end of its life. The initial outlay of the project is Rs 20 lakhs. The annual net profit is $Rs.3{,}50{,}000$. Find the rate of return for the new business. 08

7	a	What are the elements of cost? Explain with the help of a block diagram.	08
	b	The following details pertains to the manufacture of 2000 electric stoves. Direct material cost is Rs 5,36,000 Direct labour rate is Rs $2/hr$ Direct labour hour is 32,000 Factory overheads is Rs 3/direct labour hour Administrative overheads is Rs $50{,}000/-$ If the manufacturer desires a profit of 20% of selling price. Find the selling price of each electric stove	08
8	a	Hypothetical Ltd on the basis of investigation of alternative source of supply decides to purchase raw materials from the most advantageous supplier – ABC Ltd which quotes a list price of $Rs.1{,}820$/tonnes exclusive of freight charges. It also offers a quantity discount @ $Rs70$/tonne for orders exceeding 20 tonnes. It also allows 2% cash discount for payment within 15 days. Moreover freight charges are likely to be $Rs.1{,}700/-$ carload(20 tonnes of materials) and payable directly to the carrier. Compute the standard materials price for Hypothetical Ltd.	06
	b	A manufacturing company uses the following standard mix of their compound in one batch of its production line: $50kgs$ of material x at the std price of Rs 2 $30kgs$ of material y at the std price of Rs 3 $20kgs$ of material z at the std price of Rs 4 The actual mix was as follows : $60kgs$ of material $X, 40$ kgs of material Y and $10\,kgs$ of material Z. Determine $MMSV$	10

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E. Fast Track Examinations July-19

Industrial Engineering and Management

ENGINEERING ECONOMY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART A

1	1.1	New equipment costing \$5000 is expected to yield net cash inflows of \$1350 each year for the next five years. Assuming a required rate of return of 14%, should the equipment be purchased (use IRR)?	02
	1.2	What are the reasons for creating a depreciation fund for machinery and equipment?	02
	1.3	List the IRR misconceptions. (any two)	02
	1.4	Explain present worth by '72 rule'.	02
	1.5	Distinguish between ownership life, accounting life and economic life and their relevance in the determination of the equivalent annual worths.	02
	1.6	Why is the declining balance method an attractive method for determining the depreciation charge of an asset?	02
	1.7	Write down the formula linking the effective interest rate with the nominal interest rate. Explain the terms used in the formula.	02
	1.8	Find the effective rate of interest for an actual rate of interest of 10% when compounded continuously (infinitely).	02
	1.9	A project costs \$200000 and yields cash inflows of \$50,000 per year for 5 years. In this case, the payback period is _____.	02
	1.10	Define the terms: book value and Depreciation amount.	02

PART B

2	a	Discus fundamental approach to economic problem solving and decision making process through scientific method.	08
	b	With a block diagram, explain the steps in problem solving.	08
		OR	
3	a	List the types of compound interest problem and explain any one.	08
	b	Derive the formula for determining the annual equivalent of given sum P at time period 'O' using the symbols $A, P, i\%, N$ in the usual notations (determine the capital recovery factor of a present worth amount P).	08

4 a Autocon company is evaluating three robots for possible use in its assembly operations (only one robot will be purchased). Data associated with these robots are as follows. Assume a technological life of 3 years and $i = 12\%$. which robot seems to be preferable assuming all factors are equal? Use NPW method. 08

Item	*Robot A*	*Robot B*	*Robot C*
First cost	55,000/−	58,000/−	53,000/−
operating and maintenance cost per year	3000	4500	4000
Expected income per year	40,000	44,000	38,000
Estimated salvage value	4000/−	6000/−	4000/−

b Draw the cash flow diagram and determine its present worth and future worth if the following cash flow for an interest rate of 10%. 08

Year	0	1	2	3	4	5
Cash flow in Rs.	−1000	+2500& −2000	2500& −2000	2500 & −2000	2500& −2000	3000 & −2400

OR

5 a A company needs a minibus to bring its employees to work and take them home. It has two alternatives:
i) To rent a vehicle for a payment of $Rs.3$ lakhs per year for the next 5 years.
ii) To buy a second hand vehicle for $Rs.3$ lakhs with an operating and maintenance cost of $Rs.2$ lakhs per year. The salvage value of the vehicle after five year would be about 1 lakh. Select the best alternative based on present worth method of comparison using an interest rate of 13% compounded annually. 10

b How much would an investor have to set aside today in order to have \$20000 five years from now if the current rate is 15%. 06

6 a Explain the comparison of assets with equal and unequal lines by sinking fund method from an economic point of view. 06

b Following alternatives can perform the same function.

Alternative	*Initial cost* $(Rs.)$	*Life* (yrs)	*Salvage* $(Rs.)$	*Annual Maintenance cost* $(Rs.)$
A	12000	6	4000	1600
B	6000	3	2000	2000
C	4000	3	1000	2400

At an annual interest rate of 12%, rank the alternatives as per the equivalent annual cost. 10

OR

7 a A company is investing \$25,000 into a 5-year project. The project will produce a uniform annual revenue of \$8000. The salvage value of the project is \$5,000. The company has decided that they will accept all projects returning a $MARR$ of 20% or more/year. Using the annual worth method, show if the company should adopt this project. 08

	b	consider the well known engineering economy relationship: $$EAC = P\left(\frac{A}{P}, i, N\right) - S\left[\left(\frac{A}{P}, i, N\right) - i\right] = (P - S)\left(\frac{A}{P}, i, N\right) + Si$$ Bring out he intuitive meaning that an analyst can derive from this relationship. Give meaning to the two terms in which relationship and the interpretation for EAC determination.	08
8	a	Explain the meaning of the three rates of return. i) Minimum acceptable rate of return ($MARR$) ii) Internal rate of return(IRR) iii) External rate of return.	08
	b	A farm house can be purchased for $Rs.90{,}000$ and the expected resale value after 20 years is $Rs.60{,}000$. If the annual rental income is $Rs.11{,}800$ and expenses $Rs.4{,}700$, what will be the rate of return earned on this farm house?	08
		OR	
9	a	What are the uses of IRR?	03
	b	Which method of capital budgeting is better: the NPV or the IRR?	03
	c	A farm house can be purchased for $Rs.90{,}000$ and the expected resale value after 20 years is $Rs.60{,}000$. If the annual rental income is $Rs.11{,}800$ and expenses $Rs.4{,}700$. What will be the rate of return earned on this farm house?	10
10	a	A student has bought a moped whose first cost is $Rs.10{,}000$ with an estimated life of 8 years. The estimated salvage value of the moped at the end of its life time is $Rs.2000$. Determine the depreciation amount and the book value at the end of various years using straight line method of depreciation. Also find the book value at the end of 7th year as specific period.	08
	b	A copy machine is purchased for \$3,217.89. The expected life is 4 years. Using double declining balance calculate the accumulated depreciation.	08
		OR	
11	a	Explain the causes and consequences of inflation.	06
	b	A company can purchase a piece of equipment for $Rs.20{,}000/-$ and sell it for $Rs.4000/-$ (in today's rupee) at the end of its 8-year service life or can lease the unit for the same period by making beginning-of-the-year payments of $Rs.3000/-$. Compare the two alternatives, using an inflation adjusted interest rate of 15 percent. Assume that the salvage value increases at the general inflation rate of 5 percent per year.	10

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. ExaminationsFast Track July - 18

Industrial Engineering and Management

ENGINEERING ECONOMY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.
3. Use of statistical and control chart tables permitted.

PART-A

1	1.1	Define the term Time Value of Money.	02
	1.2	State the conventions used while drawing a cash flow diagram.	02
	1.3	Mention any two conditions used in comparing the present-worth of alternatives.	02
	1.4	Mention the methods used for comparing present-worth of alternatives with unequal lives. Give the meaning of any one method.	02
	1.5	Write the situations for equivalent annual worth comparisons.	02
	1.6	Differentiate between equivalent annual cost and equivalent annual worth.	02
	1.7	Define the term Minimum Acceptable (attractive) Rate of Return.	02
	1.8	Differentiate between simple (conventional) and non-simple (non-conventional) investment.	02
	1.9	List the basic methods of computing depreciation of an asset/machine/equipment.	02
	1.10	Define the term inflation. Mention any two causes of inflation.	02

PART-B

2	a	State and explain the principles of engineering economy.	08
	b	Differentiate with examples, between i) Intuition and analysis. ii) Strategy and tactics.	08
		OR	
3	a	List all the compound interest factors. Deduce an equation for single payment compound amount factor.	08

	b	An engineer received a bonus of *Rs.*12000 that he will invest now. He wants to calculate the equivalent value after 24 years, when he plans to use all the resulting money as the down payment on an island vacation home. Assuming a rate of return of 8% per year, find the maturity amount the engineer gets after 24 years if the interest is compounded: i) Annually ii) Half yearly iii) Quarterly iv) Monthly Draw the cash flow diagram in each case.	08
4	a	Explain the conditions for present worth comparison of alternatives.	08
	b	Perform a present worth analysis of equal service machines with the cash flows shown below, if the MARR is 10% per year. Revenues for all three alternatives are expected to be the same.	

	Electric powered	Gas powered	Solar powered
First cost, Rs.	2500	3500	6000
Annual operating cost, Rs.	900	700	50
Salvage value, Rs.	200	350	100
Life, years	5	5	5

Find which alternative is suitable. 08

OR

5 A project engineer of environmental department is assigned to start up a new office in a city where a 6 year contract has been finalized to collect and analyze ozone – level readings. Two lease options are available, each with a first cost, annual lease cost, and deposit return estimates shown below. The rate of interest is 15% per year.

First cost, Rs.	15000	18000
Annual lease cost, Rs.	3500	3100
Deposit return, Rs.	1000	2000
Lease term, years	6	9

Determine which location should be selected on the basis of a present worth comparison. 08

b Two sites are currently under consideration for a bridge over a small river. The north site requires a suspension bridge. The south site has a much shorter span, allowing for a truss bridge, but it would require new road construction. The suspension bridge will cost *Rs.*500 million with annual inspection and maintenance costs of *Rs.*350,000. In addition, the concrete deck would have to be resurfaced every 10 years at a cost of Rs. 10,00,000. The truss bridge and approach roads are expected to cost Rs. 250 million and have annual maintenance costs of Rs. 200,000. This bridge would have to be painted every 3 years at a cost of Rs. 400,000. In addition, the bridge would have to be sandblasted every 10 years at a cost of Rs. 1,900,000.

The cost of purchasing right-of-way is expected to be Rs. 20 million for the suspension bridge and Rs. 150 million for the truss bridge. Compare the alternatives on the basis of their capitalized cost if the interest rate is 6% per year. 08

6	a	Explain the following terms associated with the life of an asset/machine/equipment: i) Service life ii) Accounting life iii) Economic life	06
	b	The first cost of a machine is Rs. 500,000. The annual maintenance cost is Rs. 2000 in the first year and increases by Rs. 1000 every year up to 10th year. The annual income is expected to be Rs. 50,000 in the first year with an increase of Rs. 25,000 every year upto 10th year. The operating cost is Rs. 6000 per year. The salvage value is estimated to be Rs. 30,000 at the end of 10 years. Find the equivalent annual worth of the machine at an interest rate of 12% per annum.	10
		OR	
7	a	A production utility can be purchased for Rs. 10,000 and is supposed to serve a 5 year term. The annual operating expenses are estimated to be Rs. 2500. The company intends to reduce the annual operating expenses to Rs. 1000 for the next 5 years by investing in additional attachments worth Rs. 13,000 in (the beginning of) the 6th year. A similar utility of Rs. 50,000 can be taken on lease for an 8 year term for an onetime payment now , with no returns. Assuming an interest rate of 10%, comment on these proposals using EAC.	10
	b	An asset was purchased 5 years ago for Rs. 520,000. It was expected to have an economic life of 8 years, at which time its salvage value would be Rs. 40,000. If the function that the asset was serving is no longer needed, for what price it must be sold now to recover the invested capital when $i = 12\%$?	06
8	a	Explain the three rates of return that appear frequently in engineering economic studies.	06
	b	A mine is for sale for Rs. 240,000. It is believed that the mine will produce a profit Rs. 65,000 the first year, but the profit will decline Rs. 5000 every year after that, eventually reaching zero, whereupon the mine will be worthless. What rate of return would this Rs. 240,000 investment produce for the purchaser of the mine?	10
		OR	
9	a	Define the term structural analysis of alternatives. Explain with examples, the two types of alternatives.	06
	b	The table below shows the cash flows of three investment proposals. If the life of all the proposals is 10 years, select the best alternative based on IRR comparisons. Take MARR as 15% per year.	10

Proposal	Initial cost, Rs.	Annual return, Rs.
A	400,000	100,000
B	550,000	140,000
C	625,000	160,000

10	a	Explain with examples, the causes of depreciation of an asset/machine/equipment.	08
	b	A student has bought a motor bike whose first cost is Rs. 100,000. The life of the bike is estimated as 8 years at which the salvage value will be Rs. 20,000. Determine the depreciation and the book value of the bike at the end of every year by straight line method. Also find the book value of the bike at the end of 7th year as a specific period.	08
		OR	
11	a	Explain any four causes of inflation.	08
	b	Anequipment has a first cost of Rs. 9000 and an estimated life of 5 years. The salvage value at the end of its life is Rs. 700. Determine the depreciation and the book value of the equipment at the end of each year by sum of the years digits method.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Fast Track Examinations July-17

Industrial Engineering and Management

ENGINEERING ECONOMY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

6. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
7. Answer FIVE full questions from Part B.
8. Use of engineering economy tables is permitted.
9. Cash flow diagrams and steps in problem formulation should be shown clearly.

PART-A

1	1.1	A bank offers 5% compound interest calculated on half yearly basis. A customer deposits $Rs.1600$ each on 1st January and 1^{st}July of a year. At the end of the year, the amount he would have gained by way of interest.	02
	1.2	Distinguish between strategy and tactics.	02
	1.3	Depict a cash flow diagram from a borrower's and lender's viewpoint.	02
	1.4	The annual interest rate is 6% and the interest is compounded quarterly. What is the quarterly nominal interest rate? What is the effective annual interest rate if compounded quarterly?	02
	1.5	The net difference of the present costs and benefits is _____.	01
	1.6	Bring out the relationship between economic life and the equivalent annual cost and equivalent annual worth.	02
	1.7	Distinguish between internal rate of return and external rate of return.	02
	1.8	List the several methods of accounting depreciation fund.	02
	1.9	The three rate of return used in engineering economy are _____, _____ and _____.	02
	1.10	State the reasons for declining method an attractive method for determining the depreciation charge of an asset.	02
	1.11	Define inflation.	01

PART-B

2	a	Determine the effective rate of interest for a nominal annual rate of 8% that is compounded: i) Semi annually ii) Quarterly iii) Monthly and iv) Weekly.	08

	b	An electrical engineer wants to deposit an amount P now such that she can withdraw an equal annual amount of $A_1 = \$2000$ per year for the first 5 years, starting 1 year after the deposit, and a different annual withdrawal of $A_2 = \$3000$ per year for the following 3 years. How would the cash flow diagram appear if $i = 8.5\%$ per year?	03
	c	Why should engineers and managers become knowledgeable on the methods of evaluating alternatives economically? Give any five reasons to bring out the importance of engineering economy in decision making.	05
		OR	
3	a	Explain with help of a block diagram the steps in problem solving and decision making.	08
	b	A company has to replace a present facility after 15 years at an outlay of $Rs.5{,}00{,}000$. It plans to deposit an equal amount at the end of every year for the next 15 years at an interest rate of 18% compounded annually. Find the equivalent amount that must be deposited at the end of every year for the next 15 years.	08
4	a	Explain the conditions for present worth comparisons.	06
	b	Novel Investment ltd. Accepts $Rs.10{,}000$ at the end of every year for 20 years and pays the investor $Rs.8{,}00{,}000$ at the end of the 20th year. Innovative Investment ltd. Accepts $Rs.10{,}000$ at the end of every year for 20 years and pays the investor $Rs.15{,}00{,}000$ at the end of the 25th year. Which is the best investment alternative? Use present worth base with $i = 12\%$.	10
		OR	
5	a	A company must decide whether to buy machine A or machine B.	

	Machine A	*Machine B*
Initial cost	$Rs.4{,}00{,}000$	$Rs.8{,}00{,}000$
Useful life, in years	4	4
Salvage value at the end of machine life	$Rs.2{,}00{,}000$	$Rs.5{,}50{,}000$
Annual maintenance cost	$Rs.40{,}000$	0

		At 12% interest rate, which machine should be selected? (use future worth method of comparison).	10
	b	Explain the methods used to compare assets that have unequal lives.	06
6	a	Two possible routes for laying a power line are under study. Data on the routes are as follows	

	Around the lake	*Under the lake*
Length	$15\ km$	$5\ km$
First cost (Rs.)	$1{,}50{,}000/km$	$7{,}50{,}000/km$
Useful life (years)	15	15
Maintenance cost (Rs.)	$6{,}000/km/yr$	$12{,}000/km/yr$
Salvage value (Rs)	$90{,}000/km$	$1{,}50{,}000/km$
Yearly power loss (Rs)	$15{,}000/km$	$15{,}000/km$

		If 15% interest is used, should the power line be routed around the lake or under the lake?	10

	b	List the situation for equivalent annual worth comparisons and give an example for each that involve equivalent annual cost and equivalent annual worth comparison.	06
		OR	
7	a	Jothi Lakshmi has arranged to buy some home recording equipment. She estimates that it will have a five year useful life and no salvage value at the end of equipment life. The dealer who is a friend has offered Jothi Lakshmi two alternative ways to pay for the equipment. i) Pay $Rs.60{,}000$ immediately and $Rs.15{,}000$ at the end of one year. ii) Pay nothing until the end of fourth year when a single payment of $Rs.90{,}000$ must be made. If Jothi Lakshmi believes 12% is suitable interest rate, which alternative is the best for her?	10
	b	Explain the comparison of assets with equal and unequal lives by sinking fund method from economic point of view.	06
8	a	Applications of green, lean manufacturing techniques coupled with value stream mapping can make large financial differences over future years while placing greater emphasis on environmental factors. Engineers with Monarch Paints have recommended to management an investment of \$2,00,000 now in novel method that will reduce the amount of wastewater, packaging materials, and other solid waste in their consumer paint manufacturing facility. Estimated savings are \$15,000 per year for each of the next 10 years and an additional savings of \$300,000 at the end of 10 years in facility and equipment upgrade costs. Determine the rate of return.	10
	b	Explain the three rates of return that appear frequently in engineering economy studies.	06
		OR	
9	a	A company is trying to diversify its business in a new product line. The life of the project is 10 years with no salvage value at the end of its life. The initial outlay of the project is $Rs.20{,}00{,}000$. The annual net profit is $Rs.3{,}50{,}000$. Find the rate of return for the new business.	08
	b	Highlight the IRR misconceptions with suitable example.	08
10	a	Explain the causes of declining value of an asset.	10
	b	A 15 year \$50,000 bond that has a dividend rate of 10% per year, payable semiannually is currently for sale. If the expected rate of return of the purchaser is 8% per year, compounded semiannually, and if the inflation rate is expected to be 2.5% each 6 month period. What is the bond worth now i) Without an adjustment for inflation ii) When inflation is considered?	06
		OR	

11	a	A new electric saw for cutting small pieces of wood in a furniture manufacturing plant has a cost of $Rs.40{,}000$. The salvage value at the end of 10 years is zero. Tabulate the annual depreciation and cumulative depreciation and the book value at the end of each year using i) Straight line method ii) 20% declining balance method.	10
	b	Discuss the consequences of inflation.	06

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Examinations April/May-17

Industrial Engineering and Management

ENGINEERING ECONOMY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

6. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
7. Answer FIVE full questions from Part B.
8. Use of engineering economy tables is permitted.
9. Cash flow diagrams and steps in problem formulation should be shown clearly.

PART A

1	1.1	Draw the cash flow diagrams based on borrower's point of view and lender's point of view.	02
	1.2	For a deposit of $Rs.10{,}000/-$ made today, what equal amounts can be withdrawn at the end of each quarter for the next 4 years, if the nominal interest rate is 10%?	02
	1.3	Find the effective rate of interest for an actual rate of interest if 10% when compounded continuously.	02
	1.4	Write the assumptions that are made for present worth and annual worth comparison.	02
	1.5	Bring out the relationship between economic life and the equivalent annual cost and equivalent annual worth.	02
	1.6	What are the reasons for creating a depreciation fund for machinery and equipment?	02
	1.7	Define rate of return and list the three rates if return that appear frequently in engineering economic studies.	02
	1.8	Why is the declining balance method an attractive method for determining the depreciation charge of an asset?	02
	1.9	List any two basic methods of computing depreciation charges.	02
	1.10	Differentiate between nominal interest and effective interest.	02

PART B

2	a	Mention any six reasons to bring out the importance of engineering economy in decision making.	06
	b	Determine the effective interest rate for a nominal rate if 8% that is compounded i) Semiannually ii) Quarterly iii) Monthly iv) Weekly.	10
		OR	

3	a	State and explain the principles of engineering economy.	08
	b	With a block diagram, explain the steps in problem solving.	08
4	a	A loan of $Rs.1000$ Is made today under an agreement that $Rs.1400$ will be received some time in future. When should be $Rs.1400$ received if the loan is to earn interest at a rate of 10% compounded quarterly?	06
	b	Draw the cash flow diagram and determine its present worth and future worth if the following cash flow for an interest rate of 10%.	

$Year$	0	1	2	3	4	5
$Cash\ flow\ in\ Rs.$	-1000	$+2500$ & -2000	2500 & -2000	2500 & -2000	2500 & -2000	3000 & -2400

10

OR

5	a	An automobile company recently advertised its car for a down payment of $Rs.1{,}50{,}000$ alternatively the car can be taken by customers without making any payment but they have to pay an equal yearly amount of Rs. 25,000 for 15 $years$ at an interest rate of 18% compounded annually. Suggest the best alternative to the customer.	10
	b	What are the different methods to effect comparison of assets that have unequal lives based on the present worth evaluations? Mention the assumptions of the study period method.	06
6	a	Explain the situations for equivalent annual worth comparison.	06
	b	Following alternatives can perform the same function.	

$Alternative$	$Initial\ Cost\ (Rs.)$	$Life\ (yrs)$	$Salvage\ (Rs.)$	$Annual\ Maintenance\ Cost\ (Rs.)$
A	12000	6	4000	1600
B	6000	3	2000	2000
C	4000	3	1000	2400

At an annual interest rate of 12%, rank the alternatives as per the equivalent annual cost. 10

OR

7	a	Two alternatives for laying a power line are under study. Data on the routes are as follows:	

	$Details\ of\ the\ alternatives$	$Around\ the\ lake$	$Under\ the\ lake$
1	$Length\ (kms)$	15	05
2	$First\ cost\ (Rs.)$	1,50,000	7,50,000
3	$Useful\ life\ (yrs)$	15	15
4	$Maintenance\ Cost\ (Rs.)$	$6000Kms$	$12000Kms$
5	$Salvage\ value\ (Rs.)$	90000	1,50,000
6	$Yearly\ power\ loss(Rs.)$	$15{,}000Kms$	$15{,}000Kms$

If 15% interest is used, should the power line be routed around the lake or under the lake? 10

	b	The equation for the equivalent annual cost computational is given by $EAC = (P - S)\left(A/P, i, n\right) + S_i$ with the usual notations used. Explain how this equation converts the cash flows into annuities.	06

8 a An automobile company is planning to buy a robot for a forging unit. It has identified two different companies for the supply of robots. Details of cost and revenue of using robots are as follows:

Particulars	*Initial Cost (Rs)*	*Annual Revenue (Rs.)*	*Life (yr)*	*Salvage Value (Rs.)*
Company A	5,00,000	80,000	8	40,000
Company B	9,00,000	2,50,000	8	60,000

MARR for the company is 12%. Suggest the best brand of robot to the company based on rate of return method. 12

b What do you understand by dependent alternatives? Explain. 04

OR

9 a A farm house can be purchased for $Rs.90{,}000$ and the expected resale value after 20 years is $Rs.60{,}000$. If the annual rental income is $Rs.11{,}800$ and expenses $Rs.4{,}700$. What will be the rate of return earned on this farm house? 10

b Explain the *IRR* misconceptions(any two). 06

10 a Explain the causes of depreciation. 06

b *AC* device is purchased for $Rs.1{,}000$ and has no salvage value. It is expected to serve for 5 years. Calculate the book value by sum of years digit method and double declining balance method. 10

OR

11 a An automobile company has purchased a wheel alignment device for $Rs.10\ Lakhs$. The device can be used for 15 yrs. The salvage value at the end of life of the device is 10% of the purchase value. Determine the following using the double declining balance method of depreciation:

i) Depreciation at the end of the $7th$ year.
ii) Depreciation at the end of the $12th$ year.
iii) Book value at the end of the $8th$ year. 10

b Explain the causes and consequences of inflation. 06

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

III / IV Semester B. E. Fast Track Examinations July-16

Industrial Engineering and Management

ENGINEERING ECONOMY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

6. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
7. Answer FIVE full questions from Part B.
8. Use of Engineering economy tables is permitted.
9. Cash flow diagrams and steps in problem formulation should be shown clearly.

PART-A

1	1.1	The terms strategy and tactics historically are military terms associated with broad plans from the high command and _____ from the lower echelons respectively.	01
	1.2	Interest payments have been linened to the reward for waiting, but it is more appropriate for engineering economist to view interest as the _____ from the efficient use of money resource.	01
	1.3	If a loan of $Rs.1000$ is availed for 2 months at the rate of 10 percent. If the simple interest concept is applied. The amount to be repaid is _____	01
	1.4	The effective annual interest, for a 1 year loan of $Rs.1000/-$ at a nominal annual interest rate of 18 percent compounded monthly as against compounded semi annually is higher by _____ %	01
	1.5	A Rs today is not worth the same tomorrow. This statement recognizes the concept of _____ money and is the foundation for calculations in engineering economy.	01
	1.6	Net Present worth is the _____ between the present worths of benefit and costs, is the most widely used present-worth models.	01
	1.7	The 72 role indicates the approximate number of years N* at which the present worth (PW) is one half the _____ at the annually compounded interest rate i. Mathematically $72\,m : N^* = \frac{72}{(i) \times 100}$.	01
	1.8	The two prominent methods for the present worth comparisons of assets that have unequal lives are the common multiple method and the _____.	01
	1.9	A do-nothing option is always an alternative, even if it resorts only in procrastination. This option is included in the present worth computations as one of the _____ in the decision making.	01
	1.10	The payback method is an extremely simple method used to obtain rough estimate of _____ that an investment will take to pay for itself.	01

	1.11	In the annual worth method all receipts disbursements occurring over a period are converted to an _____ uniform yearly amount.	01
	1.12	The term equivalent annual costs (EAC) is used to designate comparisons involving only costs and the term equivalent annual worth (EAW) is used when costs and _____ are both present.	01
	1.13	_____ life is the time period that minimizes the assets total equivalent annual cost or maximizes its equivalent net annual income.	01
	1.14	The internal rate of return (IRR) of an investment is the rate of interest earned on the unrecovered balance of an investment where the terminal balance is _____	01
	1.15	The _____ is the rate set by an organization to designate the lowest level of return that makes an investment acceptable. This is also known as hurdle rate.	01
	1.16	For the case of a single simple investment the interest rate 'i' determined by setting the present worth (or EAW) of receipts equal to the present worth (or EAW) of disbursements. That is Find (i) so that PW (receipts) = PW (disbursements) then the interest rate 'i' is known as _____.	01
	1.17	A plot of $PW(i)$ versus 'i' gives the structure for evaluating alternatives based on IRR (Internal rate of return) for any project. When the cumulative cash flow of a project switches from negative to positive (or reverse) more than once then there could be _____ one possible rate of return.	01
	1.18	Provisions for recovering capital invested income producing assets are made by charging _____ against current income.	01
	1.19	Steam locomotives cost value rapidly as rail roads turned to diesel power. This cause of declining value of the assets is known as _____ depreciation.	01
	1.20	Inflation results in a decline over time in the _____ power of a unit of money.	01

PART-B

2	a	Explain why Engineering economy plays an important role in decision making and problem solving in the area of engineering and management. Give the general method of analyzing problems using the principles of engineering economy.	06
	b	Service records for a specific piece of production equipment indicate that a replacement machine will have first year maintenance costs of approximately $Rs.1000/-$ and there costs will increase by $Rs.200$ per year for each additional year of service. Assuming the equipment is to be in service for 10 years and using an interest rate of 15 percent. Determine the maximum amount that should be paid for a lifetime maintenance contract at the time the equipment is purchased.	08
	c	Draw the cash flow diagram of a loan situation from the Borrower's viewpoint and from the lender's viewpoint.	02
		OR	

3	a	In the context of problem solving and decision making, distinguish clearly between: i) Intuition and Analysis ii) Sensitivity and Sub-optimization iii) Tactics and Strategy.	07
	b	You have a chance to buy a new car with a list price of $Rs.12{,}00{,}000$(Twelve lakhs). You have to make a down payment of $Rs.2{,}00{,}000$ (Two lakhs), and the dealer will finance the remainder at an nominal interest rate of 6 percent, compounded monthly for 5 years. i) Determine the amount of your monthly payment. ii) How much total interest will you pay over 5 years?	07
	c	Distinguish between the concepts of Annuity due and Deferred Annuity. Show the cash flow diagrams for both the cases.	02
4	a	Autocon Company is evaluating three robots for possible use in its assembly operations. (only one robot will be purchased). Data associated with these robots are as follows.	

	Robot A	*Robot B*	*Robot C*
First Cost($)	55,000	58,000	53,000
Operating and Maintenance Costs ($)	$3000/year$	$4500/year$	$4000/year$
Expected income ($)	$40{,}000/year$	$44{,}000/year$	$38{,}000/year$
Estimated salvage valve ($)	4000	6000	4000

		Assuming a technological life of 3 years and a desired interest rate of 12 percent, which robot seems preferable? Assume all other factors are equal. Use the net present worth evaluation method.	08
	b	List the six basic conditions for present-worth comparisons. Explain the relevance of these in the present worth analysis.	04
	c	Distinguish between the patterns in present worth calculations based on present-worth equivalence and net-present worth.	04
		OR	
5	a	List the steps in the comparison of assets that have unequal lives using the common-multiple method.	03
	b	A proposed improvement in an assembly line will have an initial purchase an installation cost of $Rs.1{,}75{,}000/-$. The annual maintenance cost will be $Rs.6000/-$, periodic over hauls once every 3 years excluding the last, year of use will cost $Rs.11{,}500/-$ each. The improvement will have a useful life of 9 years at which time it will have no salvage value. What is the present worth of the 9 year costs of improvement at $i = 8$ percent.	08
	c	What are the pros and cons of the payback comparison method?	05
6	a	A standby electric power generator was purchased 6 years ago for $Rs.8000/-$. At that time it was expected that the equipment would be used for 15 years and would have a salvage value of 10 percent of the first cost. The generator is no longer needed and is to be sold for $Rs.2500/-$ using an interest rate of 15 percent. Determine the difference between the anticipated and actual equivalent annual capital costs.	08

	b	The sinking fond factor ($^{A}/_{F}$, i, N) is used to calculate the capiral covery costs, as an alternative formulation as follows: $(P-S)(A/P,i,N) + Si$ $-------$ (1) Explain the terms in the equation (1) above and the physical interpretation of each of the components in the equation.	06
	c	Give any three situations that suit the requirement for carrying out economic feasibility using the equivalent Annual-worth comparisons.	02
		OR	
7	a	Two types power converters, alpha and beta are under consideration for a specific application. An economic comparison is to be made at an interest rate of 10 percent, and the following cost estimates have been obtained.	

Particulars	Alpha	Beta
Purchase Price	$Rs.10{,}000/-$	$Rs.25{,}000/-$
Estimated service life	5 *years*	9 *years*
Salvage value	$Rs.0$	$Rs.5000/-$
Annual operating cost	$Rs.2500/-$	$Rs.1200/-$

		Determine the annual equivalent costs of the alternative systems.	08
	b	Why is it necessary to distinguish between the following aspect of the life of an asset: i) Ownership life ii) Accounting life iii) Economic life. Clearly define these terms and state their significance.	04
	c	Define perceptual life. Why is this applicable to make economic comparison involving an asset with infinite life, such as land?	04
8	a	Clearly define the following rates of return that appear frequently in engineering economy studies: i) Minimum Acceptable rate of return ($MARR$) ii) The internal rate of return (IRR) iii) The external rate of return (ERR).	04
	b	Additional parking space for a factory can be either rented for $Rs.15{,}000/-$ per year on a 10 year lease or purchased for $Rs.1{,}60{,}000/-$. The rental fees are payable in advance at the beginning of each year. Taxes and maintenance fees will be paid by the lessee. The land should be worth at least $Rs.95{,}000/-$ after 10 years. What rate of return will be earned from purchase of the lot?	08
	c	Adam Smith in his famous work on wealth of nations (1776) shared: "No law can reduce the common rate of interest below the lowest ordinary market rate at the time when the law is made". Explain this in the modern day contest.	04
		OR	
9	a	Consider the following cash flow stream:	

End of period	0	1	2
Cash flow	$-Rs.100/-$	$Rs.250/-$	$-Rs.156/-$

		Determine the present worth at 0% and 25% Guess the Internal rate of return in this case. Why do you classify this as a non simple case?	06

	b	Three mutually exclusive investments to be compared using the rate of return method are described below. The $MARR$ is 9 percent. $Alternatives$ X | Y | Z $First\ cost$ (\$) | 20,000/− | 16,000/− | 10,000/− $Annual\ Expense$ (\$) | 5,000/− | 3,000/− | 4,000/− $Annual\ Income$(\$) | 11,500/− | 11,500/− | 7000/− $Economic\ life, years$ | 4 | 2 | 4 i) Which alternative has the highest overall IRR and what is it? ii) Recommend an alternative and justify your recommendation iii) Suppose that first costs are expected to increase at 4 percent per year, due to technological instability. How will this effect your decision?	10
10	a	"Depreciation reserves are used to fund the most attractive proposals put forward to improve operations, some of which are to replace worn-out and obsolete assets" Discuss this statement and reason out with examples the need for provisioning depreciation fund to help in managing capital assets.	04
	b	An asset was purchased for $Rs.20{,}000/-$ Its useful life was estimated as ten years and the salvage value was estimated as $Rs.5000/-$ i) Use the straight line method of depreciation to work out the annual depreciation change. ii) Use the declining balance method to compute the depreciation rate in percentage terms. Also estimate the depreciation fund at the end of two years.	08
	c	Why should cash flows be adjusted to consider the effect of taxation for effective comparison of proposals? (Given that the tax rate uniformly impacts all the proposals)	04
		OR	
11	a	Distinguish clearly between the term depreciation and depletion of assets. Give an example each to clarify their significance in respect of economic value assessments.	04
	b	Computers purchased by a public utility company cost $Rs.7000/-$ each past records indicate that they should have a useful life of 5 years after which they will be disposed of with no salvage value. The company evaluates at a cost of capital of 7 percent. Compare using the straight line method, declining-balance method and the double declining balance method the following: i) The depreciation charge during year 1 and year 2 separately ii) The depreciation reserve accumulated by the end of year 3 iii) The book value of the computers at the end of year 3.	08
	c	List the causes of inflation. Inflation is a general increase in the price level. Based on your experience in life explain how inflation has affected the purchasing power of the common man during the process of buying clarity necessities such as food, vegetables and other commodities. Why inflation is to be considered during the economic feasibility study of investment proposals?	04

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Examinations April/May-16

Industrial Engineering and Management

ENGINEERING ECONOMY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

6. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
7. Answer FIVE full questions from Part B.
8. Use of engineering economy tables is permitted.
9. Cash flow diagrams and steps in problem formulation should be shown clearly.

PART-A

1	1.1	State the conventions used while drawing the cash flow diagram.	02
	1.2	Differentiate between simple and compound interest.	02
	1.3	Mention the conditions (any four) for present worth comparison.	02
	1.4	Find the effective rate of interest for an actual rate of interest of 10% when compounded continuously (infinitely).	02
	1.5	write the assumption that are made for present worth and basic annual worth comparison (any two)	02
	1.6	Write the situations for equivalent annual worth comparison(any two)	02
	1.7	List the basic methods of computing depreciation charges |(any two)	02
	1.8	Define the terms: Book value (VBV) and Depreciation amount.	02
	1.9	List the IRR misconceptions.(any two)	02
	1.10	Define rate of return and list the three rates of return that appear frequently in engineering economic studies.	02

PART-B

2	a	With a block diagram explain the steps in problem solving.	08
	b	State and explain the principles of engineering economy.	08
		OR	
3	a	List the types of compound interest problems and explain any one.	08
	b	A film star is at the height of his career. He wants to invest Rs.10 lakhs from the end of this year and follow it up with 9 lakhs, 8 lakhs and so on. For the next five years, when his income would go on diminishing. Find the maturity amount 6 years later if a film producer agrees to pay him 15% rate of interest, compounded annually.	08

4	a	Explain the conditions for present worth comparisons.	06
	b	Two holidays cottages are under consideration .Compare the present worth of the cost of 24 years service at an interest rate of 5%, when neither cottage has a realizable salvage value.	

	$Cottage\ 1$	$Cottage\ 2$
$First\ cost$	Rs 4,500	Rs 10,000
$Estimated\ life$	12 $years$	24 $years$
$Annual\ maintenance\ cost$	Rs 1,000	Rs 720

10

OR

5	a	A company needs a minibus to bring its employees to work and take them home. It has two alternatives: i) To rent a vehicle for a payment of Rs. 3 lakhs per year for the next 5 years. ii) To buy a second hand vehicle for Rs 3 lakhs with an operating and maintenance cost of Rs 2 lakhs per year. The salvage value of the vehicle after the five year would be about 1 lakhs. Select the best alternative based on present worth method of comparison using an interest rate of 13% compounded annually.	10
	b	Following are the estimates of three alternative investments made on 3 different machines in an industry. Find out which machine has the fastest payback period.	

	$Particulars$	$Machine\ A$	$Machine\ B$	$Machine\ C$
1	$Initial\ Investment$	30,000	38,000	42,000
2	$Annual\ receipts$	20,000	23,500	26,000
3	$Annual\ expenditures$	5500	6,500	7,000
4	$Economic\ life$	4 $years$	4 $years$	4 $years$

06

6	a	Explain the comparison of assets with equal and unequal lines by sinking fund method from an economic point of view.	06
	b	A publishing house wants to purchase an offset printing press. Three dealers have responded to the tender called whose particulars are given in the table. Determine the best alternative based on the annual equivalent method by assuming $i = 12.5\%$ compounded annually.	

$Manufacturer$	$Down\ payment$	$Yearly\ equal\ install, (Rs)$	$Salvage\ value$	$No.of\ years\ of\ payment\ and\ life$
X	8,00,000	2,25,000	0	10
Y	7,00,000	2,00,000	0	10
Z	5,00,000	2,50,000	0	10

10

OR

7	a	Explain the situations for equivalent annual worth comparison.	06
	b	A stand by lighting generator is required for a shop two types are available. $\begin{array}{\|c\|c\|c\|}\hline & Type-1 & Type-II \\ \hline First\ cost & Rs.5{,}000 & Rs.3{,}200 \\ Salvage\ value & Rs.1{,}000 & Nil \\ Annual\ operating\ costs & Rs.780 & Rs.950 \\ \hline\end{array}$ If both generators have a life of 4 years and the interest rate is 15 percent per year, which offers the lowest equivalent annual cost.	10
8	a	What is rate of return? Explain the three rates of return that appear frequently in engineering economic studies.	06
	b	A farm house can be purchased for $Rs.90{,}000$ and the expected resale value after 20 years is $Rs.60,000$. If the annual rental income is $Rs.11,\ 800$ and expenses $Rs.4{,}700$ what will be the rate of return earned on this farm house?	10
		OR	
9	a	Explain the IRR misconceptions.	06
	b	A company is in the process of selecting the best alternative among the following three mutually exclusive alternatives. $\begin{array}{\|c\|c\|c\|c\|}\hline Alternative & Initial\ investment & Annual\ revenue.Rs. & Life\ years \\ \hline A_1 & 5{,}00{,}000 & 1{,}00{,}000 & 10 \\ A_2 & 8{,}00{,}000 & 1{,}40{,}000 & 10 \\ A_3 & 3{,}00{,}000 & 70{,}000 & 10 \\ \hline\end{array}$ Find the best alternatives based on the rate of return method of comparisons.	10
10	a	Explain the causes of Depreciation.	06
	b	A student has bought a moped whose first cost is Rs.10,000 with an estimated life of 8 years. The estimated salvage value of the moped at the end of its life time is Rs.2,000. Determine the depreciation amount and the book value at the end of various years using straight line method of depreciation. also find the book value at the end of 7th year as a specific period.	10
		OR	
11	a	Explain the causes and consequences of inflations.	06
	b	Ac device is purchased for Rs 1,000 and has no salvage value. It is expected to serve for 5 years calculate the book value by sum of years digit method and double declining balance method.	10

Machine Design

IV Semester SEE Question Papers

Course codes: 18IM46, 16IM41

- Nov 2022 (18IM46)
- Grade Improvement Nov 2021 (18IM46)
- Grade Improvement Nov 2020 (18IM46)
- Fast Track Oct 2020 (16IM41)
- Fast Track Jul 2019 (16IM41)
- Apr/May 2019 (16IM41)
- Fast Track Jul 2018 (16IM41)
- Apr/May 2018 (16IM41)

18IM46

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E. Examinations November - 2022

Industrial Engineering and Management

STRENGTH OF MATERIALS AND MACHINE DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

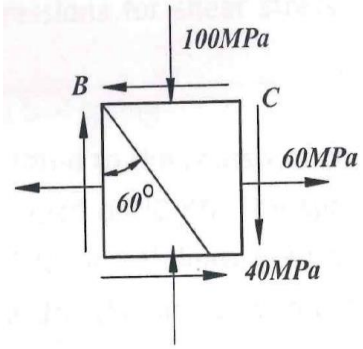
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of Design Data Handbook is permitted.

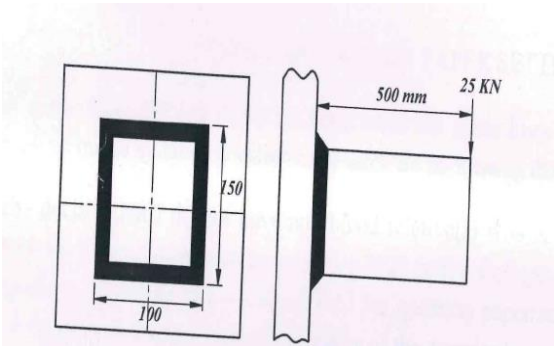
PART A

1	1.1	If an element is subjected to stresses of $60Mpa$ compressive tensile an $30\,MPa$in two mutually perpendicular directions, maximum shear stress in the element will be _____.	02
	1.2	Show the sign conventions for shear force and bending moment with sketches.	02
	1.3	For a cantilever beam of span L with uniformly distributed load of w/unit length, the shear force and bending moment at the fixed support is _____ and _____.	02
	1.4	Write down Euler- Bernoulli's bending equation and mention the dimension of each term.	02
	1.5	The polar modulus of a hollow shaft of outer diameter 100mm and ratio of diameter 0.7 is _____.	02
	1.6	Mention the ASME code for design of commercial shafting.	02
	1.7	List the three important theories of failure used in design of machine elements.	02
	1.8	Spring index of a helical spring is defined as the ratio of ____ to ____.	02
	1.9	The Lewis form factor of a spur gear is a function of ____ and _____.	02
	1.10	The size of a rivet is specified by its ____ and the size of a fillet weld by its _____.	02

PART B

2	a	Derive an expression for normal stress and tangential stress on an oblique plane in an element subjected to biaxial stress system with shear.	06
	b	A point in a machine member is subjected to stresses as shown in Fig 2a. Determine using Mohr's circle method; i. Normal and shear stress on a plane inclined at 60^0 to the plane of 60MPa stress. ii. Magnitude and orientation of principal stresses. iii. Magnitude and orientation of max shear stresses.	

		 Fig 2a	10
3	a	A beam supports a maximum bending moment of magnitude $10KN-m$. Determine the maximum flexural stress induced in the beam by considering: i) Rectangular section of width 100 mm and depth 200 mm ii) Circular section of equal area as that of rectangular section. Also determine the ratio of flexural rigidities of the two sections assuming same material and length for both.	06
	b	Obtain an expression of shear stress distribution in a beam of rectangular cross section and hence show that the maximum shear stress is 50% more than the average shear stress. Also sketch the variation of shear stress across the section.	10
		OR	
4	a	A solid shaft rotating at 500rpm transmits 30kW. Max torque is 20% more than the mean torque. Material of the shaft has an allowable shear stress 65MPa and modulus of rigidity 81 GPa. Angle of twist in the shaft should not exceed 1^0 per meter length of shaft. Determine the safe diameter of shaft.	06
	b	With usual notations, derive the torsion equation for a shaft subjected to pure torsion. State clearly all the assumptions made.	10
5	a	Define stress concentration factor. Discuss the methods of minimizing the severity of stress concentration with neat sketches.	06
	b	A steel shaft is subjected to a bending moment of 9KN-m and a twisting moment of 12KN-m. The yield strength of steel is 360MPa and factor of safety is 2. Determine the permissible diameter of the shaft by using: i. Maximum principal stress theory. ii. Maximum shear stress theory iii. Distortion energy theory.	10
		OR	
6	a	Discuss the load factor, size factor, surface factor and notch sensitivity as applied to fatigue loading.	06
	b	Determine the maximum load for a beam simply supported at its ends and cyclically loaded from W to 3W at its mid point. The diameter of the beam section is 50mm and length between supports is 400mm. the yield point in tension is 520Mpa and the endurance limit in reversed bending is 320MPa. Use factor of safety 1.25. the size and surface correction factors are 0.75 and 0.9 respectively.	10

7	a	With usual notations, derive expressions for shear stress induced and deflection of a helical spring.	06
	b	Explain equalization of stresses in a leaf spring. A multi leaf spring with camber is fitted to the chassis of an automobile over a span of 1.2m to absorb shocks due to a maximum load of 20KN. The spring material can sustain a maximum stress of 0.4Gpa. The spring has 2 full length leaves out of 8 leaves. The leaves are assembled with bolts with a central band of width 150mm. Design the spring for a maximum deflection of 50mm. Take E=207GPa.	10
8	a	Determine the size of the weld for the joint shown in Fig8a taking the allowable shear stress as 80MPa.  Fig.8a	06
	b	Discuss : i. Failure modes of riveted joints with neat sketches. ii. Merits and demerits of welded joints.	10

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E.Grade Improvement ExaminationsNov-2021

Industrial Engineering and Management

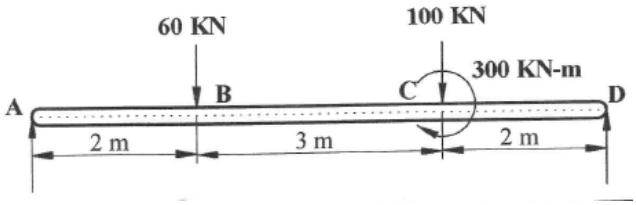
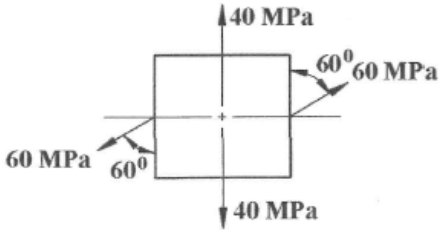
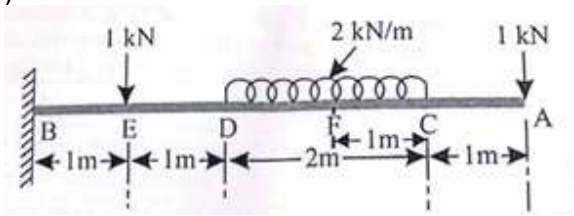
STRENGTH OF MATERIALS AND MACHINE DESIGN

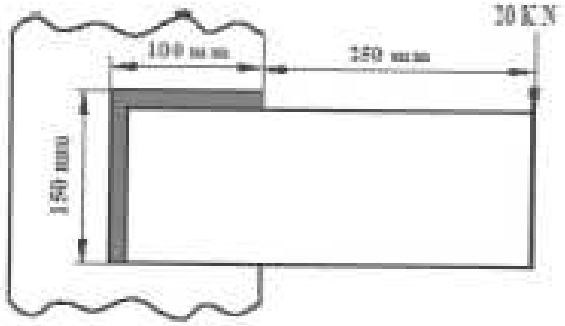
Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

Answer any FIVE full questions out of TEN. Each carries 20 marks

1	1.1	The ratio of lateral strain to linear strain in a loaded member is known as ____.	01
	1.2	In a biaxial stress system, the angle between principal planes and planes of maximum shear stress is _____.	01
	1.3	The point on a beam at which the bending moment changes its sign is known as ____.	01
	1.4	The rate of change of shear force in a beam gives____	01
	1.5	The elastic constant used in analysis of torsion is _____.	01
	1.6	The ratio of maximum stress around a discontinuity to nominal stress is known as _____.	01
	1.7	The theory of failure widely used for design of transmission shafting is _____.	01
	1.8	The surface finish factor of a highly polished component is ____.	01
	1.9	The cross section of a key which is equally strong in crushing and shearing is _____.	01
	1.10	A rivet is specified by its _____ diameter.	01
	1.11	When the diameter of a shaft is doubled, its torque carrying capacity will increase by _____ times.	02
	1.12	Define the following i. Pitch ii. Lead as applied to threaded fasteners.	02
	1.13	List the correction factors to be applied to Soderberg's equation for fatigue life.	02
	1.14	If the torque on a gear of 20^0pressure angle is 100Nm and its pitch diameter is 500mm, the tangential force on the gear is _____ N and the radial force is ____ N.	02
	1.15	If the mean coil diameter of a helical spring is 8 times its wire diameter, the Wahl's stress factor is ____.	02
2	a	Define Principal stresses and Principal planes as applied to 2D stress system. Also write down the expression for the same.	08

	b	Draw the shear force and bending moment diagrams for a simply supported beam subjected to the forces and moments as shown in fig 2b. Also locate the point of inflexion.  Fig 2b	12
3	a	A point in a strained material is subjected to stresses as shown in fig 2a. Determine analytically or graphically, the principal stresses, maximum shear stress and their orientations.  Fig 3a	10
	b	Draw the shear force and bending moment diagram for cantilever loaded as shown in the fig3 (c)  Fig 3c	10
4	a	List the assumptions made in theory of simple bending. Also write down Euler-Bernoulli equation for bending theory.	08
	b	A simply supported beam carrying uniformly distributed load is subjected to maximum bending stress of 45MPa and maximum shear stress of 4.5Mpa. The cross section of the beam rectangular with 50mm width and 100mm depth. Determine the length of the beam and the UDL that the beam can carry.	12
5	a	Derive an expression for the horizontal shear stress induced due to shear force in a loaded beam.	08
	b	Determine the diameter of solid circular shaft to transmit $192\,KW$ at 450rpm. Allowable shear stress may be taken as $70MPa$. If the solid shaft is replaced by a hollow shaft of same material and length, whose diameter ratio is 0.8, find the percentage reduction in weight achieved and the ratio of torsional stiffness of hollow and solid shafts.	12
6	a	A shaft is stepped down from 80 mm diameter of 40 mm diameter with a fillet radius of 6 mm. Determine the maximum stress induced in the shaft taking stress concentration into account if the shaft is subject to: i) A bending moment of 200 $N-m$ ii) A twisting moment of 300 $N-m$	10

	b	A machine member made of carbon steel having stress of 328.6MPa is loaded such that $\sigma_x = 60MPa, \sigma_y = -20MPa, \tau_{xy} = 30MPa$. Determine the factor of safety by using the following theories of failure: i. Rankine's theory ii. Tresca's theory iii. Von-Mises theory.	10
7	a	Define i. Endurance limit ii. Notch sensitivity iii. Low cycle and high cycle fatigue.	06
	b	A hollow shaft of marine vessel is mounted between bearings located 9.5m apart and transmits 10MW at 90rpm. The shaft weighs 66kN. The outside diameter of shaft is 450mm and its inside diameter is 300mm. Determine the shear stress induced in the shaft and angular deflection between bearings. Assume steady loads and take G=80GPa.	14
8	a	With usual notations and neat sketch, derive Lewis equation for strength of a spur gear tooth. State clearly the assumptions made.	10
	b	A railway car weighing 18KN and moving at a speed of 72m/min is brought to rest by a buffer consisting of 2 helical springs of circular cross section. In bringing the car to rest, the spring undergoes a deflection of 0.25m, The allowable shear stress for the steel wire is 0.3 GPa. Spring index =6. Rigidity modulus G=84GPa. Determine the dimensions of the spring, and number of turns of the spring.	10
9	a	Explain the following terms used in compression springs i) Solid length ii) Free length iii) Pitch	08
	b	The spring loaded safety valve for a boiler is required to blow off at a pressure of $1.3MPa$. The diameter of the valve is 65mm and maximum lift of the valve is $17.5mm$. Design a suitable compression spring for the valve, assuming spring index to be 6 and providing initial compression of $30mm$. Take $\tau = 0.45GPa$ and $G = 84GPa$	12
10	a	Discuss the modes of failure of riveted joints with neat sketches.	08
	b	Determine the size of the weld shown if permissible shear stress $\tau = 75MPa$. 	12

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E. Grade Improvement Examinations Nov-2020

Industrial Engineering and Management

STRENGTH OF MATERIALS AND MACHINE DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

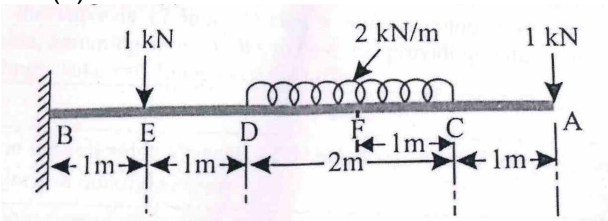
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of design data handbook is permitted.

PART A

1	1.1	The internal resistance which the body offers to meet the load is called _____.	01
	1.2	The ratio of normal stress to volume strain is called _____ modulus of elasticity.	01
	1.3	A _____ beam is one which has more than two supports.	01
	1.4	Bending moment at supports in case of simply supported beams is always _____.	01
	1.5	The strength of the beam mainly depends on _____.	01
	1.6	In case of a circular section the section modulus is given by _____.	01
	1.7	If the bending moment is consistent there will be no _____ stress	01
	1.8	The angle of twist is _____ proportional to the twisting moment	01
	1.9	Shafts are designed on the basis of _____.	01
	1.10	Stress concentration factor is defined as the ratio of _____.	01
	1.11	Failure of a material is called fatigue, when it fails _____.	01
	1.12	Rankine's theory is used for _____.	01
	1.13	When a helical compression spring is subjected to an axial compression load, the stress induced in the wire is _____.	01
	1.14	When helical compression spring is cut into two halves, the stiffness of the resulting spring will be _____.	01
	1.15	The size of a gear is usually specified by _____.	01
	1.16	Lewis equation is applied for checking the _____.	01
	1.17	A screw is specified by its _____.	01
	1.18	A rivet is specified by _____.	01
	1.19	The rivet head used for boiler plate riveting is usually _____.	01
	1.20	In a fusion welding process _____ is used.	01

PART B

2	a	Explain the terms: i) Principal planes ii) Principal stresses.	04

	b	Explain the following beams with sketches i) Over hanging beam ii) Continuous beam	04
	c	Draw the shear force and bending moment diagram for cantilever loaded as shown in the figure 2 (c)  Figure 2 (c)	08
3	a	List out the assumptions made is simple bending theory.	06
	b	Explain about modulus of rupture.	02
	c	A simply supported beam of span $5m$ has a cross section $150mm \times 250mm$. If the permissible stress is $10\ N/mm^2$, find i) Maximum intensity of uniformly distributed load it can carry ii) Maximum concentrated load P applied at $2m$ from one end it can carry.	08
		OR	
4	a	A wooden beam $150mm \times 250mm$ is simply supported over a span of $5m$. When a concentrated load W is placed at a distance 'a' from the left supported the maximum bending in beam is $11.2MPa$ and maximum shear stress is $0.7MPa$. Determine W and 'a'.	08
	b	A shaft is required to transmit $245KW$ power at $240\ rpm$. The maximum torque may be 1.5 times the mean torque. The shear stress in the shaft should not exceed $40MPa$ and the twist 1° per meter length. Determine the diameter required if: i) The shaft is solid ii) The shaft is hollow with external diameter twice the internal diameter. Take modulus of rigidity = $80000\ MPa$.	08
5	a	Derive Soderberg's equation when a member is subjected to fatigue axial load.	06
	b	Determine the maximum stress in the following cases taking stress concentration into account. i) A rectangular plate of $50mm \times 80mm$ with a hole of $10mm$ diameter in the center is loaded in axial tension of $10KN$, Thickness of the plate is $10mm$. ii) A circular shaft of $45mm$ diameter stepped down to $30mm$ diameter having a fillet radius of $6mm$ subjected to a twisting moment of $150Nm$.	10
		OR	

6		A shaft is supported by two bearings placed 1 meter apart. A $500mm$ diameter pulley is mounted at a distance of $200mm$ to the right of left hand bearing and this drives a pulley directly below it with the help of belt having maximum tension of $3000N$. The pulley weighs $1000N$. Another pulley $300\ mm$ diameter is placed $300\ mm$ to the left of right hand bearing is driven with the help of electric motor and the belt which is placed horizontally to the right when viewed from the left bearing. This pulley weighs $500N$. The angle of contact for both the pulleys is 180° and $= 0.24$. Determine suitable diameter for a solid shaft, assuming torque on one pulley is equal to torque on the other pulley. Choose $C15$ steel ($\sigma_y = 234.4\ MPa$, $\sigma_u = 425MPa$) as the shaft material and use $ASME$ code for the design of shaft. Assume minor shock condition.	16
7	a	Explain the following terms used in compression springs i) Solid length ii) Free length iii) Pitch	06
	b	The spring loaded safety valve for a boiler is required to blow off at a pressure of $1.3MPa$. The diameter of the valve is 65mm and maximum lift of the valve is $17.5mm$. Design a suitable compression spring for the valve, assuming spring index to be 6 and providing initial compression of $30mm$. Take $\tau = 0.45GPa$ and $G = 84Gpa$	10
8	a	Explain the following elements of threaded screw terminology: i) Lead ii) Pitch iii) Major and minor diameter.	06
	b	A double riveted chain type lap is to be made between $9mm$ plates. If the safe working stresses in tension crushing and shear are $80\ N/mm^2$, $120\ N/mm^2$ and $60\text{N}/mm^2$ respectively, design the riveted joint.	10

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E. Fast track Examinations Oct-2020

Industrial Engineering and Management

BASICS OF MACHINE DESIGN AND DRAWING

Time: 03 Hours ***Maximum Marks: 100***

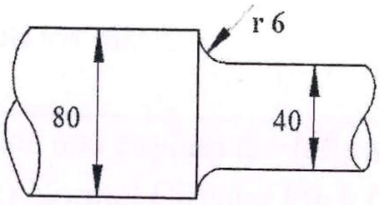
Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of design data handbook permitted.

PART A

1	1.1	Mathematically Factor of Safety (FOS) is given by $FOS =$ _____.	02
	1.2	When a beam is subjected to a load in the transverse direction, it results in _____ stresses on the convex side and _____ stresses on the concave side.	02
	1.3	Name any two design criteria for infinite life of a machine member subjected to fluctuating stresses.	02
	1.4	A shaft is required to transmit $1MW$ at $240rpm$. The shaft must not twist more than $1\,degree$ over a length of 15 diameters. If the modulus of rigidity of shaft material is $80\ KN/mm^2$, find the diameter of the shaft.	02
	1.5	For the same material, length and outside diameter, the weight of a hollow shaft with ratio of diameters 2 is _____ times the weight of a solid shaft.	02
	1.6	The magnitudes of tangential load and radial load on a shaft due to a gear of pitch diameter $400mm$ and pressure angle 20^0 transmitting $200Nm$ are _____ and _____.	02
	1.7	The stresses in full length leaves of a semi elliptic laminated springs are _____ times that of the graduated leaves and the process of equalizing the stresses is called _____.	02
	1.8	The circular pitch and Lewis form factor of a 20^0 FDI spur gear whose pitch circle diameter is $120mm$ and the number of teeth is 30 is _____ and _____.	02
	1.9	The relation between normal circular pitch $'p_n'$ and a transverse pitch $'p'$ of a helical gear is given by _____.	01
	1.10	To obtain leak proof joints, the edges of the plates to be jointed and the rivet heads are forced together by _____ operations.	01
	1.11	Mention any two failure modes of riveted joints.	02

PART B

2	a	A shaft is stepped down from $80mm$ diameter to $40mm$ with a fillet radius of $6mm$. Determine the maximum stress induced in the shaft when subjected to: i) Bending moment of $200Nm$ ii) Twisting moment of $400Nm$. 	06
	b	A hot rolled steel shaft is subjected to a torsional moment that varies from $330Nm$ (clockwise) to $110Nm$ (counterclockwise) as an applied bending moment at the critical section varies from $+440Nm$ to $-220Nm$. The shaft is of uniform cross section and no keyway is present at the critical section. Determine the required shaft diameter. The material has an ultimate tensile strength of $550MPa$ and yield strength of $410MPa$. Factor of safety=1.5	10
3	a	A steel shaft is supported between two bearings $1.5m$ apart. A keyed gear $20\ degree$ involute gear of $175mm$ diameter is located $400mm$ to the left of the right bearing and is driven by a gear directly behind it. A $600mm$ diameter pulley is keyed to the shaft $600mm$ to the right of the left bearing and drives a pulley with a horizontal belt directly behind it. The ratio of tensions of the belt drive is $3:1$. The shaft rotates clockwise when viewed from the right side bearing. The drive transmits $45KW$ at $330rpm$. Assuming that $C_m = C_t = 1.5$, calculate the necessary shaft diameter of the shaft. Use allowable shear stress $40MPa$.	12
	b	Prove that a square key is equally strong in crushing and shearing.	04
		OR	
4	a	Name two types of rigid coupling and two types of flexible coupling.	04
	b	Design a protected type cast iron flange coupling for a steel shaft transmitting $30KW$ at $200\ rpm$. The allowable shear stress in the shaft and key material is $40MPa$. The maximum torque transmitted to be 20% greater than the full load torque. The allowable shear stress in the bolt is $60MPa$ and allowable shear stress in the flange is $25MPa$	12
5	a	With a neat diagram, derive Modified Soderberg's equation for designing a helical spring subjected to fatigue loading.	06
	b	The load on a steel helical compression spring varies from $500N$ to $1200N$. The spring index is 6 and the desired design factor of safety is 1.3. Determine the required wire size by taking the yield shear stress as $690\ MN/m^2$ and the endurance limit stress as $300\ MN/m^2$.	10
		OR	

6	a	A locomotive spring has an overall length if $1100mm$ and a laod of $75KN$ at its center. The spring has 3 full length leaves and 15 graduated leaves with a central band $100mm$ wide. All the leaves are to be stresses to $0.4GPa$ when fully loaded. The ratio of total spring depth to width is 2. Determine: i) Width and thickness of leaves ii) Nip or clearance to be provided iii) Load exerted on the band by the clip bolt after assembly Take $E = 206.8\ GPa$.	08
	b	Derive expressions for bolt load $'Fb'$, nip $'c'$ and theequalized stress $'\sigma'$ in a semi elliptic laminated leaf spring.	08
7	a	With a neat sketch, derive Lewis equation for beam strength of a spur gear tooth. State clearly all the assumptions made.	08
	b	For a helical gear define and explain the following with appropriate sketches. i) Helix angle ii) Normal circular pitch iii) Transverse circular pitch iv) Formative number of teeth.	08
8	a	Explain different stresses that occur due to static loading in threaded fasteners.	06
	b	A double riveted lap joint (chain type) is to be made of $10mm$ plates. Design the riveted joint if the safe working stresses in tearing of plates, shearing and crushing are $124MPa, 93MPa$ & $165MPa$ respectively.	10

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RV COLLEGE OF ENGINEERING®

(Autonomous Institution affiliated to VTU)

IV Semester Fast Track Examinations July-19

Industrial Engineering and Management

BASICS OF MACHINE DESIGN AND DRAWING

Time: 03 Hours ***Maximum Marks: 100***

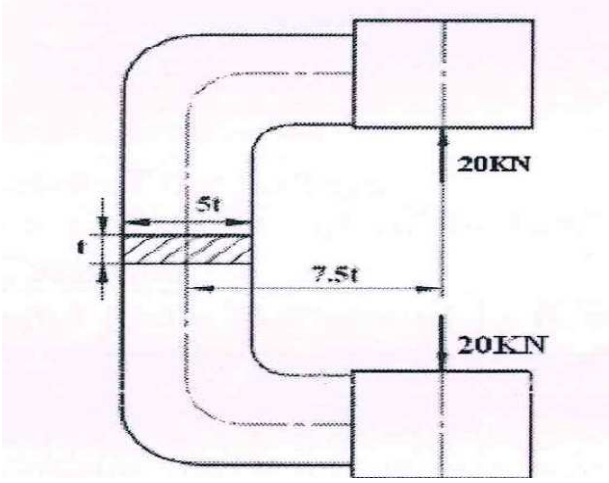
Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of design data handbook permitted.

PART A

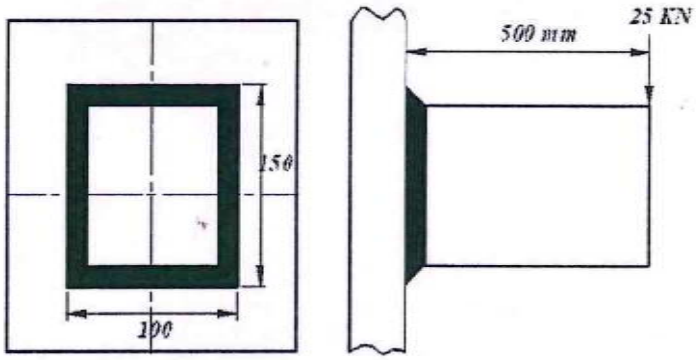
1	1.1	A steel rod of circular section and length 200 mm is subjected to a compressive load of 40 KN. If the safe stress in rod material is 80 MPa, Take the modulus of elasticity of the material as 207 GPa. The diameter of the rod is _____	02
	1.2	A short cantilever beam is of width 20 mm and depth 40 mm supports a transverse load of 2 KN at a distance of 200 mm from its fixed end. Maximum bending stress in the beam is _____MPa.	02
	1.3	Couplings can be classified as _____and _____	02
	1.4	Compute Torsion applied on a shaft which is used to Transmit power of 30KW and running at a speed of 600 rpm.	02
	1.5	Two commonly used profiles of gear teeth are the _____&_____	02
	1.6	If d=Major diameter of bolt in mm, and d_1=Minor diameter of bolt in mm, Pitch diameter of bolt d_2= Tensile stress area A_c=	02
	1.7	In Helical gears Helix Angle(β) is defined as _____ and its value ranges from _____-	02
	1.8	Structural rivets have diameter in range of _____and small rivets have diameter in range of _____	02
	1.9	Stiffness of material is defined as _____	01
	1.10	High carbon Steels have _____% of Carbon	01
	1.11	In cyclic loading, the effect of stress concentration is more serious in case of _____	01
	1.12	The maximum stress concentration factor for a rectangular plate with a transverse hole loaded in tension or compression is _____	01

PART B

2	a	A C frame is subjected to a force of 20 KN as shown in fig. It is made of grey cast iron FG 300 and the factor of safety is 2.5. Determine the dimensions of the cross section of the frame.  	10
	b	Explain Rankine's theory and Coulomb's theory.	06

3	a	A centrifugal pump for a steam turbine condenser is to be connected to the motor through a short shaft & a pair of flexible couplings. The pump delivers 30000 liters of water at 900 rpm against a head of 8m. the pump efficiency is 80%. Assuming a permissible shear stress of 50 MPa, determine the diameter of the shaft.	08
	b	A rectangular key 14 x 10 x 60 mm is used to transmit 15 KW at 1200 rpm. Taking the permissible stresses as 56 MPa in shear and 135 MPa in crushing, check whether the length of key is sufficient or not.	08
		OR	
4	a	In a flange coupling used to connect two coaxial shafts of diameter 80 mm to transmit 60 KW at 200rpm, 6 bolts of M 14x1.5 on a bolt circle diameter of 210 mm are used. If the flange thickness is 20 mm and the hub diameter is 145 mm, determine; i)Shear stress induced in the shaft ii)Shear stress induced in the bolt iii)Shear stress induced in the flange.	08
	b	Design a rectangular key to transmit 30 KW at 1200 rpm for a motor shaft. Assume both shaft & key to be made of C40 steel and select a factor of safety 2.5 based on yield strength.	08

5	a	Design a helical compression spring to support an axial load of 3000 N, the deflection under the load being limited to 60mm. The spring index is 6. The spring is to be made of chrome-vanadium steel & a factor of safety is 2.	08
	b	Design a spring for an elevator shaft at the bottom of which 8 identical springs are set in parallel to absorb the shock of the elevator in case of failure. The weight of elevator is 60 KN & the counter weight is 20 KN. The elevator has a free fall of 1.5 m from rest. The spring is made of 25 mm diameter rod. Determine the maximum stress in each spring, if the spring index is 6 & each spring has 15 active turns. Take G=84GPa.	08

		OR	
6		In an open belt drive system smaller pulley is of diameter d and runs at 1440 rpm. Larger pulley is of diameter 'D'and runs at 480 rpm. The center distance between the pulleys is twice the diameter of the bigger pulley. The belt operates at a velocity of 20 m/s approximately and the stress in the belt should not exceed 2.25 N/mm^2. The density of the belt is 0.95 gm/cc and the coefficient of friction is 0.35. the thickness of the belt is 5 mm. Calculate: i) The diameter of the pulleys ii) The Length and the Width of the belt and iii) The belt Tensions.	16
7		Design a pair of spur gears to transmit 12 KW at 1000 rpm of pinion with a velocity ratio 2.5. The design should be as compact as possible.	16
8	a	A generator weighing 60 KN is to be provided with an eye bolt for lifting purposes. The nominal diameter of eye bolt is 30 mm and pitch is 3.5 mm. The length of engaged portion of screw is 45 mm. Determine the direct tensile stress & shear stress in the threaded portion of the bolt.	08
	b	Determine the size of the weld for the joint shown in fig taking the allowable shear stress as 80MPa. 	08

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RV COLLEGE OF ENGINEERING®

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IV Semester B. E. Examinations April/May-19

Industrial Engineering and Management

BASICS OF MACHINE DESIGN AND DRAWING

Time: 03 Hours ***Maximum Marks: 100***

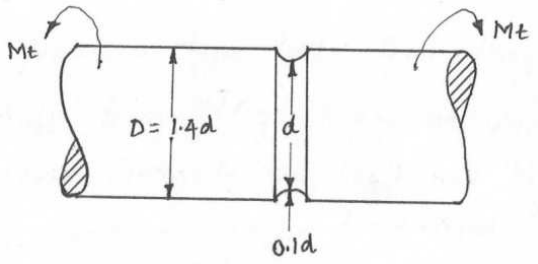
Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of design data handbook permitted.

PART A

1	1.1	Polar moment of inertia "J" for a solid circular shaft is given by _____.	01
	1.2	An SAE 1045 annealed Carbon steel ($\sigma_y = 309.9MPa$) is subjected to tensile load of $60KN$. Calculate the diameter of the rod using factor of safety = 3.	02
	1.3	Write down the expression for modified Soderberg's equation for bending.	01
	1.4	List out four properties of shaft material.	02
	1.5	Keys are mainly classified into _____ and _____.	01
	1.6	_____ coupling is used to connect two shafts having intersecting axes.	01
	1.7	In helical springs, the Wahl's factor takes into account the effect of _____ and _____.	01
	1.8	The stress in full length leaves is _____ greater than the stress in graduated leaves.	01
	1.9	What are the advantages of 20° pressure angle over $14^{1/2}$° pressure angle in the design of gears?	02
	1.10	The Lewis equation for tangential tooth load in terms of circular pitch is _____.	01
	1.11	In wear strength equation of gear tooth the ratio factor Q is given by _____.	01
	1.12	Write down two advantages of Threaded Fasteners.	02
	1.13	_____and _____ are operations done to prevent leakages in Riveted Joints used in Boilers.	02
	1.14	_____ type of Rivet head is used in ship building.	01
	1.15	Write down one advantage of Welded Joint over Riveted Join.	01

PART B

2	a	A grooved shaft as shown in the fig 2a is to transmit $5KW$ at $120rpm$. Determine the diameter of the shaft at the groove if it is made of $C15$ steel ($\sigma_y = 235.4\,MPa$), FOS is 2.  Fig 2a	08
	b	A piston rod is subjected to a maximum axial load of $110KN$. It is made of steel having an ultimate stress of $900\,N/mm^2$ and the surface is machined. The average endurance limit is 50% of the ultimate strength. Take the size correction factor as 0.85 and factor of safety = 1.75. Determine diameter of the rod.	08
3		A solid steel shaft running at $600rpm$ is to support on bearing $600\,mm$ apart. The shaft receiver $40KW$ through a $400mm$ diameter pulley weighing $400N$ located $300mm$ to the right of left bearing by a vertical flat belt drive. The power is transmitted from a shaft through another pulley of diameter $600mm$ weighing $600N$ located $200mm$ to the right of right bearing. The belt drives are at right angles to each other and ratio of belt tension is 3. Determine the size of the shaft necessary if the allowable shear stress in the shaft material is $40MPa$ and loads are steady.	16
		OR	
4	a	Differentiate between a rigid coupling and a flexible coupling.	04
	b	Design a protected type cast iron flange coupling for a steel shaft transmitting $30KW$ at $200rpm$, The allowable shear stress in the shaft and key material is $40MPa$. The maximum torque transmitted to be 20% greater than the full load torque. The allowable shear stress in the bolt is $60MPa$ and the allowable shear stress in the flange is $40MPa$.	12
5	a	With a neat sketch derive an expression for stresses in helical springs of circular wire.	06
	b	A loaded narrow guage Car weighs $18KN$ and moving with a velocity of $80\,m/min$ is brought to rest by a buffer consisting of two helical springs. In bringing the car to rest the spring undergoes a compression of $200mm$. The allowable shear stress is $0.3GPa$ and the spring index is 8. Design a suitable spring with $G = 84GPa$.	10
		OR	

6	a	What is equalized stress [Nipping] in spring leaves? Explain clearly.	06
	b	A locomotive spring has an overall length of $1100mm$ and sustain a load of $75KN$ at its centre. The spring has 3 full length leaves and 15 graduated leaves with a central band of $100mm$. All the leaves are to be stressed at $0.4GPa$ when fully loaded. The ratio of total spring depth to width is 2. Determine: i) Width and thickness of leaves. ii) Initial space that must be provided between full length and graduated leaves before the band is applied iii) What load is exerted on the band after the spring is assembled. Take $E = 206.8GPa$.	08
	c	Write down few merits and Demerits of a flat belt drive.	02
7		Design a pair of spur gear to transmit $20KW$ from a shaft rotating at $1000\ rpm$ to a parallel shaft which is to rotate at $310rpm$. Assume number of teeths on Pinion 31 and 20° full depth tooth form. The material for Pinion $C40$ steel ($\sigma = 206.81MPa$) untreated and for gear cast steel $20\%\ C$ untreated ($\sigma = 137.34MPa$).	16
8	a	With neat sketch explain different failure modes of Riveted Joints.	06
	b	Design a double riveted butt joint with two cover plates for a longitudinal seam of a boiler shell $1.5m$ in diameter subjected to a steam pressure $0.95\ N/mm^2$ Assume an efficiency of 75%. Allowable tensile stress in the plate of $90\ N/mm^2$, allowable compressive stress of $140N/mm^2$ and an allowable shear stress in the rivet of $56N/mm^2$.	10

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Fast Track ExaminationsJuly-18

Industrial Engineering and Management

BASICS OF MACHINE DESIGN AND DRAWING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

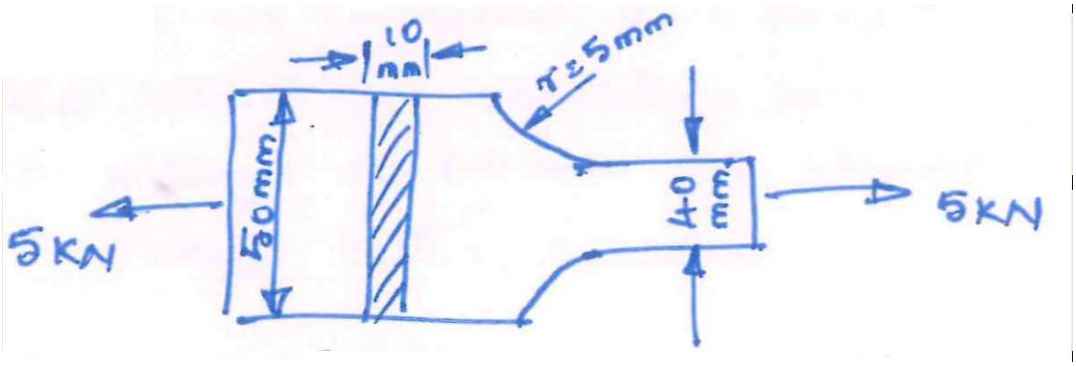
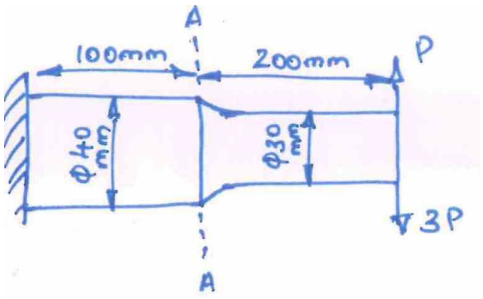
4. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
5. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Max. Normal stress theory is used for _____materials.	01
	1.2	Stress concentration factor is defined as the ratio of _____.	01
	1.3	The resistance of fatigue of a material is measured by _____.	01
	1.4	The stress which vary from one value of compressive to the same value of tensile or vice versa, are known as _____ stresses.	01
	1.5	Two shafts A and B are made of the same material. The diameter of the shaft A is twice as that of shaft B. The power transmitted by the shaft A will be ____ times of shaft B.	01
	1.6	When a pulley or other mating piece is required to slide along the shaft _____ sunk key is used.	01
	1.7	Oldham coupling is used to connect two shafts which have _____ misalignment.	01
	1.8	_____ spring is used in a mechanical wrist watch.	01
	1.9	In leaf springs, the longest leaf is known as _____.	01
	1.10	The stresses produced in a belt are _____type.	01
	1.11	A shaft rotating at $200rpm$ having allowable shear stress as $42Mpa$ is to transmit $20kW$.Determine the diameter.	02
	1.12	For a spur gear system, the distance between the centres are 660mm, the ratio of pitch circle diameter of gear to pinion is 10, the module is _____.	02
	1.13	For a given helical gear pair the number of teeth of gear are 24 with 20° full depth system. Find the form factor.	02
	1.14	A bolt of $M24X2$ means _____.	02
	1.15	The no. of rivets per pitch length for a double riveted lap joint with zigzag riveting and triple riveted lap joint with zigzag riveting are _____ and _____ respectively.	02

PART-B

2	a	State Maximum normal stress theory and Maximum shear stress theory.	04

	b	A filleted flat plate as shown in figure 2(b) is subjected to a pull of $5kN$. Determine the maximum stress induced in the member taking stress concentration into account.  fig 2(b)	04
	c	A cantilever beam shown in figure 2 (c) is subjected to load varying from P to 3P. Determine the value of P. The material of the beam has $\sigma_u = 620.8MPa, \sigma_y = 400.1MPa$ and $\sigma_{-1} = 345.2MPa$. The stress concentration factor may be taken as 1.4 . Analyze the member at the change of c/s A-A. Use Fos=3.  Fig. 2 (c)	08
3	a	A commercial steel shaft =1m long supported between bearings has a pulley of diameter $600mm$ weighing $1kN$ driven by a horizontal belt keyed to the shaft at a distance of 400mm to the left of he right bearing and receives $25kW$ at $1000rpm$. Power from the shaft is transmitted from a 20°spur pinion of pitch circle diameter $200mm$ which is mounted =200mm to the right of the left bearing to a gear such that the tangential force on the gear acts vertically upwards takes the rate of belt tension=3, determine the size of shaft based on maximum shear stress theory.	16
		OR	
4	a	With a neat sketch explain the purpose of using a Gib head key [also mention the properties].	06

	b	Two $35mm$shafts are connected by a flanged coupling. The flanges are fitted with 6 bolts on 125mm bolt curve. The shafts transmit a torque of 800Nm at 350rpm. For safe stresses mentioned below. Calculate i. Diameter of both, ii. Thickness of flanges, iii. Key dimensions, iv. Hub length, v. Power transmitted. $\tau_{all}\ for\ shaft = 63MPa$ $\tau_{all}\ for\ bolt = 56MPa$ $\tau_{all}\ for\ C.J\ Coupling = 10MPa$ $\tau_{all}\ for\ key = 46MPa$	10
5	a	A loaded narrow gauge car weighs $18kN$ and moving at a velocity of $80m/min$ is brought to rest by a buffer consisting of two helical springs. In bringing the car to rest the spring undergoes a compression of 200mm. The allowable shear stress is $0.3GPa$ and the spring index is 8. Design suitable spring. Take $G = 84GPa$.	16
		OR	
6	a	For a flat belt drive the following data is given: $Speed\ of\ motor = 1500rpm,$ $Speed\ of\ driven\ pulley = 300rpm$ $Power\ to\ be\ transmitted = 10kW$ $weight\ density\ of\ leather = 10 \times 10^{-6} N/mm^3$ $smaller\ pulley\ diameter\ to\ thickness\ of\ belt\ ratio = 20$ $center\ distance = 3m$ $allowable\ stress\ in\ the\ belt = 2.5MPa$ $velocity\ of\ belt\ drive = 15\ m\ /sec$ $load\ factor = 1.2$ $coefficient\ of\ friction = 0.3$ Design a suitable cross belt drive.	16
7	a	Design a pair of spar gears to transmit a power of $18kW$ from a shaft running at $1000rpm$ to a parallel shaft to be run at $250rpm$ maintaining a distance of $160mm$ between the shaft centers. Suggest suitable surface hardness for the gear pair.	16
8	a	Design a double riveted lap joints with chain riveting for MS plates 20mm thick taking allowable values of stresses in shear, tension and compression equal to 60,90 and $120MPa$ respectively.	10
	b	A MS plate 15mm thickness is welded to another plate by two parallel welds to carry a load of $50kN$. Determine the length of weld required when: i. Load is static, ii. Load is dynamic.	06

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Examinations April/May-18

Industrial Engineering and Management

BASICS OF MACHINE DESIGN AND DRAWING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of design data handbook permitted.

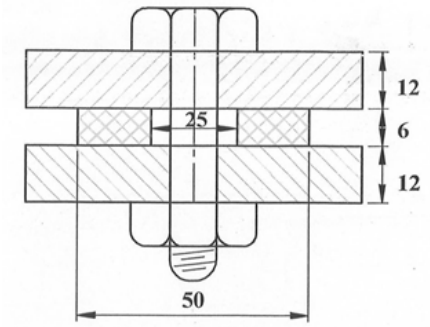
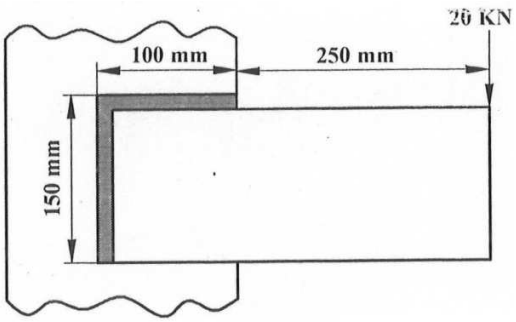
PART A

1	1.1	The type of design based on practice and past experience is known as _____.	01
	1.2	Materials which can undergo only small plastic deformation before fracture are known as _____ materials.	01
	1.3	The designation of steel $C-40$ indicates that the percentage of carbon in steel is _____.	01
	1.4	The section modulus of a square cross section of a beam of side $30mm$ is _____ mm^3.	01
	1.5	The theory of failure widely used for design of transmission shafting is _____ theory.	01
	1.6	The maximum value of the theoretical stress concentration factor for a plate with a circular hole subjected to axial load is _____	01
	1.7	When the diameter of a shaft is doubled, its torque carrying capacity will increase by _____ times.	01
	1.8	If an open belt is used to connect two pulleys of equal size, the angle of contact is _____.	01
	1.9	If the stress amplitude in a machine member subjected to repeated loading is 50 MPa, the maximum stress is _____.	01
	1.10	As per $ASME$ code, the permissible shear stress for commercial shafting with keyway is _____ MPa.	01
	1.11	The surface finish factor of a highly polished component is _____.	01
	1.12	A screw is said to be self-locking if its efficiency is _____ than 50%.	01
	1.13	The ratio of mean coil diameter to wire diameter in a spring is known as _____.	01
	1.14	Leaf springs are subjected to _____ stress.	01
	1.15	Transverse fillet welds are designed to support _____ loads.	01
	1.16	A rivet is specified by its _____ diameter.	01
	1.17	The distance between the edge of the plate and the center line of rivets in the nearest row is called _____.	01
	1.18	The cross section of a key which is equally strong in crushing and shearing is _____.	01

	1.19	The ratio of pitch circle diameter in mm of a gear to the number of teeth is called _____.	01
	1.20	Two helical gears mounted side by side with opposing hands of helix are called _____.	01

PART B

2	a	A shaft is stepped down from $80mm$ diameter to $40mm$ diameter with a fillet radius of $6mm$. Determine the maximum stress induced in the shaft taking stress concentration into account if the shaft is subjected to: i) A bending moment of $200\ Nm$ ii) A twisting moment if $300\ Nm$.	08
	b	a machine member made of carbon steel having a yield stress of $328.6\ MPa$ is loaded such that $\sigma_x = 60\ MPa, \sigma_y = -20\ MPa, \tau_{xy} = 30\ MPa$. Determine the factor of safety by using the following theories of failure: i) Rankine's theory ii) Tresca's theory iii) Von Mises theory.	08
3		A commercial steel shaft is supported on bearings $1m$ between centers. A cast iron pulley of $0.6m$ diameter weighing $1kN$ is located $0.3m$ to the right of the right bearing and receives $25\ kW$ power at $1000\ rpm$ from a motor pulley by horizontal belt drive directly behind it. The ratio of belt tensions is 3. A 20°spur pinion of pitch circle diameter $0.2m$ weighing $200N$ is located $0.2m$ to the left of the left bearing. The pinion delivers power to another gear mounted directly behind it such that the tangential force on the pinion acts vertically upwards. Assuming minor shock loads on the shaft, determine the necessary diameter of the shaft if the allowable shear stress is limited to $60\ MPa$.	16
		OR	
4	a	In a flange coupling used to connect two coaxial shafts of diameter $80mm$ to transmit $60kW$ at $200\ rpm, 6\ bolts$ of $M14 \times 1.5$ on a bolt circle diameter of $210\ mm$ are used. If the flange thickness is $20\ mm$ and the hub diameter is $145\ mm$, determine: i) Shear stress induced in the shaft ii) Shear stress induced in the bolt iii) Shear stress induced in the flange.	10
	b	A rectangular key $14 \times 10 \times 60\ mm$ is used to transmit $15\ kW$ at $1200\ rpm$. Taking the permissible stresses at $56\ MPa$ in shear and $135\ MPa$ in crushing, check whether the length of key is sufficient or not.	06
5	a	With usual notations, derive expressions for shear stress induced and deflection in a helical spring.	08
	b	A multi leaf spring with camber is fitted to the chassis of an automobile over a span of $1.2\ m$ to absorb shocks due to maximum load of $20kN$. The spring material can sustain a maximum stress of $0.4\ GPa$. The spring has 2 full length leaves out of 8 leaves. The leaves are assembled with bolts with a central band of width $150\ mm$. Design the spring for a maximum deflection of $50\ mm$. Take $E = 207\ GPa$.	08
		OR	

6	a	A flat belt $100\,mm$ wide and $10\,mm$ thick is transmitting power at $1000\,m/min$. The net driving tension is 1.8 times the tension on the slack side. If the stress in the belt is $2\,MPa$ and the density of leather as $1000\,kg/m^3$, calculate: i) The maximum power that can be transmitted at this speed ii) The absolute maximum power that can be transmitted by the belt and the speed at which it can be transmitted iii) Percentage increase in power.	10
	b	Mention the advantages of chain drives over belt drives. List the types of chains.	06
7		Design a pair of spur gears to transmit $9\,kW\ at\ 2000\,rpm$ of cast steel (untreated) pinion. The gear is made of high grade cast iron (heat treated) and is to rotate at $500\,rpm$. The pinion has 20 teeth of $14\frac{1}{2}°$ involute form.	16
8	a	A steel bolt of size $M\,20 \times 2.5$ is used to connect two plates as shown in figure 8a. A copper gasket is placed in between the plates. Determine the resultant load on the bolt if it is subjected to an external load of $10\,kN$. Take E for steel bolt as $206\,GPa$ and for copper as $121\,GPa$.  Fig 8a	06
	b	Determine the size of the weld shown if permissible shear stress $\tau = 75\,MPa$.  	10

Work Systems Design

IV Semester SEE Question Papers

Course codes: 16IM45

- Fast Track Oct 2020 (16IM45)
- Fast Track Jul 2019 (16IM45)
- Apr/May 2019 (16IM45)
- Fast Track Jul 2018 (16IM45)
- Apr/May 2018 (16IM45)

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E. Fast Track Examinations Oct-2020

Industrial Engineering and Management

DESIGN OF WORK SYSTEMS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Productivity is defined by _____.	01
	1.2	Identify the five symbols commonly used in process charts.	02
	1.3	Distinguish between Flow Process Chart (FPC) and Flow Diagram.	02
	1.4	What are the two techniques of work study and what is their relationship?	02
	1.5	What are the categories under principles of motion economy?	02
	1.6	What is meant by work sampling?	02
	1.7	An element which does not occur in every cycle of job is known as _____.	01
	1.8	Expand the acronyms $MOST$ and MTM.	02
	1.9	What is takt time?	02
	1.10	What is performance rating?	02
	1.11	Identify the two pillars of Toyota Production System.	02

PART-B

2	a	Distinguish between increase in production and increase in productivity with relevant examples.	08
	b	Elaborate on the different factors which increase the basic work content.	08
3	a	What is method study? Explain the steps "Select" and "Record" involved in conducting method study.	08
	b	A surface grinding operation comprises of the following elements.	

Load and start the machine	04 *min*
First machining operation	08 *min*
Reset table for next operation	02 *min*
Second machining operation	10 *min*
Unload machines	04 *min*

Assuming that the nature of operation allows the operator to attend two machines. Construct a multiple activity chart. Determine cycle time and the percentage utilization of man and machines. 08

OR

4	a	Identify the various types of flow process charts. Mention an area of application for each of the type of Flow Process Charts.	06
	b	Construct a two handed process chart for assembly of washer with a nut and bolt arrangement.	10
5	a	An initial work sampling study showed that a machine was idle 75 times out of 300 observations. If 95% confidence level and ∓5% accuracy is assumed, calculate the number of observations necessary.	06
	b	What is work measurement? Provide a brief note on various techniques of work measurement.	10
		OR	
6	a	Discuss the different elements identified in time study with examples.	08
	b	The following information were collected during the work sampling study: Total time of study = 8*hours* Number of pieces produced during the period= 420 The % of working during study = 85% The average performance index noted = 110% Personal and other allowances = 15% Determine the standard time per piece.	08
7	a	"Various allowances are considered in building up the standard time of a job". What are these allowances? Provide a brief note on these allowances.	08
	b	The following readings were observed for 2 *cycles* using continuous timing method.	

		Cycle I		*Cycle II*	
Sl. No	*Description of element*	Rating (%)	*Watch reading (cms)*	*Rating* (%)	*Watch reading (cms)*
1	*Pick up casting or positions it*	100	0.36	90	3.05
2	*Drills the first*	110	0.86	100	3.60
3	*Moves the casting*	110	1.06	100	3.82
4	*Drills the casting*	100	1.61	110	4.32
5	*Moves the casting*	100	1.83	110	4.52
6	*Drills the third hole*	100	2.38	110	5.02
7	*Removes the casting and keeps it aside*	100	2.65	120	5.22

		Assuming an allowance of 20%, calculate the standard time.	08
8	a	Elaborate on the seven forms of waste in production as identified by Taiichi Ohno.	10
	b	Differentiate between push and pull production system.	06

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RV COLLEGE OF ENGINEERING®

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IV Semester B. E. Fast Track Examinations July-19

Industrial Engineering and Management

DESIGN OF WORK SYSTEMS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART A

1	1.1	Define Productivity.	02
	1.2	What are the objectives and scope of work study?	02
	1.3	Distinguish between flow process chart and flow diagram.	02
	1.4	What are THERBLIGS? List any two therbligs with appropriate symbol.	02
	1.5	Distinguish between cycle graph and chronocyclegraph.	02
	1.6	Define a qualified worker.	02
	1.7	Distinguish between cumulative and fly-back timing.	02
	1.8	Give reasons for providing allowances in computing standard time for a job.	02
	1.9	Define lean manufacturing.	02
	1.10	List the principles of lean manufacturing.	02

PART B

2	a	Define industrial engineering as given by American Institute of Industrial Engineering. What are the functions of industrial engineering?	08
	b	Discuss the reaction of workers, supervisors and management towards the application of work study in their organization.	08

3	a	Explain the steps ”Select” and “Record” in conducting method study.	06
	b	What is critical examination? Explain how the critical examination is carried out?	10
		OR	
4	a	What are the principles of Motion Economy? List the principle of motion economy w.r.t use of human body.	08

	b	In a continuous production process, the following is the sequence of operations of a man-machine interface for one cycle. i. Draw a man-machine chart and find the cycle time and utilization of man and machine. ii. Apply method study principles and arrive at an improved method for the operation.	08

Sl No	Elements	Time(min)
1	Casting picked up and brought to the machine	0.09
2	Job positioned and fastened to machine.	0.12
3	Machine stated	0.04
4	Feed changed and machining begun	0.07
5	Casting is machined automatically.	0.35
6	Machine stopped	0.04
7	Job unfastened and removed	0.10
8	Casting taken back and deposited	0.09

5	a	Explain briefly the steps to be followed in carrying out stopwatch time study.	08
	b	What are the reasons for breaking down the operations into elements? State the rules.	08
		OR	
6	a	What is work sampling? Describe the procedure followed in conducting a work sampling study.	08
	b	The following readings were observed for 2 cycles using continuous timing method. Calculate the standard time for the above work assuming a fatigue allowance of 8% for 1 and 7 and 6% for 3 and 5. the personal needs allowance for all elements is 7%.	08

Sl No	Description of element	Cycle-I		Cycle-II	
		Rating (%)	Watch reading(min)	Rating (%)	Watch reading(min)
1	Picks up the casting and positions it	100	0.36	90	3.05
2	Drills the first hole	110	0.86	100	3.60
3	Moves the casting	110	1.06	100	3.82
4	Drills the second hole	100	1.61	110	4.32
5	Moves the casting	100	1.83	110	4.52
6	Drills the third hole	100	2.38	110	5.02
7	Removes the casting and keeps it aside.	100	2.65	120	5.22

7	a	Give a brief account of various allowances used in establishing standard time.	08
	b	Discuss the specific applications of standard data and MTM in standard time determination.	08

8	a	List lean tools and explain any two of them.	08
	b	What is value stream mapping? How is it different from charts and diagrams used in work study?	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B. E. Examinations April/May-19

Industrial Engineering and Management

DESIGN OF WORK SYSTEMS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART A

1	1.1	If a potter has been producing 30 plates an hour and improved methods of work enable him to produce 40 plates an hour, the productivity of that man has increased by how much?	02
	1.2	What are the two techniques of work study and what is their relationship to one another?	02
	1.3	Write the five symbols commonly used in process charts.	02
	1.4	What are *SIMO* charts? When are they used?	02
	1.5	What do you mean by performance rating?	02
	1.6	Differentiate between snapback timing and continuous timing.	02
	1.7	What is meant by work sampling?	02
	1.8	Expand the acronyms *MOST* and *MTM*.	02
	1.9	Name the two pillars of the Toyota production system.	02
	1.10	What is takt time?	02

PART B

2	a	What is meant by work study? List and explain the basic procedure of work study.	08
	b	During the base year in a small steel mill 326000 tons of steel were produced using 203000 labour hours. In the next year, the output was 341,000 tons using 246000 labour hours. Determine: i) The labour productivity ratio for the base year ii) The labour productivity ratio for the second year iii) The labour productivity index for the second year.	08
3	a	List the various types of flow process charts. Mention an area of application for each of the type of *FPC's*.	06
	b	Enumerate the principles of motion economy in respect of "Rules concerning use of human body".	10
		OR	

4	a	Explain briefly the steps involved in the conduction of method study.	08
	b	Identify the 17 Therbligs with its symbols used in a micro motion study.	08
5	a	A total of 500 observations were made at random times during a one week period(a total of 40hrs) in the automatic lathe section. The subjects consists of 10 identical machines. In each observation, the activity of the machine was identified and classified into one of the three categories: i) Being set up ii) Running production iii) Idle. The number of observations in each category is summarized in the table. Estimate how many hours/week an avg m/c spent in each of the three categories. $category$ | $Number\ of\ observations$ 1. $Being\ setup$ | 75 2. $Running\ production$ | 300 3. $M/c\ Idle$ | $\frac{125}{500}$ And determine the 95% confidence interval for the proportion of time spent setting up the machine.	08
	b	Discuss the various reasons for breaking a task into element.	08
		OR	
6	a	A direct time study was taken on a manual work cycle using the snap back timing method. The regular work cycle consisted of 3 elements identified as a, b and c in the following table. Element d is an irregular element performed every 5 cycles. Observed times and performance ratings of the elements are also given. Determine : i) The normal time ii) The standard time for the work cycle using an allowance factor of 15%. $Work\ element$ | a | b | c | d $Observed\ time$ | $0.56min$ | $0.25min$ | $0.5min$ | $1.10min$ $Performance\ rating$ | 100% | 80% | 110% | 100%	08
	b	Identify at least eight advantages of work sampling over time study.	08
7	a	Enumerate the steps necessary to use MTM technique.	08
	b	Differentiate between general move and controlled move in $MOST$.	08
8	a	Name and briefly explain the seven forms of waste in production as identified by Taiichi Ohno.	08
	b	Differentiate between push system and pull system in production control.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Fast Track ExaminationsJuly-18

Industrial Engineering and Management

DESIGN OF WORK SYSTEMS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

11. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
12. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Productivity is the ratio of _____.	01
	1.2	The prime objective of lean manufacturing practice is _____.	01
	1.3	The real founder of motion study was _____.	01
	1.4	The work study is done by means of _____.	01
	1.5	The time taken by a trained worker to perform an operation, while working a steady pace is known as _____.	01
	1.6	Chart used in work study is _____.	01
	1.7	Write the arrangement in the sequence of operation for simplification in a motion study (Method study).	01
	1.8	A diagram showing the path followed by men and materials while performing a task is known as _____.	01
	1.9	Gantt chart gives information about _____.	01
	1.10	Work study is most useful where _____.	01
	1.11	The symbol ◯ used to represent _____.	01
	1.12	Study used to find a simple, easier and better way of performing a job is called _____.	01
	1.13	Average time recorded by work study man for an operation is called ____.	01
	1.14	Define a qualified worker.	02
	1.15	Define RATING FACTOR.	02
	1.16	What is work simplification?	01
	1.17	What are the types of stop watches?	01
	1.18	Expand PMTS.	01

PART B

2	a	Define Industrial engineering as given by American Institute of Industrial Engineering. What are the functions of Industrial Engineering?	10
	b	Discuss the factors affecting productivity.	06

3	a	Explain the steps “select” and “record” in conductivity method study.	08
	b	List the principles of motion economy w.r.t arrangement of facilities within the workplace.	08
		OR	
4	a	Explain the step “critical examination” of method study.	08
	b	List the tools used to record activities on a factory floor and elaborate.	08
5	a	What is work sampling? Describe the procedure followed in conducting work sampling study.	08
	b	Check whether the number of observations is sufficient for ±5% accuracy and 95% confidence limit, if Number of observations of machine working=3000 Number of observations of machine idle=1000 Total observations=4000	08
		OR	
6	a	Explain briefly the steps to be followed in carrying out stop watch time study.	08
	b	What are the reasons for braking down the operations into elements? State the rules.	08
7	a	Explain any two of the type of allowances with examples.	08
	b	In a time study for job by a worker whose rating is 90, and data obtained as follows: observed time=15 minutes Personal needs allowance=4% of basic time Fatigue allowance=2.5% of basic time Contingency work allowance=2% basic time contingency delay allowance=1% of basic time Find: i) Basic time. ii) Work content. iii) Standard time.	08
8	a	Discuss the core concepts of Lean manufacturing systems with one simple area of application for lean principles.	08
	b	List any four symbols used during the construction of a value stream Map and give the interpretation of these symbols.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Examinations April/May-18

Industrial Engineering and Management

DESIGN OF WORK SYSTEMS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

9. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
10. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART A

1	1.1	List the three basic ways to increase productivity.	02
	1.2	Identify the appropriate methodstudy symbol used to represent following activity: "Physical count of incoming material to tally quantity actually received with the quantity mentioned on the delivery challan"	02
	1.3	Differentiate between "methodstudy" and "work measurement".	02
	1.4	Distinguish between "flow Diagram and "flow chart".	02
	1.5	List the different techniques of work measurement.	02
	1.6	What is meant by the terms 'Element' and 'Breakpoint' in Time study?	02
	1.7	Switching on and off of the machine is a constant element. Justify.	02
	1.8	Expand the acronym '*MOST*'.	02
	1.9	What is meant by Lean Manufacturing?	02
	1.10	List the methods of measuring 'Interference Allowance'.	02

PART B

2	a	Discuss management techniques to reduce ineffective time due to (a) product and (b) process.	10
	b	How does workstudy help to improve productivity of an organization?	06
3	a	Briefly describe different recording aids used in method study.	10
	b	"Critical examination forms the basis for methods improvement" Justify.	06
		OR	
4	a	Enumerate the principles of motion economy in respect of: i) Arrangement of workplace and ii) Design of tools and equipment.	10
	b	"Human factor plays an important role in the successful implementation of proposed changes" Discuss.	06
5	a	"The task to be studied in time study in broken into small tiny details called elements". Why is it done? What rules govern such a break up?	10

	b	What are the two most commonly used timing methods? What factors influence their choice?	06
		OR	
6	a	Explain the different methods of performance rating used/followed in Time Study?	10
	b	The end results of a work sampling study are as follows: - Observations of machine working = 3500 - Observations of machine idle = 1500 - Total observations = 5000 i) Calculate the limit of accuracy. ii) If the accuracy desired is $\mp 5\%$ have sufficient observations been made?	06
7	a	Turning of gear blanks on centre lathe involved the following five elements and related stop watch data. Assuming rest and personal allowances as 13% and the contingencies as 2%, find the standard time.	

Element No.	*Description*	*Observations*				
		1	2	3	4	5
1	*Pickup part and secure*	0.20	1.46	5.22	6.49 (b)	14.25
2	*Start machine and approach tool*	0.30	1.55	5.30	13.10	14.35
3	*Turn diameter*	1.05	2.31	6.05	13.84	15.10
4	*Withdraw tool and stop machine*	1.13	2.38	6.14	13.92	15.17
5	*Release part and keep it aside*	1.28	2.54(a)	6.29	14.06	15.32

Foreign elements are:
2.54 to 5.02 – Talking to another operator
6.49 to 12.98 – Away for personal work
Estimated rating factors:
i) 90% for element 1
ii) 110% for elements 2 and 4
iii) 100% for element 3 since auto cycle
iv) 95% for element 5 — 10

	b	Give a brief note on MTM.	06
8	a	Do time and motion studies have a place in modern lean environment? Give your view point along with suitable justification.	10
	b	Review the guiding principles of lean manufacturing.	06

Applied Ergonomics

IV Semester SEE Question Papers

Course codes: 12IM46

- Fast Track Jul 2018 (12IM46)
- Apr/May 2017 (12IM46)
- Fast Track Jul 2016 (12IM46)
- Apr/May 2016 (12IM46)

12IM46

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VISemester B. E. ExaminationsFast Track July-18

Industrial Engineering and Management

APPLIED ERGONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART A

1	1.1	The two key parameters for design under the principles of applied antropometry are _____ and _____.	01
	1.2	The human body posture fully free of stress is called as _____.	01
	1.3	Which organ in human body detects the muscle tension?	01
	1.4	What is proprioceptive feedback?	01
	1.5	The thoracic spine is _____ anteriorly.	01
	1.6	Name the bones that makes ring shaped pelvis.	01
	1.7	Static loading in the muscles and tissues in the neck can be avoided if the visual component of task is kept within a cone from _____ and _____	01
	1.8	Jobs falling into _____ category require maximum flexibility to be built into the workspace.	01
	1.9	Expand the term NIOSH.	01
	1.10	Human information process acts as _____ rather than _____.	01
	1.11	Write the equation for the condition when probabilities of various alternatives are not equally likely. And indicate the terms used in the equation.	01
	1.12	The bandwidth of human ear is between _____ and _____.	01
	1.13	When the Human Machine interface can be called as transparent?	01
	1.14	What are the two ways in which errors lead to fail the system?	01
	1.15	What is mode?	01
	1.16	What does EMG and EEG stand for?	01
	1.17	Under which type of error detection you categorize the following analogy? A person placing a teabag into a cup, intending to pour boiling water over it and pored coffee over it instead. The error was detected only when the bag floated on the surface.	01
	1.18	Expand and define the term GOM.	01
	1.19	What are normative and descriptive models?	02

PART B

2	a	With the help of a block diagram, explain the Leamon's human machine system.	08
	b	Describe the anatomy of the spine and pelvis related to posture. Support your answer with the help of neat diagram.	08
		OR	
3	a	Highlight some of the salient features with respect to historical development of ergonomics. Explain the characteristics of the three domains of human factors and ergonomics.	08
	b	What are the functions of skeletal and muscular systems? Illustrate an analogy comparing human skeletal and muscular systems with an articulated tent.	08
4	a	Highlight the tasks of ergonomics in the design of mass produced items and with the help of suitable diagrams elaborate the types of anthropometric data.	08
	b	Explain the principles of applied anthropometry in ergonomics.	08
		OR	
5	a	What are the different types of antropometric data? Explain in detail the relevance of measuring these data. Support your answer with the suitable diagrams.	08
	b	Explain in detail the applications of anthropometry in design.	08
6	a	Illustrate the health hazards and antropometric mismatches of the seated workstations design.	08
	b	Propose a conceptual model for an ergonomically designed study table keeping work study, human factors into consideration. Illustrate the model through sketches and ergonomics aspects in your model.	08
		OR	
7	a	With the help of neat diagrams explain the task requirements towards workstation design. Support your answer with suitable examples.	08
	b	Summarize the important considerations in the design of the VDU workspace. Draw a neat diagram exhibiting essential furniture considerations in VDU seating and posture for typical office tasks.	08
8	a	Illustrate the decision making process under certainty, uncertainty and risk. Write the formulae used to compute the information when probabilities of alternatives are likely, unlikely, average information and also percentage of redundancy.	08
	b	What is coding of information? List the characteristics of good coding system. Explain the types of compatibility.	08
		OR	

9	a	Draw the wickens model of information process and explain the stages of human information processing and hypothesized relation between them.	10
	b	List different types of information categories through displays to access the information with direct and indirect methods and explain any two of them.	06
10	a	Give an overview of the research outcomes carried out by various researchers on considering the human error and medical device design.	08
	b	Explain in detail the different types of measures of mental workloads.	08
		OR	
11	a	Illustrate the categorization of errors and their methods of detection. Explain the cognitive aspects of decision making.	08
	b	List and explain the steps under systems design methods, considering ergonomic approach to design, drafted as per ISO/DIS 6385 – Ergonomic principles in the design of work systems.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Examinations April/May-17

Industrial Engineering and Management

APPLIED ERGONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
2. Answer any FIVE full questions from Part B.

PART A

1	1.1	What is meant by ergonomic entropy?	02
	1.2	Identify the possible interactions in a simple system comprising of one person, one machine and an environment.	02
	1.3	Mention the type of anthropometric data used in mechanical analysis of the loads on the human body.	02
	1.4	List the functions of human skeletal system.	02
	1.5	Analyze the following interaction between the components of a work system and identify the design issues-"Display: Environment".	02
	1.6	What are the two senses of human being that are relevant to the ergonomic approach to system design?	02
	1.7	Mention the data capture methods for requirements specifications as a part of ergonomic approach to system design.	02
	1.8	Distinguish between FMJ and FJM approaches to job design.	02
	1.9	Distinguish between sensation and perception in the context of the Cognitive ergonomics.	02
	1.10	What is the impact of "Veiling effect" in the context of the design of Visual displays terminals in a work system?	02

PART B

2	a	Illustrate a human-machine system along with a detailed description of the same.	10
	b	List the contribution of modern ergonomics in system design and management.	06
		OR	
3	a	Categorize the risk factors for musculoskeletal disorders in the work place. Also provide a brief note on each of them.	10
	b	Distinguish between "designing – in" and "designing-out" approaches towards the design of systems.	06

4	a	Critically review the application of various types of anthropometric data in the design of work spaces and equipment design.	10
	b	Justify how mannequins can help potential problems of anthropometric fit to be visualized.	06
		OR	
5	a	Explain different ways to solve the problem of designing to suit a wide range of users.	10
	b	Write a brief note on "Psycho-anthropometry".	06
6	a	Analyze the three components considered for postural analysis and workspace design.	10
	b	Provide a brief note on the participative approach towards avoiding anthropometric mismatches.	06
		OR	
7	a	Illustrate any three common anthropometric mismatches in seated work. Also bring out the negative consequences these mismatches will result in human beings.	10
	b	Evaluate the effect of Anti fatigue mats on the comfort and fatigue levels of standing workers.	06
8	a	Explain the heuristics generally followed for simplification of mental operations required to make decisions.	10
	b	Elaborate on any one approach to the study of human decision making.	06
		OR	
9	a	Explain several verbal methods of data capture for modeling the human operator cognitive control environment.	10
	b	What are the stages involved in reviewing biases in human decision making process.	06
10	a	Describe the steps proposed by ISO/DIS 6385 for an ergonomic approach to systems design.	10
	b	Human-Machine systems are termed as "Socio technical Systems". Elaborate.	06
		OR	
11	a	Provide an overview of the principles derived from socio-technical systems theory for the design of jobs in organizations.	10
	b	Analyze the role of ergonomics interventions in enhancing the employee productivity in organizations.	06

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Fast Track Examinations July-16

Industrial Engineering and Management

APPLIED ERGONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

10. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
11. Answer any FIVE full questions from Part B.

PART-A

1	1.1	The central and fundamental concept in human factors is _____.	01
	1.2	Mechanical systems are also known as _____.	01
	1.3	In descriptive studies two basic classes of variables measured are _____ & _____.	02
	1.4	The variables being manipulated by the experimenter are called _____.	01
	1.5	Psychometric requirements should be _____.	02
	1.6	The activity through which information can be gathered is known as _____.	01
	1.7	What is Participatory Ergonomics?	01
	1.8	According to information theory, information is considered as _____ and _____.	01
	1.9	The amount of information required to decide between tow likely alternatives is measured by _____.	01
	1.10	Occurrence of highly unlikely events conveys _____ information.	01
	1.11	The maximum possible information is always obtained when the alternatives are _____.	01
	1.12	Probability of successful performance is expressed as _____.	01
	1.13	Which type of judgment is made when there is no opportunity to make companions?	01
	1.14	How does dynamic information work?	01
	1.15	What are the advantages and disadvantages of a VDT?	02
	1.16	What deals with the behaviors of human during task performance?	01
	1.17	Which criteria usually reflect essentially engineering aspects of entire system?	01

PART-B

2	a	Explain an ergo system and their basic interactions in a work system and with the help of schematic diagram explain the human machine system.	08
	b	With the help of root diagram explain in detail the three main parameters to be studied, when human is considered as subject in a human machine interface.	08
		OR	

3	a	Propose a conceptual model for an ergonomically designed study table keeping work study, human factors into consideration. Illustrate the model through sketches and ergonomic aspects in your model.	08
	b	Show the importance of ergonomics in various disciplines.	08
4	a	Taking a product (any tool, equipment or machine) as an example, discuss application in the work system.	08
	b	What is modern ergonomic? With the help of suitable live examples list and explain the steps involved in it.	08
		OR	
5	a	What do you mean by anthropometric data and what are the principles of applied anthropometry in ergonomics?	08
	b	With the help of neat diagrams explain the different types of anthropometric data.	08
6	a	Keeping in view, the application of anthropometry in design, carry out a anthropometric evaluation of domestic or commercial kitchen and comment on the following, i) Height of all major work surfaces including sink. ii) The reach requirement of all major work and storage area. The amount of space for movement and foot position.	08
	b	From ergonomist's point of view, write a report which explain to design engineer why anthropometric data should be used in human-machine system. Give examples from everyday life to support your report.	08
		OR	
7	a	With the help of schematic diagram for an ergonomic approach to work station design and the basic requirement for a good work posture, list the steps to avoid anthropometric mismatch.	08
	b	List and explain the key features for the following and support your explanation with diagrams, i) Design for standing workers ii) Design for seated workers iii) Work surface design iv) Visual display terminals.	08
8	a	List and explain the breakdown reasons and improving techniques of workers ability in solving the problems.	08
	b	Explain in detail modeling of human operator control system.	08
		OR	
9	a	Discuss the human operator as the decision maker and with the necessary examples explain the four stages involved in human decision making.	08
	b	Draw flow chart of systems approach proposed by Hunchingson and briefly explain the different stages involved.	08

10	a	What is job design? List and explain the do's and don'ts required to improve the job design.	08
	b	What is the role of trends in work systems design? List the characteristics of a good job.	08
		OR	
11	a	What is ergonomic risk? With the help of symbolic representation, explain the standards and notations used to construct operational sequence diagram.	08
	b	Represent the six major stages in system design process with the help of neat block diagram.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Examinations April/May-16

Industrial Engineering and Management

APPLIED ERGONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
2. Answer any FIVE full questions from Part B.

PART-A

1	1.1	How does ergonomics helps in improving work system?	02
	1.2	Distinguish between controls and displays.	02
	1.3	Explain tent analogy.	02
	1.4	Explain the use of anthropometric data in ergonomics.	02
	1.5	List different types of human adoptions.	02
	1.6	List any four advantages of the standing work position.	02
	1.7	List any two keyboard and screen features for enhancing VDU workstation flexibility.	02
	1.8	The data obtained on the height of population indicated $\bar{X} = 166\,cms$ and $S.D = 5.5\,cms$, what should be the heights of the door for 95th percentile.	02
	1.9	List any four data capturing methods for requirements specifications.	02
	1.10	Distinguish between hygiene factors and motivators.	02

PART-B

2	a	Explain the basic interactions in a work system and their evaluation.	08
	b	Give a brief account of history of ergonomics.	08
		OR	
3	a	Discuss the various functions of skeletal and muscular systems, of human body.	08
	b	List and explain the risk factors for musculoskeletal disorders at work place.	08
4	a	List any eight anthropometric variables and their measurements as specified by $ISO/DIS\ 7250$.	08
	b	Distinguish between structural and functional anthropometric data. Give few examples of use of these data.	08
		OR	

5	a	Illustrate with examples, the use of normal distribution in designing products to a range of people.	08
	b	What is designing for minimum and maximum dimensions? Give three examples for each.	08
6	a	Explain the three sets of tasks requirements that influence workspace design.	08
	b	Discuss the benefits of using various standing aids at workplace.	08
		OR	
7		Discuss the key features of a modern office chair suitable for both visual display unit (VDU) and Non-VDU office work.	16
8	a	What is mental model? How this model helps in decision making?	08
	b	Human-machine systems should be analyzed, conceived and designed at a cognitive level. Do you agree with the statement? Justify your stand.	08
		OR	
9		Explain wickens' general model of Human Information Processing.	16
10		Explain macro-ergonomic interventions which resulted in productivity improvements.	16
		OR	
11		Discuss the psychosocial factors that are to be taken into account in work organization, over and above fitting the job to the worker variety.	16

Digital Metrology / Metrology and Measurements

IV Semester SEE Question Papers

Course codes: 12IM43 / 12ME43

- Fast Track Jul 2017 (12IM43 / 12ME43)
- Apr/May 2017 (12IM43 / 12ME43)
- Fast Track Jul 2016 (12IM43 / 12ME43)
- Apr/May 2016 (12IM43 / 12ME43)

12IM43 / 12ME43

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Fast Track Examinations July-17

Common to IM / ME

MEASUREMENTS AND METROLOGY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

4. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
5. Answer FIVE full questions from Part B.

PART-A

1	1.1	When airy points are used to support a length standard at two points they will be apart be a distance of _____.	01
	1.2	List any two advantages of wavelength standards.	02
	1.3	Recall Taylor's principle of Gauge design.	02
	1.4	The maximum angle that can be set using a sine bar is limited to _____.	01
	1.5	Determine the tolerance on the hole and the shaft for the following limits $Hole = 25^{+0.03}_{-0.00}\ mm$ and $shaft = 25^{-0.006}_{-0.020}\ mm$.	02
	1.6	Write the expression for vacuum pressure using McLeod gauge.	02
	1.7	List the sources of errors in measurement.	02
	1.8	List the expression for Best Size Wire for screw thread measurement.	02
	1.9	Describe Seebech effect.	02
	1.10	List and describe any two errors in screw thread measurement.	02
	1.11	Build the dimension 58.975 mm using $M112$ set of slip gauges.	02

PART-B

2	a	Discuss the characteristics of line standards and end standards.	04
	b	Describe airy points and points of miinimum deflection of a simply supported beam.	04
	c	Distinguish between inter-changeability and selective assembly.	08
		OR	
3	a	List and explain the different types of gauges with neat sketches.	06
	b	Design the general type of "GO" and "$NO\ GO$" gauges for component having 25 H_7f8 fit. Given: • $i = 0.45\sqrt[3]{D} + 0.001D$ • Upper deviation of shaft = $-5.5\ D^{0.41}$ • $25m$ falls in the diameter step of 18 and 30. Take wear allowance as 10 % of the gauge tolerance.	

		Also determine: i) type of fit ii) allowance for the above fit.	10
4	a	Illustrate the working principle of a linear variable differential transformer and list its applications.	08
	b	Explain the working principle of autocollimator and illustrate the procedure for straightness measurement using autocollimator.	08
		OR	
5	a	List the essential requirements for accuracy in the construction of a sine bar. Explain how a sine bar may be employed to determine the inclined angle.	08
	b	Classify the different types of comparators. Illustrate the construction and working principle of pneumatic comparators.	08
6	a	Recommend a suitable laser interferometer for measuring linear positioning straightness in two planes.	08
	b	Explain the working principle of a Michelson interferometer and justify the inclusion of the compensating plate.	08
		OR	
7	a	Discuss the purpose of Machine tool testing. List the various tests conducted in machine tools.	10
	b	Classify and explain different configurations of CMM used for inspection purpose.	06
8	a	Explain the concept of generalized measurement system with suitable examples.	10
	b	Classify the various types of errors in measurements. Discuss the various sources of these errors.	06
		OR	
9	a	Identify the type of transducers used to measure liquid level and pressure pick up based on variations in capacitance.	06
	b	Draw a block diagram to explain telemetering transmitting system and list its application.	10
10	a	Recommend a suitable temperature measuring device to measure very high temperature. Explain its working principle.	08
	b	Discuss the various methods of compensating lead resistance errors in resistance thermometers.	08
		OR	
11	a	Identify and explain the pressure measurement device that works on the principle of Boyle's law. Derive an expression for vacuum pressure using the same.	12
	b	Discuss the treatment regarding preparation and mounting of strain gauges.	04

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Examinations April/May-17

Common to IM / ME

MEASUREMENTS AND METROLOGY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

4. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
5. Answer FIVE full questions from Part B.

PART A

1	1.1	Define Metrology.	02
	1.2	What is meant by $F.P.S$ system?	02
	1.3	What is meant by $C.G.S$ system?	02
	1.4	What are the terms related to interchangeability?	02
	1.5	What is a sine bar? How a sine bar can be checked for its correctness?	02
	1.6	State the uses of following gauges: i) Screw pitch gauge ii) Snap –gauge.	02
	1.7	Explain how will you calculate the least count of a vernier micrometer.	02
	1.8	Define sensitivity and hysteresis with reference to a measuring instrument.	02
	1.9	List the sources of errors in measurement.	02
	1.10	Describe see beck effect.	02

PART B

2	a	Discuss the following standards of measurement: i) Working standards ii) Line standard iii) Wave-length standard iv) End standard.	08
	b	Discuss the procedure for the calibration of End bars.	08
		OR	
3	a	Define clearance fit. Give the detail classification of plain gauges.	08
	b	Calculate the limits of tolerances and allowances for $25mm$ shaft and hole pair designated by $H_8 d_9$ using the following data: i) $25mm$ lies in $(18-30)$mm diameter step. ii) $i = 0.45\sqrt[3]{D} + 0.001D$, microns iii) $IT8 = 25i$ iv) $IT9 = 40i$ v) Fundamental deviation for 'H' hole = 0 vi) Fundamental deviation for 'd' shaft = $-16D^{0.44}$ Sketch and name the above fit. Also determine the minimum and maximum clearance.	08

4	a	Describe with a neat sketch of back pressure type pneumatic comparators.	10
	b	What are the angle gauges? Compute the following angles using angle gauges: i) $37°16'42''$ ii) $57°34'9''$	06
		OR	
5	a	Illustrate the principle of interferometry with sketches.	08
	b	What is the sine bar? Describe with neat sketches.	08
6	a	Explain varieties of measurements with laser interferometers.	08
	b	With the help of neat sketch illustrate the functioning of a laser interferometer.	08
		OR	
7	a	Explain the uses of computer aided inspection.	08
	b	What is the coordinate measuring machine? What are its uses? Explain the working principle.	08
8	a	Elaborate the significance of the following terms used with reference to measurement: i) Sensitivity ii) Repeatability iii) Hysteresis iv) Loading effect.	08
	b	List the classification of errors encountered during measurement. Identify the causes for each type of error.	08
		OR	
9	a	List the primary detector transducers elements and the operation they perform.	08
	b	Define a transducer and "transfer efficiency", how do you classify first stage devices? Give example for each.	08
10	a	Define the following: i) Static pressure ii) Dynamic pressure iii) Absolute pressure iv) Gauge pressure v) Vacuum vi) Atmospheric pressure.	08
	b	Derive the expression for the gauge factor for the strain gauge in terms of poisson's ratio.	08
		OR	
11	a	State and explain the laws of thermocouple.	08
	b	With neat sketch explain the working principle of optical pyrometer.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Fast Track Examinations July-16

Common to IM / ME

MEASUREMENTS AND METROLOGY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

4. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
5. Answer FIVE full questions from Part B.

PART-A

1	1.1	The cross section of international prototype meter is _____.	01
	1.2	Parallax effect is a disadvantage of _____ standards.	01
	1.3	An M 112 set of slip gauges will have _____ number of gage blocks in stems of $0.5mm$.	01
	1.4	The process of grading of parts according to the size and matching to obtain close tolerances is called _____.	01
	1.5	The intentional difference between hole dimension and shaft dimension is known as _____.	01
	1.6	In the calculation of standard tolerance unit, the value 'D' is calculated as _____ of the step size.	01
	1.7	According to Taylor's principle, a $No\ GO$ gauge should correspond to _____ material condition.	01
	1.8	The limit gauges used to check non-cylindrical work are known as _____.	01
	1.9	The type of mechanical comparator which uses a perforated and twisted metal strip is _____.	01
	1.10	The displacement transducer which uses three symmetrically placed AC coils is known as _____.	01
	1.11	The type of optical flat which has only one flat surface is called _____.	01
	1.12	An optical device which is combined 'infinity telescope' and a 'collimator' is known as _____.	01
	1.13	The ratio of 'information' delivered by a pickup to the 'information' received by the pickup is known as _____.	01
	1.14	In a floating carriage micrometer, V-shaped prisms are used for measurement of _____ of screw thread.	01
	1.15	The ability of an instrument to give consistent or repetitive readings is known as _____.	01
	1.16	The type of transducer which does not require an external power source is known as _____.	01
	1.17	A Pirani gauge is based on variation of _____ with pressure.	01
	1.18	The operation of McLeod gauge is based on _____ law of ideal gas.	01
	1.19	The ratio of change of resistance to the change of gauge length is of a conductor is called _____.	01
	1.20	The % composition of constantan used in the thermocouple is _____.	01

PART-B

2	a	Discuss the following standards of measurement with suitable examples: i) Line standard ii) End standard iii) Wavelength standard.	12
	b	Build up a length of 35.4875 mm using $M112$ set. Use two protector slips of 2.5mm each.	04
		OR	
3	a	Distinguish between: i) Interchangeability and selective assembly ii) Hole basis and shaft basis system of limits and fits	08
	b	Determine the tolerances on the hole and the shaft for a precision running fit designated by $50\,H_7g_6$, given $50mm$ lies between $30-50\ mm, I\ (in\ microns) = 0.45(D)1/3 + 0.001D$, fundamental deviation for H hole $= 0$, fundamental deviation fo g shaft = $-2.5D^{0.34}$, $IT7 = 16i$ & $IT6 = 10i$.	08
4	a	Explain with a schematic diagram, the method of measuring the major diameter of a screw thread using bench micrometer.	08
	b	With a neat diagram, explain the working of a Tool maker's microscope.	08
		OR	
5	a	With a neat sketch, explain the construction and working of $LVDT$.	10
	b	Distinguish between sine bar and sine center with the help of schematic diagrams.	06
6	a	What is an interferometer? Name the different types of interferometer and the different light sources used for interferometer.	08
	b	List the various geometrical checks made on machine tools.	04
	c	What is the principle of Laser?	04
		OR	
7	a	Define machine vision. What are the four basic types of machine vision system?	04
	b	Mention any three advantages, disadvantages and applications of CMM.	09
	c	Mention the features of a flexible inspection system.	03
8	a	With a block diagram, explain the generalized layout of a measurement system.	08
	b	Define 'error' in measurements. Classify errors and briefly explain with examples.	08
		OR	

9	a	Distinguish between: i) Active and passive transducers ii) Primary and secondary transducers	08
	b	What is an oscillograph? Mention the types and explain any one type with a neat sketch.	08
10	a	With a neat sketch, explain the working of a McLeod gauge for measurement of low pressures.	10
	b	Explain the principle of strain gauge. List the different types of strain gauges.	06
		OR	
11	a	What is a thermocouple? Explain the laws of thermocouples with neat schematic diagrams.	08
	b	Sketch and explain the working of an optical pyrometer.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Examinations April/May-16

Common to IM / ME

MEASUREMENTS AND METROLOGY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

4. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
5. Answer FIVE full questions from Part B.

PART-A

1	1.1	Define line standard and end standard.	02
	1.2	Define Tolerance and its types with example.	02
	1.3	Distinguish between True Value and Indicated Value.	02
	1.4	What is a Gauge? Explain the use of Plug Gauge and Snap Gauge.	02
	1.5	Differentiate between Error and Discrepancy.	02
	1.6	Define Sensitivity. What is its significance in a measuring instrument?	02
	1.7	Define Precision and Repeatability with reference to a measuring instrument.	02
	1.8	Define Measurement. What are the conditions to be satisfied for measured result to be meaningful?	02
	1.9	Build the following dimensions using $M112$ slip gauge set.	
		i) 30.4875	02
		ii) 27.0075	02

PART-B

2	a	Describe and distinguish the:	
		i) Interchangeable Manufacturing	
		ii) Selective Assembly Manufacturing	08
	b	Using neat sketches distinguish between the Hole basis system and Shaft basis system.	08
		OR	
3	a	What is a Fit? Explain the various types of fits using neat sketch.	08
	b	Calculate the dimensions of Shaft and Hole pair $50H_7d_8$ using the following data:	
		i) $50mm$ falls in the diameter step of $30 - 50mm$.	
		ii) Upper deviation of 'd' shaft is $-16D^{0.44}$	
		iii) $IT_7 = 16i$ and $IT_8 = 25i$	
		Determine which type of fit it is?	08

4	a	What is a Comparator? Explain the working of Solex Pneumatic Comparator to determine the inner dimension of a component.	10
	b	What are angle gauges? Compute the following angles using angle gauges. i) $57^{\circ}34'9"$ ii) $12^{\circ}56'36"$.	06
		OR	
5	a	Explain with help of neat sketch, working of tool makers microscope.	08
	b	Using neat sketches illustrate the use of sine bar for setting a required angle and to measure the slope/angle of a component.	08
6	a	Explain the use of laser for angular measurement.	08
	b	With help of neat sketch explain laser interferometer.	08
		OR	
7	a	What are the uses of computer aided inspection.	08
	b	What is coordinate measuring machine? What are its uses?	08
8	a	With help of a neat sketch explain the various stages of a measuring system.	08
	b	List the classification of errors encountered during measurement. Identify the causes for each type of error.	08
		OR	
9	a	What is a transducer? Identify and explain a suitable transducer which can be used to determine the liquid level of liquid O_2, N_2, etc.	08
	b	With help of neat sketch explain the working of light beam oscillograph.	08
10	a	Recommend a suitable pressure measuring instrument which can be used to measure very low pressures. Explain its working using a neat sketch.	08
	b	Recommend an instrument which can be used to measure very high temperatures. Explain how it works using a neat sketch.	08
		OR	
11	a	Predict a suitable pressure measuring instrument which can be used to measure very high pressures. Explain its working using a neat sketch.	08
	b	State the laws of thermocouple using appropriate line diagram. Interpret how a thermocouple can be used to measure temperature using a neat sketch.	08

Manufacturing Processes

IV Semester SEE Question Papers

Course codes: 12IM45

- Fast Track Jul 2017 (12IM45)
- Apr/May 2017 (12IM45)
- Apr/May 2016 (12IM45)

12IM45

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Fast Track Examinations July-17

Industrial Engineering and Management

MANUFACTURING PROCESS – II

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
2. Answer FIVE full questions from Part B.
3. Draw neat sketches wherever necessary, Assume suitable data if missing.

PART-A

1	1.1	In machining metal, cutting force at the cutting edge is measured by a _____.	01
	1.2	In order to achieve a specific surface finish in single point turning the most important factor to be controlled is _____.	01
	1.3	The vector sum of cutting velocity and chip velocity is _____ shear velocity.	01
	1.4	The tool made of cemented carbide wear out faster at _____ speeds.	01
	1.5	It is required to cut screw threads of 2 mm pitch on a lathe. The lead screw has a pitch of 67 mm. if the spindle speed is 60 rpm, then the speed of lead screw will be _____ rpm.	01
	1.6	The maximum production of small and slender parts is done by _____.	01
	1.7	What is the name of the tool used to remove the drill check from spindle of the drill press?	01
	1.8	Reciprocation of the cutting tool in shaping machine is accomplished by oscillating _____ mechanism.	01
	1.9	The size of shaper is given by _____ length	01
	1.10	In a planer the horizontal feed to the tool is given by moving the tool head along the _____.	01
	1.11	For tool setting in a planer, a clearance of _____ mm is sufficient between the highest point on the work and lowest point of the tool head.	01
	1.12	An arbor is used to hold the milling _____.	01
	1.13	The type of tool used on milling machine and broaching machine is _____ cutting tool.	01
	1.14	The operation of producing grooves around the periphery of a cylindrical or conical work-piece is called _____ milling.	01
	1.15	The hardness of a grinding wheel is determined by the ability of the bond to retain _____.	01

	1.16	Crack in grinding wheel is developed due to _____speed of wheel.	01
	1.17	Based on the control feature system, classify NC machines.	01
	1.18	In an NC machine programmed instructions are stored in _____.	01
	1.19	The instructions on the tape of the NC machine are prepared in _____ coded decimal form.	01
	1.20	The NC machines are controlledby a tape whose width is _____mm.	01

PART-B

2	a	With a neat sketch explain the nomenclature of a single point cutting tool.	06
	b	An experiment was conducted on a MS tube $200\,mm$ diameter and $3\,mm$ thick. An orthogonal cut was taken with cutting speed of $80\,m/min$ and $0.15\,mm$ per revolution, feed with a cutting tool having back rake angle of - 10°. It was determined that cutting force $= 150\,kg$, feed force $= 40\,kg$, net horse for cutting was 3 hp and chip thickness was $0.25\,mm$. calculate the shear strain and strain energy per unit volume.	10
		OR	
3	a	Prove that the specific cutting pressure in an ideal orthogonal cutting is $2S\cot\phi$ if $2\phi + \beta - \alpha = \pi/2$.	10
	b	Identify the different variables that affect the tool life.	06
4	a	Illustrate with a neat sketch a tool layout for producing a hexagonal headed bolt in capstan lathe.	10
	b	Explain the working principle of a collet chuck.	06
		OR	
5	a	Discuss with neat sketch the geometry of twist drill.	08
	b	A hole with $40\,mm$ diameter and $590\,mm$ depth is to be drilled in mild steel component. The cutting speed can be taken as $65\,m/min$ and the feed rate as $0.25\,mm/rev$. calculate the machining time and the metal removal rate.	08
6	a	Discuss with a neat sketch the hydraulic mechanism of a shaper.	08
	b	A cast iron surface $300\,mm$ long and $180\,m$ wide is to be machined on a shaper with a cutting to return ratio $3:2$. Cutting speed, feed and clearance are $24.6\,m/min, 2\,mm$ per double stroke and $30n\,mm$ respectively. The available ram strokes on the shaper are $28, 40, 60$ and 90 strokes/min. if the depth of cut is $3.5\,mm$, determine: i) Time required to machine the surface ii) Material removal rate.	08
		OR	
7	a	Classify planing machine according to the following: i) According to drive ii) According to general construction.	08

	b	A shaping machine is operated once at a maximum length of stroke and then at the shortest length of strokes. The two stroke lengths respectively are $600\,mm$ and $200\,mm$ and the corresponding cutting speeds obtained are 45 meters per min and 12 meters per min respectively. The shaper carries a slotted link if $1000\,mm$ length. Estimate the maximum and minimum number of double stroke (cycles) per minute.	08
8	a	Explain the nomenclature of a milling cutter with a neat sketch.	08
	b	State clearly, the plate no. and the indexing movements to be made using a universal dividing head, to divide the periphery of a work into 87 equal divisions. Following indexing plates are available: $Plate\ No\ 1$: 15, 16, 17, 18, 19, 20 $Plate\ No\ 2$: 21, 23, 27, 29, 31, 33 $Plate\ No\ 3$: 37, 38, 41, 43, 47, 49	08
		OR	
9	a	Explain with sketch the common shapes of a grinding wheel.	08
	b	Write a note on dressing and truing grinding wheel.	04
	c	State the advantages and limitations of center-less grinding.	04
10	a	State the major advantages of NC machines over the conventional machines.	10
	b	State the different steps used to prepare a part programming.	06
		OR	
11	a	Explain the different types of numerical control systems.	06
	b	Name the function of G and M codes for the following: $G00$, $M04$ $G21$, $M06$ $G32$, $M09$ $G90$, $M30$ $G91$, $M99$	10

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Examinations April/May-17

Industrial Engineering and Management

MANUFACTURING PROCESS – II

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

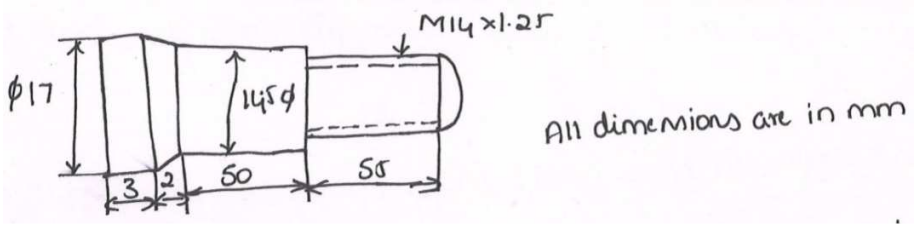
10. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
11. Answer FIVE full questions from Part B.
12. Draw neat sketches wherever necessary, Assume suitable data if missing.

PART A

1	1.1	Internal thread cutting: Tapping:: External thread cutting : _____.	01
	1.2	The flat portion of rectangular cross section provided at the end of tapered shank is known as _____.	01
	1.3	_____ grinding wheels have good strength and more elastic in nature.	01
	1.4	The acronym of CBN is _____.	01
	1.5	State the difference between absolute and incremental programming.	02
	1.6	Give an example for non chip removal processes.	02
	1.7	The two common shapes produced by form milling are _____ and _____.	02
	1.8	Define metal cutting.	02
	1.9	Define orthogonal cutting and oblique cutting.	02
	1.10	List the factors affecting the tool life.	02
	1.11	Define cutting ratio of the shaper.	02
	1.12	What is meant by the term indexing?	02

PART B

2	a	In an orthogonal cutting process, the following data were obtained: Chip length $= 96mm$, uncut chip length $= 240mm$, rake angle used $= 20^{\circ}$, depth of cut $= 0.6mm$, horizontal component of cutting force $= 2400N$ and vertical component of cutting force $= 240N$. Calculate: i) Shear plane angle ii) Chip thickness iii) Friction angle iv) Resultant cutting force.	12
	b	Briefly describe different forms of tool wear.	04
		OR	
3	a	List the types of chips. Explain with the help of a neat sketch.	10
	b	What are the factors that are affecting machinability of materials?	06

4	a	Differentiate between capstan and turret lathe.	08
	b	It is required to drill four holes of $16mm$ diameter in an $30mm$ thick plate using a standard drill. Find the machining time, when the cutting speed is $22mpm$ and feed $= 0.2mm/rev$.	08
		OR	
5	a	Explain with a neat sketch, a tool layout for producing a threaded adopter as shown in figure below. 	12
	b	Explain counter boring operation with relevant sketch.	04
6	a	A shaping machine is used to machine a rectangular piece of $18cm$ long and $35cm$ width, with cutting speed $26m/\text{min}$. Feed $= 0.8mm/\text{cycle}$, cutting stroke adjusted to $20cm$, cutting ratio $= 3:2$. Find the time required for machining the whole surface.	08
	b	Sketch a planing machine indicating major parts with explanation.	08
		OR	
7	a	Explain with a neat sketch, the following work holding devices in a planer. i) Angle plate ii) Step block.	08
	b	Explain with a neat sketch, the crank and slotted lever mechanism of a shaper.	08
8	a	A milling cutter of $100mm$ diameter having 15 teeth is used to machine steel part. The cutting speed is $25m/\text{min}$ and the feed rate is $0.075mm/\text{tooth}$. Determine the cutting speed of the spindle(rpm), the feeds per revolution and per minute.	06
	b	List and explain different types of abrasives.	10
		OR	
9	a	Differentiate between up milling and down milling.	06
	b	Write a short note on the following: i) Abrasives ii) Bonding process.	10
10	a	Explain the different types of NC machine control systems.	06
	b	Briefly explain different words used in NC programming required for NC part program.	10
		OR	
11	a	Explain with the help of an example. i) Linear interpolation ii) Circular interpolation.	06
	b	State the merits and demerits of CNC machines.	10

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Examinations April/May-16

Industrial Engineering and Management

MANUFACTURING PROCESS – II

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
2. Answer FIVE full questions from Part B.
3. Draw neat sketches wherever necessary, Assume suitable data if missing.

PART-A

1	1.1	When cutting ductile material, the type of chip produced is _____.	01
	1.2	$VT^n = C$ is known as _____ tool life equation.	01
	1.3	In order to prevent tool from rubbing the work _____ on tools are provided.	01
	1.4	Drilling is an example of _____ cutting.	01
	1.5	In terms of grinding wheels, what does bond indicate?	01
	1.6	What does the grade of a grinding wheel refer to and how is it specified?	01
	1.7	The cutting tool in a milling machine is mounted on _____.	01
	1.8	Velocity of tool along the tool face is known as _____.	01
	1.9	The tool life due to crater wear varies from _____ to_____.	02
	1.10	Define merchant circle diagram.	02
	1.11	Match the standard machine tool with respect to the tool movement.	02
	1.12	If the horizontal and vertical components of cutting forces are 2400 N and 240 N, what is the resultant cutting force?	02
	1.13	A drill of diameter 20 mm makes a hole on a steel part at a cutting speed of 25 m/min. Find out the spindle speed.	02
	1.14	What is machining program?	02

Group A		*Group B*	
a	*Forward Translation*	*i*	*Drilling*
b	*Reciprocating*	*ii*	*Lathe*
c	*Rotary*	*iii*	*Planning*
d	*Rotary as well as translation*	*iv*	*Shaping*
		v	*Milling*

PART-B

2	a	Differentiate between orthogonal and oblique cutting.	04
	b	Explain the mechanism of chip formation with a neat sketch.	04
	c	In an orthogonal turning operations on a lathe the following data were obtained: Cutting force $= 120\,kg$, feed force $= 30\,kg$, back rake angle $= 15°$, feed rate $= 0.2\,mm/rev$, chip thickness $= 0.3\,mm$, cutting speed $= 100\,m/min$, work piece diameter $= 120\,mm$, depth of cut $= 0.4\,mm$. Calculate chip thickness ratio, shear angle, coefficient of friction, friction angles, shear stress, shear strain, strain energy and chip flow velocity.	08
		OR	
3	a	Prove that the specific cutting pressure in an ideal orthogonal cutting is $2Scot\,\Phi\ if\ 2\Phi + \beta - \alpha = \pi/2$	08
	b	The following data were recorded while turning a work piece on a lathe: Cutting speed $= 25\,m/min$ Depth of cut $= 2.0\,mm$ Feed rate $= 0.3\,mm/rev$ Tool life $= 100\,min$ The following tool life equation is given for this operation: $$VT^{0.12}\ f^{0.7}\ d^{0.3} = C$$ If the cutting speeds, feed and depth of cut are all increased by 25% each, and also collectively, what will be their effect on the tool life.	08
4	a	Explain with a neat sketch, a tool layout for producing a hexagonal headed bolt in capstan lathe.	12
	b	Explain the working principle of a collet chuck.	04
		OR	
5	a	State different types of drilling operations and explain any two with a neat sketch.	08
	b	A hole with $40\,mm$ diameter and $50\,mm$ depth is to be drilled in mild steel component. The cutting speed can be taken as $65\,m/min$ and the feed rate as $0.25\,mm/rev$. Calculate the machining time and the metal removal rate.	08
6	a	Explain with a neat sketch the quick return motion in a shaper.	06
	b	A mild steel plate $400\,mm \times 800\,mm \times 30\,mm$ is to be shaped along its wider face. The ratio of return to cutting time is $2:3$ and the feed per cycle is $2\,mm$. Tool approach and the over travel respectively are $50\,m$ each. Assuming a cutting speed of $24\,m/min$. calculate the machining time required for machining the plate with HSS tools.	10
		OR	

7	a	Explain the different driving mechanisms of a planar.	08
	b	Explain with a neat sketch the friction disc feed mechanism.	08
8	a	Differentiate between up milling and down milling.	08
	b	For a given milling operation, it was decided to switch from HSS cutter to carbide cutter, changing the speed from 35 m/min to 110 m/min. The other parameters of the cutting operation in the two cases are:	

	$Carbide\ Cutter$	$HSS\ Cutter$
$Cutter\ Diameter\ (mm)$	150	125
$Feed\ (mm/tooth)$	0.0425	0.0375
$Number\ of\ teeth$	12	10

Calculate the following for each of the cutting tools:

i) Cutter rpm.
ii) Feed in mm/min.
iii) Time required to take 200 mm long cut including approach and over travel.
iv) Percentage saving in time by changing from HSS to carbide tool. 08

OR

9	a	Explain with sketch the common shapes of a grinding wheel.	08
	b	Write a note on dressing and Turing of grinding wheel.	04
	c	State the advantages and limitations of center-less grinding.	04
10	a	Explain the elements of NC machine tool system.	08
	b	State at least four advantages and disadvantages of using NC machines.	08

OR

11	a	With the help of a neat sketch, explain the following motion control system: i) Point to point system. ii) Contouring path system.	08
	b	Define part programming. Explain the manual part programming and computer aided part programming.	08

Bridge Course Mathematics (lateral entry)

IV Semester SEE Question Papers

Course codes: 18DMA48

- Nov 2022 (18DMA48)

18DMA48

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RV COLLEGE OF ENGINEERING

(An Autonomous Institution Affiliated to VTU)

IV Semester B. E. Examinations November-2022

Common to CV/BT/ET/CH/ME/EE/EC/IM/EI/AS

BRIDGE COURSE MATHEMATICS

Time: 02 Hours ***Maximum Marks: 50***

Instructions to candidates:

Answer FIVE full questions. Question numbers 1, 6, 7 are compulsory. Answer any one full question from 2 and 3 & one full question from 4 and 5.

1	a	Using Maclaurin's series, expand $\sin x$ upto the term containing x^5.	05
	b	If $u = x^3 + y^3 - 3axy$, then find $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial^2 u}{\partial x^2}, \frac{\partial^2 u}{\partial y^2}$ and $\frac{\partial^2 u}{\partial x \partial y}$.	05
2	a	A particle moves along the curve $\vec{r} = (t^3 - 4t)\hat{\imath} + (t^3 + 4t)\hat{\jmath} + (8t^2 - 3t^3)\hat{k}$, where $'t'$ denotes time. Find the magnitudes of velocity and acceleration at time $t = 2$.	05
	b	If $\Phi = xy^3 + yz^3$ then find $\nabla\Phi$ and $\lvert\nabla\Phi\rvert$ at $(2,-1,1)$.	05
		OR	
3	a	If $\vec{F} = x^2yz\,\hat{\imath} + xy^2z\,\hat{\jmath} + xyz^2\,\hat{k}$, then find $\nabla.\vec{F}$ at $(1,1,-1)$.	04
	b	A vector filed is given by $\vec{F} = (x^2 - y^2 + x)\hat{\imath} - (2xy + y)\hat{\jmath}$. Show that the field is irrotational and find its scalar potential ϕ such that $\vec{F} = \nabla\phi$.	06
4	a	Solve $y''' - 3y'' + 3y' - y = 0$.	05
	b	Solve $(D-1)^2 y = 2\sinh x$.	05
		OR	
5	a	Solve $(D^2 + 3D + 2)y = 4\cos 2x$.	05
	b	Solve $(D^2 + 2)y = x^2 + 1$.	05
6	a	Apply Newton-Raphson method to find an approximate root of the equation $xe^x - \cos x = 0$ which lies near $x = 0.5$.	05
	b	Evaluate $\int_0^1 \frac{dx}{1+x^2}$, by using Simpson's $\frac{3}{8}$ rule by taking $h = \frac{1}{6}$.	05
7	a	Evaluate $\int_{y=0}^{1}\int_{x=1}^{2}(x^2 + y^2)\,dx\,dy$.	04
	b	Evaluate $\int_0^1\int_1^2\int_0^3(x + y + z)\,dz\,dy\,dx$.	06